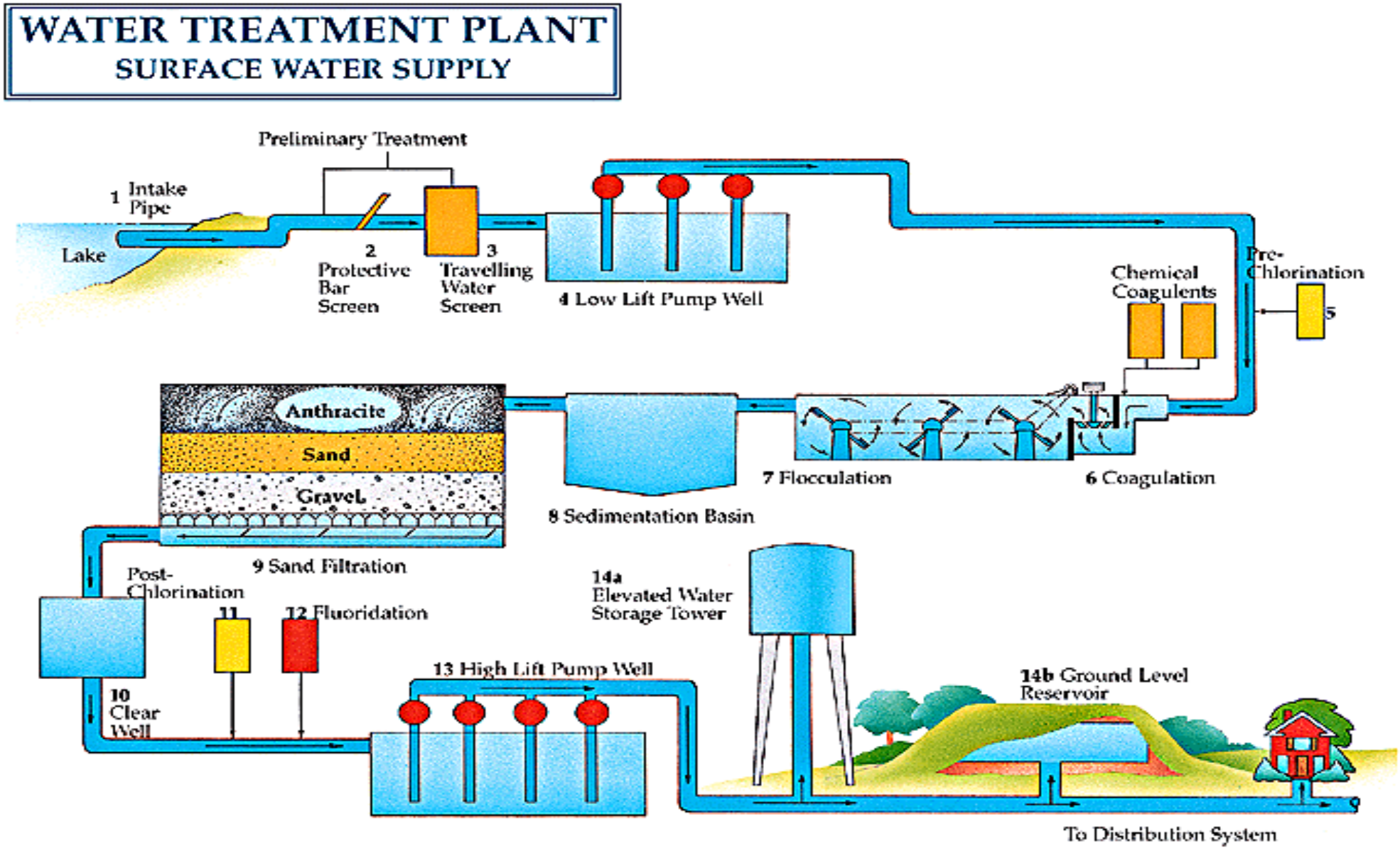
Conventional Water Treatment Process

Water treatment processes:

- Screening
- Aeration
- Sedimentation with or without coagulation
- Filtration
- Disinfection
- Miscellaneous – water softening, desalination etc.

TYPICAL URBAN WATER TREATMENT SCHEME WITH SURFACE WATER SOURCE



Conventional Water Treatment

- **Screening** (for the removal of large floating and suspended materials).
  - Mostly used at intake site
- **Aeration** (for the oxidation of iron and manganese, removal of dissolved gasses and VOCs).
  - Optional unit, and may not be provided if target impurities are not present in water
- **Sedimentation** (for the removal of suspended sediments of specific gravity >1).
  - Plain (or primary) sedimentation may not be provided, as in most cases, settling units are provided after coagulation and flocculation for chemical assisted settling.
  - In many conventional water treatment systems settling unit is combined with flocculation unit, named as clariflocculator

Screening

Definition:

- *Screening* is a unit operation that separates large floating materials in and/or on water (found in different sizes) from water and from entering water treatment/ Wastewater treatment facilities and mains.
- The unit involved is called a *screen*.

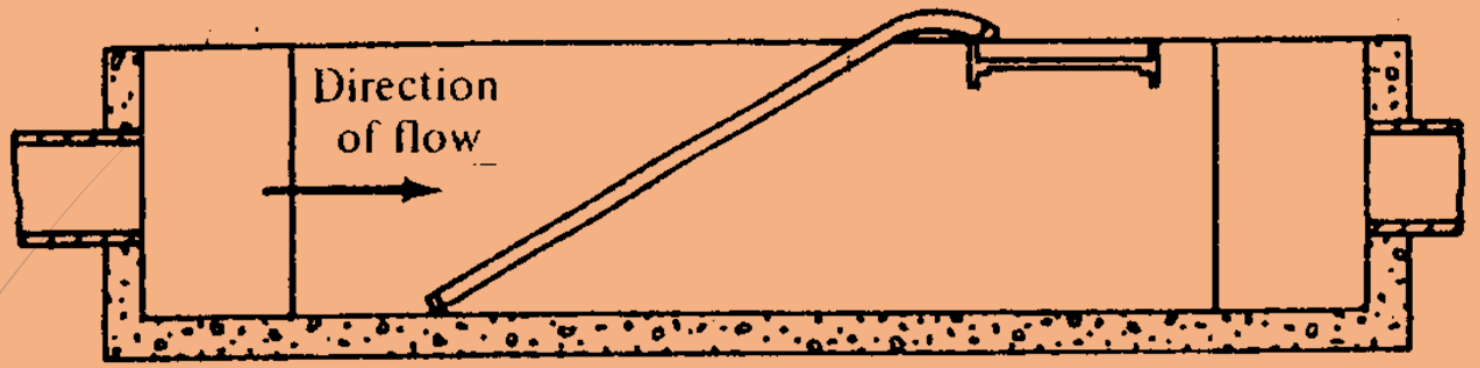
Objective of providing screens

- Screens are provided to protect:

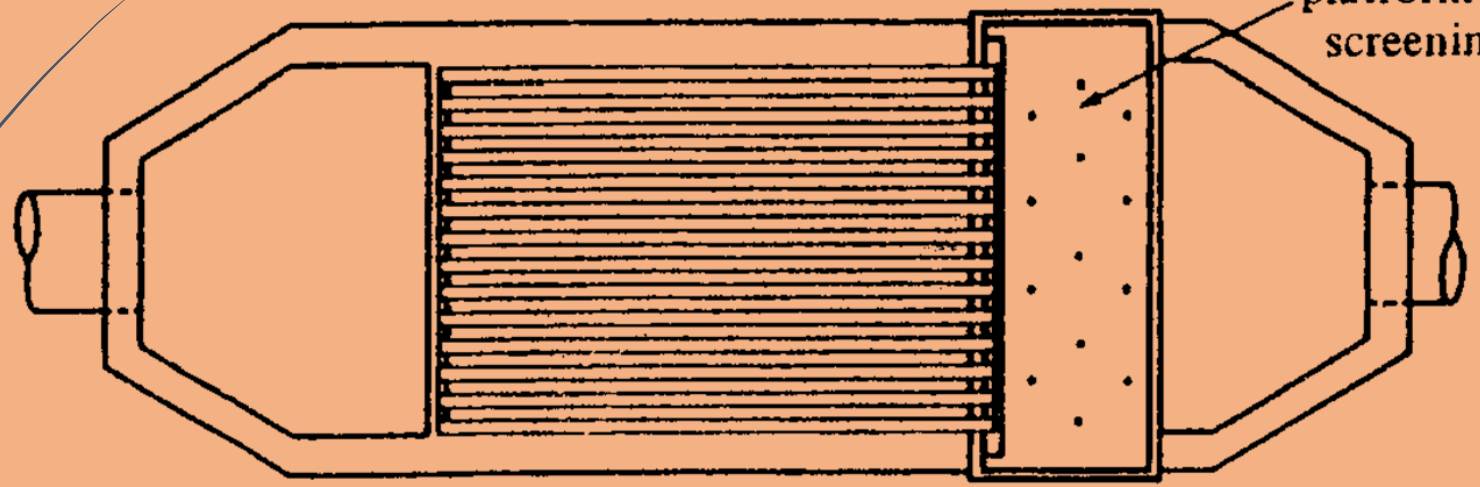
i. Pumps

ii. Valves

iii. Pipe line and other appurtenances from damage or clogging by rags and other large objects



Section



Plan

(*a*)





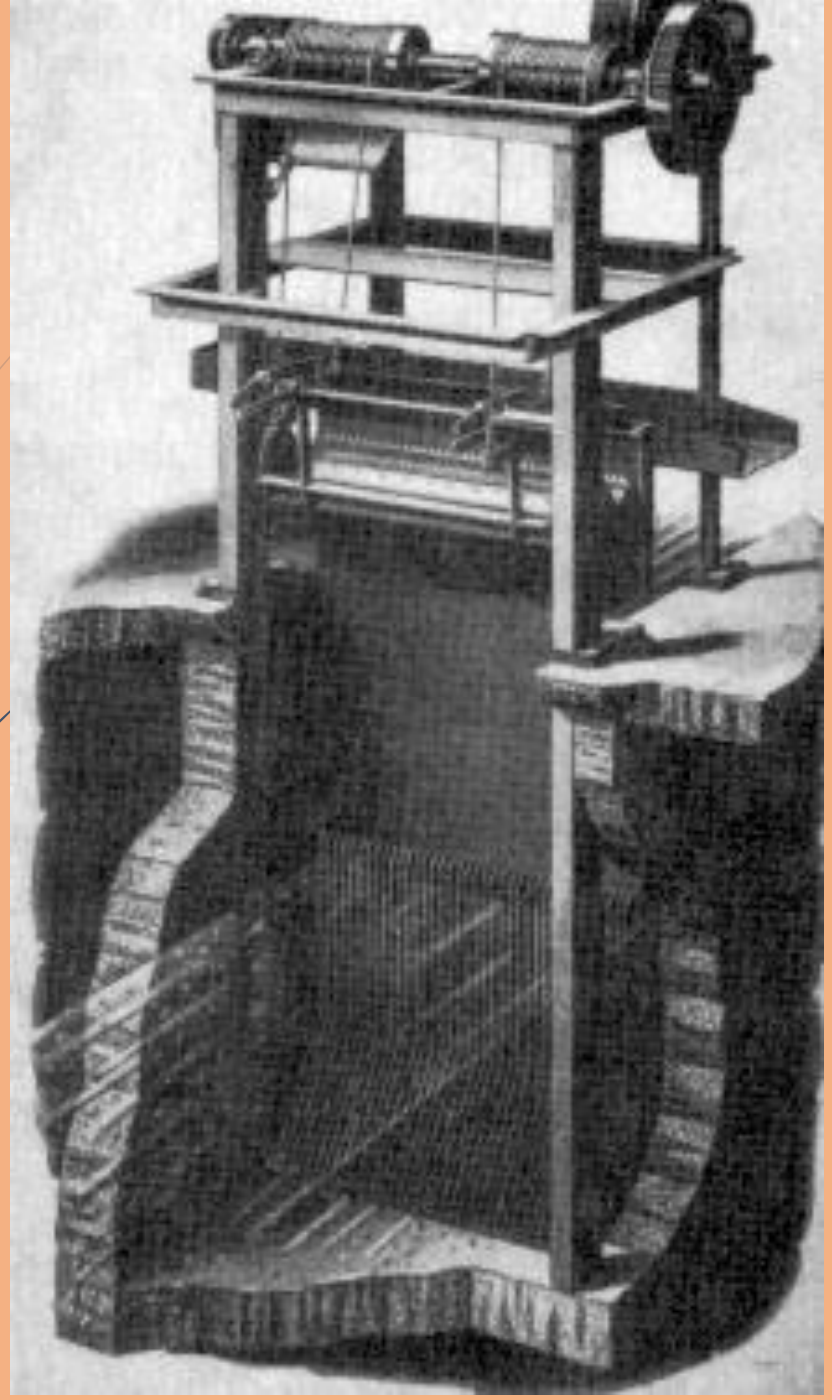




Bar Screen

Mechanically Cleaned

- More frequently used because labor and overflowing are greatly reduced.
- A by-pass channel with a hand cleaned bar screen must also be provided. A second mechanically cleaned bar screen can also be provided.
- The purpose of the by-pass channel is to provide treatment in case of a mechanical failure.
- Screens are either front or back cleaned.



Bar Screen

Classification of Screens

Point of Comparison	Coarse screen	Medium screen	**Fine screen**
Angle of Inclination	@ 45^0	30^0 to 60^0	30^0 to 60^0
Openings	50 mm or more	6-40 mm	1.5 to 3 mm
Cleaning	Mechanically or manually cleaned		

Disposal of screenings	By Burial or dumping or incineration		
Screenings collection	6 lit of solids/ mLd	30 to 90 lit solids/mLd	20% of SS from sewage
Frequency of clogging	less	moderate	Often or more

Velocity

- The velocity of flow ahead of and through the screen varies and affects its operation.
- The lower the velocity through the screen, the greater is the amount of screenings that would be removed from sewage.
- However, the lower the velocity, the greater would be the amount of solids deposited in the channel.

- Hence, the design velocity should be such as to permit 100% removal of material of certain size without undue depositions.
- Velocities of ***0.6 to 1.2 m/s through the open area for the peak flows*** have been used satisfactorily.
- Further, the velocity at low flows in the approach channel should ***not be less than 0.3 m/s*** to avoid deposition of solids.

Head loss

- Head loss varies with the quantity and nature of screenings allowed to accumulate between cleanings.
- The head loss created by a clean screen may be calculated by considering the flow and the effective areas of screen openings, the latter being the sum of the vertical projections of the openings.

- The head loss through clean flat bar screens is calculated from the following formula:
- $$\mathbf{h = 0.0729 (V^2 - v^2)}$$
- where, h = head loss in m
- V = velocity through the screen in m/s
- v = velocity before the screen in m/s

- Another formula often used to determine the head loss through a bar rack is Kirschmer's equation:
- $\mathbf{h = \beta(W/b)^{4/3} h_v \sin \theta}$
- where h = head loss,m

- β = bar shape factor (2.42 for sharp edge rectangular bar, 1.83 for rectangular bar with semicircle upstream, 1.79 for circular bar and 1.67 for rectangular bar with both u/s and d/s face as semicircular).
- W = maximum width of bar u/s of flow, m
- b = minimum clear spacing between bars, m
- h_v = velocity head of flow approaching rack, m = $V^2/2g$
- θ = angle of inclination of rack with horizontal

Number of bars in screen *chamber*

[clear spacing x (n+1)] +[size of bar x n] =B

Where,

N= number of bars

B= width of screen chamber or channel

Clear spacing and size of bars is expressed in m

Other formulae used

- Gross area
- **$A_g = A_{net}$ x (c/c spacing/ clear spacing)**
- Velocity of flow above screen
- **v = Velocity through the screen**
- **(V) x (clear spacing/c/c spacing)**

Disposal of Screenings

- **Screenings is the waste materials collected from screens.** Screenings should be properly disposed.
- Various methods of screening disposal were used such as:
  - **burning,**
  - **burying,**
  - **digestion,**
  - **dumping into large bodies of water,**
  - **and shredding and returning it to wastewater collection or treatment system.**

- Inland burying is efficient in small treatment plants, while burning is best for medium and large treatment plants.
- Other methods cause problems and may need subsequent treatment.
- Digestion is used for large systems and in combination with the treatment of the organic portion of municipal solid waste.

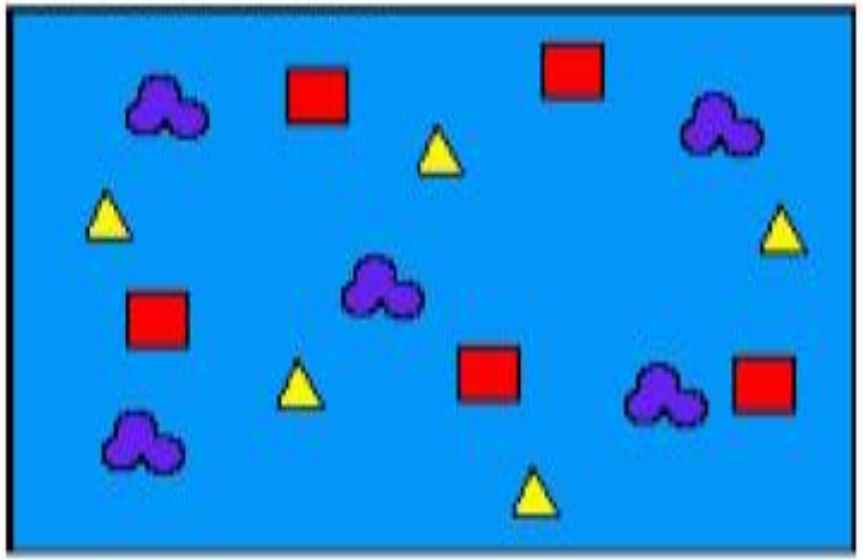
Aeration

1. Aeration is used to remove tastes and odours
2. to remove minerals such as iron and manganese from water
3. to remove carbon dioxide from the water.
4. Increases dissolved oxygen content
5. Removes Hydrogen Sulphide and odour due to it.
6. Decreases CO_2 content and raises pH.
7. Removes Iron and Manganese.
8. Due to agitation bacteria may get killed.
9. Can be used for mixing chemicals

HOW DOES AERATIONWORK?

- **Aeration** is the intimate exposure of water and air.
- It is a way of thoroughly mixing the air and water so that various reactions can occur between the components of the air and the components of the water.

1. Raw water:



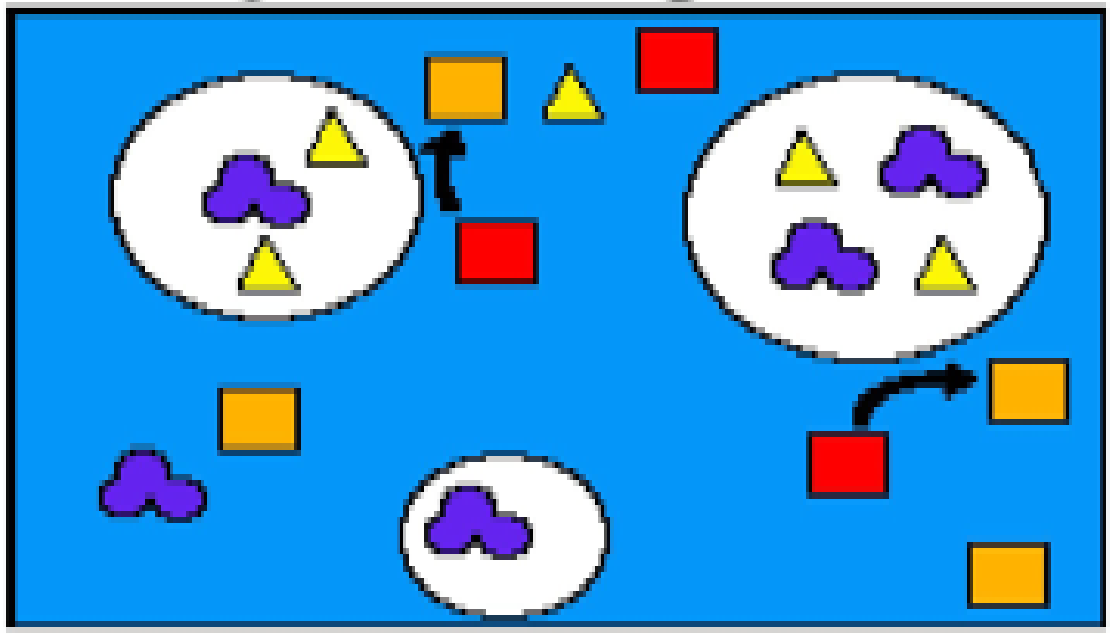
Carbon dioxide

Hydrogen sulfide

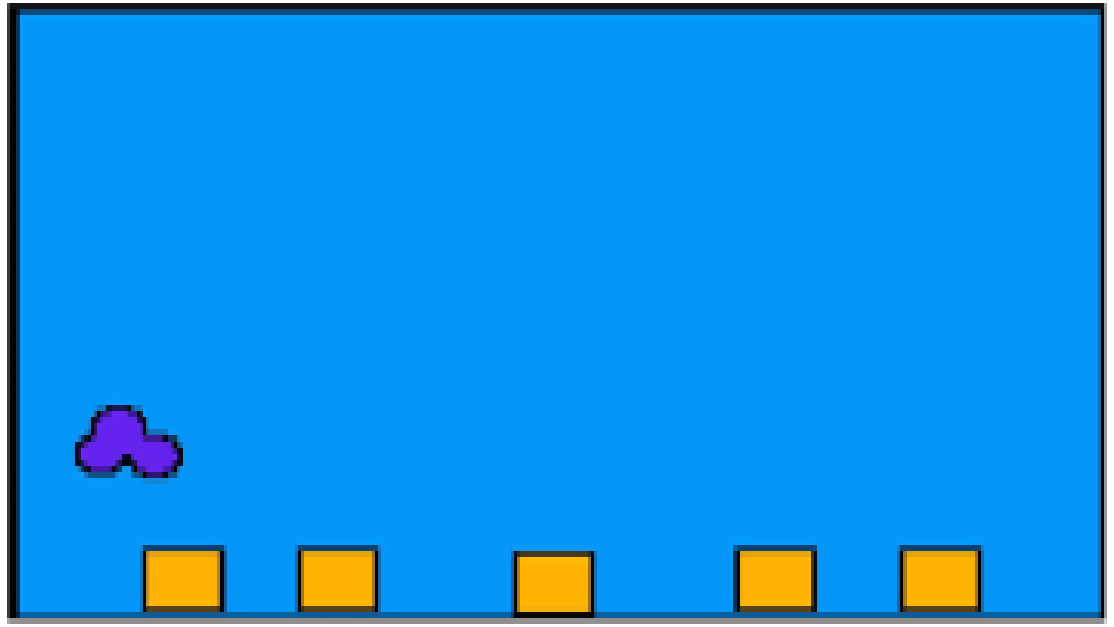
Iron

Iron precipitate

2. Air passes through the water:



3. Aerated water:



- Aeration removes or modifies the constituents of water using two methods - scrubbing action and oxidation.

Scrubbing action is caused by turbulence which results when the water and air mix together.

The scrubbing action physically removes gases from solution in the water, allowing them to escape into the surrounding air.

- Oxidation is the other process through which aeration purifies water.

When air is mixed with water, some impurities in the water, such as iron and manganese, become oxidized.

- Once oxidized, these chemicals fall out of solution and become suspended in the water.
- The suspended material can then be removed later in the treatment process through filtration.

EFFICIENCY

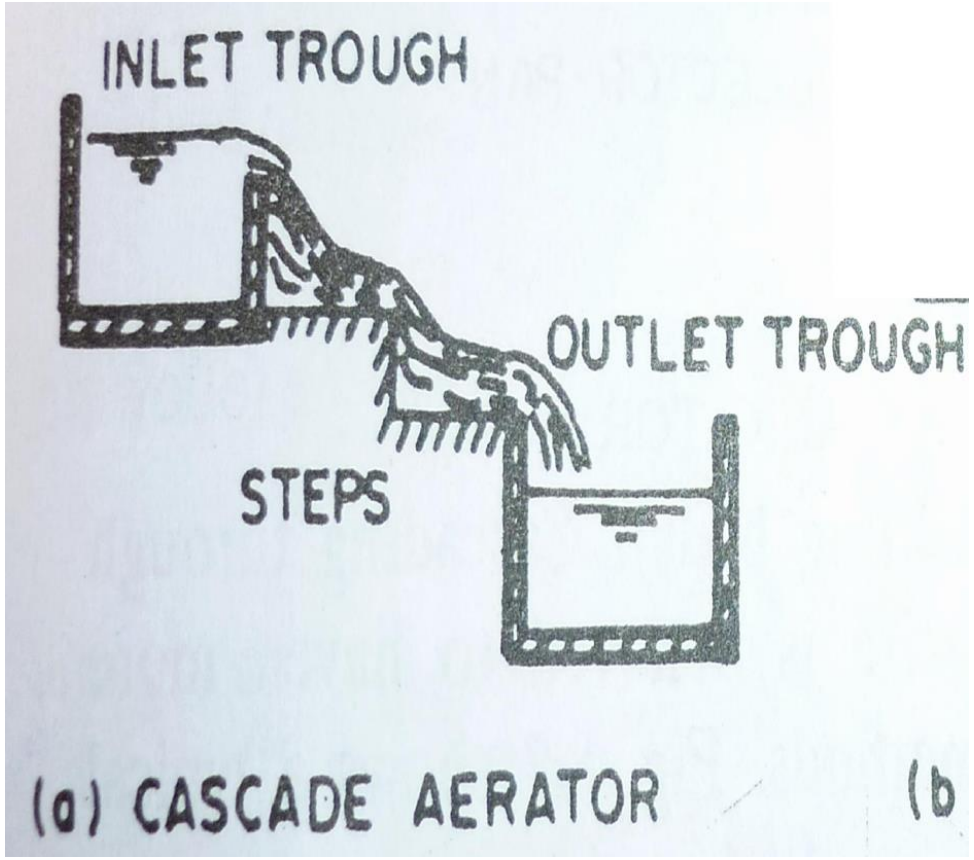
- The efficiency of the aeration process depends almost entirely on the amount of surface contact between the air and water.
- This contact is controlled primarily by the size of the water droplet or air bubble.
- The goal of an aerator is to increase the surface area of water coming in contact with air so that more air can react with the water.

TYPES OF AERATOR

1. Free fall or gravity aerator
   i. Cascade aerator
   ii. Inclined apron aerator
   iii. Slat tray aerator
   iv. Gravel bed aerator
2. Spray aerators
3. Air diffusers
4. Mechanical aerators

1. CASCADE AERATOR

- Simplest of free fall aerator
- Can be constructed of concrete or metal
- Height is 1 to 3 m.
- 50 to 60% CO_2 reduction





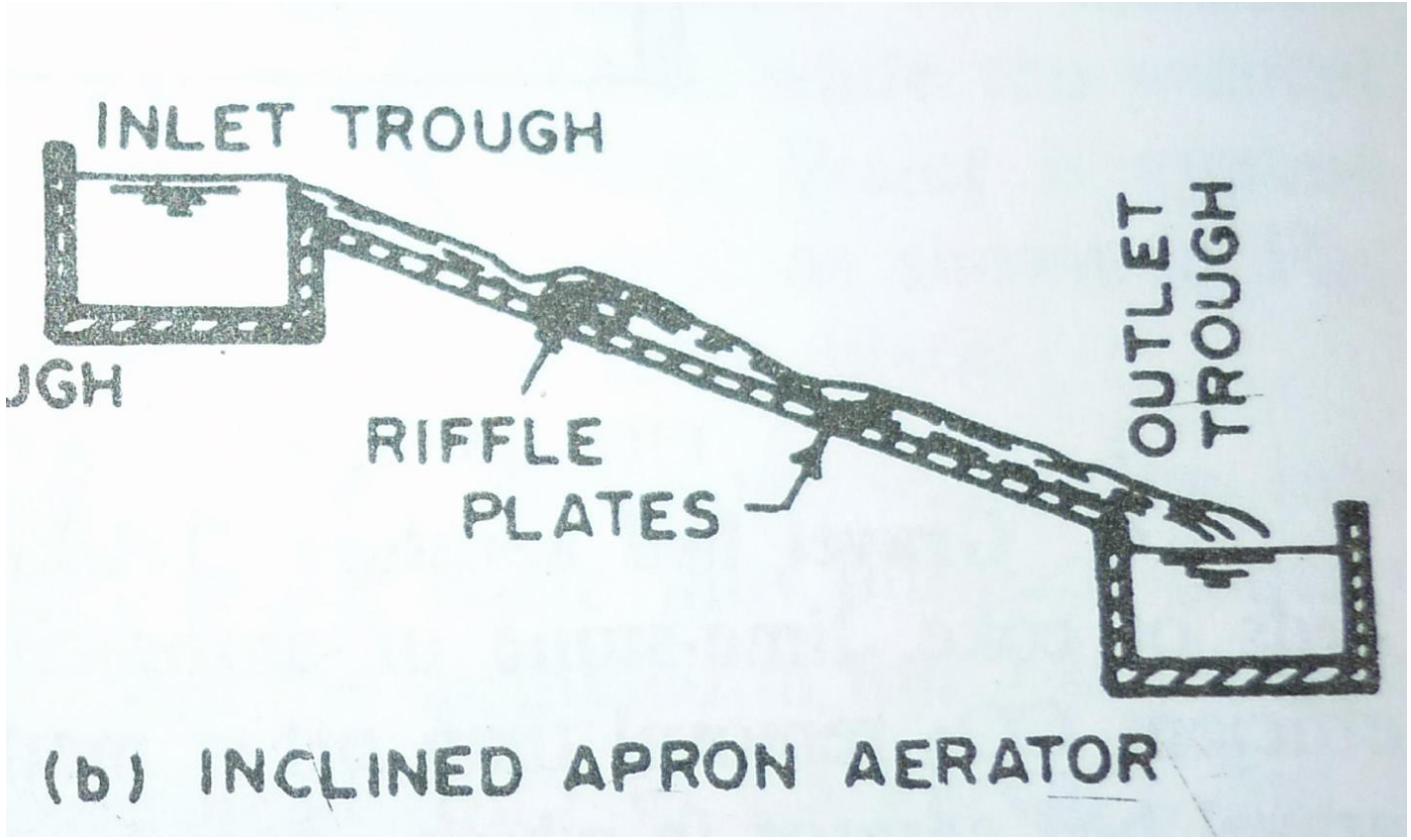





17.02.2007

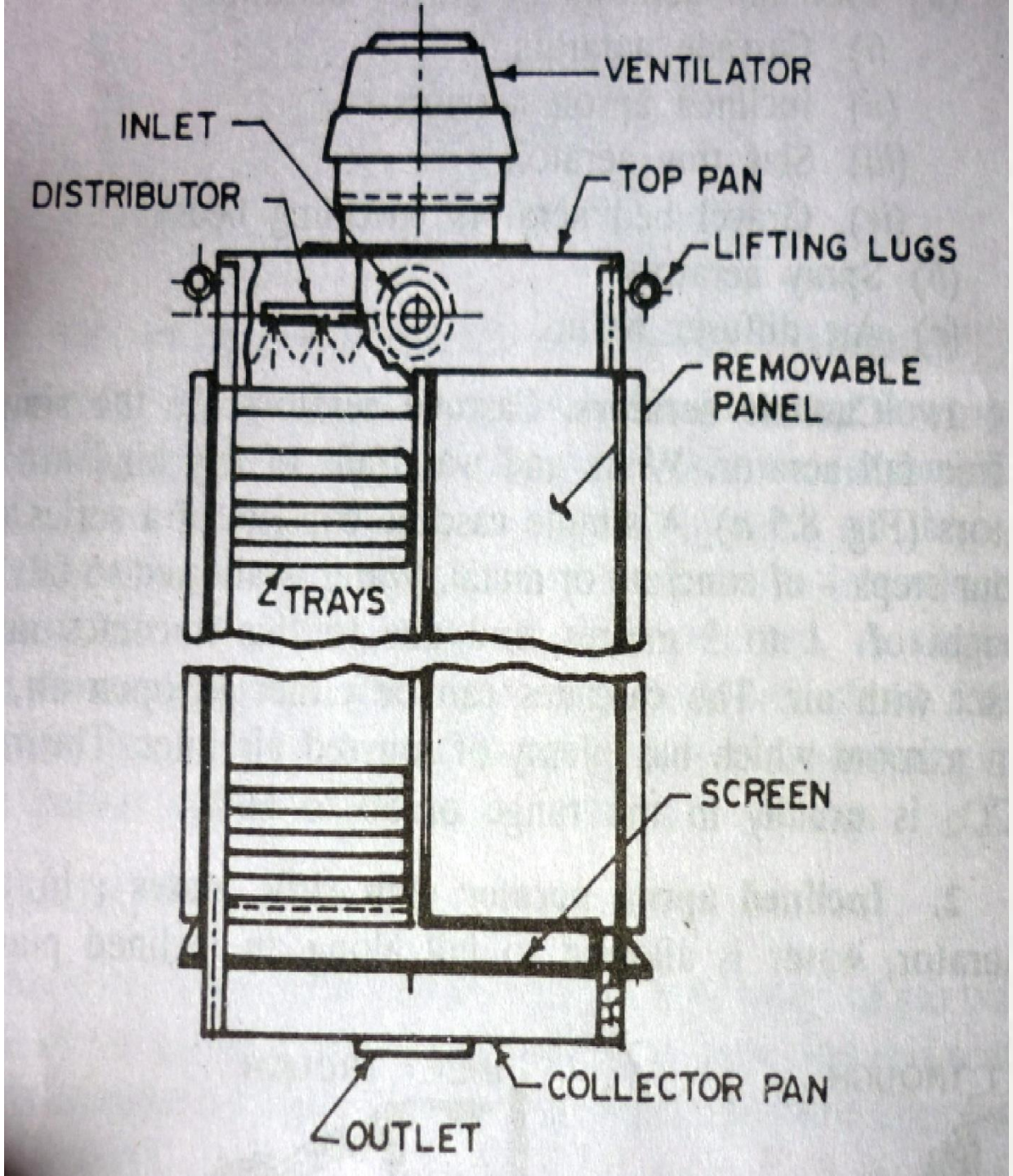
2. INCLINED APRON AERATOR WITH RIFFLE PLATES

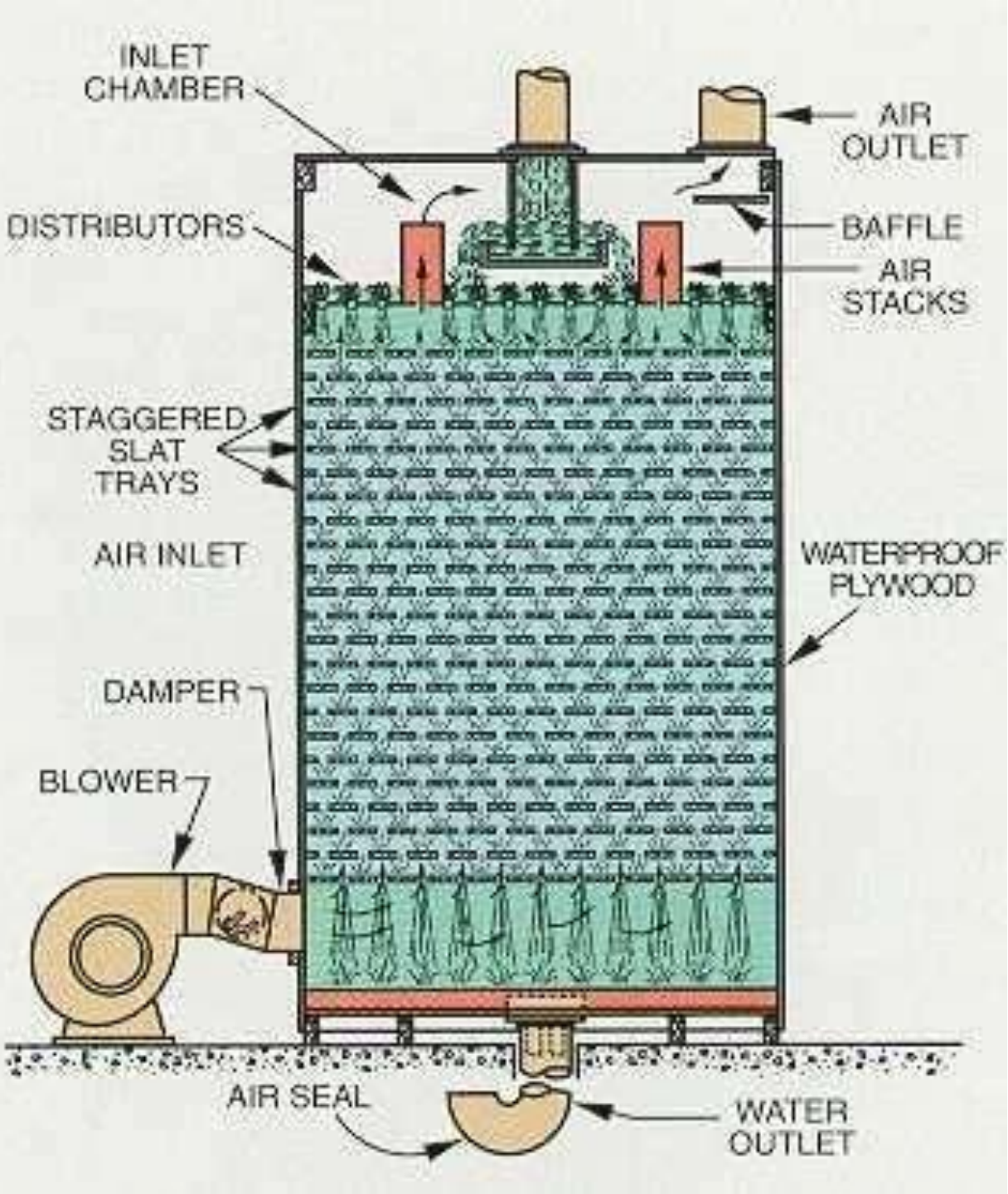
- Water falls along inclined plane/apron.
- Breaking up of water dueto aprons cause agitation
- Agitation increases contact of air and water



3. SLAT TRAY AERATOR

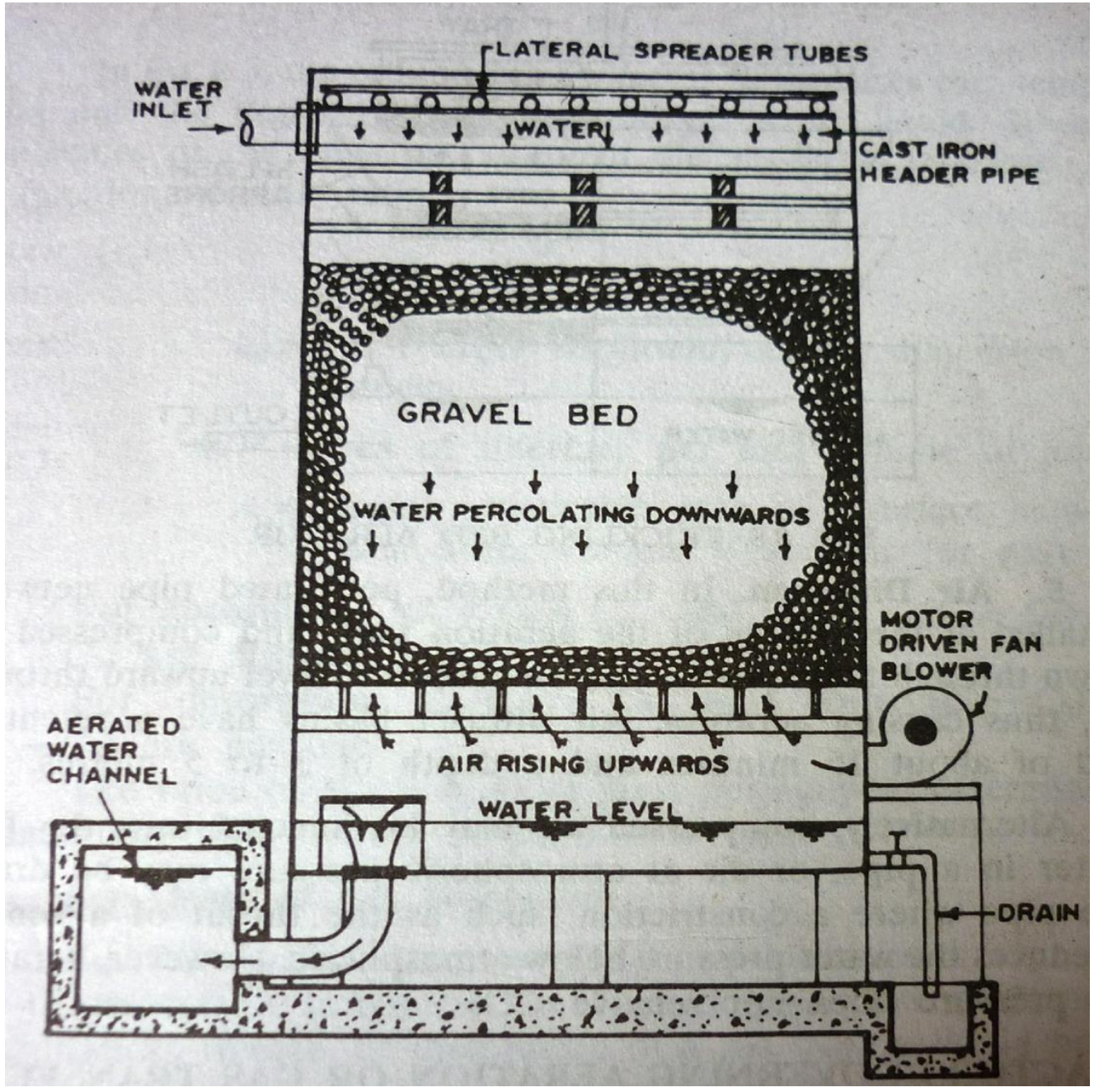
- Consists of closely stacked super imposed wood slat-trays
- Water is sprayed evenly on top tray.
- Water trickles from one tray to the other.
- Air is supplied at the bottom with the help of blower (Air flow is in upward direction).
- Ventilator discharges air and gases


VENTILATOR
INLET
DISTRIBUTOR
TOP PAN
LIFTING LUGS
REMOVABLE PANEL
TRAYS
SCREEN
COLLECTOR PAN
OUTLET


INLET CHAMBER
AIR OUTLET
DISTRIBUTORS
BAFFLE
AIR STACKS
STAGGERED SLAT TRAYS
AIR INLET
WATERPROOF PLYWOOD
DAMPER
BLOWER
AIR SEAL
WATER OUTLET

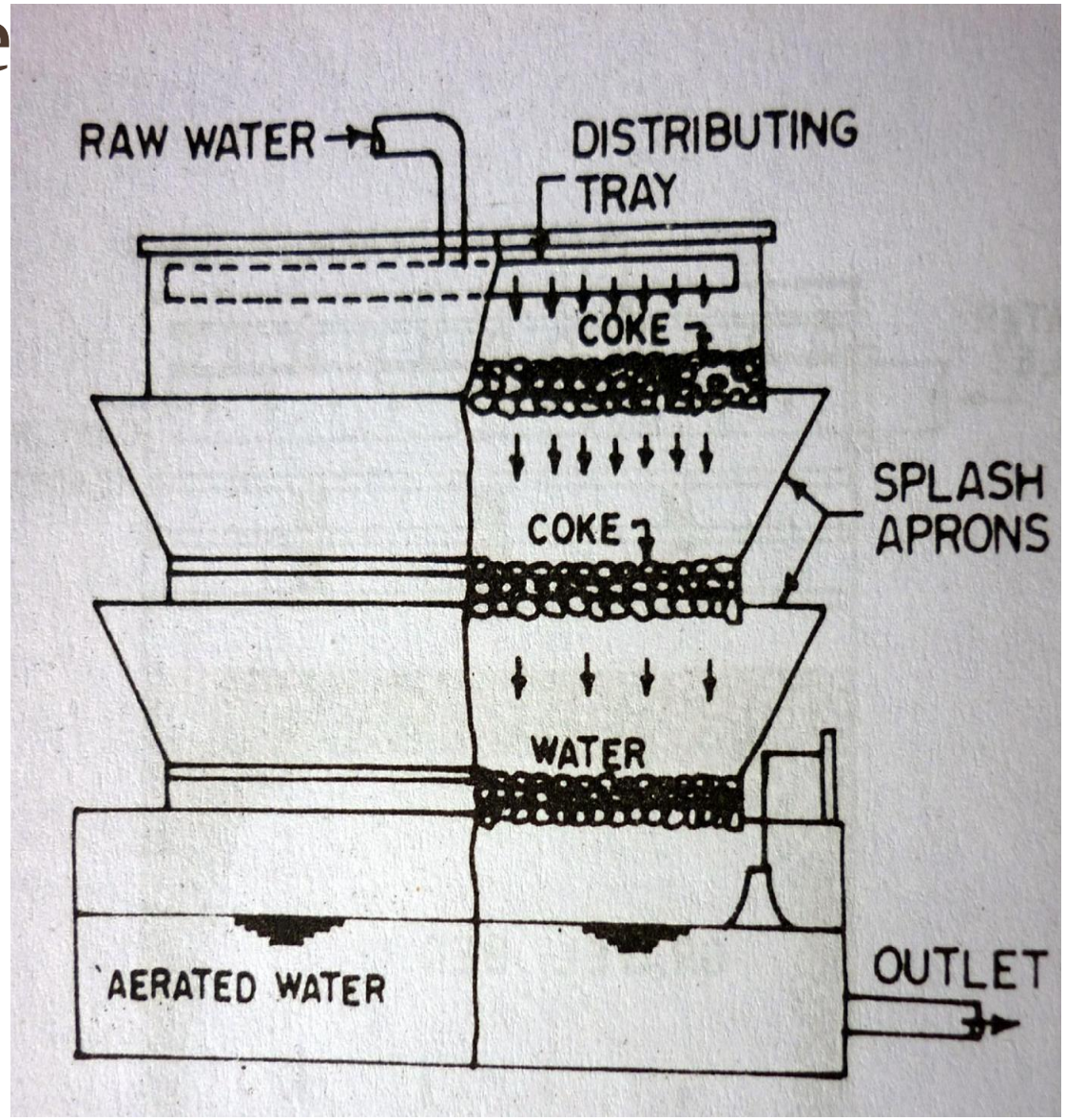
4. GRAVEL BED AERATOR

- Water is sprayed at the top and it trickles down the bed of coke or anthracite.
- Thickness of bed is 1 to 1.5 m

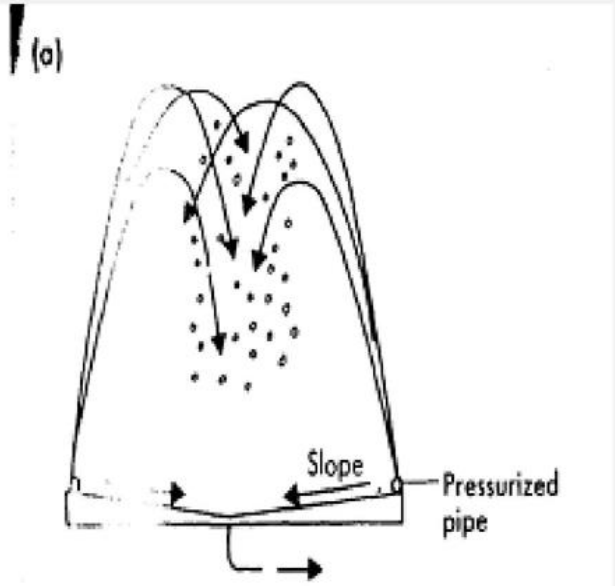


TRICKLING BEDS

- 3-4 trays are kept filled with coke or stone
- Thickness of bed is – 0.5 to 0.6 m
- Distance between trays is 0.5m
- Water is sprayed from distribution pipe at the top.



Spray Aerator



→applied in the course of water treatment for absorption of oxygen and/or desorption of gases

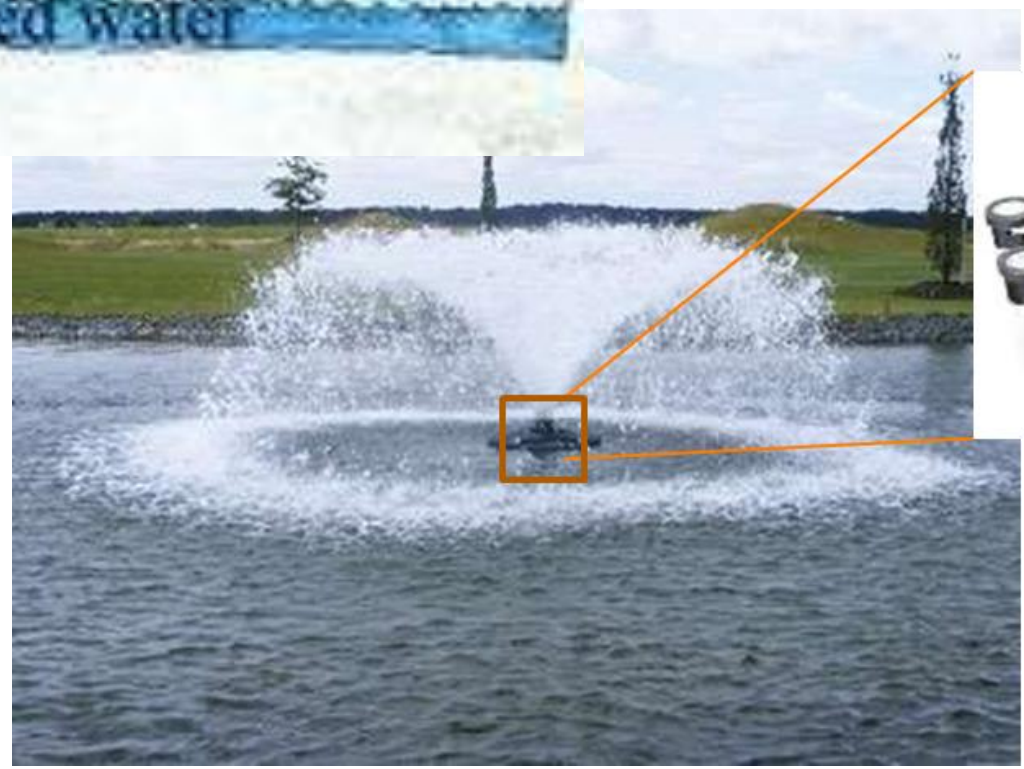
→water is distributed into air in the form of small droplets by means of

- orifices
- nozzles

mounted on a stationary pipe system

Spray Nozzle Aerator







Diffused – Air Aeration Systems

→air is introduced into liquid being aerated in the form of bubbles which typically rise through the liquid

→common device for ;

transferring oxygen in aerobic biological treatment systems

air stripping of volatile organics



Ref: http://www.brightwaterfli.com/diffused_aeration_systems.htm

FINE BUBBLE

→the size of bubbles varies from coarse to fine

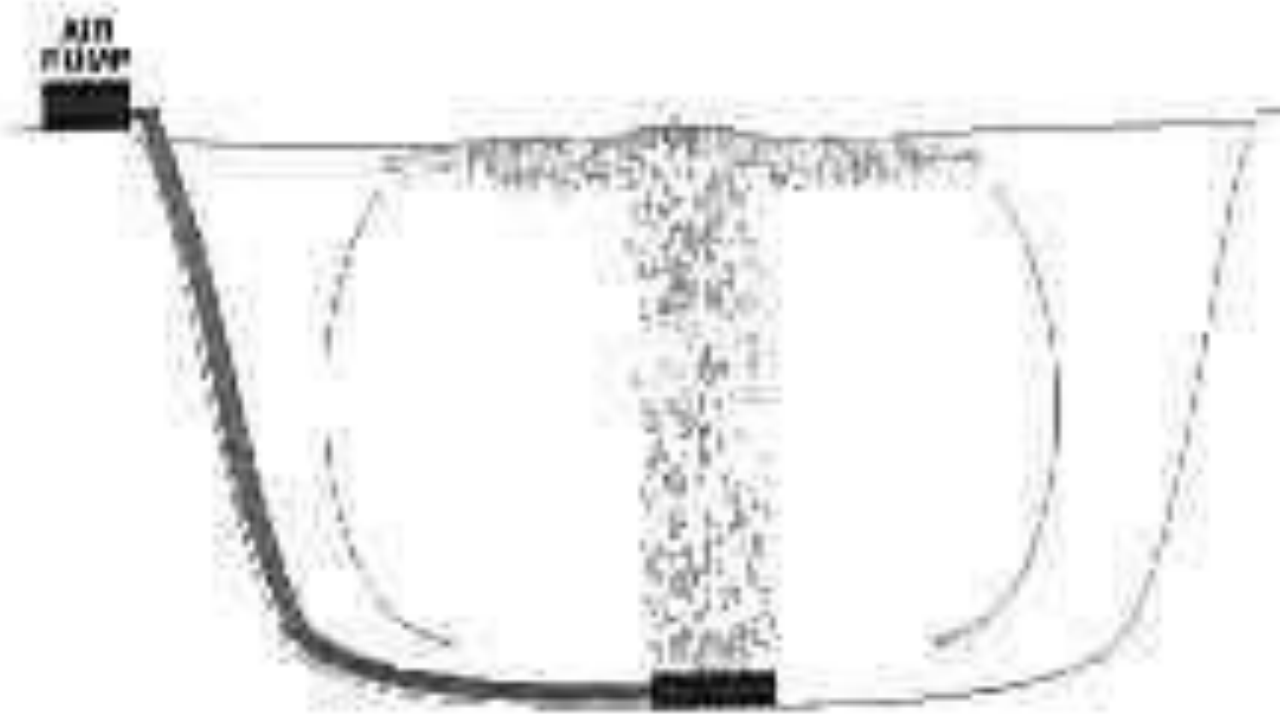
- fine-bubble diffusers
- coarse bubble diffusers



Ref:http://www.hellotrade.com/diffused-gas-technologies-incorporated/ss-series-plenum-coarse-bubble-diffusers.html

COARSE BUBBLE




AIR
PUMP

MECHANICAL AERATORS

- By producing a large air-water interface the transfer of oxygen from atmosphere is enhanced

- Can be VERTICAL SHAFT or HORIZONTAL SHAFT





Ref: http://www.waterandwastewater.com/www_services/newsletter/april_18_2011.htm

SEDIMENTATION

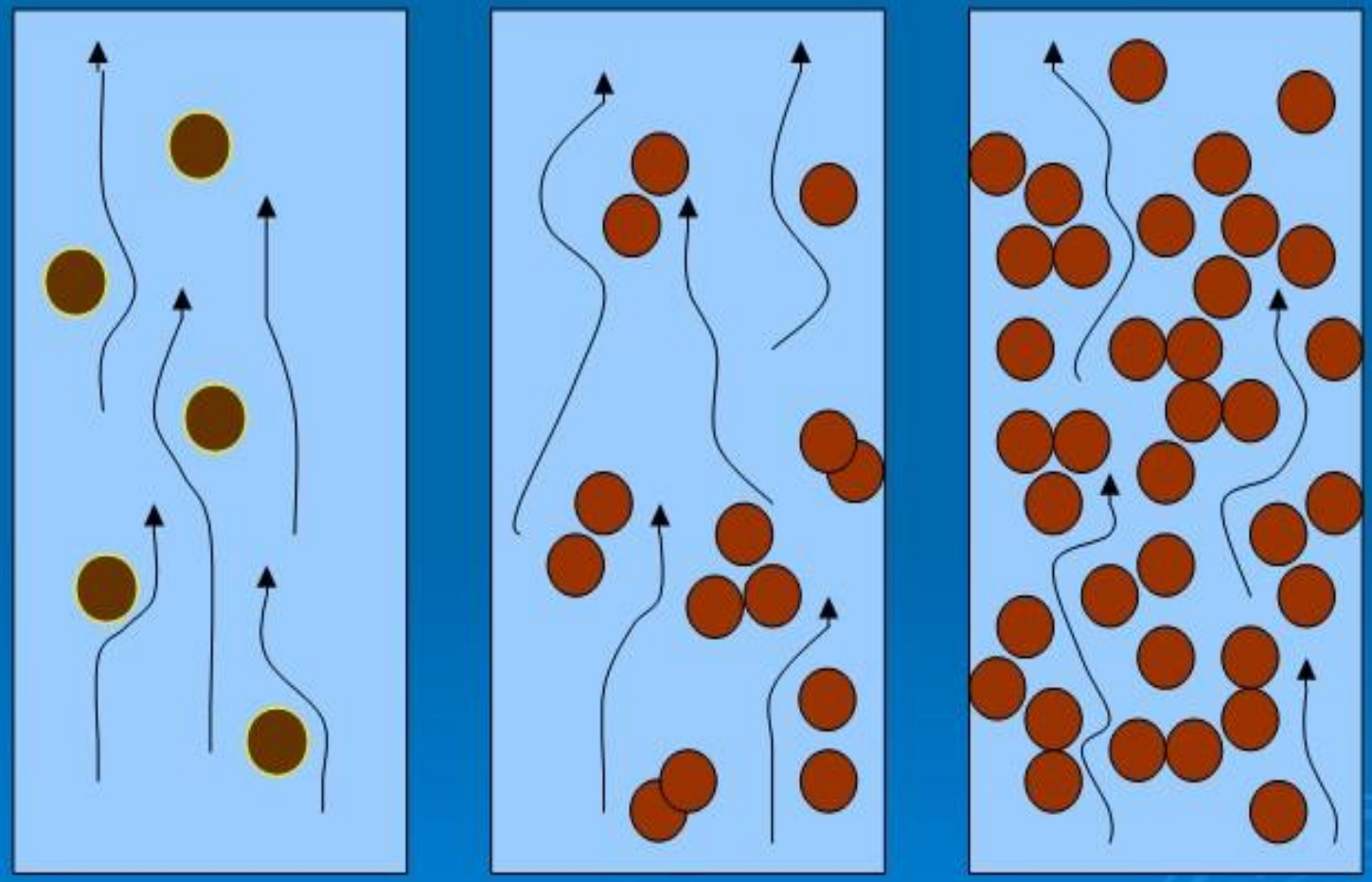
- Plain sedimentation is first unit operation to apply to catch up suspended particles from water
- All particles whose specific gravity greater than specific gravity of water has tendency to settle but turbulence and flow velocity resist particle settlement.
- Plain sedimentation is method of removing suspended particle from water by controlling , turbulence and flow velocity , thereby particle settlement by virtue of their own gravity force.

Types of Settling

Four types of sedimentation:

- Discrete settling
- Flocculant settling
- Hindered settling
- Compression

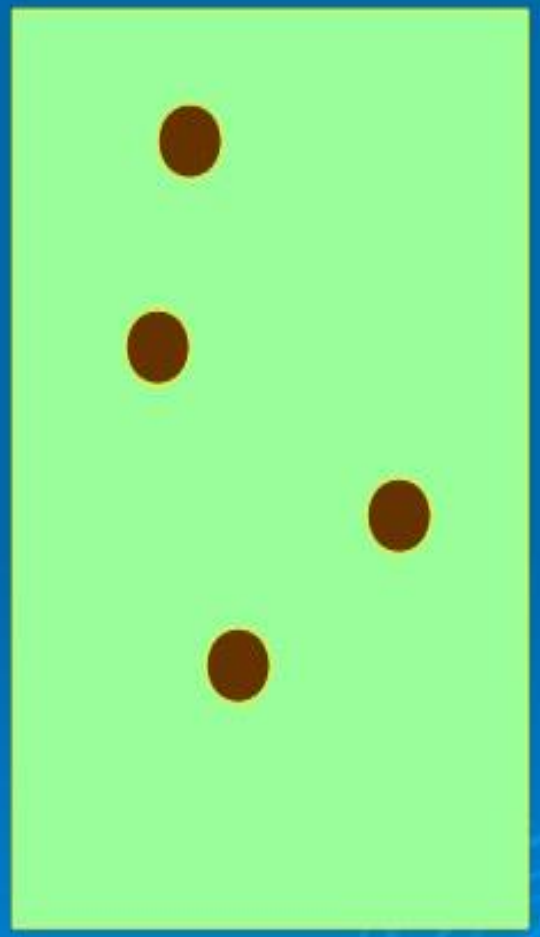
Examples of Settling Types



Discrete Flocculant Hindered

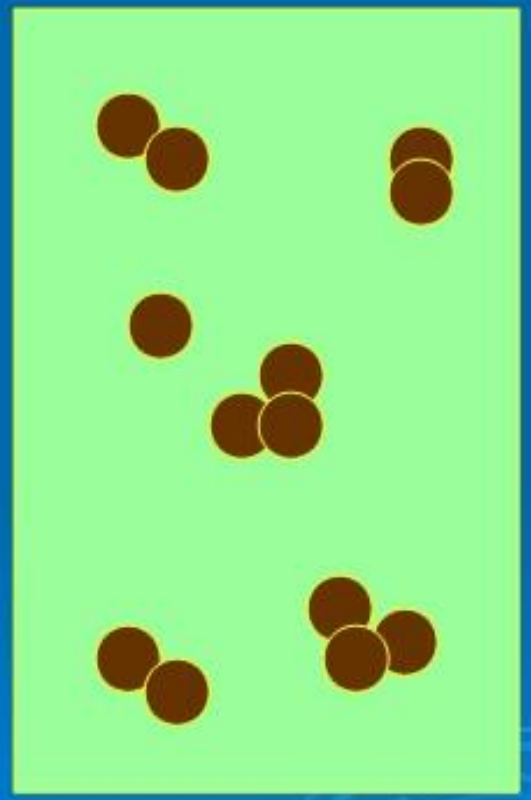
Types of Sedimentation

- In **discrete settling** individual particles settle independently
- It occurs when there is a relatively low solids concentration



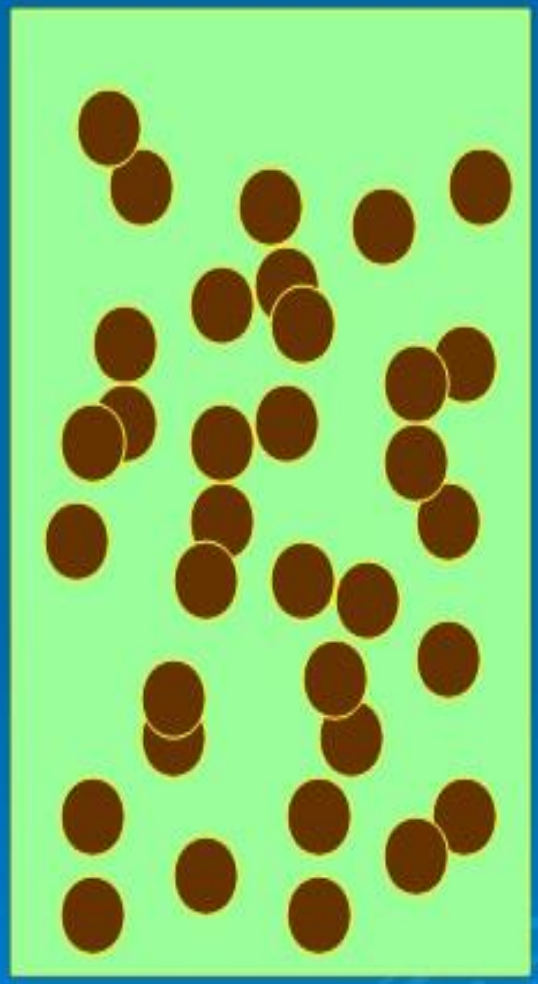
Types of Sedimentation

- In **flocculant settling**, individual particles stick together into clumps called flocs
- This occurs when there is a greater solids concentration and chemical or biological reactions alter particle surfaces to enhance attachment



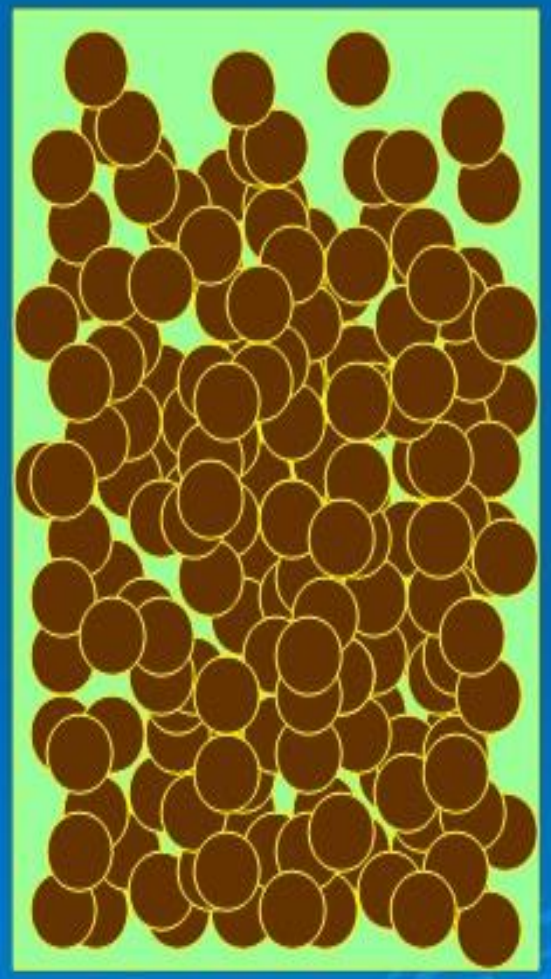
Types of Sedimentation

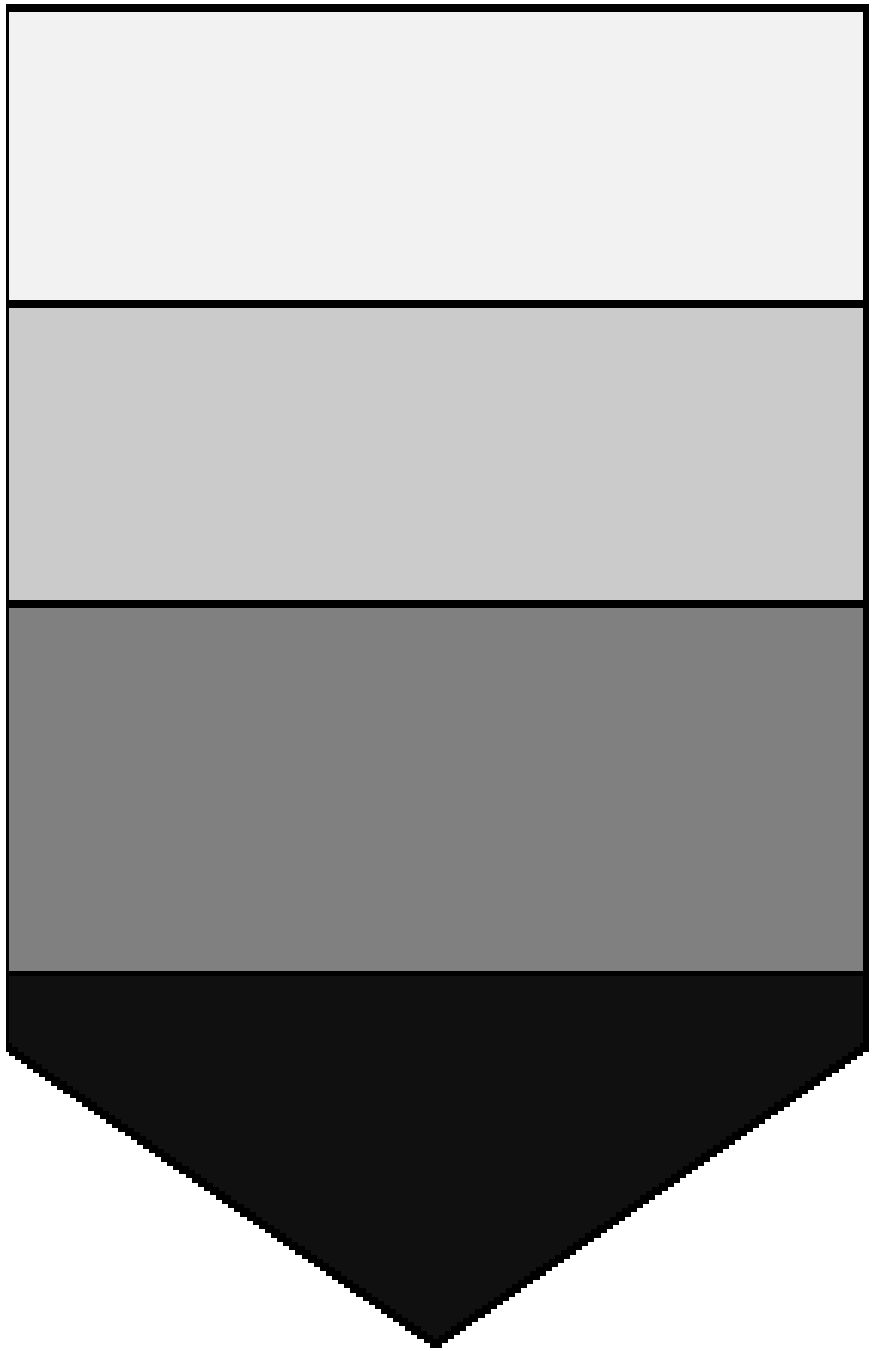
- In **hindered settling**, particle concentration is great enough to inhibit water movement
- Water must move in spaces between particles



Types of Sedimentation

- Compression settling occurs when particles settle by compressing the mass below





Discrete particle settling

Flocculent settling

Hindered (or zone) settling

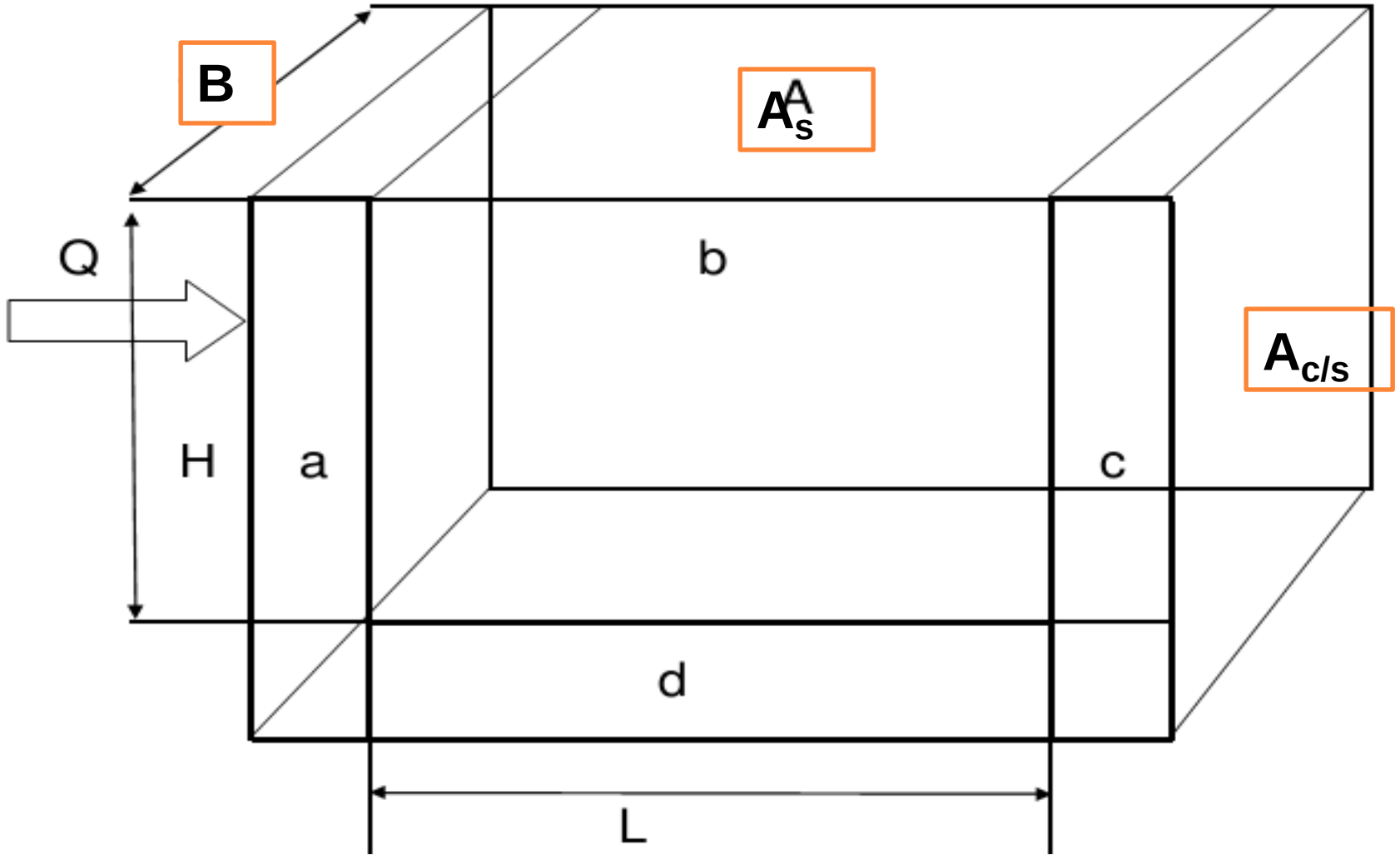
Compression

ZONES IN SETTLING TANK

$A_{c/s} = B \times H$

$A_s = B \times L$

$Vol = B \times L \times H$



a = Inlet zone; b = Settling zone;
c = Outlet zone; d =Sludge zone

a. Inlet Zone

- Incoming flow is uniformly distributed over the cross section of the tank.

b. Settling Zone

- The concentration of each size particle is uniform throughout the cross section (2D).

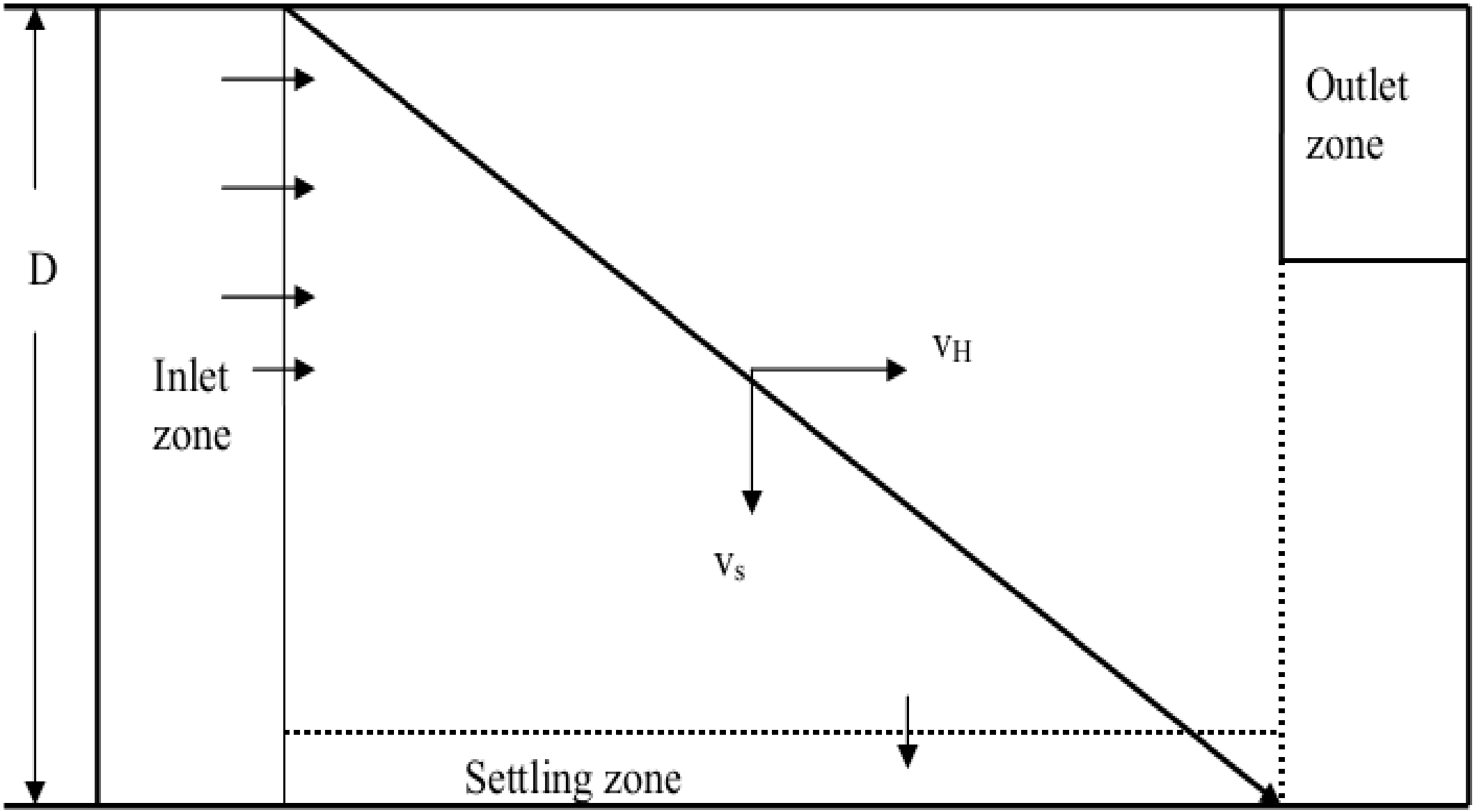
c. Outlet Zone

- Clarified effluent is collected and discharged through an outlet weir.

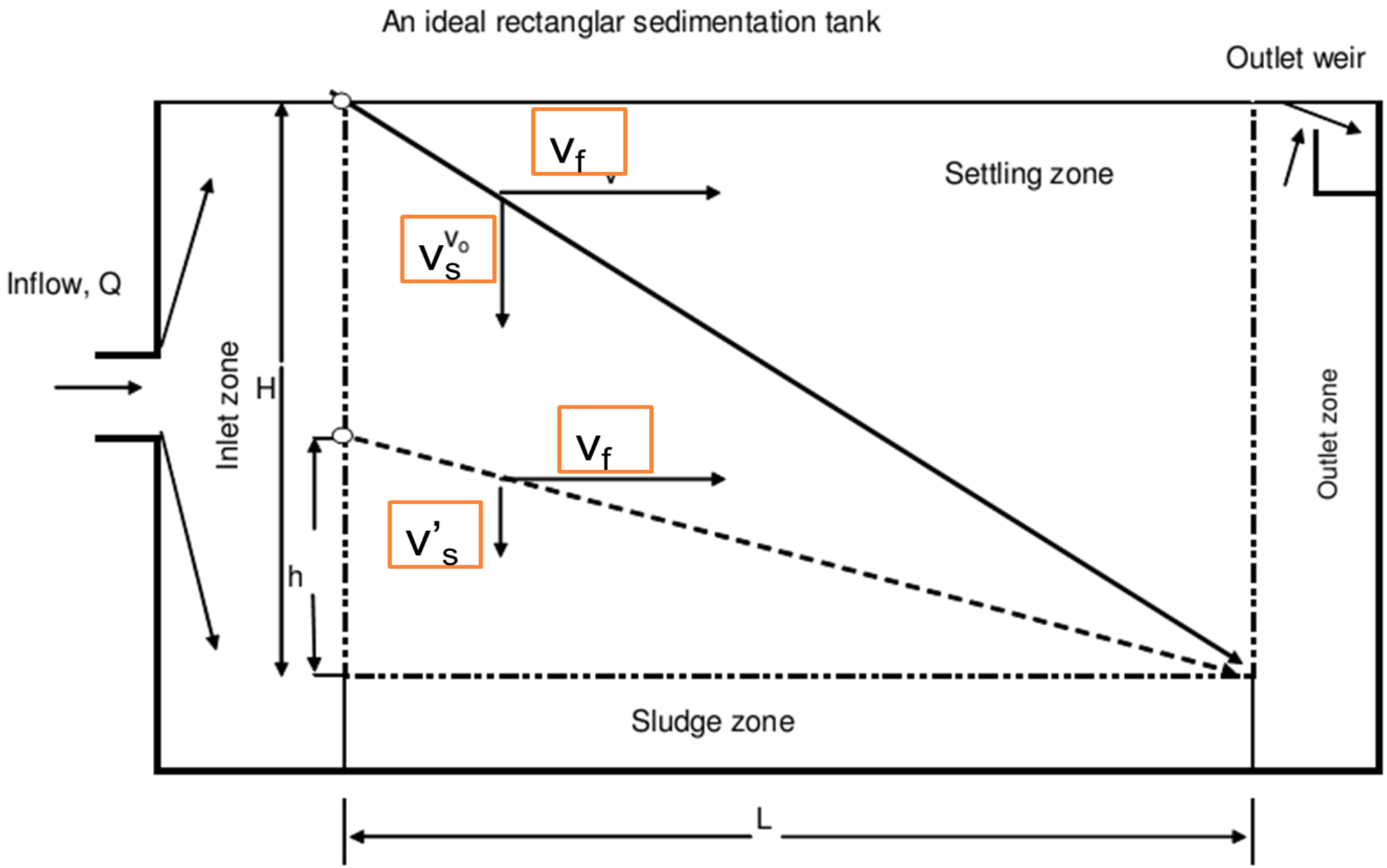
d. Sludge Zone

- Provides for the collection of particles removed from suspension.

Rectangular basins



Ideal Settling



v_f = flow velocity

v_s= terminal settling velocity of a particle that is just removed when it enters at the water surface (H).

Note:

1)All particles with terminal settling velocity ≥ v_s are removed.

2)Only part of particles with settling velocity < v_s are removed.

Terminal settling velocity of particle :-

$$V_s^2 = \frac{4/3\, g\,[\rho_p - \rho_w]\, d}{C_D \rho_w}$$ — Stokes law for settling velocity.

$C_D = \frac{24}{Re}$ → for Laminar flow

$C_D = \frac{24}{Re} + \frac{3}{\sqrt{Re}} + 0.34$ → Transitional flow

$C_D = 0.4$ → for turbulent.

$Re = \frac{\rho V d}{\mu}$

$Re < 1$ — laminar
$Re = 1 - 10^4$ — Transitional
$Re > 10^4$ — Turbulent

$$\therefore C_D = \frac{24}{Re} = \frac{24}{\frac{\rho_w V d}{\mu}} = \frac{24\mu}{\rho_w V d}$$

$$\therefore V_s^2 = \frac{\frac{4}{3} g\,[\rho_p - \rho_w]\, d}{\frac{24\mu}{\rho_w V_s d} \times \rho_w} = \frac{4/3\, g\,[\rho_p - \rho_w]\, V_s d^2}{24\mu}$$

$$= \frac{g \rho_w \left[\frac{\rho_p}{\rho_w} - 1\right] d^2}{18\mu}$$

$$= \frac{g(S-1)d^2}{18\, \mu/\rho_w}$$

$$= \frac{g(S-1)d^2}{18\nu}$$

Particle whose 'd' $< 10^{-1}$ mm :-

1) $$V_s = \frac{g\,[\rho_p - \rho_w]\, d^2}{18\mu}$$

2) $$V_s = \frac{g\,[S-1]\, d^2}{18\nu}$$

Hazen's eqn fo 'Vs'

3) $$V_s = 418\,[S-1]\, d^2 * \frac{9T+70}{100}$$ d - mm, T = °C, Vs = mm/sec

dia of particle = 1mm - 0.1mm :

Hagen's formulae for Vs.

$$V_s = 418\,[S-1]\, d * \frac{9T+70}{100}$$

dia. of particle > 1mm

$$V_s = 1.8 \sqrt{g(S-1)d^2}$$

eg ① particle of dia. 0.18mm, sp. gravity 2.65 settling from water whose $\mu = 1\times10^{-3}$ pa-sec. find settling velo. = ?

m^2/s
N/m^3
s

$$\rightarrow \quad \nu = \frac{\mu}{\rho} = \frac{1\times10^{-3}}{2.65\times1000} \quad \frac{\text{pa-sec}}{kg/m^3} \quad \frac{N/m^2\times s}{kg/m^3} = m^2/s$$

$$V_s = \frac{g(s_p - s_w)d^2}{18\mu} = \frac{g(s-1)d^2}{18\nu}$$

$$\therefore V_s = \frac{9.81(2.65-1)\times(0.18\times10^{-3})^2}{18\times3.77\times10^{-7}}$$

$$= 0.291 \text{ m/sec.}$$

② Find settling velo. of particle of dia 0.02mm
$s = 2.7$, $\nu = 1\times10^{-2}$ stokes $\quad 1 c.s = 10^{-2}$ stokes

$$= \frac{1\times10^{-2}}{100} \text{ cm}^2/s = 1\times10^{-2}\times10^{-4} m^2/s$$

$$= 1\times10^{-4} mm^2/s$$

$$\therefore V_s = \frac{g(s-1)d^2}{18\nu} = \frac{9.81\times(2.7-1)(0.02\times10^{-3})^2}{18\times1\times10^{-6}}$$

$$\therefore \boxed{V_s = 3.706\times10^{-4} \text{ m/sec}}$$

(8) particle of dia. 0.02mm, S=2.7, settling at temp. 20°C. Find terminal settling velo. of particle

→ $V_s = 418 [S-1] d^2 * \frac{9T+170}{100}$

$= 418 [2.7-1] \times (0.02 \times 10^{-3})^2 * \frac{9 \times 20 + 170}{100}$

$\therefore \boxed{V_s = 0.869 \text{ mm/sec}}$

4-11-14

$d = 0.025 \text{ mm}$, $S = 2.65$

$\nu = 0.01 \text{ cm}^2/\text{sec} = 0.01 \times 10^{-4} \text{ m}^2/\text{sec}$

$V_s = \frac{g}{18} [S_s - 1] \frac{d^2}{\nu}$

$= \frac{9.81 [2.65-1] \times (0.025 \times 10^{-3})^2}{18 \times 0.01 \times 10^{-4}}$

$V_s = 5.62 \times 10^{-4} \text{ m/sec} = \frac{5.62 \times 10^{-4} \times 10^2}{} \text{ cm/sec}$

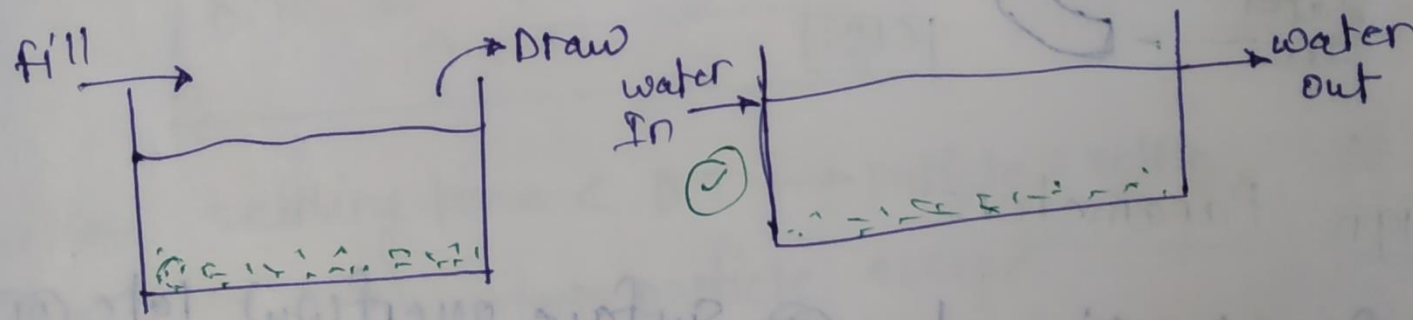
$\boxed{V_s = 0.0562 \text{ cm/sec}}$

* Sedimentation tanks : (Settling (or) Clarifiers) Tank

The place where settling is carried out by controling turbulent and flow velocity is known as settling tank / sedimentation tank.

i) Draw & Fill type ii) continuous flow type.

↳ obselete
(out dated not in use)

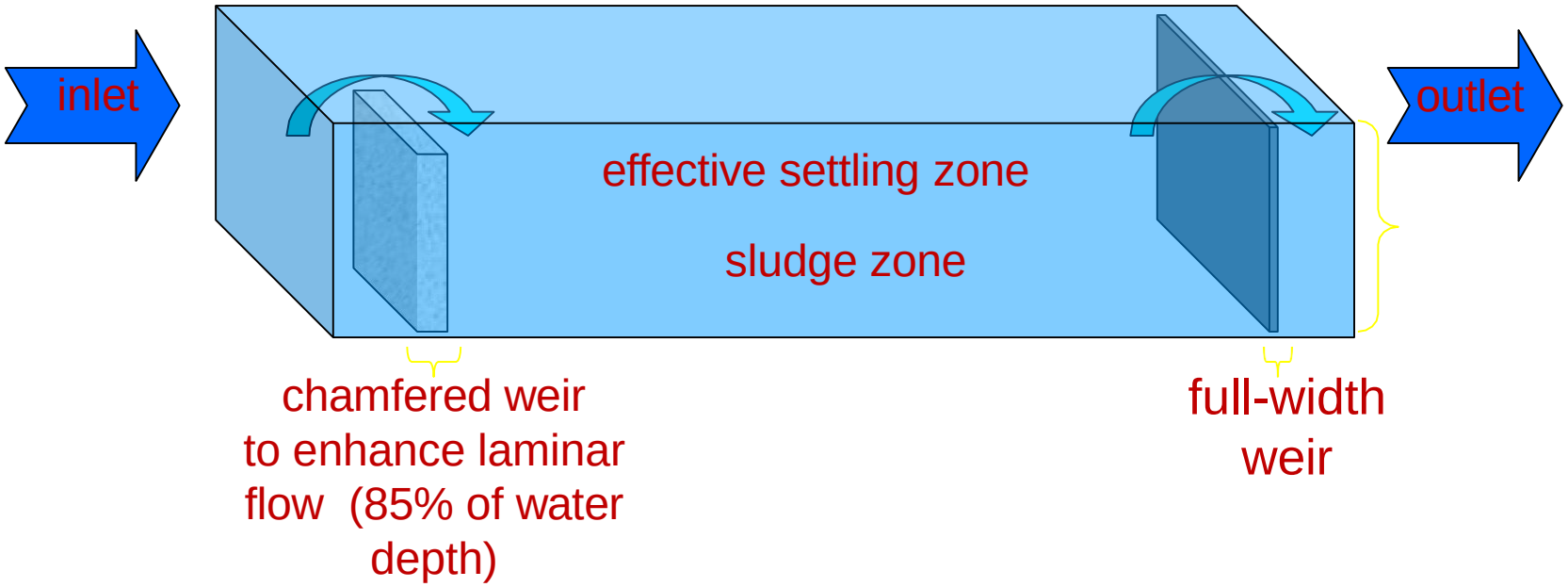


Continuous flow types:

1. Horizontal flow ~~irregular~~ rect. setting tanks
2. Radial flow circular setting tanks

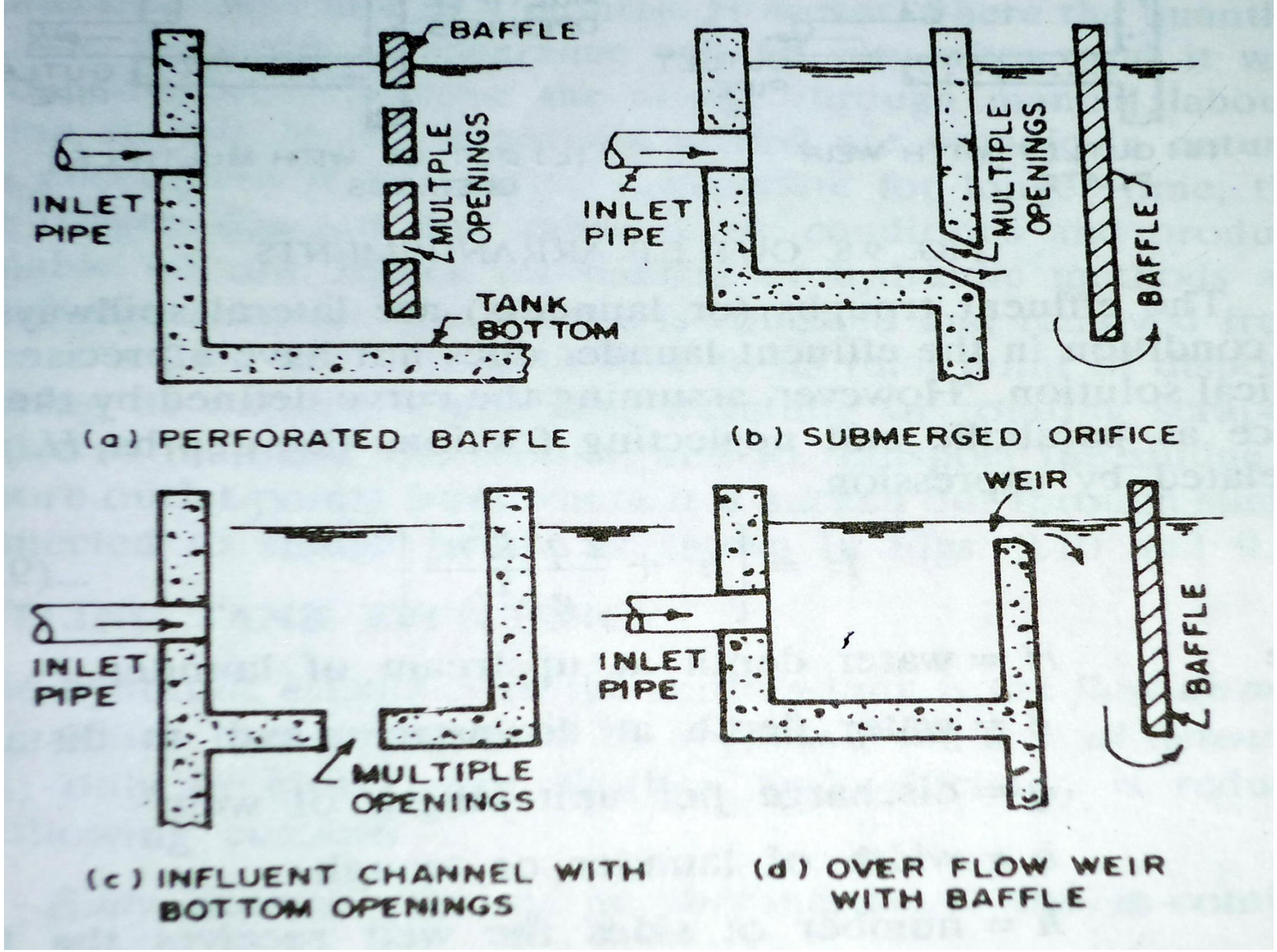
Settling Basins

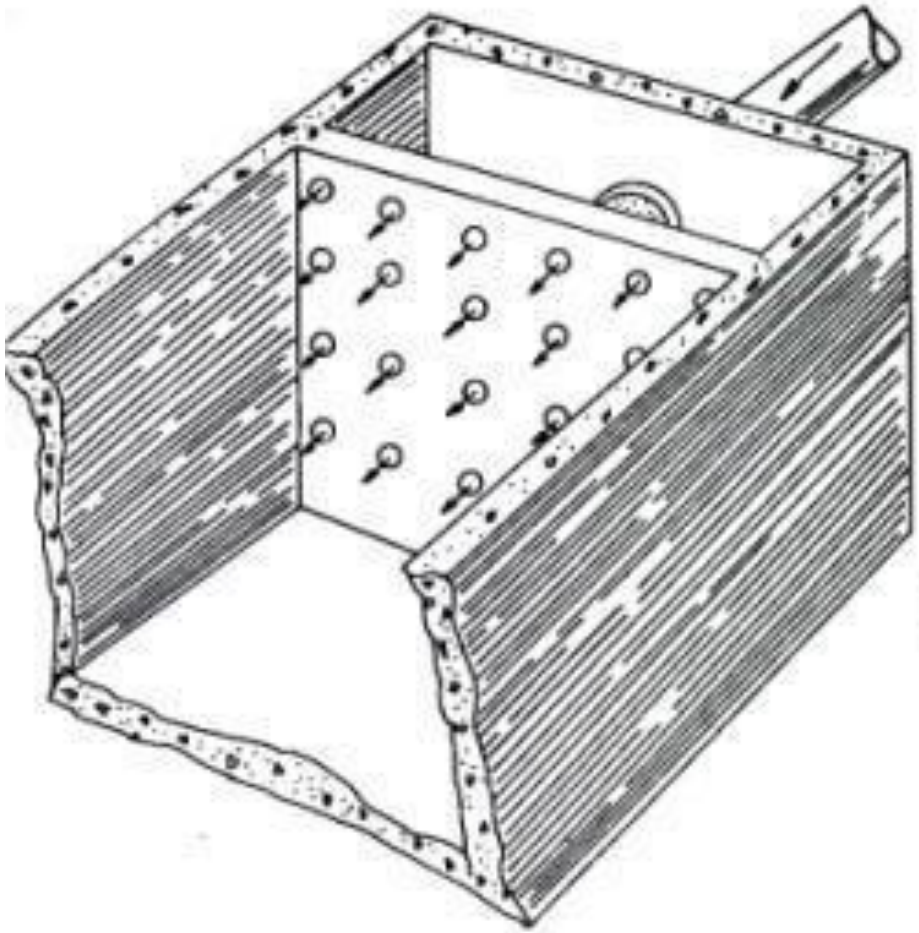
- Design to minimize turbulence:



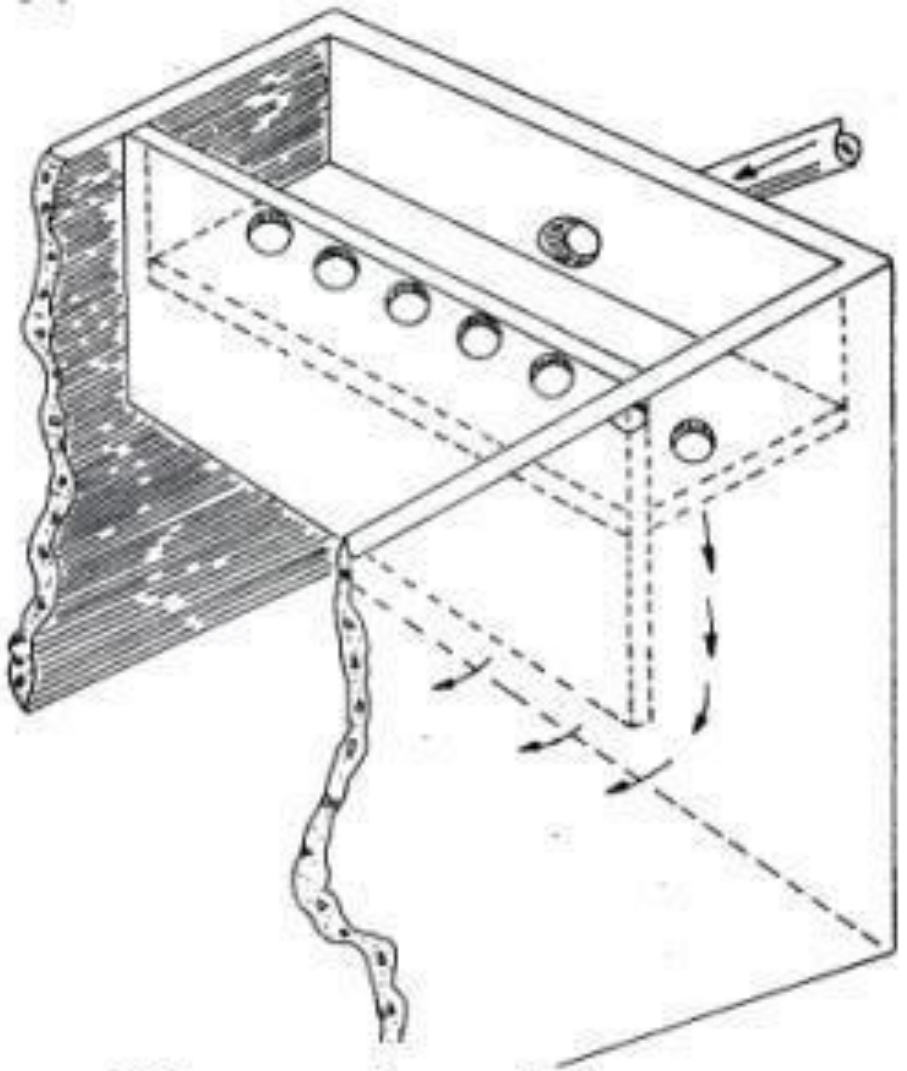
Inlet and Outlet arrangements
Types of settling tanks

Inlet Arrangements


BAFFLE
INLET PIPE
MULTIPLE OPENINGS
TANK BOTTOM
(a) PERFORATED BAFFLE
INLET PIPE
MULTIPLE OPENINGS
BAFFLE
(b) SUBMERGED ORIFICE
INLET PIPE
MULTIPLE OPENINGS
(c) INFLUENT CHANNEL WITH BOTTOM OPENINGS
WEIR
INLET PIPE
BAFFLE
(d) OVER FLOW WEIR WITH BAFFLE

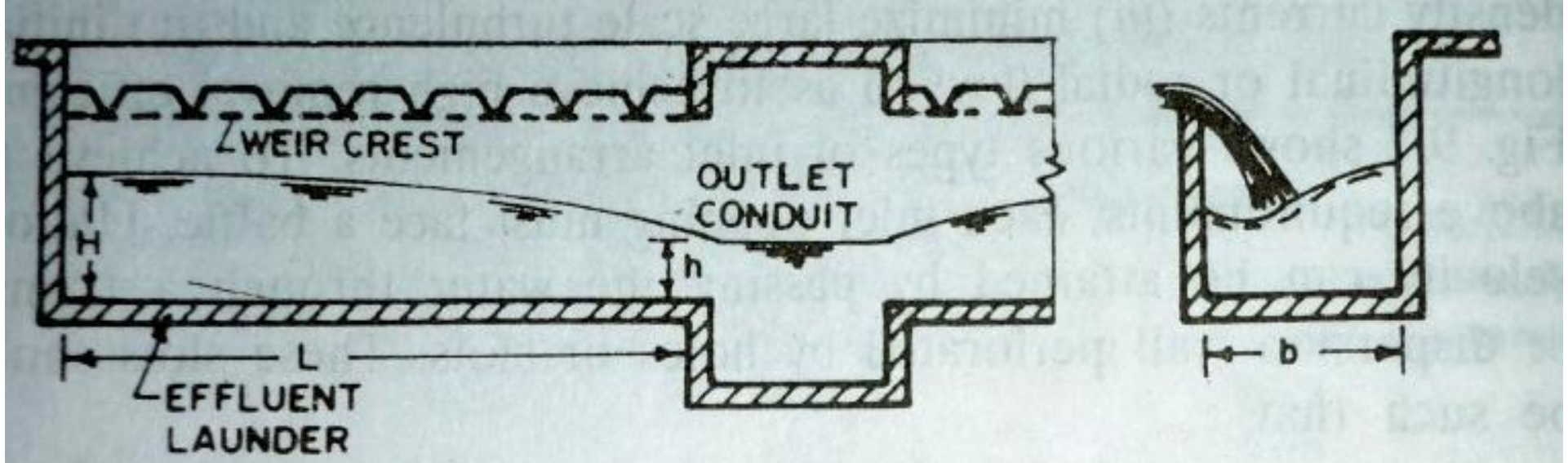


Stilling Wall

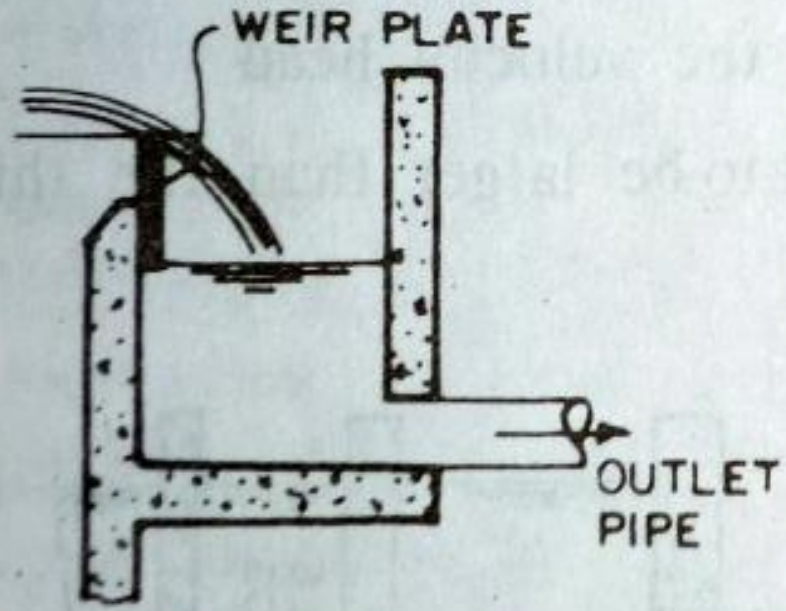


Channel or Flume

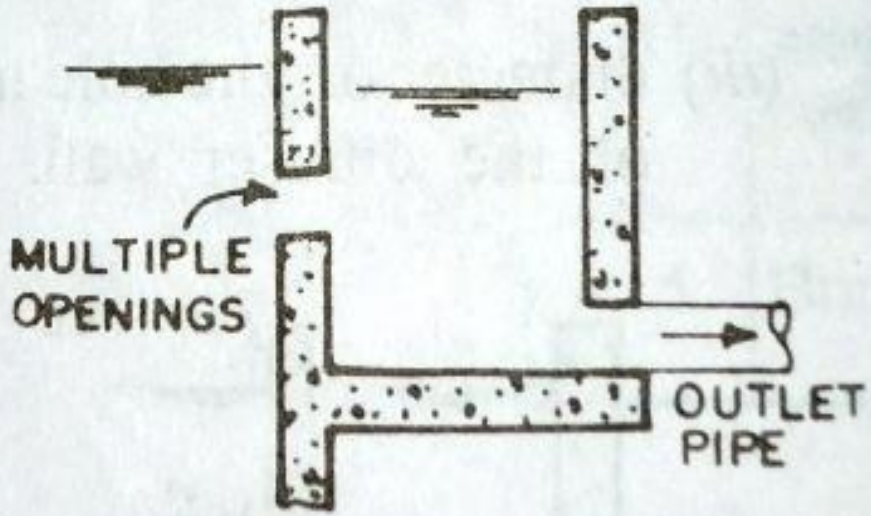
Outlet Arrangements



(a) OUTLET WITH EFFLUENT LAUNDER



(b) OUTLET WITH WEIR PLATE



(c) OUTLET WITH MULTIPLE OPENINGS

Types of Settling tanks

Types of settling tanks

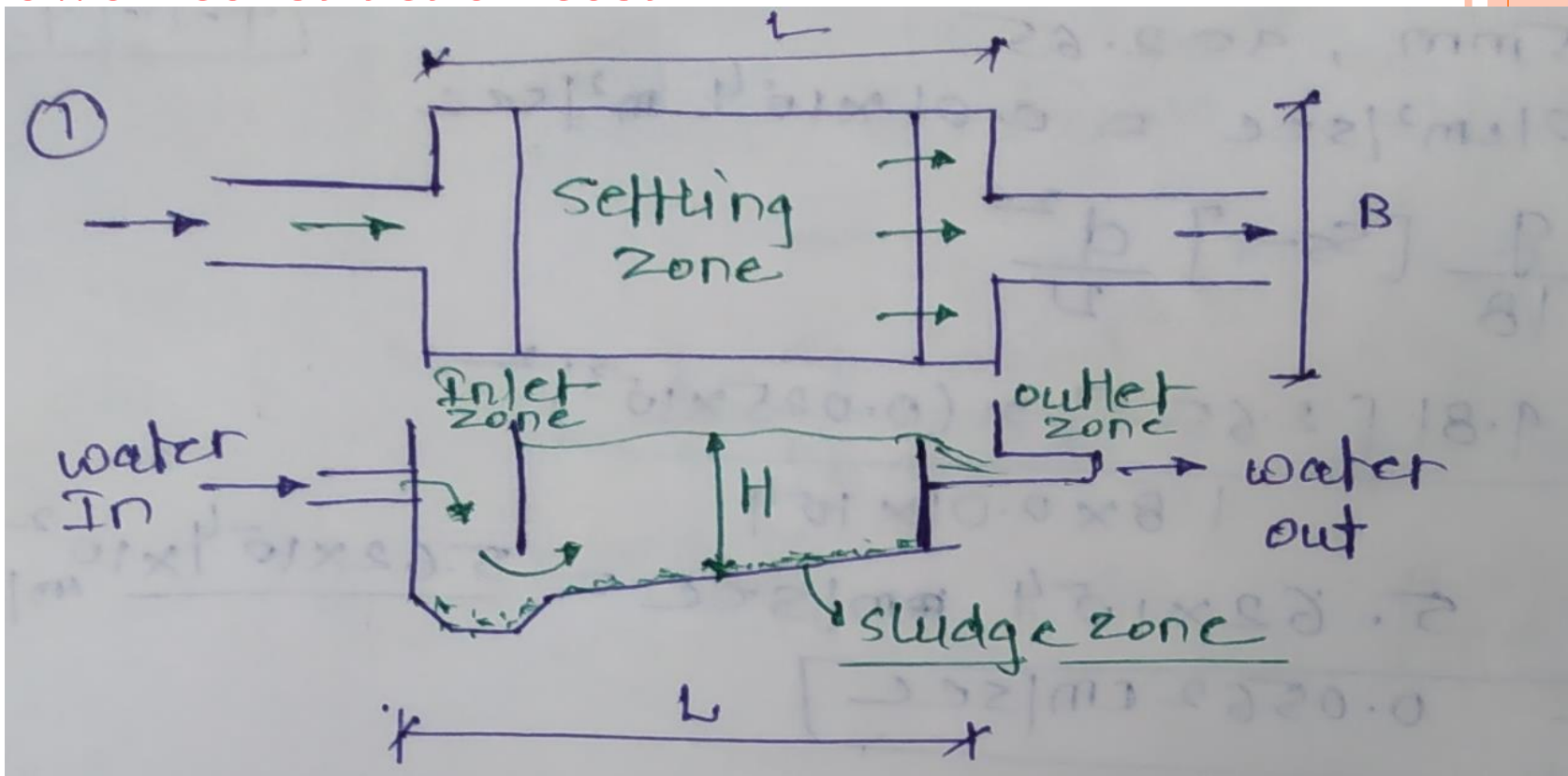
1. Fill and draw
2. Continuous flow type



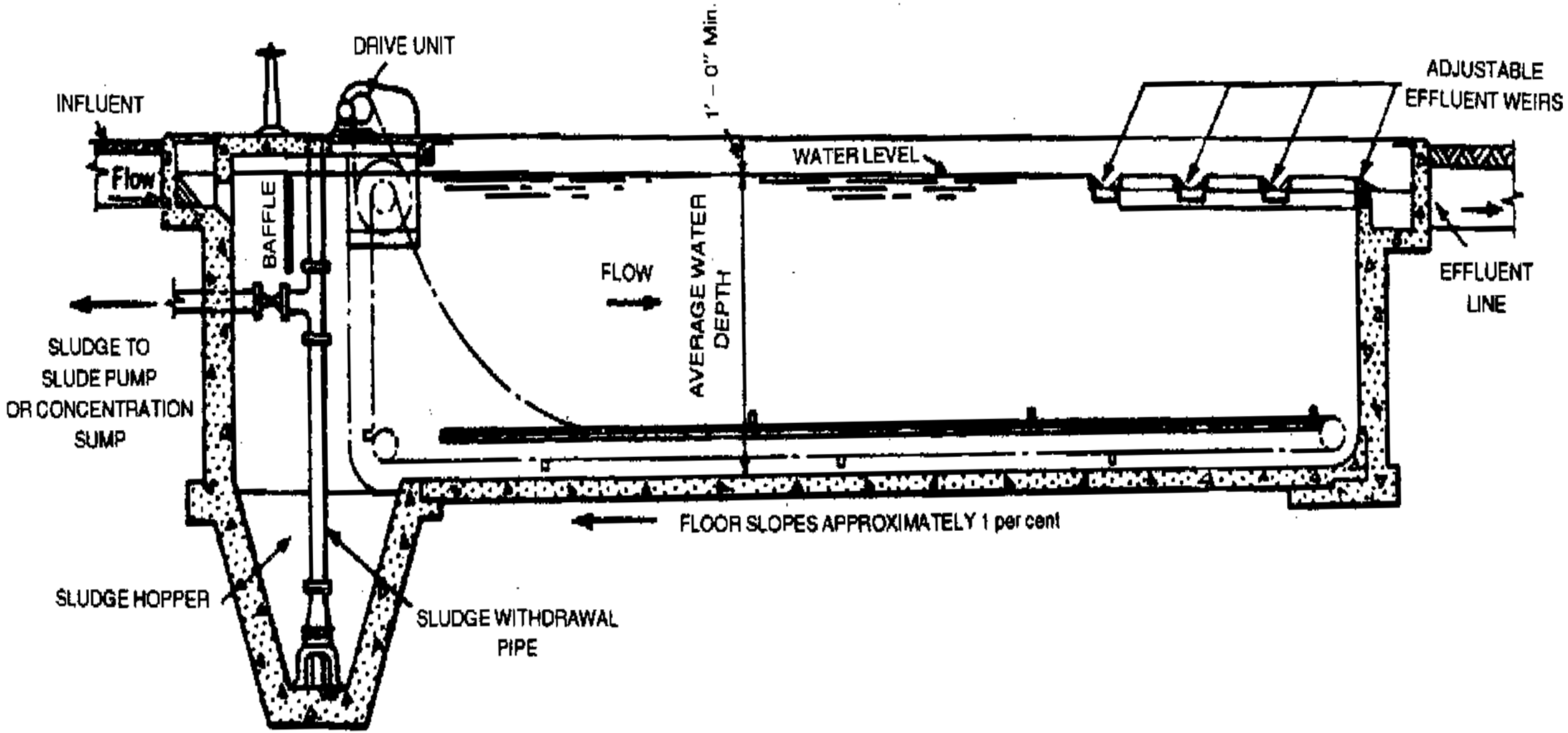
RECTANGULAR BASINS

- Rectangular basins are commonly found in large- scale water treatment plants.
- Rectangular tanks are popular as they tend to have:

- High tolerance to shock overload
- Predictable performance
- Cost effectiveness due to lower construction cost
- Lower maintenance
- Minimal short circuiting



SECTION OF RECTANGULAR TANK



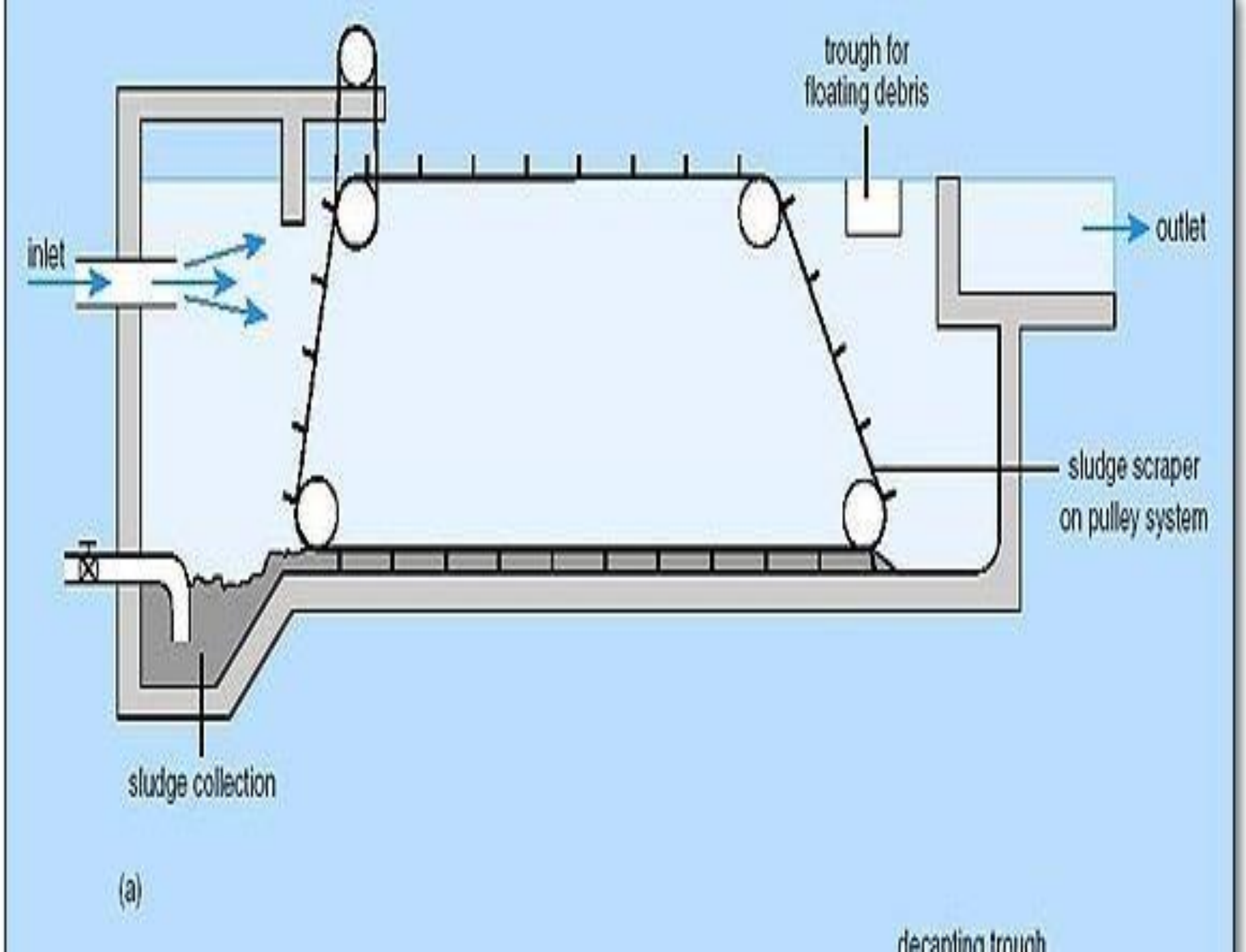

trough for
floating debris
inlet
outlet
sludge scraper
on pulley system
sludge collection
(a)
decanting trough



Figure 5 : Rectangular sedimentation Tank

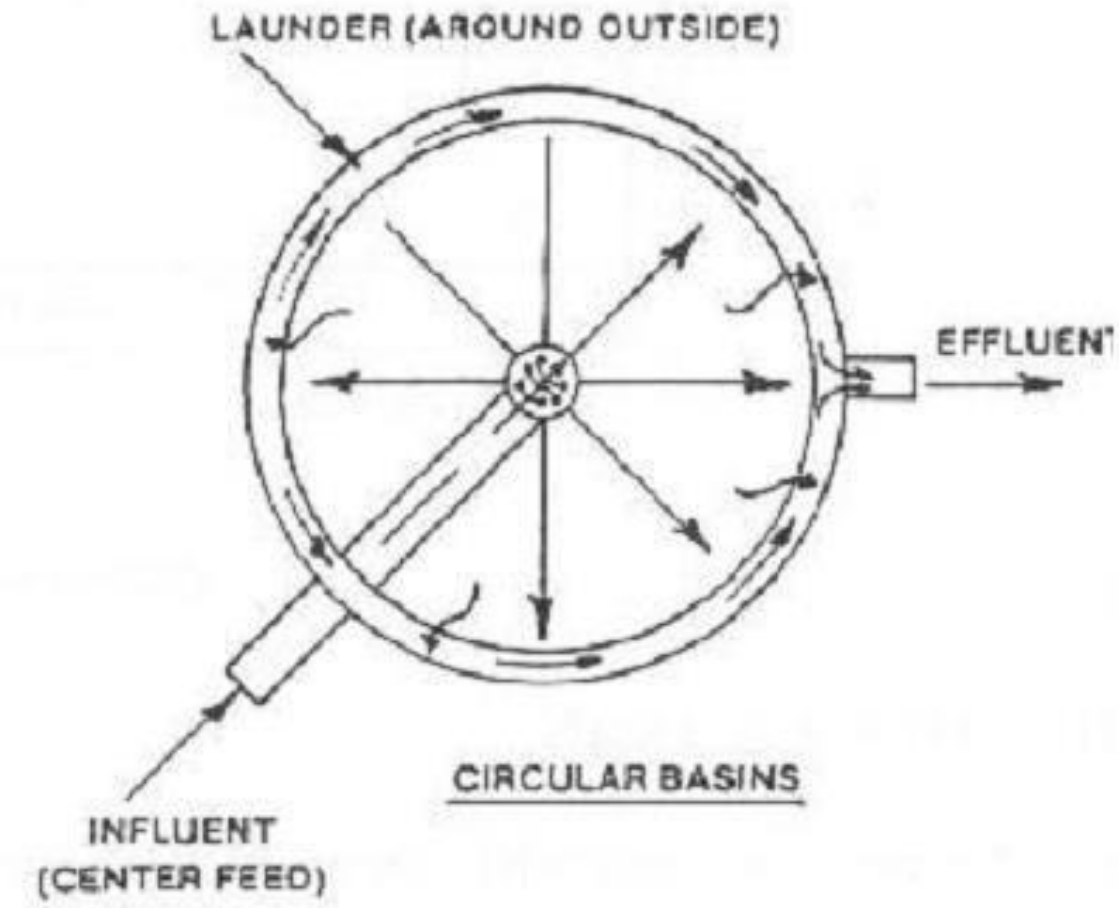


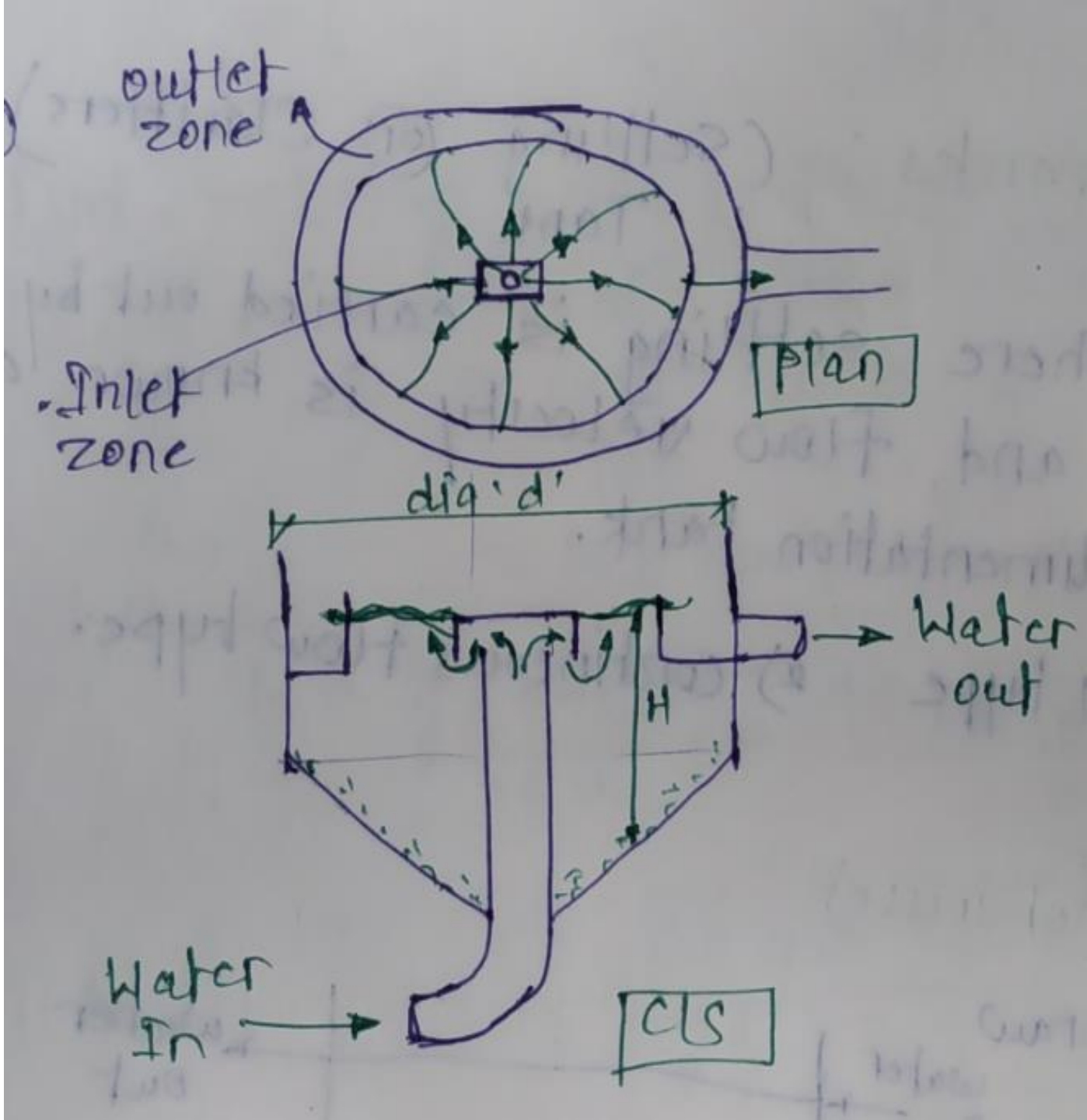
Figure 6 : Rectangular sedimentation Tank

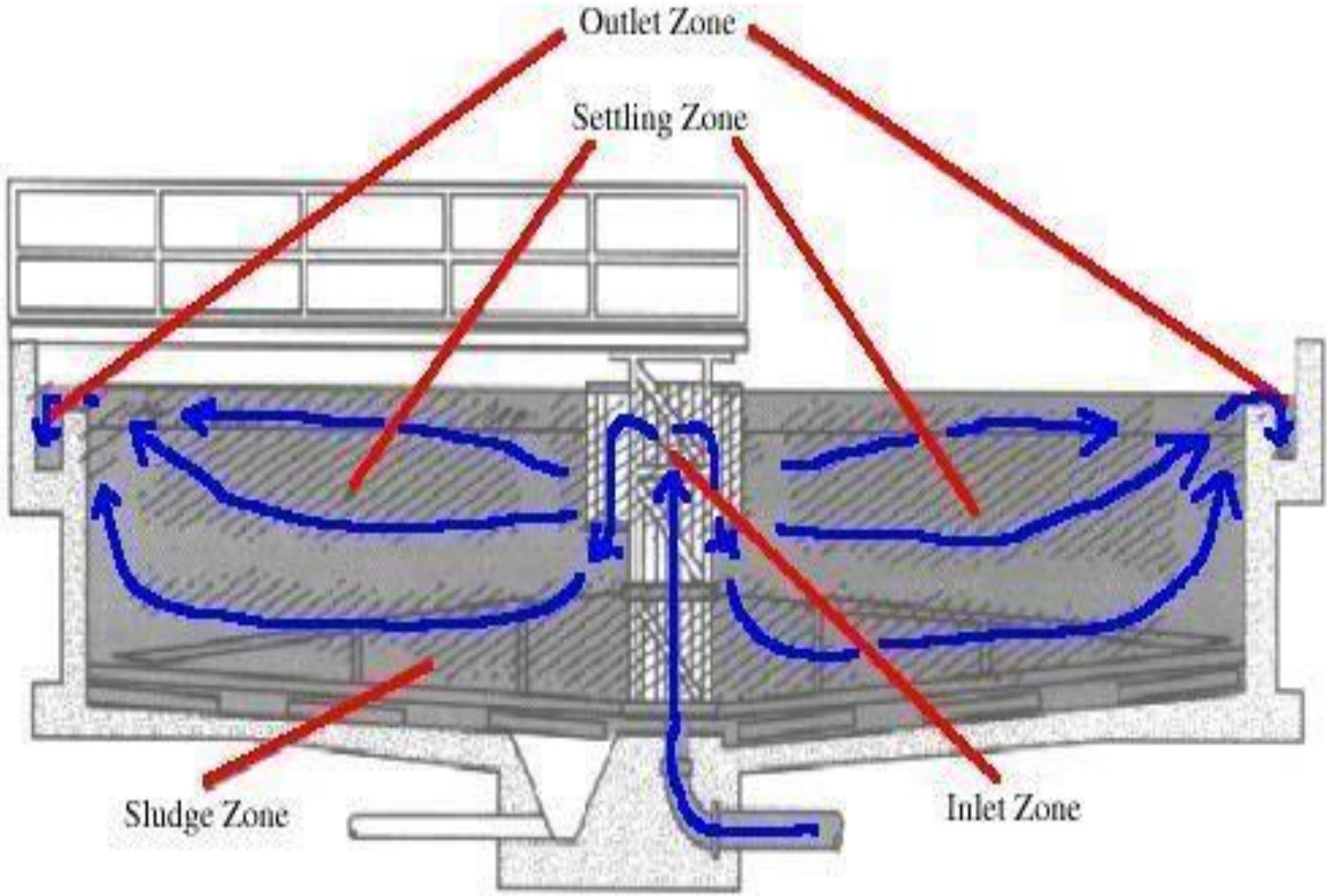
CIRCULAR BASINS

- Circular basins are often referred to as **clarifiers.**
- These basins share some of the performance advantages of the rectangular basins, but are generally more prone to short circuiting and particle removal problems.

CENTRAL INLET PERIPHERAL OUTLET






Outlet Zone
Settling Zone
Sludge Zone
Inlet Zone

Circular tank with central feed radial flow

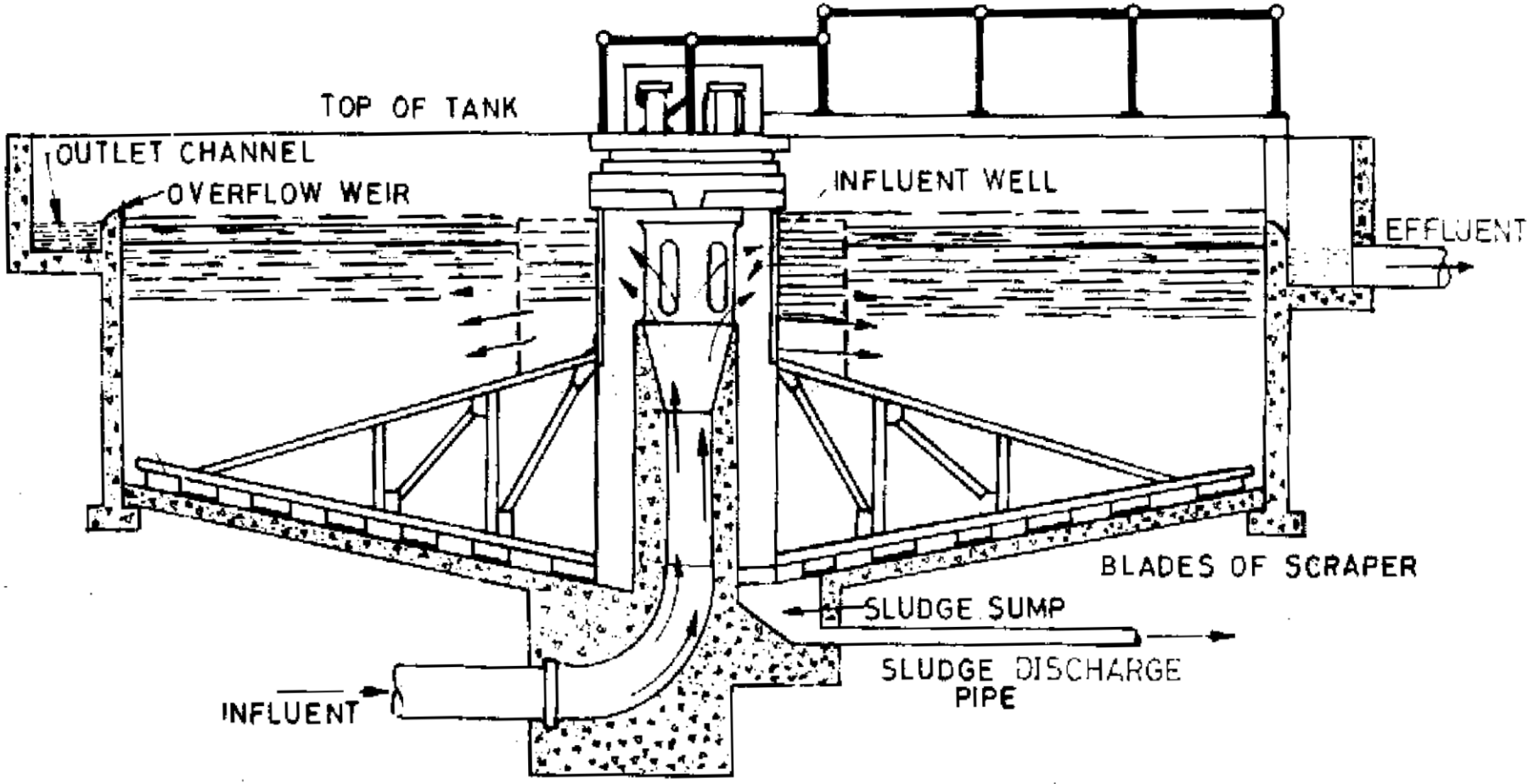


Fig. 9.4. Circular sedimentation tank (central feed) with radial flow.

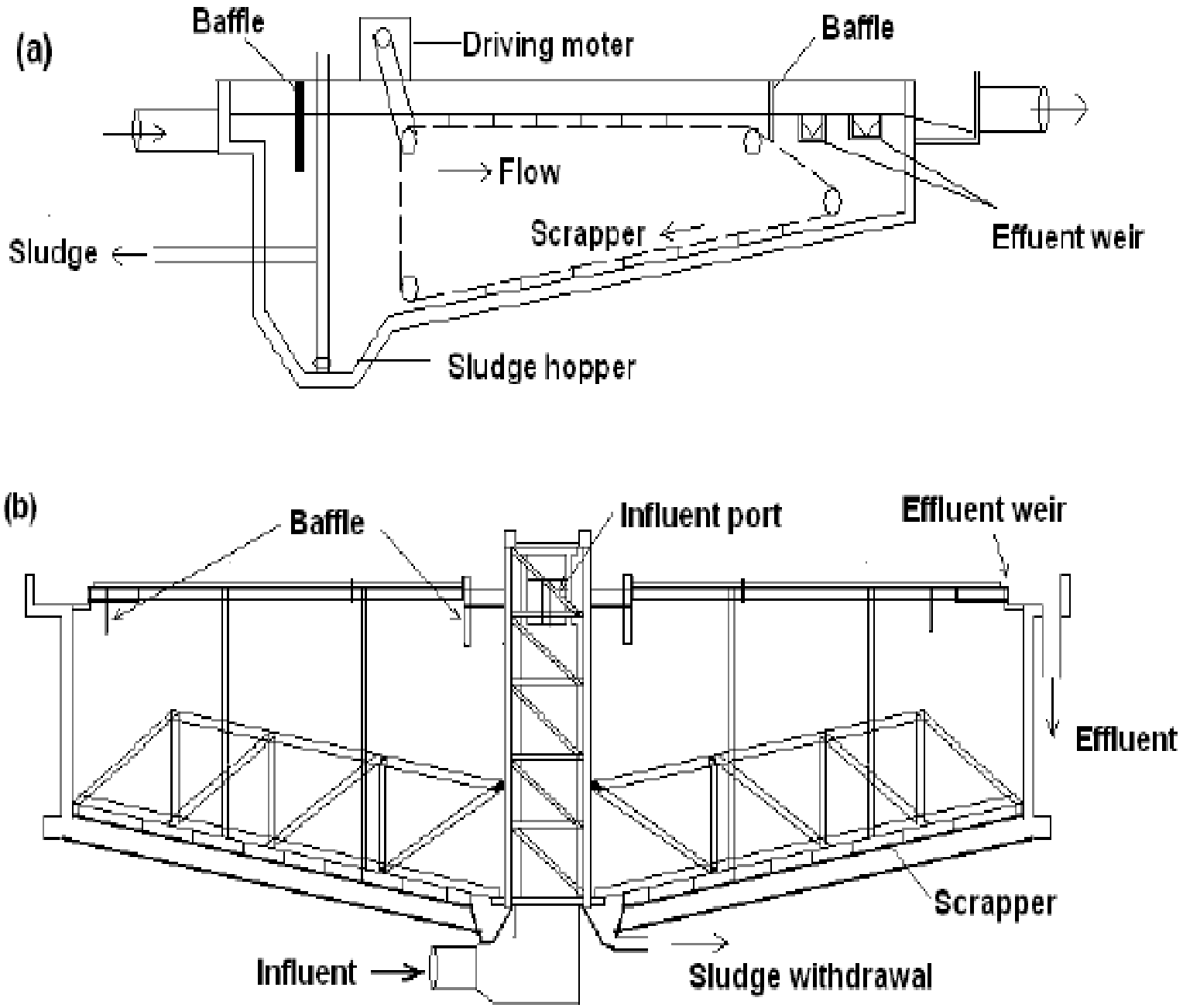
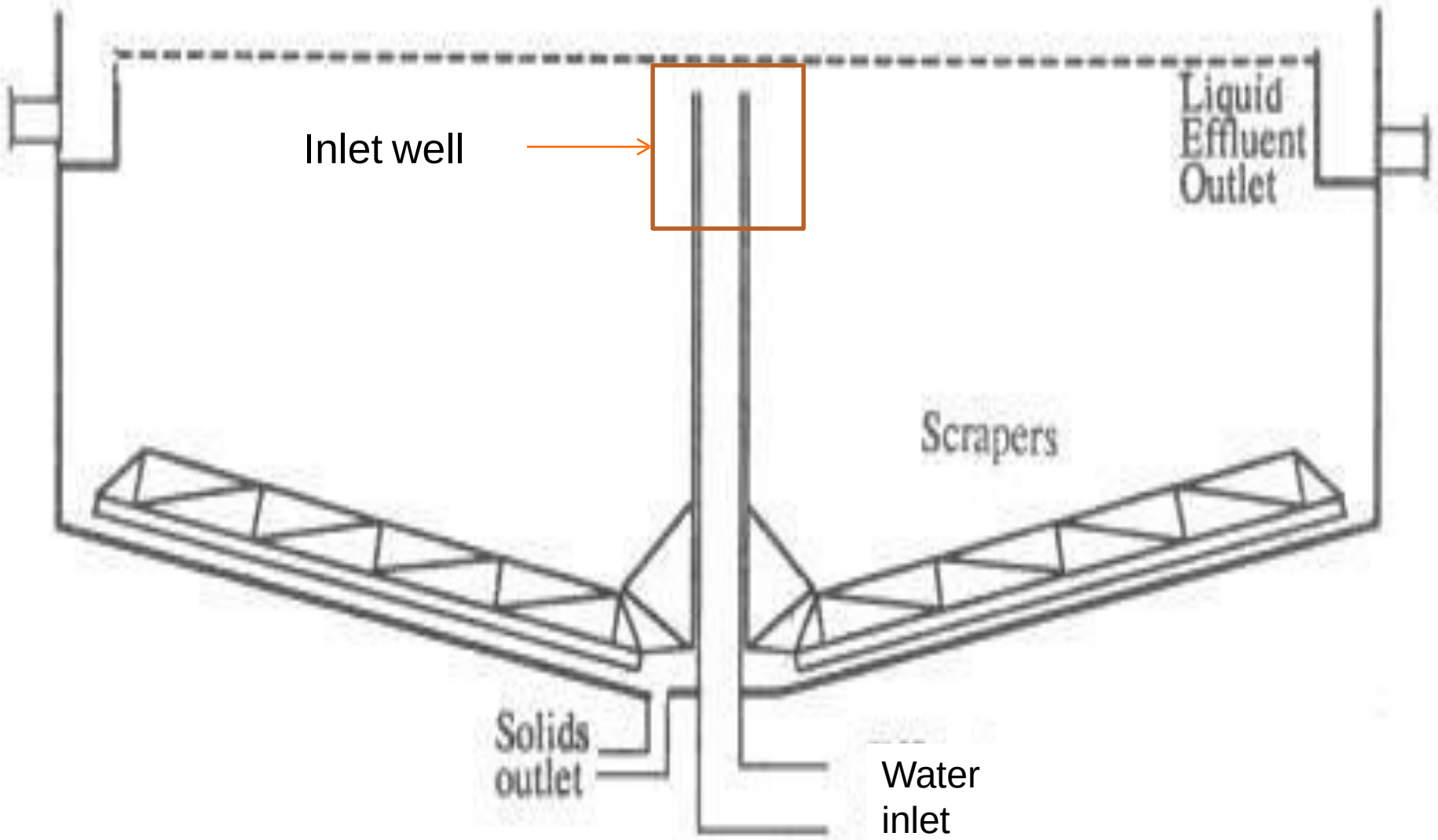


Figure 16.2 (a) Rectangular and (b) Circular primary sedimentation tank

CIRCULAR TANK



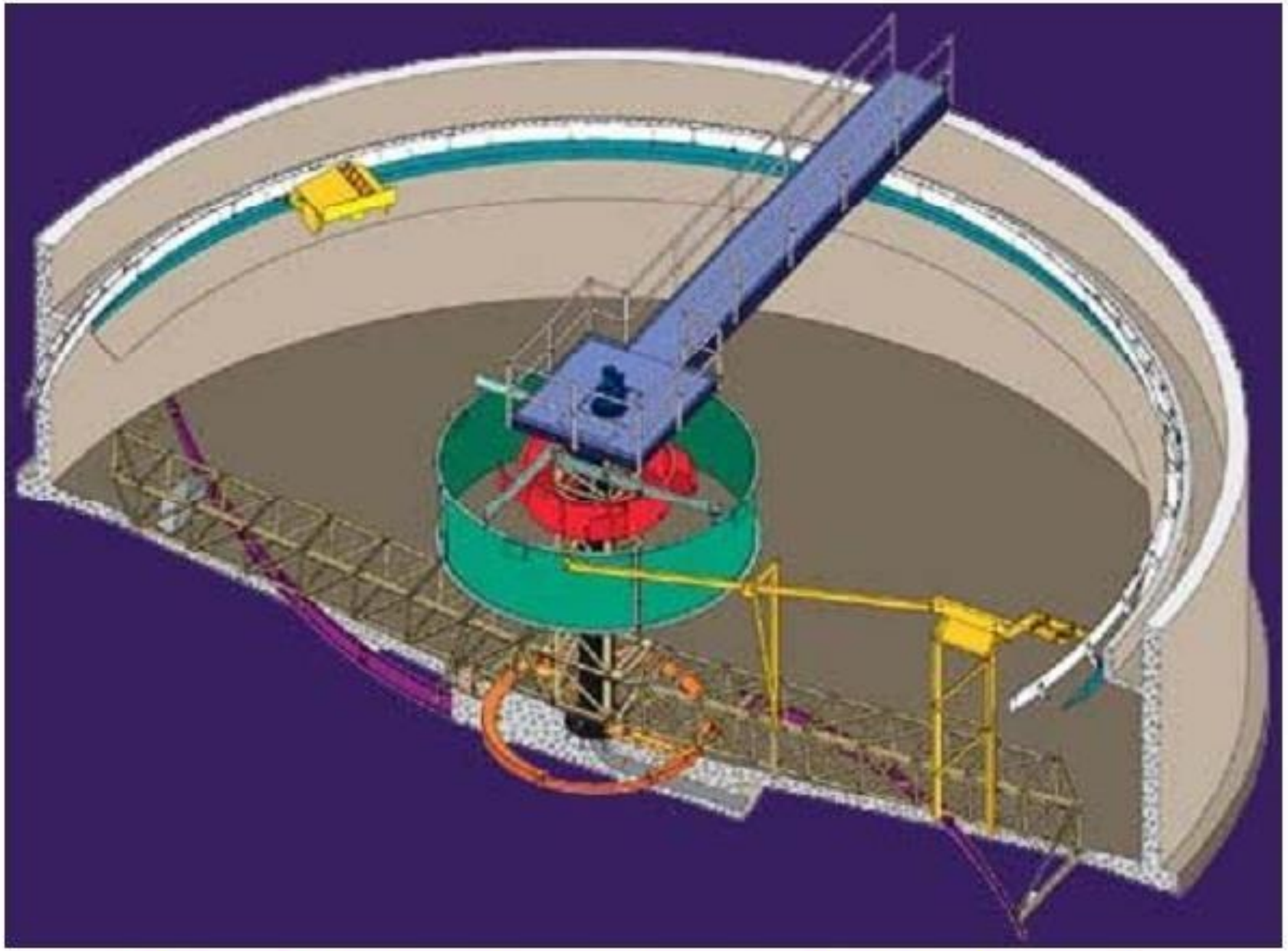


Figure 8 : Circular sedimentation Tank



Figure 7 : Circular sedimentation Tank









Fig. 1: Rectangular horizontal flow sedimentation tank

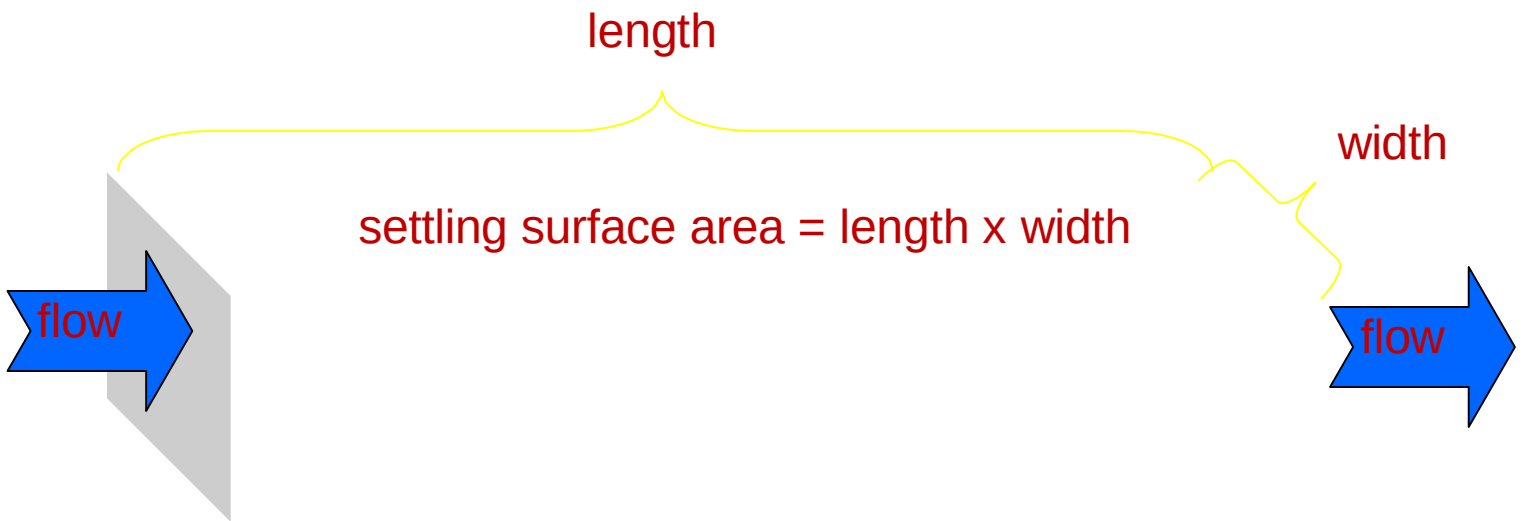


Fig. 3 Typical inboard weir arrangement to increase the weir length

Settling Basins

- Overflow rates are used for design: V_o

$$Overflow\ Rate = \frac{Flow\ Rate\ (m^3/s)}{settling\ \ surface\ area\ (m^2)}$$



Design Parameters

Surface Loading rate (or) Surface overflow rate (or) Hydraulic Loading rate (or) overflow velocity "V_o"

V_o = $\frac{\text{Q Water applied in unit time}}{\text{unit surface area of settling basin}}$

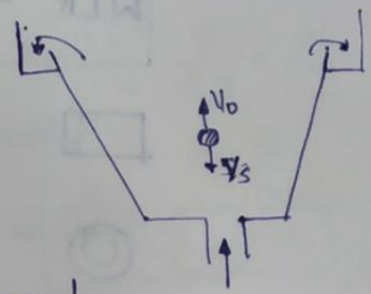
$V_o = \frac{Q}{\text{Surface area}}$

$\text{Surface Area} = \frac{Q}{\text{SLR (or) SOR}} = \frac{Q}{V_o}$

Surface area = plan Area

$\square^{\text{ler}} = L * B$

$\odot^{\text{lar}} = \frac{\pi}{4} * d^2$



$V_s > V_o$ → particle is removed

$V_s < V_o$ → particle escape

SLR (or) SOR V_o = 500 to 750 lit/hr/m^2

(12 to 18 m^3/day/m^2)

≈ 10 to 20

② Detention time: (DT or θ or t)

It is avg. theoritical time taken by particle, water travel from inlet to outlet of S.Tank.

$$D.T. = \frac{\text{Volm of setting tank}}{Q} = \frac{V}{Q}$$

particle settling time < D.T → paticle settle

P.S.T > D.T → particle escape.

③ Flow through Velocity: (Horizontal flow velocity) V_H:

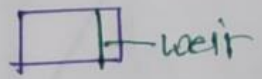
Speed with which water travel from inlet to outlet of setting time.

$$V_H = \frac{Q}{\text{cross area of setting tank}} = \frac{Q}{B \times H}$$

$$D.T = \frac{L}{V_H} = \frac{\text{Distnce travelled}}{\text{velo. of flow}}$$

④ Weir Loading rate (WLR):

$$WLR = \frac{Q}{\text{unit length of weir}}$$

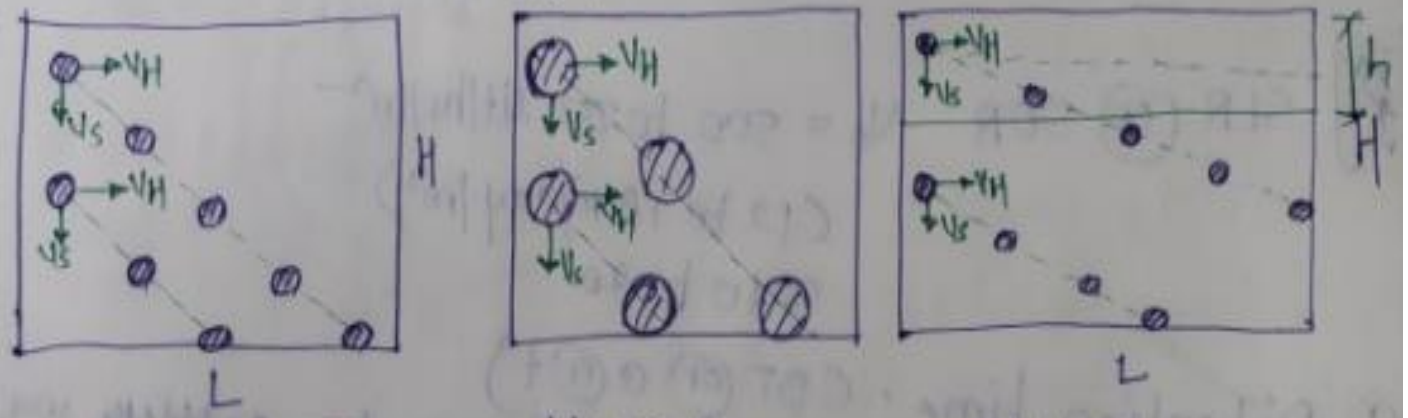


$$W.L.R = \frac{Q}{B}$$



$$W.L.R = \frac{Q}{\pi \cdot d}$$

d_p, V_{sp}, $V_o < V_{sp}$ — $d_i > d_p$, V_o, $V_{si} > V_{sp}$ — $d_i < d_p$, V_o, $V_{si} < V_{sp}$



100% Remove

$\therefore V_{sp} > V_o$

$V_{si} \gg V_o$

100% removal

$V_{si} < V_o$

$\therefore$ partial removed

$V_H \propto \frac{1}{H}$

particle removal efficiency $\eta = \frac{V_{si}}{V_o} * 100$

$\uparrow \eta \propto \frac{1}{V_o} \downarrow$

$\uparrow V_o \propto \frac{1}{\text{surface area}} \downarrow$

$\eta \propto$ surface area

Note:

* particle removal depends on surface area of settling tank & it is independent of depth * Depth of setting does not influence particle removal.

If η of particle removal > 100% then it is equated to 100% only.

* For 100% removal (effective removal)

(w)

$$V_{si} \geq V_o$$

Settling time $\leq$ Detention time

$$\frac{H}{V_s} \leq \frac{L}{V_H}$$

$$V_s = V_H * \frac{H}{L}$$

$$= \frac{Q}{B.H} * \frac{H}{L}$$

$$= \frac{Q}{B.L}$$

$$\therefore \boxed{V_s = \frac{Q}{B.L} = V_o}$$

Overall removal of Particle:

Overall efficiency $\eta_{overall} = \Sigma P_i \eta_i$
$\eta_{overall} = \Sigma P_i * \frac{V_{si}}{V_o} * 100$
P_i = % of ith particle in water

Ex.1

$Q = 1 \text{ lakh } m^3/day = 1 \times 10^5 \ m^3/day$

$V_s = 20 \ m/day$

$A = ?$

$\therefore$ Area of tank $= \frac{1 \times 10^5}{20} = 5000 \ m^2$

(For 100% removal $V_s \geq V_o$ $\therefore$ $V_o = V_s = 20 \ m/day$)

1. A sedimentation tank is to be designed to treat one lakh cumec of water per day. If the settling velocity of particle to be removed is 20 m/day then area of the tank required is

Ex.2

$Q = 1.8 \text{ MLD} = 1.8 \times 10^6 \times 10^{-3} \ m^3/day$

$D.T = 4 \text{ hr} = \frac{4}{24} \text{ day}$

$\therefore$ Vol^m of settling tank $= D.T \times Q$

$= \frac{1.8 \times 10^6 \times 4}{10^3 \times 24}$

$= 300 \ m^3$

2. A rectangular sedimentation tank is treating 1.8 MLD of raw water with a detention period of 4 hours then volume of tank required is

Ex.3

$q = 12000 \text{ lit/hr/m}^2$

a) $V_s = \frac{g}{18}(S_s - 1)\frac{d^2}{\nu}$

$= \frac{9.81}{18}(1.65) \times \frac{(0.03 \times 10^{-3})^2}{0.897 \times 10^{-6}}$

$V_s = 9.022 \times 10^{-4}$ m/sec

$V_0 = \frac{Q}{A} = \frac{12000 \times 10^{-3}}{60 \times 60} = 3.33 \times 10^{-3}$ m/sec

$\eta = \frac{V_s}{V_0} \times 100 = \frac{9.022 \times 10^{-4}}{3.33 \times 10^{-3}} \times 100 = 27.09\%$

b) $d = 12$ mm, $d > 1$ mm

$\therefore V_s = \frac{9.81}{18}(1.65) \times \frac{(12 \times 10^{-3})^2}{0.897 \times 10^{-6}} = 144.86$

$\rightarrow V_s = 1.8\sqrt{g(S-1)d} = 0.7993$ m/sec

$\therefore \eta = \frac{V_s}{V_0} \times 100 = \frac{0.7993}{3.33 \times 10^{-3}} \times 100 > 100\%$

$\therefore \eta = 100\%$

3. A Rectangular sedimentation tank is designed for a surface overflow of 12000 lit/hr/m2.What percent of suspended particle of diameter

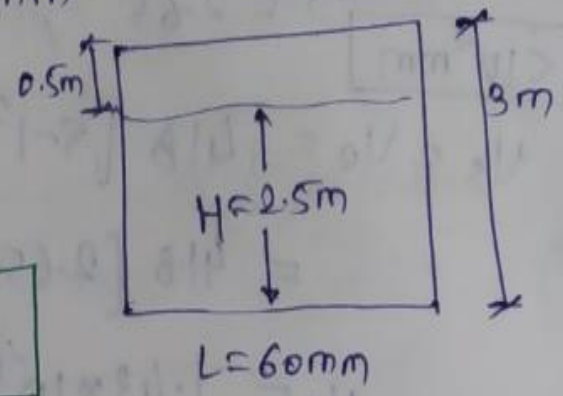
a)0.03 mm and

b) 12 mm will be removed in the tank . Dynamic viscosity is 0.897 mm2/sec and specific gravity of particle 2.65

Ex.4

$L = 60\,m$, $D = 3\,m$, $d = 0.025\,mm$

$S = 2.65$ $\quad \nu = 0.01\,cm^2/sec$



- effective removal $= 100\%\ \eta$

$\therefore V_s \geq V_o$, $\quad \frac{H}{V_s} \leq \frac{L}{V_H}$

$$\therefore V_s = \frac{g}{18}(S_s - 1)\frac{d^2}{\nu}$$

$$= \frac{9.81}{18}(2.65-1) \times \frac{(0.025 \times 10^{-3})^2}{0.01 \times 10^{-4}}$$

$$V_s = 5.62 \times 10^{-4}\ m/sec$$

$$\therefore \frac{H}{V_s} = \frac{L}{V_H}$$

$$\therefore V_H = \frac{L \times V_s}{H} = \frac{60 \times 5.62 \times 10^{-4}}{2.5} = 0.01348\ m/sec$$

4. In a continuous flow settling tank 3 m deep with free board of 0.5 m and 60 m long. What flow velocity would you recommend for effective removal of 0.025 mm particels.The specific gravity is 2.65 and Dynamic viscosity is 0.01 Cm2/sec

Ex.5

$d = 50\,\mu m$, $G = 2.3$, $Q = 100$ MLD, $H = 3m$

D.T = ? L:B = 3:1

$\nu = 1.01 \times 10^{-6}\ m^2/s$ 100% → $V_o = V_s$

→

1) $D.T = \frac{\text{Vol of tank}}{Q} = \frac{44.53 \times 14.84 \times 3}{\left(\frac{100 \times 10^6 \times 10^{-3}}{24}\right)} = 0.47\,hr$

2) $V_s = \frac{g}{18}(S_s - 1)\frac{d^2}{\nu} = \frac{9.81}{18} \times 1.3 \times \frac{(50 \times 10^{-6})^2}{1.01 \times 10^{-6}} = 1.75 \times 10^{-3}$ m/sec

$V_o = V_s = \frac{Q}{A}$ ∴ $\frac{1.75 \times 10^{-3}}{} = \frac{100 \times 10^6 \times 10^{-3}}{3B \times B \times 24 \times 3600}$

∴ $B = 14.84m$ ∴ $L = 14.84 \times 3 = 44.53m$

5.Design a rectangular sedimentation tank to remove particles of size equal to and greater than 50 micro-meter with G=2.3 from 100 MLD of water . Also determine the detention time. L:B::3:1 , depth of tank =3m and Dynamic viscosity is 1.01 x 10-6 m2/sec

Ex.6

$Q = 10\ MLD$

$\frac{B}{L} = \frac{1}{3}$, $d = 0.04mm$ (100% removal)

$H = 3.5m$, $D.T = ?$

$S = 2.65$, $T = 20°C$

$d < 10^{-1}\ mm$

d - mm

$$V_s = V_o = 418\,[S-1] \times d^2 \times \frac{3T+70}{100}$$

$$= 418\,[2.65-1] \times (0.04)^2 \times \frac{3\times 20+70}{100}$$

$V_o = 1.43 \times 10^{-6}\ m/sec = 1.43\ mm/sec$

$$V_o = \frac{Q}{A} \quad \therefore A = \frac{Q}{V_o}$$

$$= \frac{10\times 10^6 \times 10^{-3}}{24\times 3600}$$

$$\therefore A = \frac{10\times 10^6 \times 10^{-3}}{24\times 3600 \times 1.43\times 10^{-3}}$$

$A = 80.93\ m^2$

$$\therefore B^2 = \frac{80.93}{3} = 26.97$$

$\therefore B = 5.19m$ $\quad \therefore L = 15.58m$

$$\therefore D.T = \frac{V}{Q} = \frac{5.19\times 15.58\times 3.5}{\left(\frac{10\times 10^6\times 10^{-3}}{24}\right)} = 0.679\ hr$$

6.A rectangular sedimentation tank is to handle 10 MLD of raw water . A detention basin of width to length ratio 1/3 is proposed to trap all particles larger than 0.04mm in size. Compute the basin dimensions .If the depth of tank is 3.5 m ,calculate detention time . Assume relative density as 2.65 for the particles and 20 degree C as average temperature.

Ex.7

$H = 3m$, $A = 900 m^2$, $Q = 8000 m^3/d$

$T = 20°C$, $\mu = 10^{-3}$ kg/m.s, $\rho = 1000$ kg/m^3

$d = 0.01$ mm, $S = 2.65$

$$\nu = \frac{\mu}{\rho} = \frac{10^{-3}}{1000} = 1 \times 10^{-6} \text{ m/s}$$

$$V_s = V_o = \frac{g}{18}(SG-1)\frac{d^2}{\nu}$$

$$= \frac{9.81}{18}(2.65-1) \times \frac{(0.01 \times 10^{-3})^2}{1 \times 10^{-6}} = 8.99 \times 10^{-5} \text{ m/sec}$$

$$V_o = \frac{Q}{A} = \frac{\frac{8000}{24 \times 3600} \text{ m}^3/\text{s}}{900 \text{ m}^2} = 1.028 \times 10^{-4} \text{ m/s}$$

$$\eta = \frac{V_s}{V_o} \times 100 = \frac{8.99 \times 10^{-5}}{1.028 \times 10^{-4}} \times 100 = 87.5$$

Ex.7

An ideal horizontal flow settling basin is 3m deep having surface area 900m^2. Water flows at the rate of 8000m^3/d, at water temperature20^0 C ($\mu = 10^{-3}$ kg/m-s and ρ =1000kg/m^3). Assuming Stoke's law to be valid, the proportion (percentage) of spherical sand particles(0.01mm in diameter with specific gravity 2.65), that will be removed, is

Ex.8

$V_o = \frac{Q}{A} = \frac{Q}{L*B} = \frac{100000}{100 \times 50}$ $\frac{m^3/day}{m^2}$

$\therefore$ $V_o = 20 m^3/day/m^2$

For 100% removal: $V_s = V_o = 20 m^3/day/m^2 = 20 m/day$

$\therefore V_s = \frac{20}{24 \times 3600}$ m/sec.

$\rho_w = 1 gm/cc$
$= 1000 kg/m^3$
$= 1 t/m^3$

$\therefore \rightarrow S = \frac{\rho_p}{\rho_w} = \frac{2.65}{1} = 2.65$

$V_s = \frac{g}{18}(S-1)\frac{d^2}{\nu}$

$\frac{20}{24 \times 3600} = \frac{9.81}{18} * \frac{1.65 * d^2}{1.02 \times 10^{-2} \times \left(\frac{1}{100}\right)^2}$

$\therefore$ $d = 0.0162 mm$

Ex.8

. A plain sedimentation tank has dimensions 100 x 50 x 3 m and receives flow 1,00,000 m^3/d. Calculate the surface over flow rate in $m^3/m^2/d$ and dia of the smallest particle, which will be removed 100%. Take density of particle = 2.65g/cc, $\gamma = 1.02 \times 10^{-2} cm^2/sec$.

Ex.9

$L = 20\,m,\ B = 10\,m,\ H = 3\,m$

$Q = 4\,MLD,\ T = 20^{\circ}C,\ \mu = 1.002 \times 10^{-3}$

$\rho_w = 998.2\ kg/m^3,\ S = 2.65$

100% removal η.

$$\nu = \frac{\mu}{\rho} = 1 \times 10^{-6}\ m^2/s$$

$$\therefore V_0 = \frac{Q}{A} = \frac{4 \times 10^6 \times 10^{-3}}{20 \times 10}\ \frac{m^3/day}{m^2} = 20$$

(19) $V_s = V_0 = 20\ m^3/m^2/day$

$$20 = \frac{g}{18}(S_s - 1)\frac{d^2}{\nu}$$

$$20\ m^3/m^2/day = \frac{9.81}{18}(1.65) \times \frac{d^2}{1 \times 10^{-6}}\ \frac{m^2}{m^2/s}$$

$$\frac{20}{24 \times 3600} = \frac{9.81 \times 1.65 \times d^2}{18 \times 1 \times 10^{-6}}$$

$$d = 1.6 \times 10^{-5}\ m = 16 \times 10^{-3}\ mm$$

A plain sedimentation tank with a length of 20m, width of 10m, and a depth of 3m is used in a water treatment plant to treat 4 million litres of water per day (4MLD). The average temperature of water is 20^0C. The dynamic viscosity if water is 1.002×10^{-3} N.s/m^2 at 20^0 C. Density of water is 998.2 kg/m^3. Average specific gravity of particles is 2.65.

18. What is the surface overflow rate in the sedimentation tank?
   (a) 20m^3/m^2/day (b) 40m^3/m^2/day
   (c) 67m^3/m^2/day (d) 133m^3/m^2/day

19. What is the minimum diameter of the particle which can be removed with 100% efficiency in the above sedimentation tank?
   (a) 11.8×10^{-3}mm (b) 16.0×10^{-3}mm
   (c) 50×10^{-3}mm (d) 160×10^{-3}mm

Ex.10

$$\eta = \Sigma p_i * \frac{V_s}{V_o} \times 100$$

$V_o = 43.2 m^3/m^2.d$

$= 43.2 m/d$

$= \frac{43.2 \times 1000}{24 \times 3600}$

$V_o = 0.5 mm/sec$

Ex.10

. A horizontal flow primary clarifier treats wastewater in which 10%, 60% and 30% of particles have settling velocities of 0.1 mm/s, 0.2 mm/s, and 1.0 mm/s respectively. What would be the total percentage of particles removed if clarifier operates at a Surface Overflow Rate (SOR) of $43.2 m^3/m^2.d$?

5-11-14

Particle	% of particle	V_s (mm/sec)	V_o	$\eta_i = \frac{V_{si}}{V_o} \times 100$
1	10	0.1	0.5	20
2	60	0.2	0.5	40
3	30	1	0.5	100%

(200/100)

$$\therefore \eta_{overall} = \frac{10}{100} \times 20 + \frac{60}{100} \times 40 + \frac{30}{100} \times 100$$

$= 56\%$

Ex.12

$V_o = 30$ m/day
$S = 2.65$
$\mu = 0.001$ Ns/m^2
$\rho = 1000$ kg/m^3
$\therefore \nu = \frac{\mu}{\rho} = 1\times10^{-6}$ m/s

$\therefore V_o = V_s = \frac{g}{18}(S_s-1)\frac{d^2}{\nu}$

$\frac{30}{24\times3600} = \frac{9.81}{18}\times1.65\times\frac{d^2}{1\times10^{-6}}$

$\therefore d = 1.96\times10^{-5}$ m
$= 0.0196 \approx 0.02$ mm

Ex.12

A setting tank in a water treatment plant is designed for a surface overflow rate of 30 $\frac{m^3}{day.m^2}$. Assume specific gravity of sediment particles = 2.65, density of water (ρ)=1000 kg/m^3, dynamic viscosity of water(μ)= 0.001 N.s/m^2, and Stokes' lawisvalid. The approximate minimum size of particles that would be completely removed is:

Ex.13

$d = 0.10$ mm
$g = 9.8$ m/s^2
$S = 2.65$
$\nu = 1.0105\times10^{-2}$ cm^2/s
$= 1.0105\times10^{-2}\times10^{-4}$ m^2/s
$= 1.0105\times10^{-6}$ m^2/s

($\eta = 100\%$) $Q = 0.01$ m^3/s
$\rightarrow d = 0.06$ mm

$V_o = V_s = \frac{0.01}{1.B} = \frac{g}{18}(1.65)\times\frac{(0.06\times10^{-3})^2}{1.01025\times10^{-6}}$ m^3/sec, m/sec

$\therefore A = 3.121$ m^2

Ex.13

A suspension of sand like particles in water with particles of diameter 0.10 mm and below is falling into a sedimentation tank at 0.01 m3/s, g=9.8 m/s2. Specific gravity of particles=2.65, and kinematic viscosity of water = 1.0105×10–2 cm2/s. The minimum surface area (in m2 required for this setting tank to remove particles of size 0.06 mm and above with 100% efficiency is _____

(GATE-14)

Ex.14

Ex.14

A sedimentation tank (6m wide, 15m long and 3 m water depth) is treating 2 MLD of water.

1. The surface overflow rate is
   (a) 858 lit/hr/m^2 (b) 926 lit/hr/m^2
   (c) 1028 lit/hr/m^2 (d) None

2. Detention time is
   (a) 4.26 hr (b) 3.84 hrs
   (c) 4.8 hr (d) 3.24 hrs

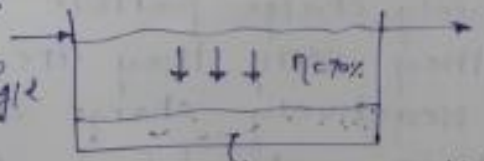
3. If 70 mg/L is the concentration of solids present in the turbid water, the amount of dry solids deposited in the tank per day with 70% tank efficiency is
   (a) 98 kg (b) 140 kg
   (c) 70 kg (d) 49 kg

(1) D=6m, L=15m, H=3m
Q=2 MLD

$V_o = \frac{Q}{A} = \frac{2\times10^6}{24\times15\times6}\ \frac{lit/hr}{m^2} = 926\ lit/hr/m^2$

(2) $D.T = \frac{V}{Q} = \frac{6\times15\times3\times24}{2\times10^6\times10^{-3}} = 3.24\ hr$

(3) $\eta = 70\%$



Conen settle = conen removed
$= C_{in}\times\eta$

Conen deposited = conen removed
$= C_{in}\times\eta$
$= 70\times\frac{70}{100} = 49\ mg/l$

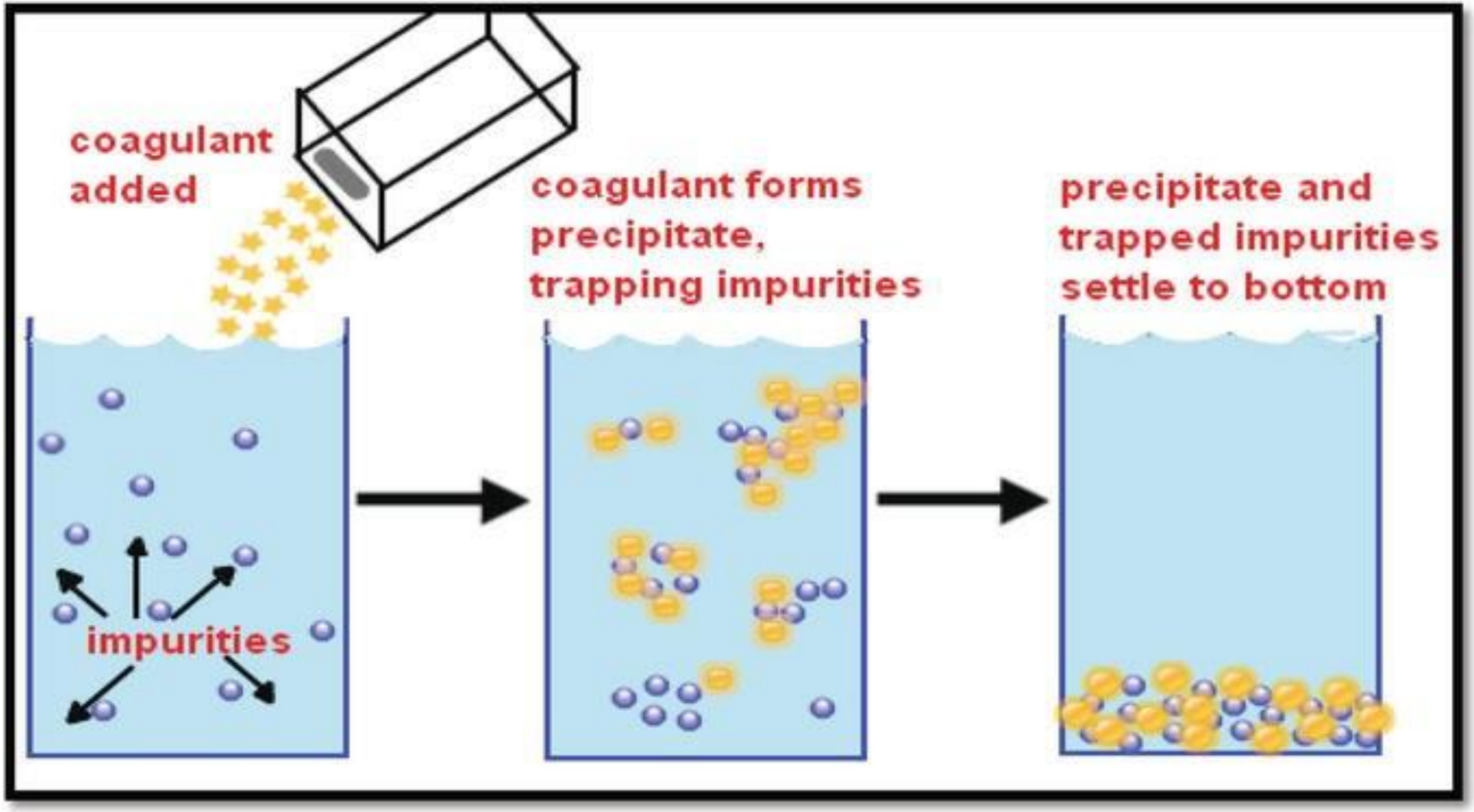
Total amount of any substance 'x' (kg/day)
$= Q \times$ conc. of 'x' $= 2\times49 = 98$ kg/day
(MLD) × (mg/L)
kg/day

kg/day = Q × dose of x
(MLD) × (mg/L)



Coagulation and Flocculation

Successive steps intended to overcome the forces stabilizing the fine suspended or colloidal particles, allowing particle collision and growth of floc.



Particles in Water

- *Dissolved Solids: < 1 nm (10^{-6} mm) in size*

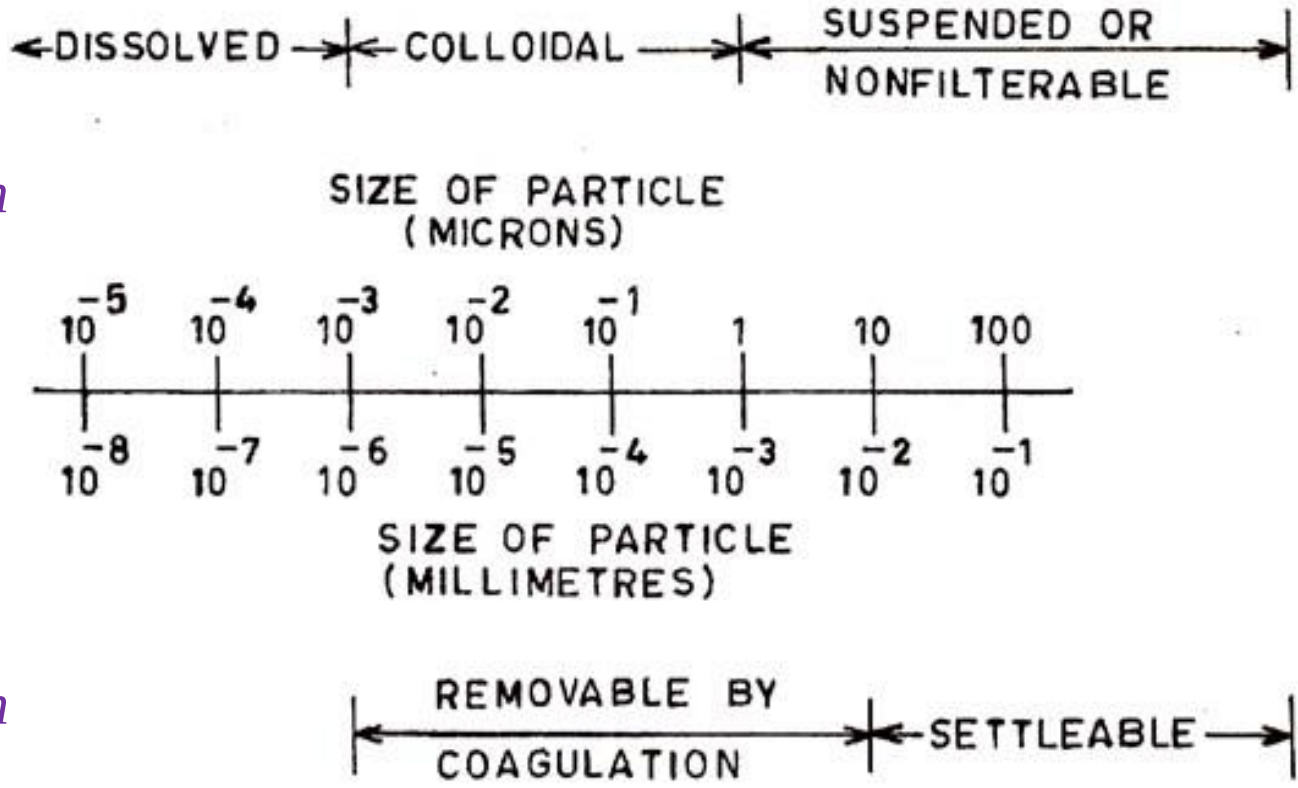
  Electrically charged and can interact with the water, so they are completely stable and will never settle out of the water. Not visible even with microscope.

- *Colloidal solids or Non-settleable solids: 1-1000 nm in size*

  Do not dissolve in water although they are electrically charged. Still, the particles are so small that they will not settle in water and cannot be removed by filtration alone. Can be seen only with a high-powered microscope.

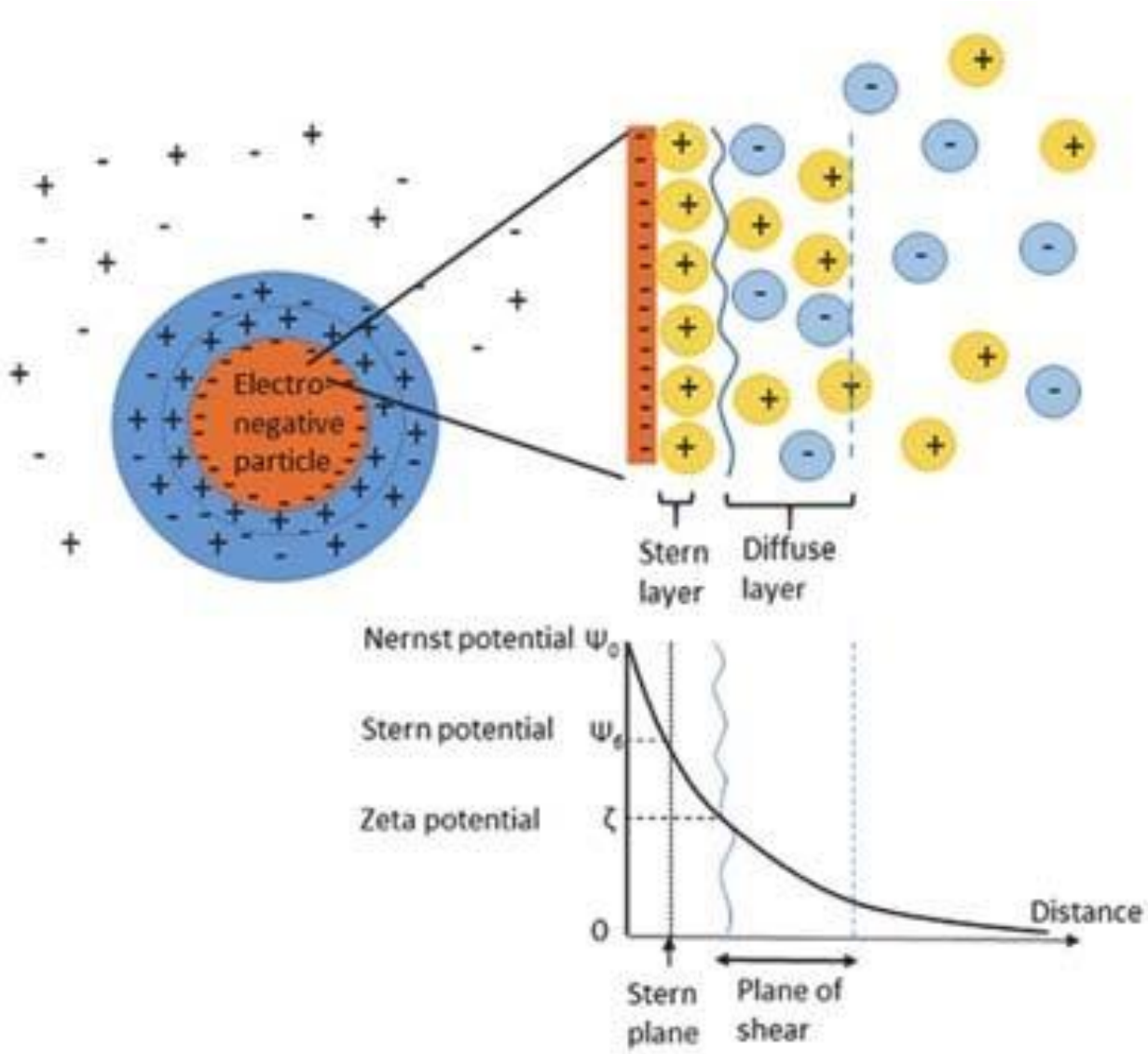
- *Suspended or settleable solids: > 1000 nm (10^{-3} mm) in size*

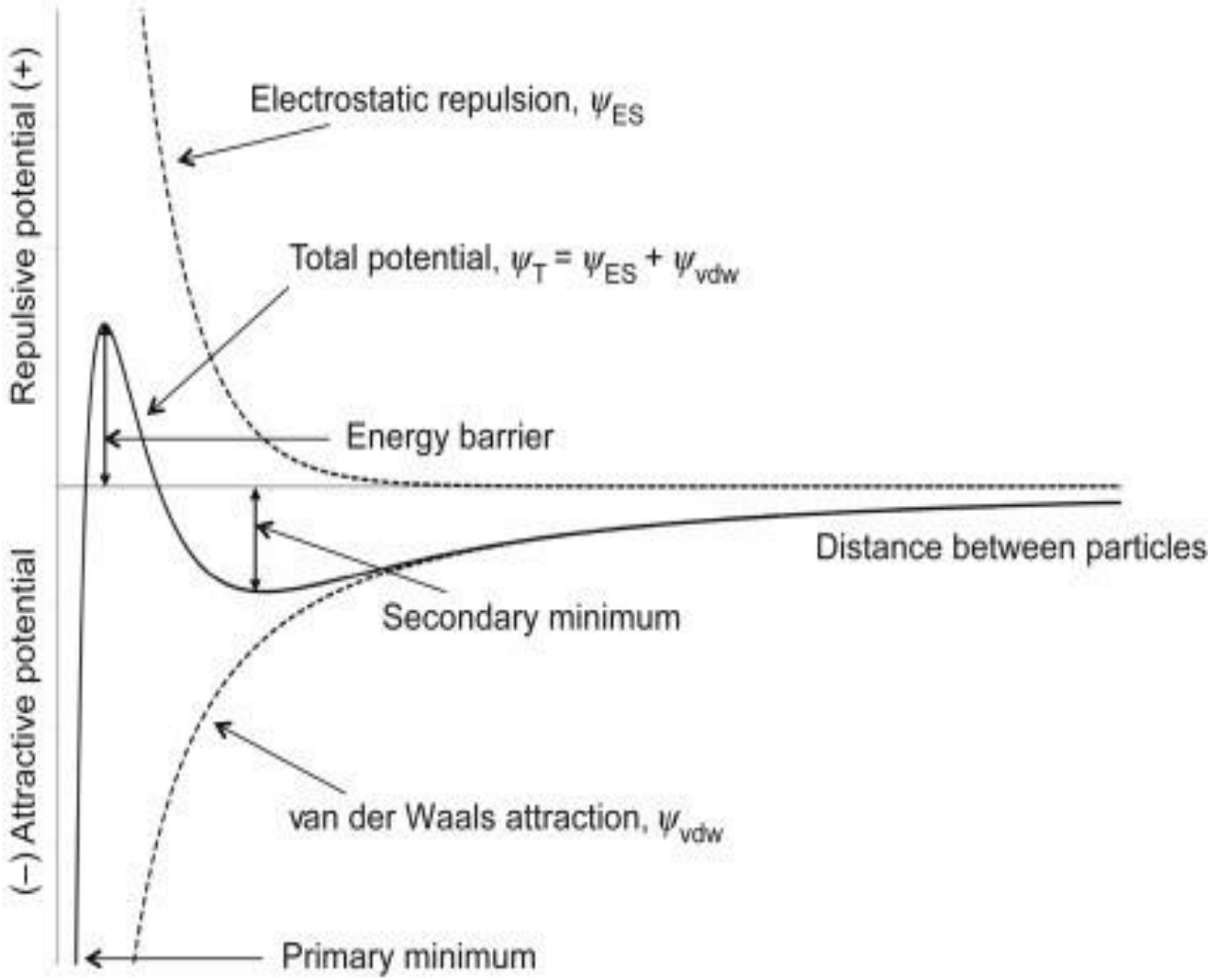
  Larger particle that can be seen through eyes. These are usually supported by buoyant and viscous forces in water and may settle (or float) in non-flowing water. Also, these can be removed by simple filtration.



Stability of Colloids

DLVO Theory





Settling Behavior of Particles in Water

- Colloidal impurities in surface waters cause the water to appear turbid or may impart colour.
- As, colloidal particles do not settle by gravity, they are difficult to separate from water through simple settling or filtration.

Diameter of Particle	Type of Particle	Settling time through 1 m. of water
10mm	Gravel	1 seconds
1mm	Sand	10 seconds
0.1mm	Fine Sand	2 minutes
10 micron	Protozoa, Algae, Clay	2 hours
1 micron	Bacteria, Algae	8 days
0.1 micron	Viruses, Colloids	2 years
10 nm	Viruses, Colloids	20 years
1 nm	Viruses, Colloids	200 years

- The most common approach adopted is to aggregate the colloids so that they grow in size and settle, which is commonly achieved through *coagulation and flocculation*.

Coagulation and Flocculation

Destabilization (or Coagulation)

Reduce the forces acting to keep the particles apart after they contact each other (i.e., lower repulsion forces).

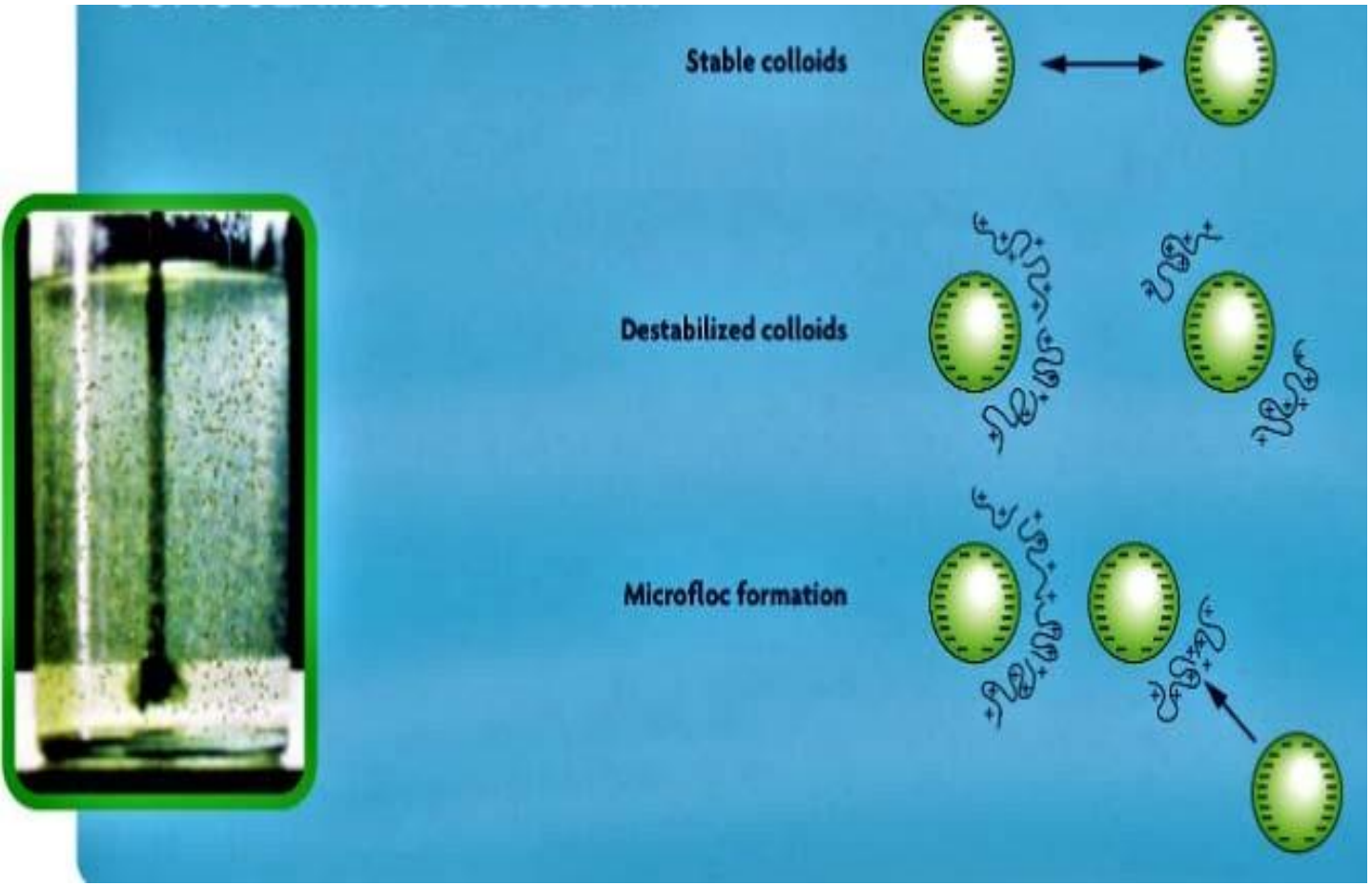
Chemical Addition, Rapid Mixing, “Pin-point” Floc Formation

Flocculation

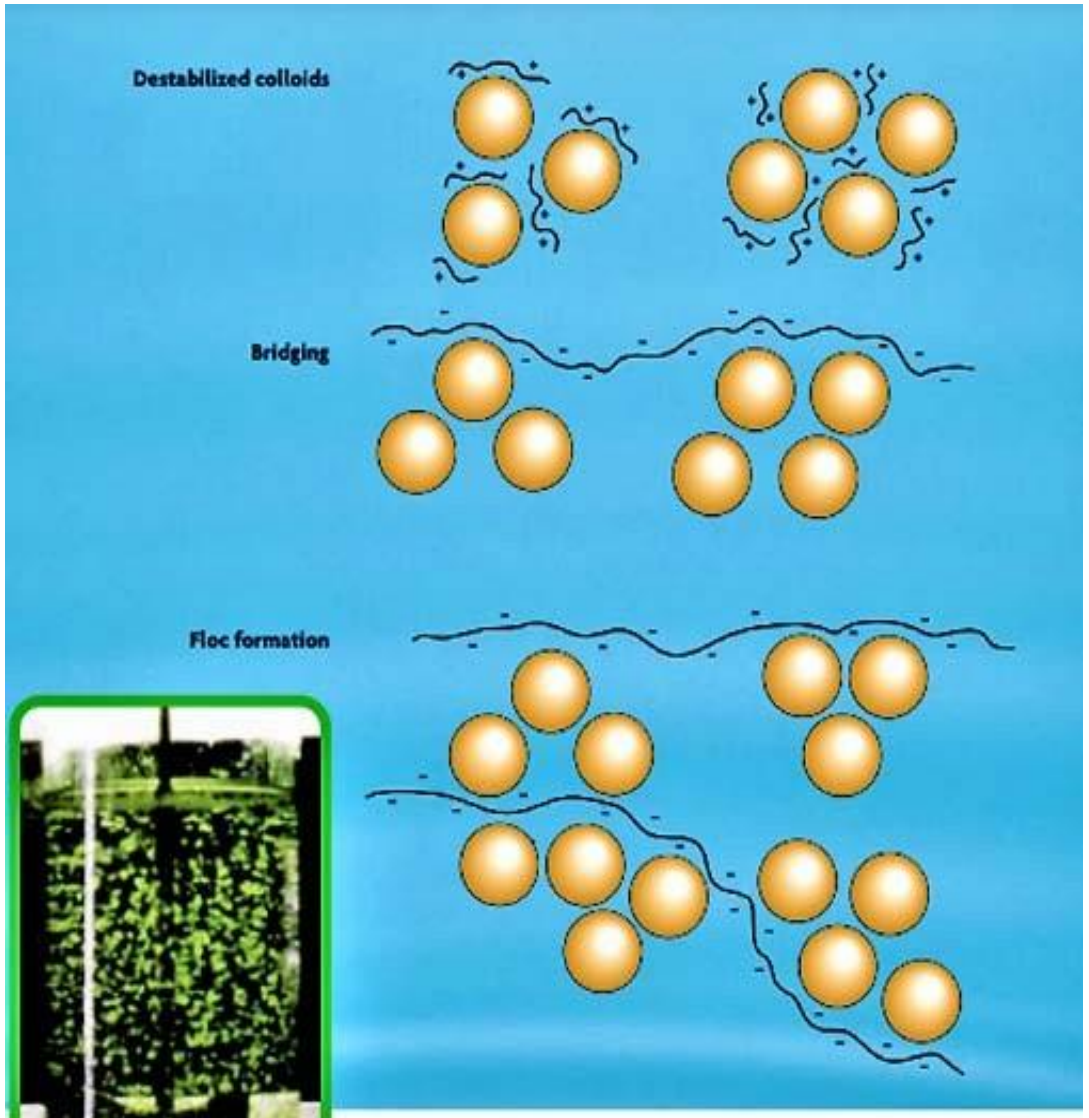
Process of bringing destabilized colloidal particles together to allow them to aggregate to a size where they will settle by gravity.

Slow Mixing, Floc Growth, Increased Diameter

Coagulation and Flocculation Theory



Coagulation

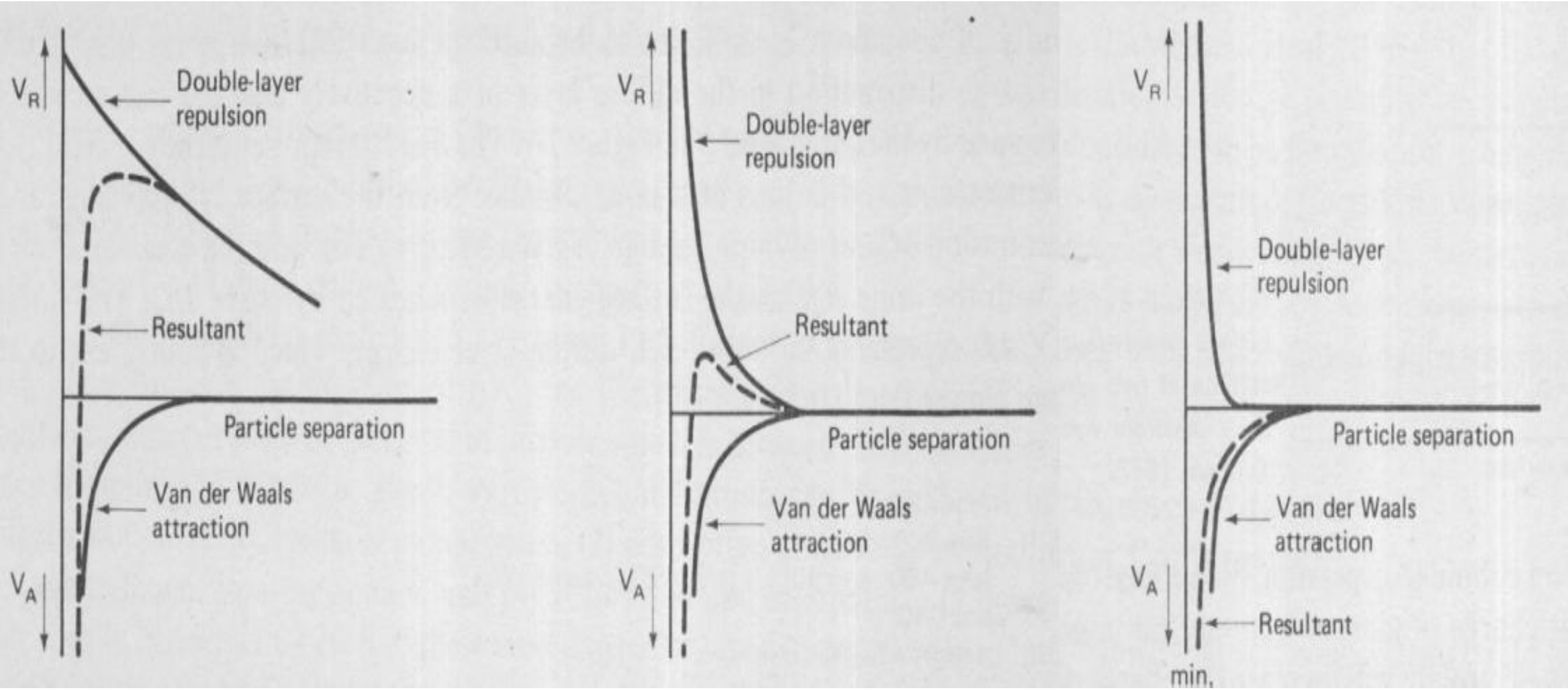




Flocculation

Coagulation and Flocculation: Mechanisms of Action

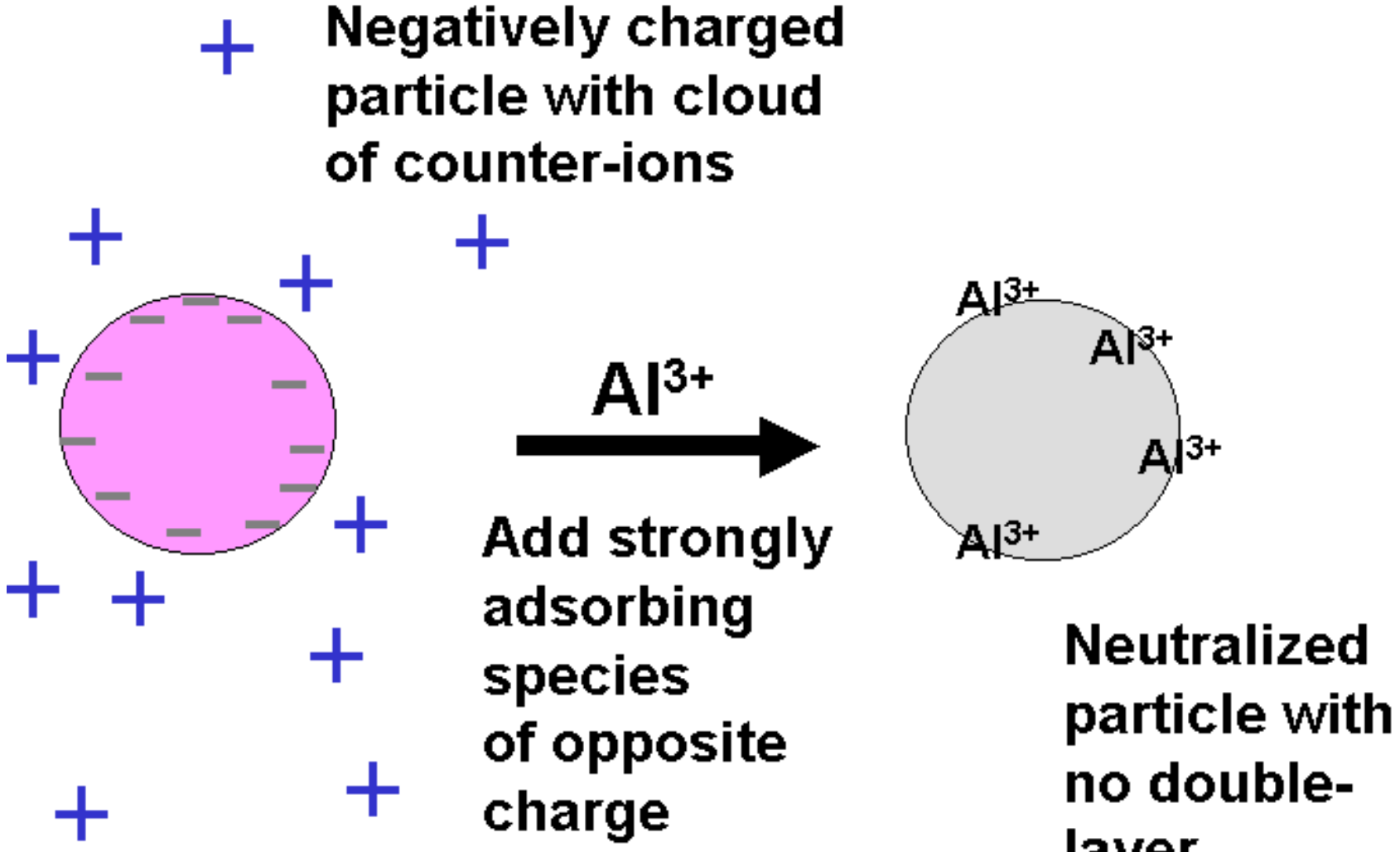
Electrostatic Charge Reduction through Double Layer Compression



Normal Double Layer *Intermediate Electrolyte dose* *High Electrolyte does*

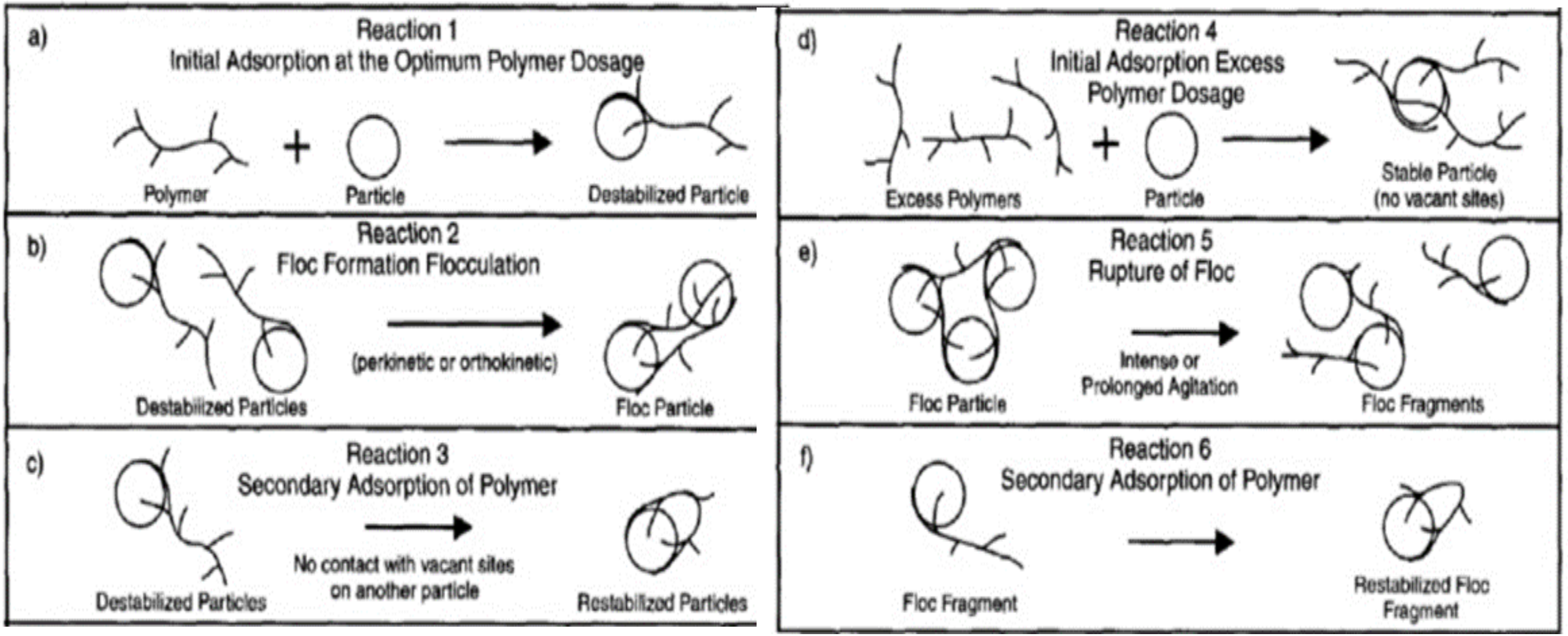
Coagulation and Flocculation: Mechanisms of Action

Adsorption and Charge Neutralization



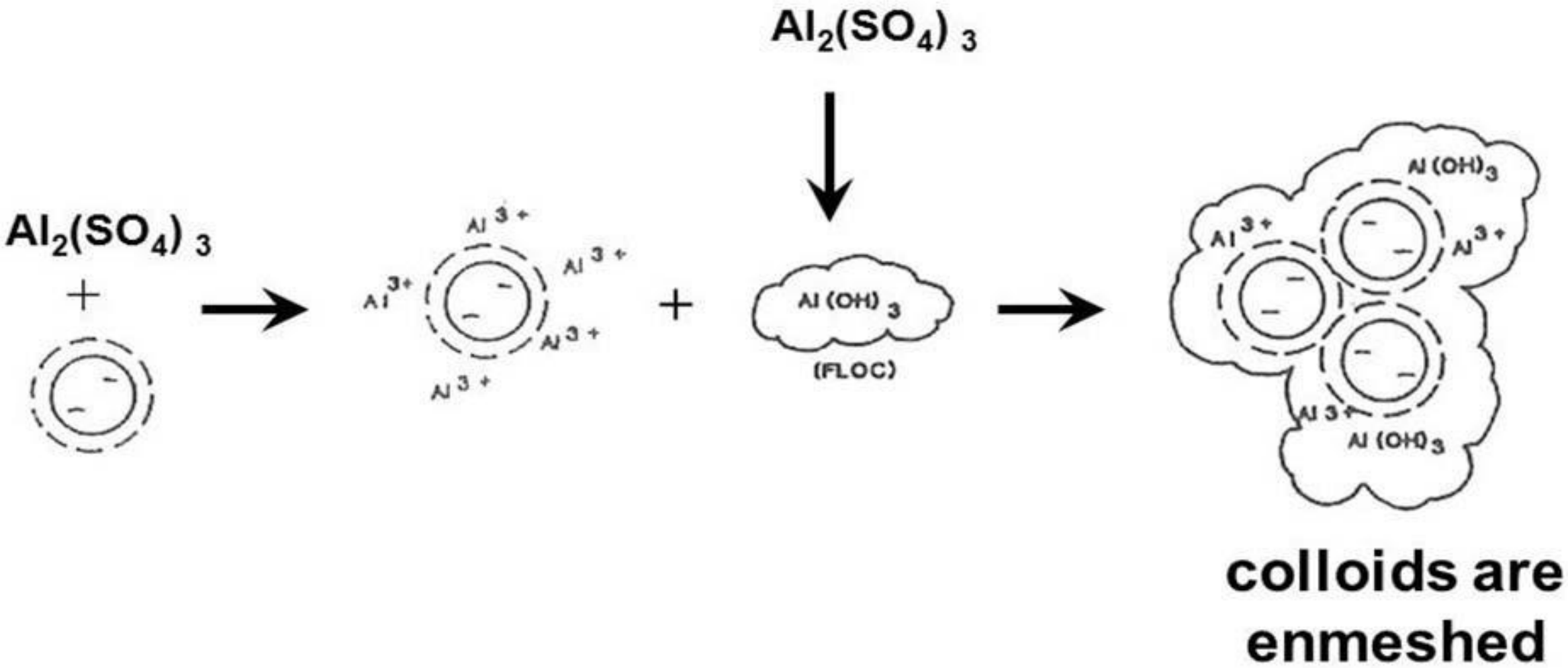
Coagulation and Flocculation: Mechanisms of Action

Adsorption and Inter-particle Bridging



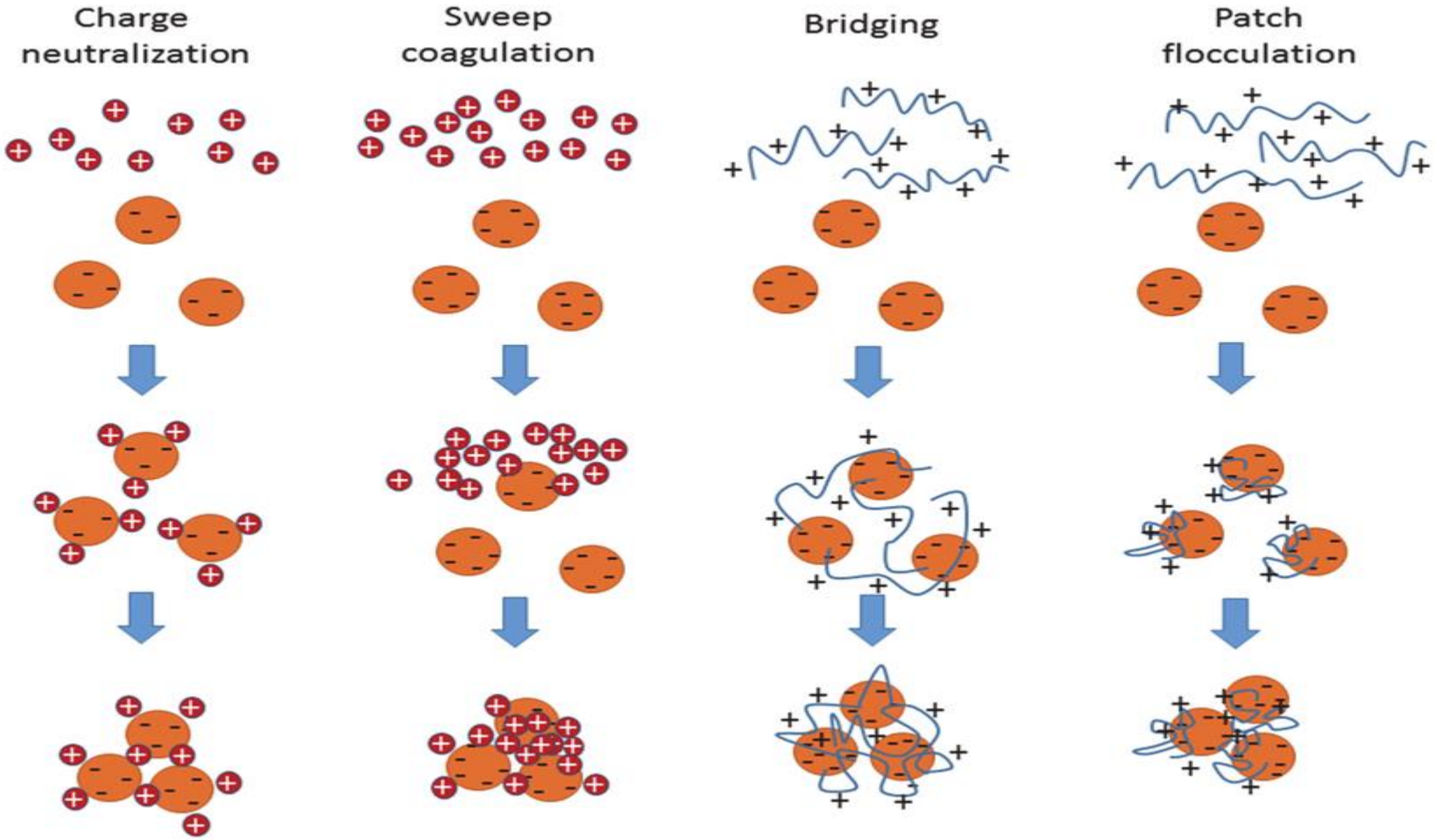
Coagulation and Flocculation: Mechanisms of Action

Entrapment in a Precipitate (Sweep Floc Formation):



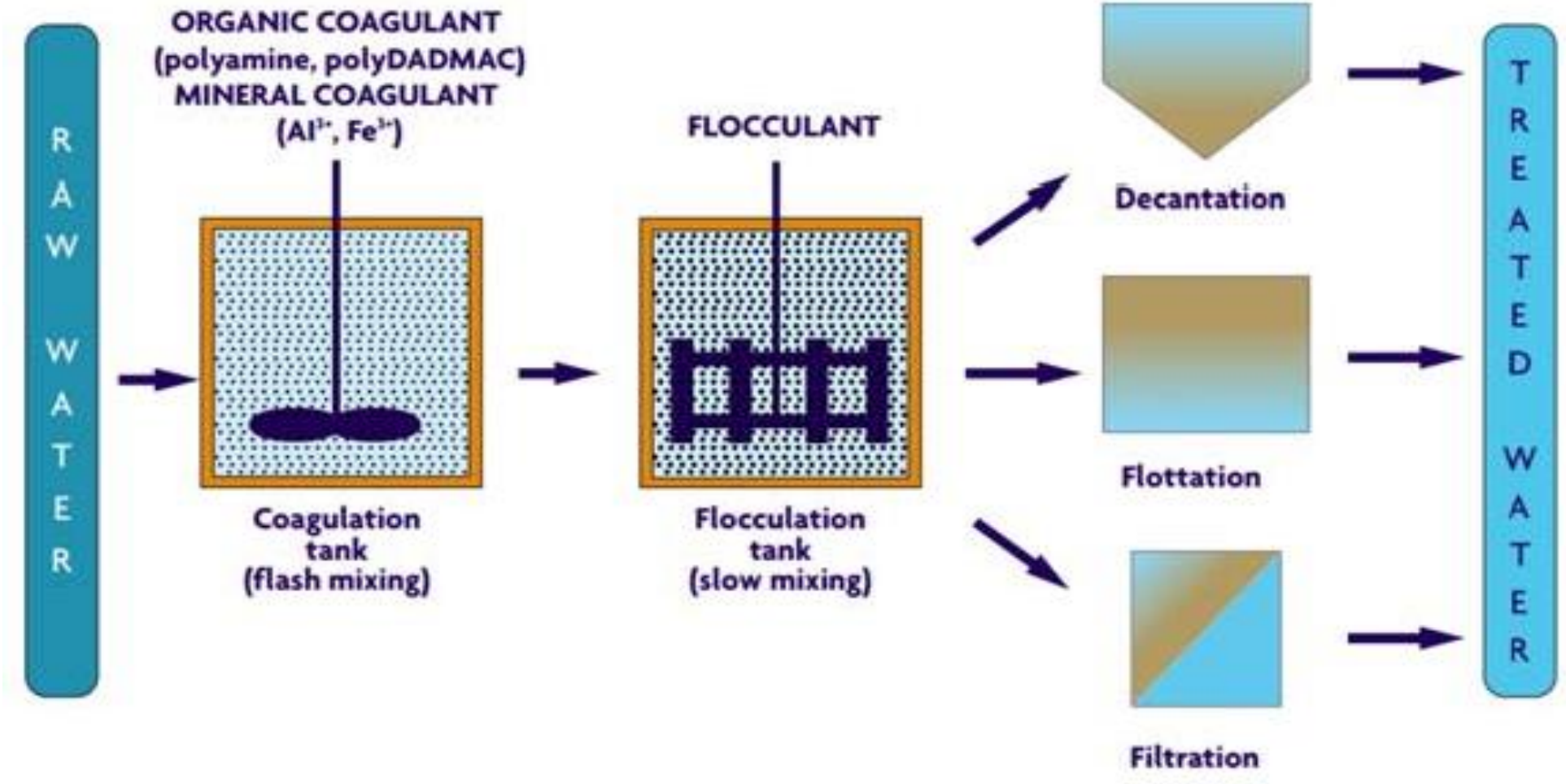
colloids are enmeshed

Coagulation and Flocculation: Mechanisms of Action



Coagulation and Flocculation Steps

1. Selection of Suitable Coagulant
2. Finding Optimum Dose of Coagulant
3. Addition of Coagulant and rapid mixing
4. Allowing floc formation through slow mixing
5. Separation of flocs from water through settling/flotation/filtration



Selection of Coagulant

Required Basic Characteristics:

Nontoxic at the working dosage; **High charge density;** **Insoluble in the neutral pH range**

Aluminum and iron salts are the most commonly used coagulants in water treatment:

Aluminium coagulants include:

Aluminium sulfate (Alum)
Aluminium chloride
Sodium aluminate
Polyaluminum Chloride (PAC)

Iron coagulants include:

Ferric sulfate
Ferrous sulfate
Ferric chloride
Ferric chloride sulfate

Other coagulants:

Organic coagulants, polyelectrolytes, hydrated lime, magnesium carbonate and various polymers etc.

Organic Coagulants *vs* Inorganic Coagulants

Organic coagulants

Generally used for solids & liquids separation and sludge generation. Polyamines function by charge neutralization alone, and are **effective at treating higher turbidity raw water** and wastewater. Melamine Formaldehydes and Tannins coagulate the colloidal material in the water, **as well as absorb organic materials such as oil and grease**. These are ***particularly well suited to operations that generate hazardous sludge***.

Inorganic coagulants

These are mostly Al or Fe based, and are **both cost-effective and applicable for a broad variety of water and wastewater**. Inorganic coagulants are **particularly effective on raw water with low turbidity** and will often treat this type of water when **organic coagulants cannot**.

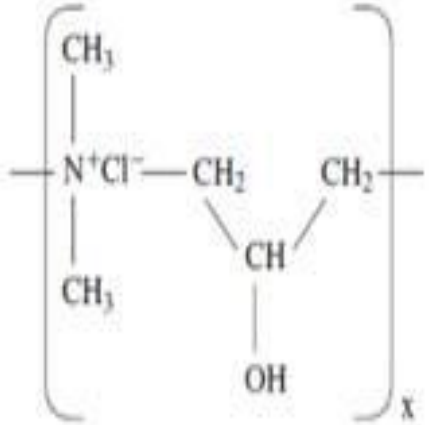
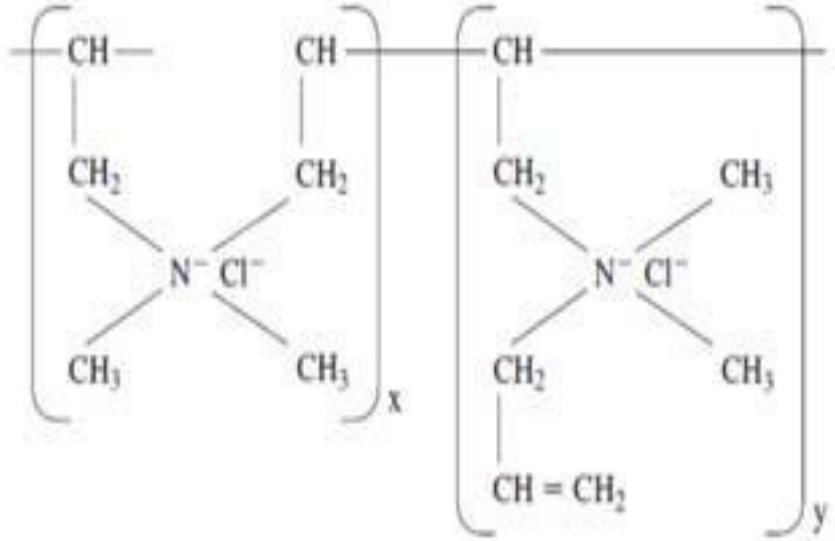
Advantages of alum are that it readily dissolves with water, and does not cause the unsightly reddish brown staining of floors, walls and equipment like ferric sulphate. However, it is effective only at certain pH range, and good flocculation may not be possible with alum in some waters. With ferric sulphate, coagulation is possible at pH values as low as 4.0, and the floc formed is heavier than alum floc, as well as does not redissolve at high pH values.

Some Common Organic and Inorganic Coagulants

Popular Inorganic Coagulants:

Coagulant	Chemical formula	Molecular weight, g/mole	Remarks
Aluminum sulfate	$Al_2(SO_4)_3 \cdot 14H_2O$	594	Hg contamination may be of concern
Sodium aluminate	$Na_2Al_2O_4$	164	Provides alkalinity and pH control
Aluminum chloride	$AlCl_3$	133.5	Used in blends with polymers
Polyaluminum chloride	$Al_w(OH)_x(Cl)_y(SO_4)_z$	Variable	"PACl" used when Hg contamination is a concern
Polyaluminum sulfate	$Al_w(OH)_x(Cl)_y(SO_4)_z$	Variable	"PAS" used when Hg contamination is a concern
Polyiron chloride	$Fe_w(OH)_x(Cl)_y(SO_4)_z$	Variable	
Ferric chloride	$FeCl_3$	162.5	
Ferric sulfate	$Fe_2(SO_4)_3$	400	

Popular Organic Coagulants:

Coagulant	Chemical formula
Epichlorohydrin dimethylamine (epi-DMA)	
Polydiallyl dimethyl ammonium chloride (poly-DADMAC)	

Advantages and Disadvantages of Common Inorganic Coagulants

Name	Advantages	Disadvantages
Aluminiumsulphate (Alum) $Al_2(SO_4)_3 \cdot {}_{18}H_2O$	Easy to handle and apply; most commonly used; produces less sludge than lime; most effective between pH 6.5 and 7.5	Adds dissolved solids (salts) to waster; effective over a limited pH range.
Sodium Aluminate $Na_2Al_2O_4$	Effective in hard waters; small dosage usually needed	Often used with alum; high cost; ineffective in soft waters
Polyaluminium Chloride (PAC) $Al_{13}(OH)_{20}(SO)_4.Cl_{15}$	In some applications, Floc, formed is more dense and faster settling than alum	Not commonly used; little full scale data compared to other aluminium derivatives
Ferric Sulfate $Fe_2(SO_4)_3$	Effective between pH 4-6 and 8.8-9.2	Ads dissolved solids(salts) to water; usually need to add alkalinity
Ferric Chloride $FeCl_3.6H_2O$	Effective between pH 4 and 11	Adds dissolved solids (salts) to water; consumes twice as much alkalinity as alum
Ferrous Sufate (Copperas) $FeSO_4 .7H_2O$	Not as pH sensitive as lime	Ads dissolved solids(salts) to water; usually need to add alkalinity
Lime $Ca(OH)_2$	Commonly used; very effective; may not add salts to effluent	pH dependent; produces large quantities of sludge; overdose can result in poor effluent quality



Coagulant Aids

- In some waters, even large doses of primary coagulant fails to produce a satisfactory floc. In such cases, the coagulation process is often enhanced through the use of coagulant aids. Coagulant aids also help to create satisfactory coagulation over a broader pH range.

- Insoluble particulate materials such as clay, sodium silicate, pure precipitated calcium carbonate, diatomite, and activated carbon are common coagulant aids. They are used in waters that have low concentrations of particles (few nucleating sites). Because their density is higher than most floc particles, floc settling velocity is increased by the addition of such coagulant aids.

- Polymeric coagulant aid those help in bridging small floc to agglomerate rapidly into larger and denser floc are also used to reduce the amount of primary coagulant required. These are usually slightly anionic polyacrylamides with very high-molecular weights. In some studies, non-ionic or cationic types have also been proven effective. Synthetic organics such as anionic polyelectrolyte, and natural organics such as starch, starch derivatives, proteins, and tannins have been used as coagulant aids.
- The coagulant add dosage must be carefully controlled to avoid lowering the water quality.

Coagulant Doses: Zones of Effectiveness

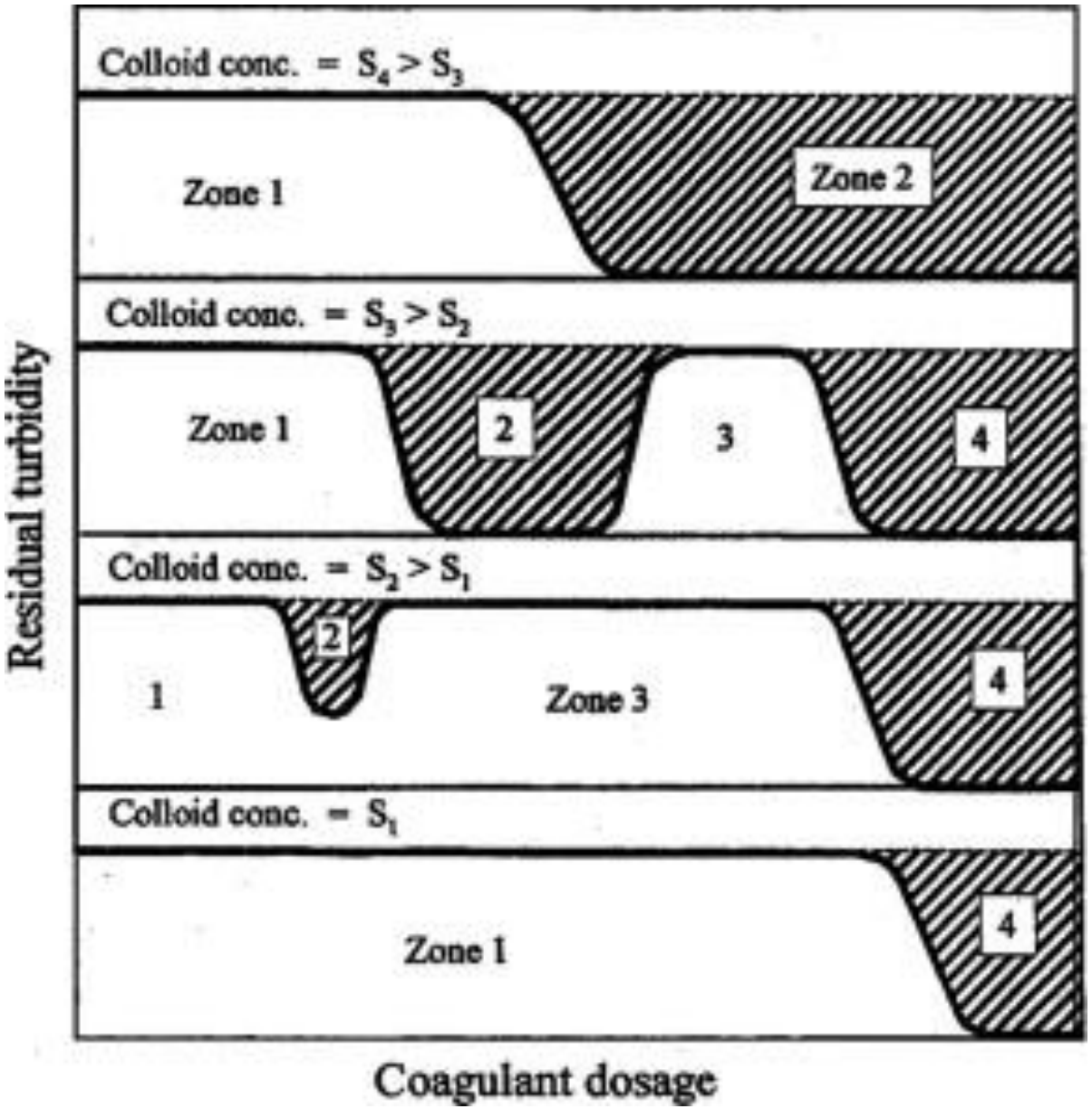
Zone 1: Low dosage, insufficient coagulant added to produce destabilization.

Zone 2: Dosage sufficient to cause efficient and rapid destabilization

Zone 3: Dosage high enough to cause restabilization (charge reversal or polymer –foldback)

Zone 4: Dosage high enough to get sweep floc which results in good destabilization.

Colloid concentration expressed in terms of surface area $S_1 < S_2 < S_3 < S_4$.

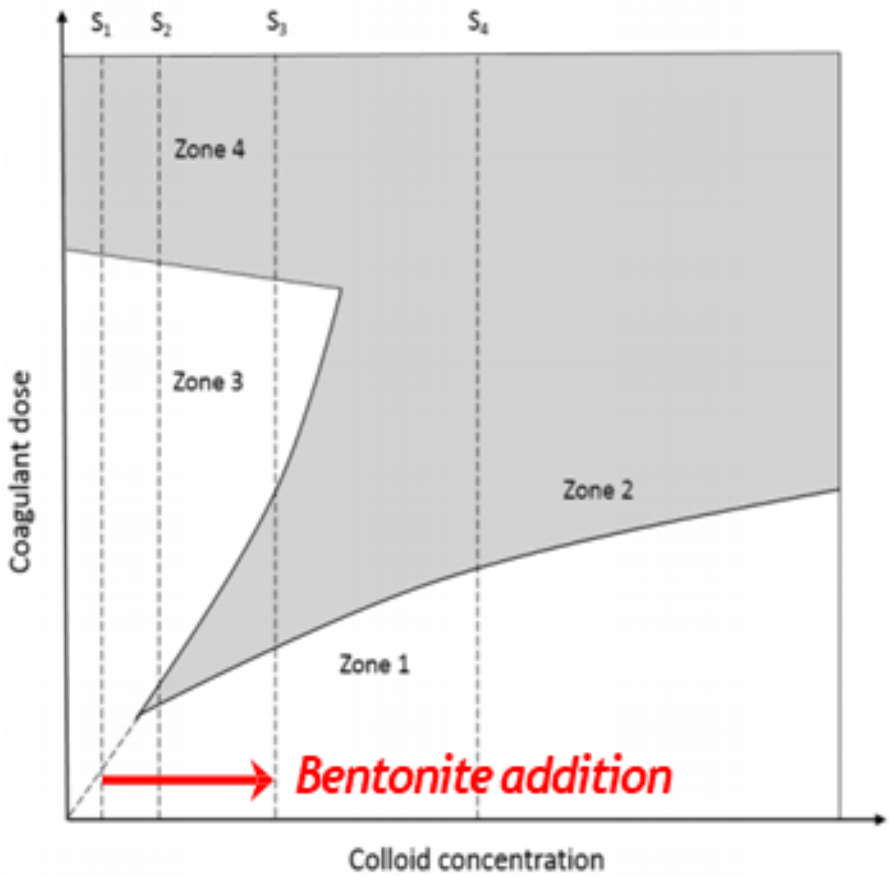


Coagulant Doses based on Colloid Concentrations

At very low S, there is not enough particle-particle interaction to produce good destabilization and provide nuclei for the sweep floc to form. Therefore need high dosages. Restabilization is not a concern.

At S2 and S3 levels, there is sufficient colloid concentration for destabilization before sweep floc is reached. It's easier to restabilize at lower S.

At very high S, lots of coagulant is needed for colloids. Generally, don't get sweep floc because much of the coagulant is used by the colloids.



Zone 1 – Insufficient coagulant dose applied to suspension.

Zone 2 – Destabilization through charge neutralization.

Zone 3 – Restabilization due to coagulant overdose. Particle surfaces become positive.

Zone 4 – Sweep flocculation due to oversaturation of coagulant.

Image Source: J.C. Crittenden, R.R. Trussell, and D. W. Hand, MWH's Water Treatment: Principles and Design, 3rd ed, Wiley, 2012

Coagulation Practices based on Colloids and Alkalinity Levels

1. *High Colloid, low alkalinity:* The strategy here is to add coagulant without worrying about pH. The lower pH is better because destabilizing is by charge neutralization. Generally, there is no concern with overdosing because the colloidal surface area is too large.
2. *High colloid concentration, high alkalinity:* The choices are to destabilize by adsorption/charge neutralization at neutral pH (a larger dose at higher pH), or add acid to lower pH. Economics dictate choices.
3. *Low colloid concentration , high alkalinity:* For this case we can either destabilize by high dosage to give sweep floc or we can add coagulant aid such as bentonite (aluminium phyllosilicate clay) to get destabilization at lower dosage.
4. *Low colloid concentration, low alkalinity:* This is the most difficult case and generally requires added alkalinity or collides. Sweep floc is difficult to form as pH drops and it's easy to overdose at low pH and low colloid concentration.

Coagulant Dose Optimization in Laboratory: *Jar Test*

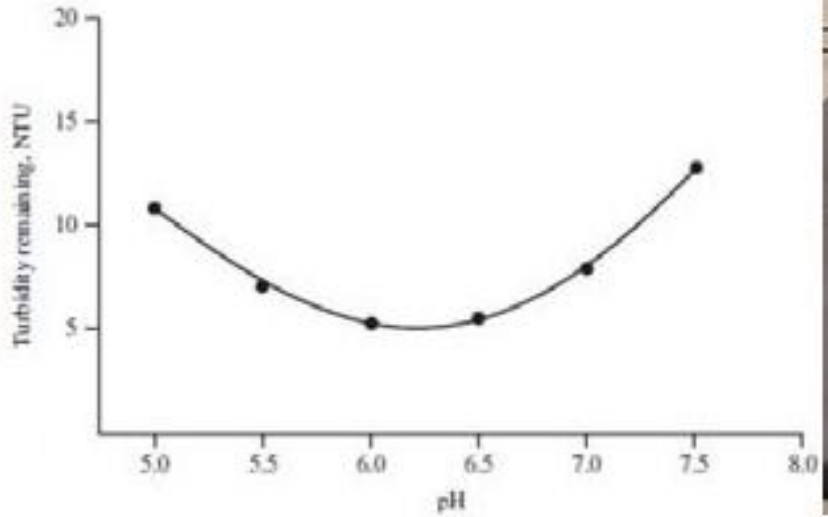
In practice, irrespective of what coagulant or coagulant aid is used, the optimum dose are usually determined by a *Jar Test*. A typical Jar Test apparatus consists of four to six beakers of 1-2 L volume, provided with a variable-speed stirrer.

[Procedure] Beakers filled with the raw water and varying amounts of coagulant dose are administered. The contents are rapidly mixed for about a minute and then allowed to flocculate at a slower pre-worked speed (*usually 20-30 rpm*) for desired time (*usually 15-30 mins*). Thereafter, the contents are allowed to settle for desired time (*usually 20-40 mins*), and the optimum dose is determined based on the measured turbidity of supernatant water (alternatively, judgement may be made based on visual inspection).



Image Source: https://www.researchgate.net/figure/Jar-test-for-high-turbidity-river-water_fig3_242313845

Coagulant Dose Optimization in Laboratory: *Jar Test*



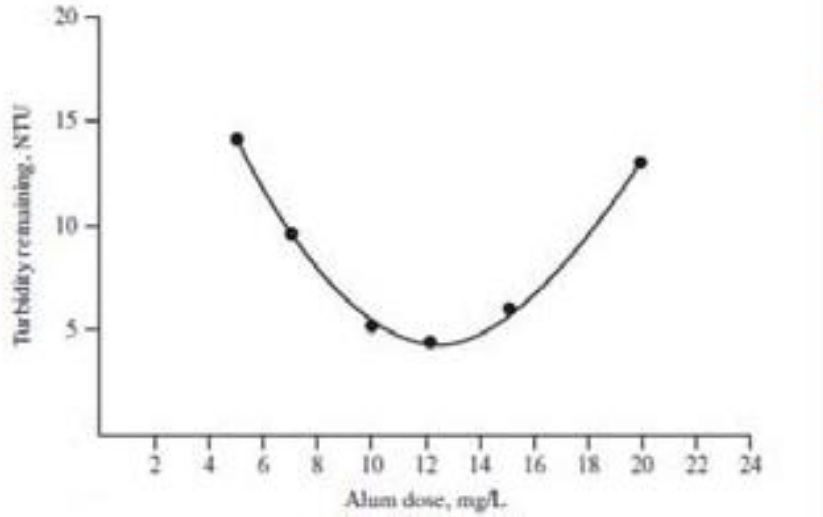




Image Source: https://readcivil.com/determination-optimum-coagulant-content-jar-test/; http://hlth.bpums.ac.ir/UploadedFiles/CourseFiles/COAGULATION_AND_FLOCCULATION-5_a6eff65e.pdf

Jar Test may also be used to optimize:

- *pH*
- *Mixing Speed for Flocculation*
- *Flocculation time*

Importance of Optimum Coagulant Dosing

Coagulant over-dosing may leads to

- Increased treatment costs
- Restabilization of colloids
- Increases sludge mass
- Public health concerns

Coagulant under-dosing may leads

- Lesser degree of removal
- Failure to meet the water quality targets

Coagulation and Flocculation Units

Coagulation:

Rapid Mix / Flash Mix

Flocculation:

Flocculation Basin

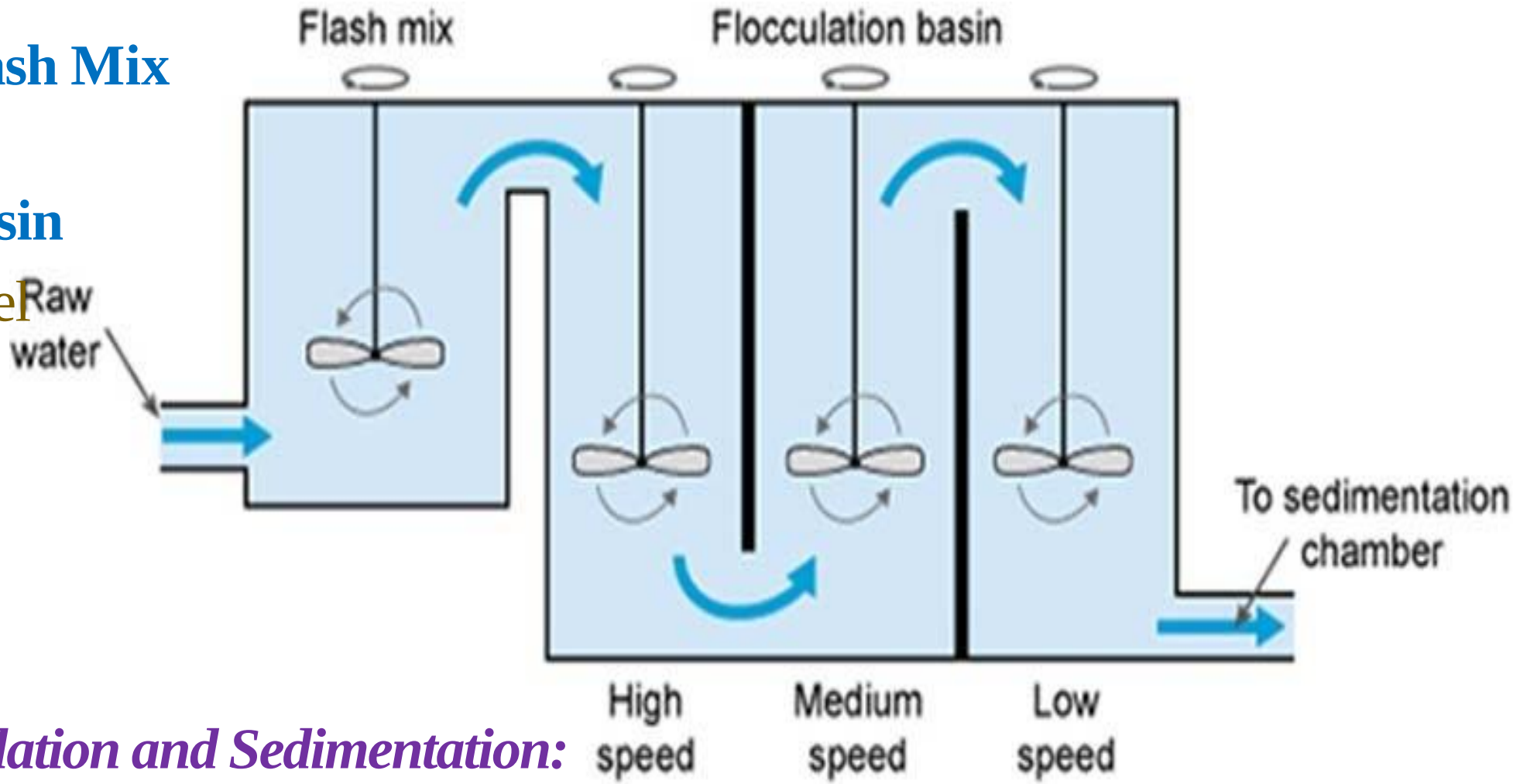
Baffled Channel

Flocculators

Paddle

Flocculators

(mechanical)



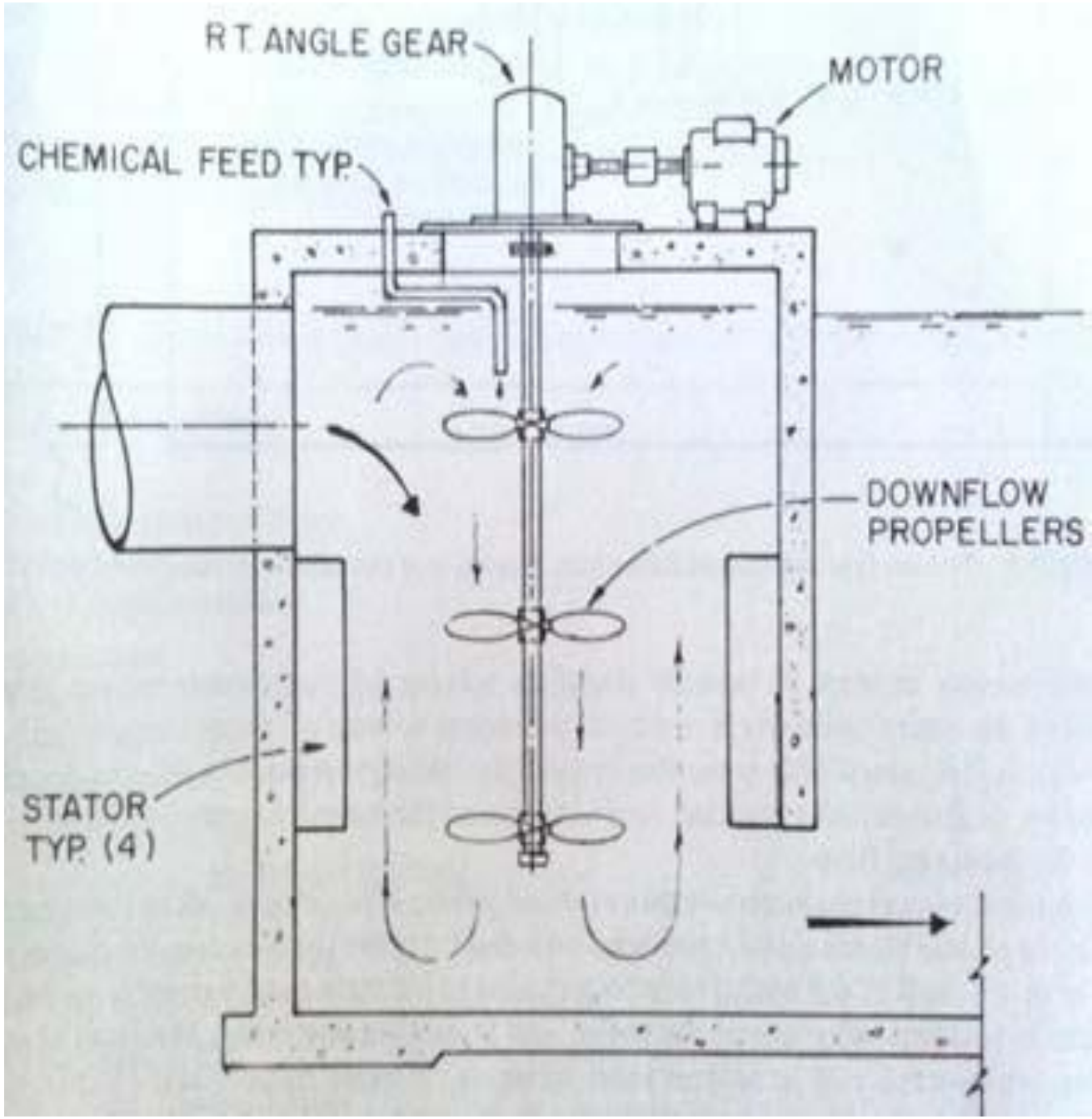
Combined Flocculation and Sedimentation:

Clariflocculator

Rapid (Flash) Mixing Unit

- The coagulation requires the addition of a coagulant, which should be very well mixed with water to produce homogeneous mixture of the influent water and the coagulant.
- This mixing is achieved in a tank called Rapid Mixer (or Flash Mixer).
- The most common mixers used in the coagulation tank are mechanical (turbine, propeller, or paddle) type mixers.
- It usually has a square or circular cross section

to achieve **best mixing efficiency.**



Design of Rapid (Flash) Mixing Unit

- Tank Volume, $V\,(m^3) = Q.t$; where, Q is design flow in m^3/S, and t is detention time in seconds.
- The detention time in the rapid mixer is in the range of 20-60 seconds. This short time is enough to achieve complete mixing of the coagulant and to complete the coagulation process.
- Water depth is usually taken as 1.5 times the width (for square) or diameter (for circular) of the tank.
- Power Requirements: $P = \mu V G^2$

  where, P = power transmitted by the mixer, N.m/s (Watt)
  V = tank volume, m^3; G = velocity gradient, S^{-1}
  μ = dynamic viscosity of water, $N.s/m^2$
- The velocity gradient G is defined as the relative velocity between two particles in water divided by the distance between them. For example, if two particles are 1 cm apart and the relative velocity between the is 10 m/s, then G = 10 (m/s)/0.01(m) = 1000 S^{-1}

Detention Time, Velocity Gradient and Power Required in Flash Mixer

Detention Time	Velocity gradient	Net Power input per unit volume	Net Power input per unit discharge
S	s^{-1}	watts/m^3 of volume	watts/m^3 of flow/hr
60	300	72	1.2
50	360	104	1.4
40	450	162	1.8
30	600	288	2.4
25	720	415	2.9
20	900	648	3.6

Note: Power calculations are based on water temperature of 30°C ($\mu = 0.8 \times 10^{-3}$ N.S./m^2)

As per CPHEEO

Source: CPHEEO Manual on on Water Supply and Treatment (1999)

Detention Time, S	Velocity Gradient, S^{-1}
20	1000
30	900
40	800
50 or more	700

Other Sources

Slow Mixing Unit: Flocculator

- The objective of flocculation is to provide increase in the number of contacts between coagulated particles by gentle and prolonged agitation to increases the particle size from microfloc to larger, visible flocs.
- The microflocs are brought into contact with each other through the process of slow mixing. Thus, the flocculation requires slow mixing which is typically achieved in gravitational, hydraulic or mechanical flocculators.
- The common types of flocculator are *horizontal or vertical flow baffled flocculators* or *horizontal or vertical shaft paddle flocculators.*

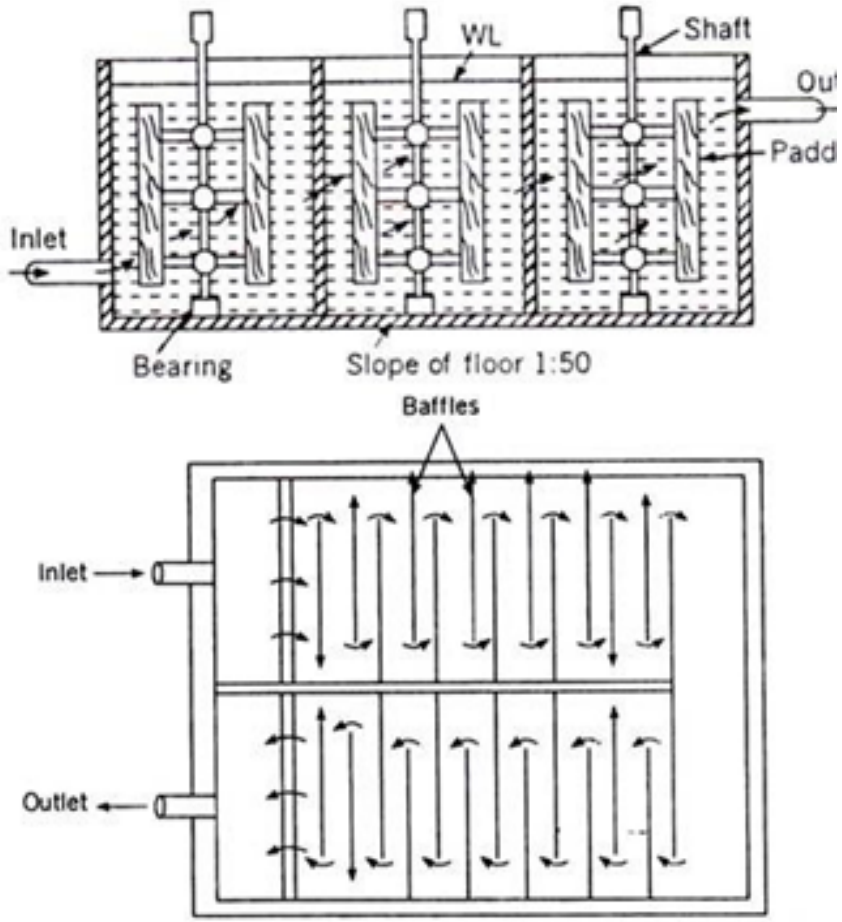


Image Source: http://www.engineeringenotes.com/essay/water-engineering/water-treatment-water-engineering/essay-on-sedimentation-with-coagulation-water-treatment-water-engineering/16487

Design Criteria for Flocculation Units

- Design procedure is similar to rapid mix unit. Velocity gradient (G) and time of mixing (t) are the important factors in design of flocculation tank, as the total number of particle collision is indicated as a function of the product of the velocity gradient and detention time (Gt).

- Gt = 10^4- 10^5 (higher the Gt, larger the number of collisions)

- G= 10-75 s^{-1} [*Too Low G adequate collision will not occur, thus, proper floc will not formed*]
  (light / dense floc) [*Too HighG shear forces will prevent the formation of large floc*]

- t= 10-30 min *(contact opportunity in the basin)*

- Mixing in a flocculator basin targets hydraulic flow regime approaching complete mix condition. Plug- flow conditions are desirable to minimize short-circuiting (a portion of the incoming flow traverses the chamber in a much shorter time than the nominal detention period).

Tapered Flocculation

- Generally: *Large G at low t → small but dense flocs*, whereas *Low G at high t → larger but lighter flocs*
- However, objective is to get *large and dense flocs*. Tapered flocculation helps in that.
- In Tapered Flocculation, two or more basins are used in series promoting plug flow and allowing G value to be decreased from one compartment to next along the length.
  - Higher G at inlet → Better mixing at inlet to enhance aggregation → Small, dense flocs
  - Lower G at outlet → Promotes larger flocs by reducing mixing and shear
    → Small dense flocs combine to form larger flocs

In Tapered Flocculation:

- ❑ Typical series of G: 80, 40, 20 s^{-1}
- ❑ The compartments are often separated with a baffle.
- ❑ Variable speed motors are used for mechanical mixing flocculators
- ❑ Baffles spacing are changed (increased) in gravity flocculators

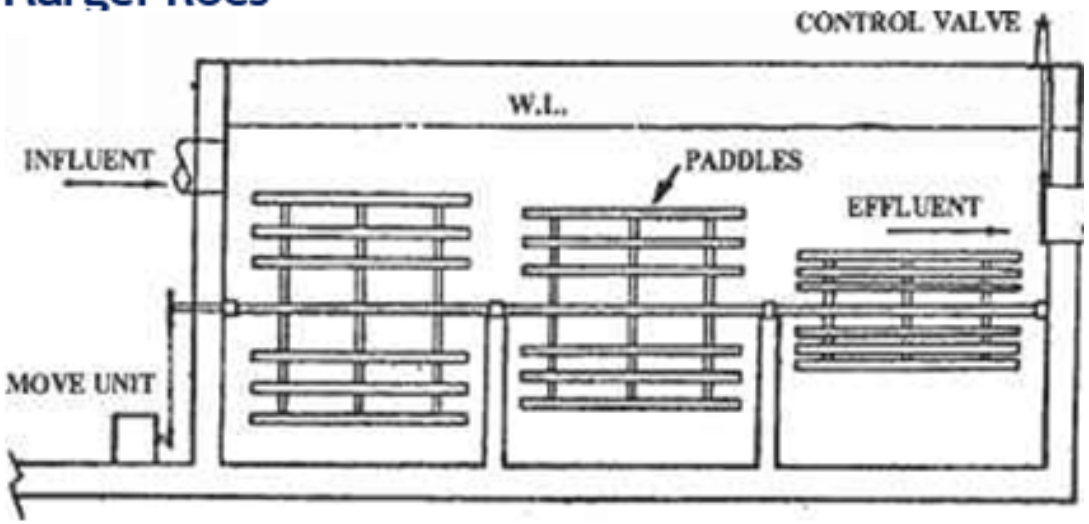


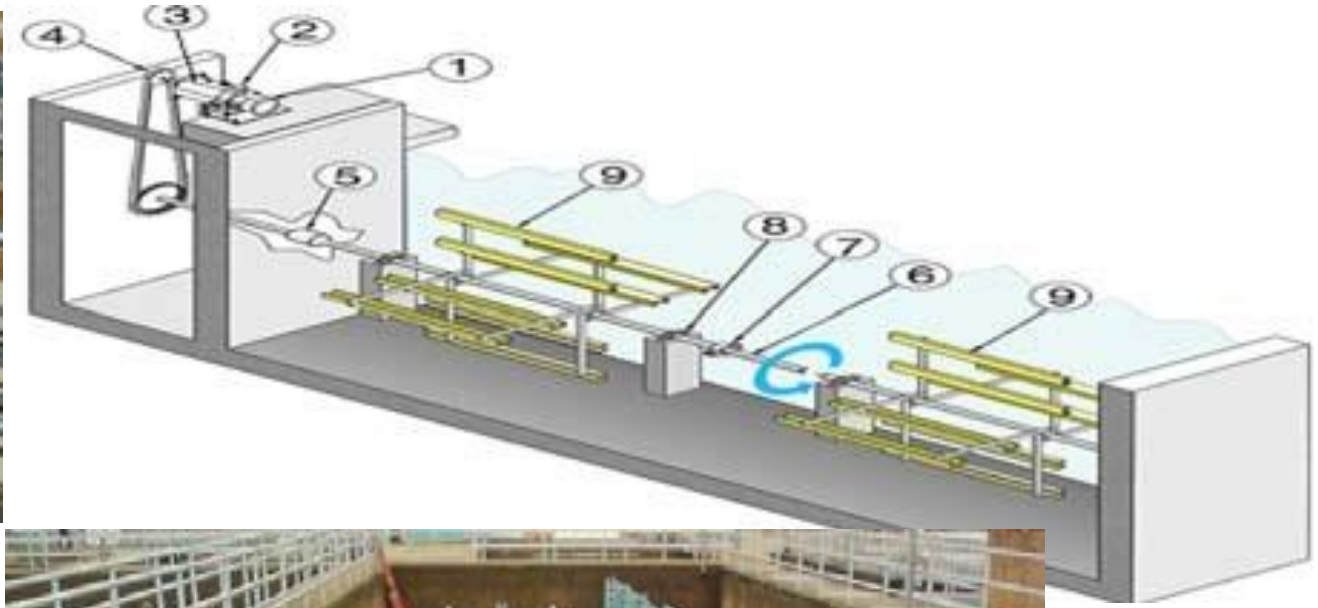
Image Source: https://www.climate-policy-watcher.org/drinking-water/chemical-mixing.html

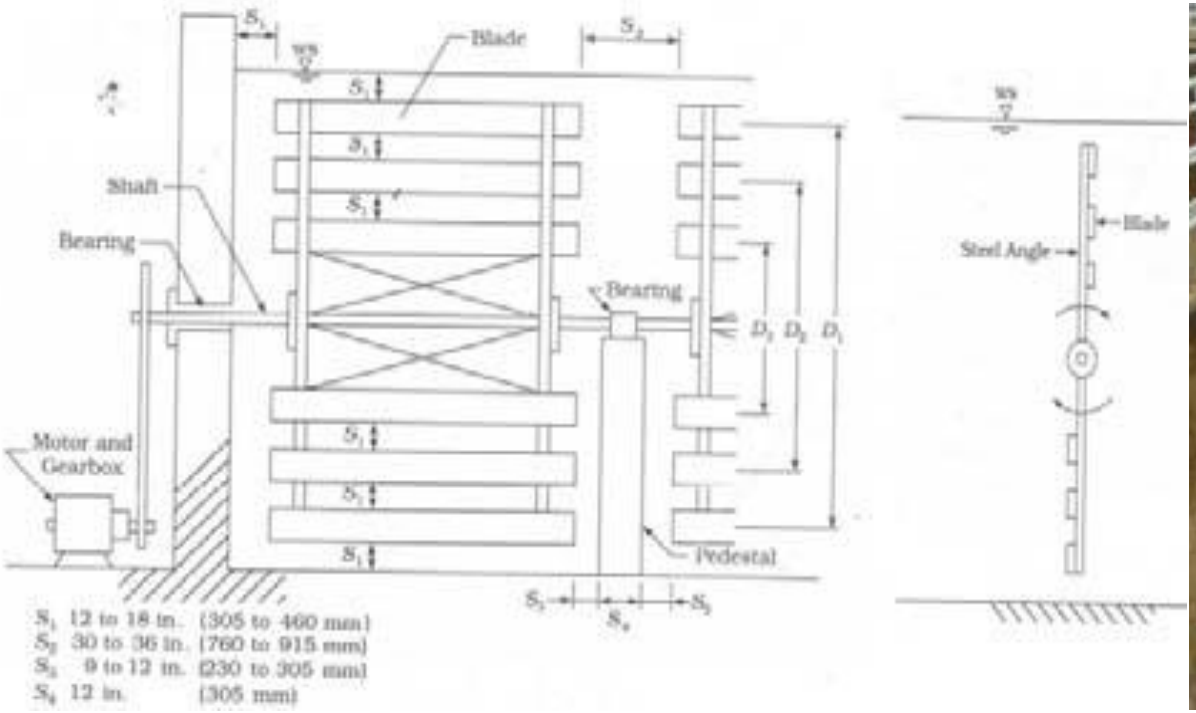
Types of Flocculation

- ***Gravitational and Mechanical Flocculation:***
  - Mixing achieved through revolving paddles with horizontal or vertical shafts, or paddles suspended from horizontal oscillating beams, moving up & down.
  - Mixing ensured by gravity flow regime through baffles provided in the basins which induce the required velocity gradients for achieving floc formation.
- ***Perikinetic and Orthokinetic Flocculation***
  - Contacts or collisions of colloidal particles resulting from bulk fluid motion, such as stirring.
  - Flocculation (contact or collisions of colloidal particles) due to Brownian motion of colloidal particles.

Horizontal Shaft Paddle Flocculators



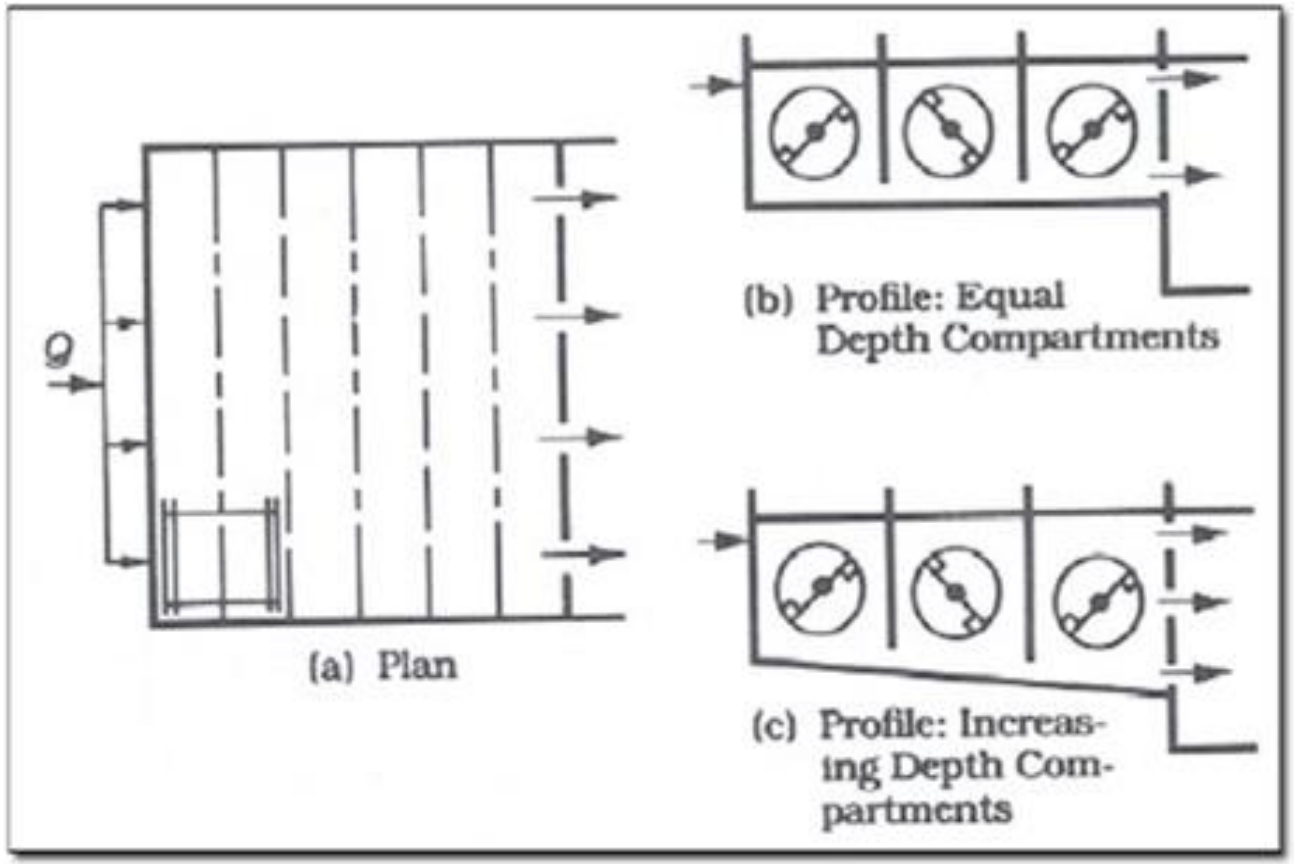




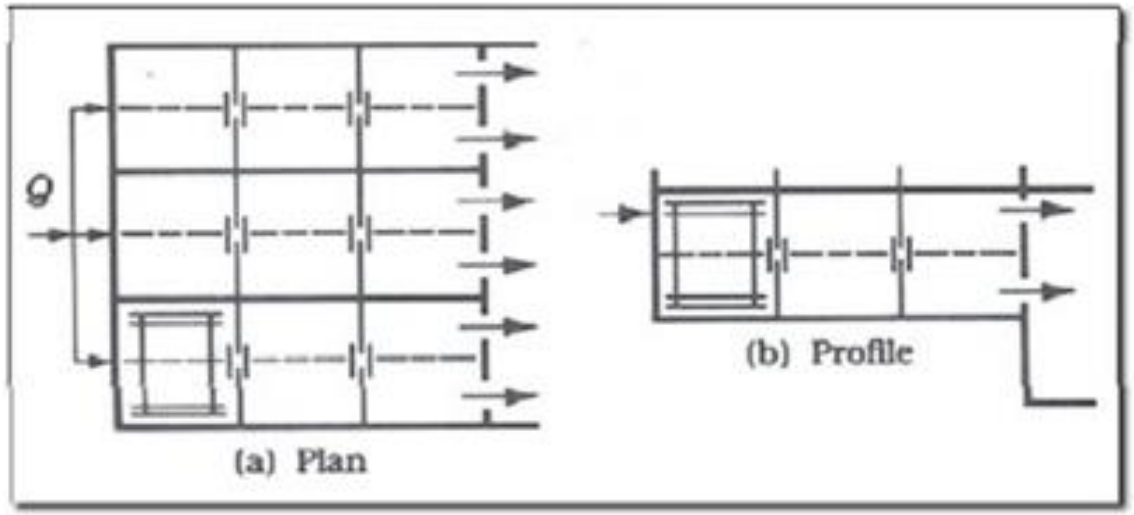


Horizontal Shaft Paddle Flocculators: Flow Patterns

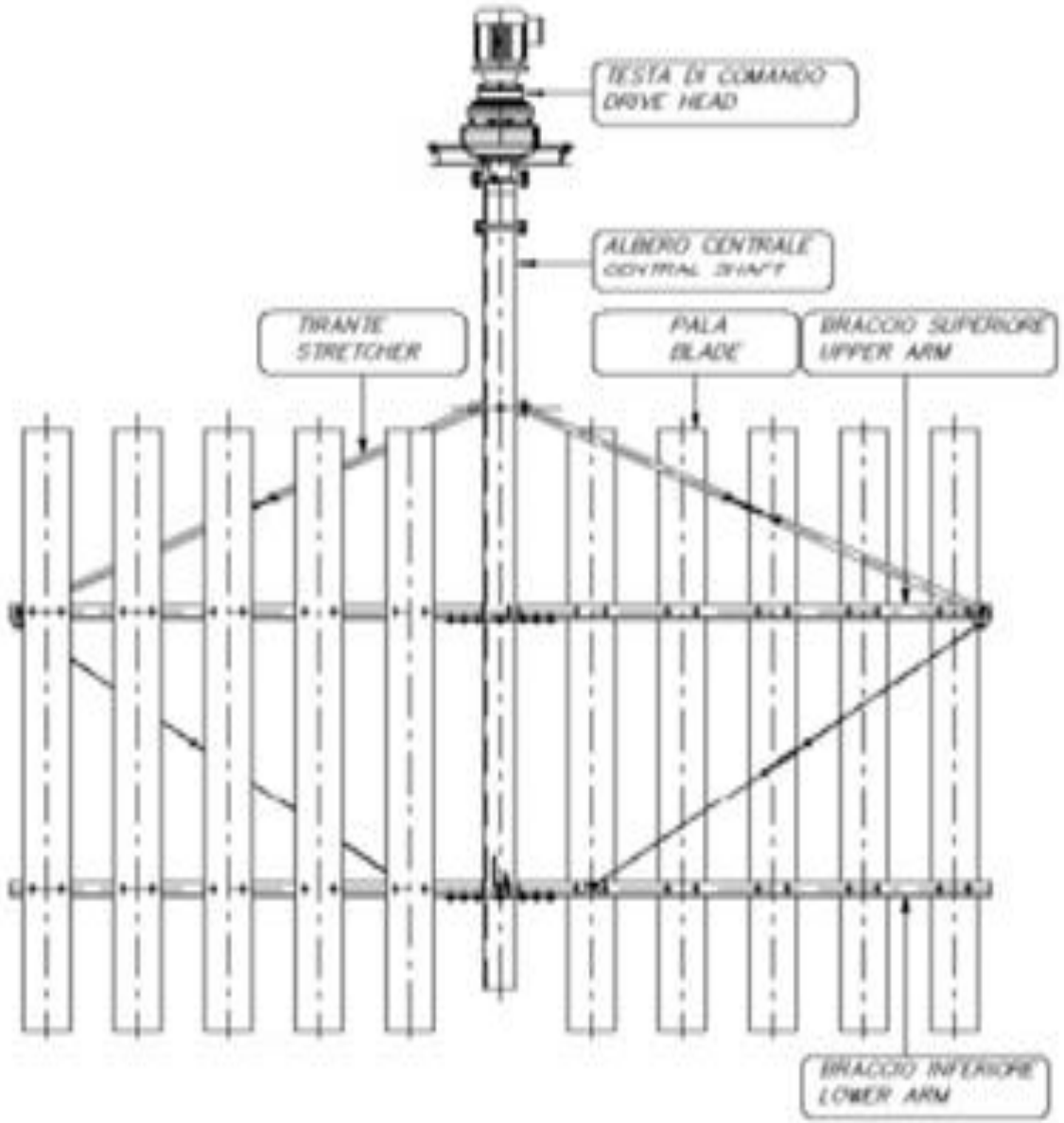
Cross-Flow Pattern

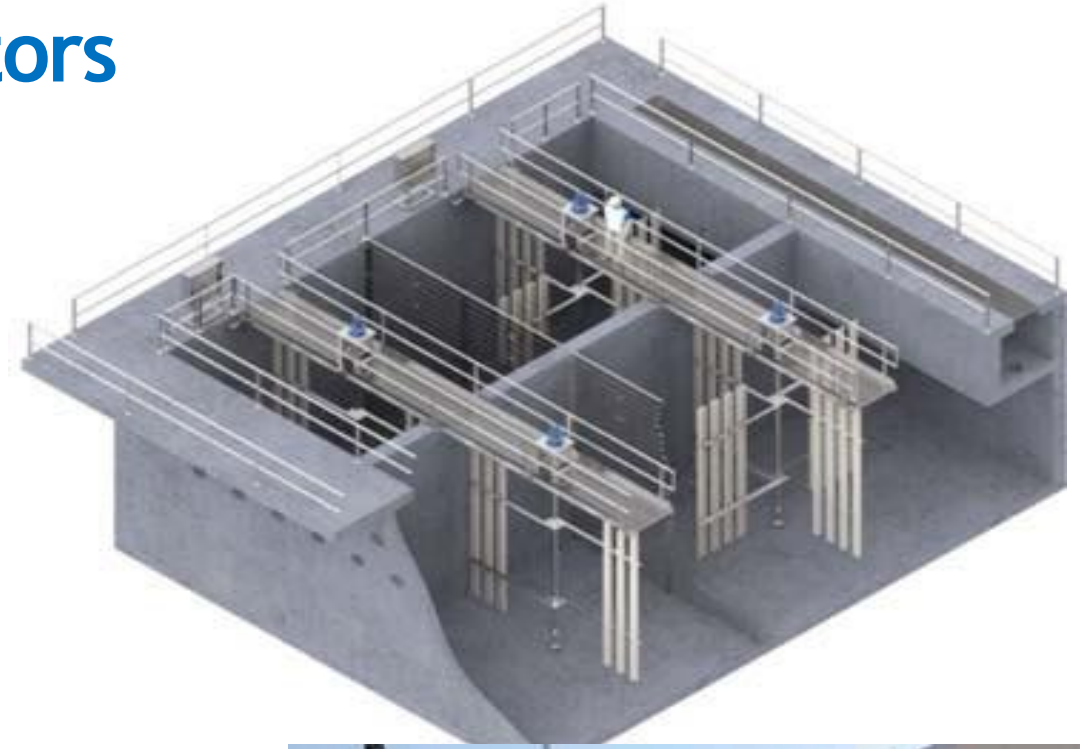


Axial-Flow Pattern



Vertical Shaft Paddle Flocculators







Design Recommendations for Paddle Wheel Flocculators

Parameter		Recommendation
G		$< 50\ s^{-1}$
Basin		
Depth		1 m > wheel diameter
Clearance between wheel and walls		0.3–0.7 m
Wheel		
Diameter		3–4 m
Spacing between wheels on same shaft		1 m
Spacing between wheel "rims" on adjacent shafts		1 m
Paddle board		
Width		10–15 cm
Length		2–3.5 m
Area of paddles/tank cross section		0.10–0.25
Number per arm		3
Spacing		at 1/3 points on arm
Tip speed		0.15–1 m/s
C_D	$L/W = 5$	1.20
	$L/W = 20$	1.50
	$L/W >> 20$	1.90
Motor		
Power		1.5–2 × water power
Turn down ratio		1:4

Sources: Kawamura, 2000; MWH, 2005; Peavy et al., 1985

The Power Input to the Water $$P = \frac{C_D A_p \rho v_p^3}{2}$$

where P = power imparted, W
C_D = coefficient of drag of paddle
A_p = cross-sectional area, m^2
v_p = relative velocity of paddles with respect to fluid, m/s

The velocity of the paddles may be estimated as

$$v_p = 2\pi k r n$$

where k = constant = 0.75
r = radius to centerline of paddle, m
n = rotational speed, rps

If more than one blade is used on the paddle the power is expressed as:

$$P = \frac{C_D \rho \left(A_{p1} v_{p1}^3 + A_{p2} v_{p2}^3 + A_{p3} v_{p3}^3 \right)}{2}$$

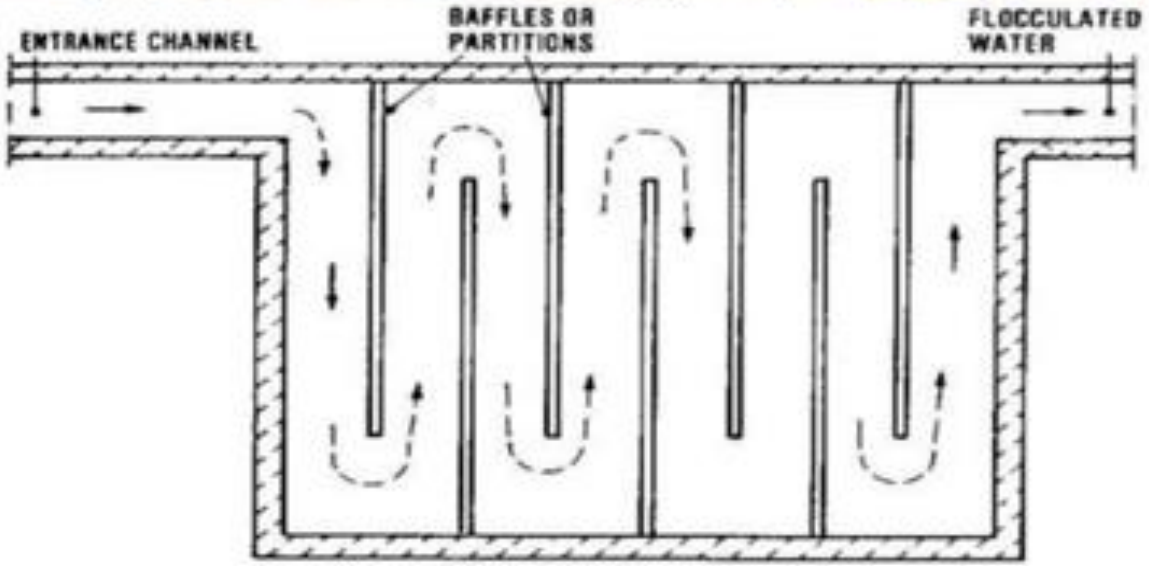
Length to Width ratio (L/W)	Drag Coefficient (C_D)
5	1.2
20	1.5
∞	1.9

Image Source: Davis M.L. Water and Wastewater Treatment: Design Principles and Practice. 2010. McGrawHill

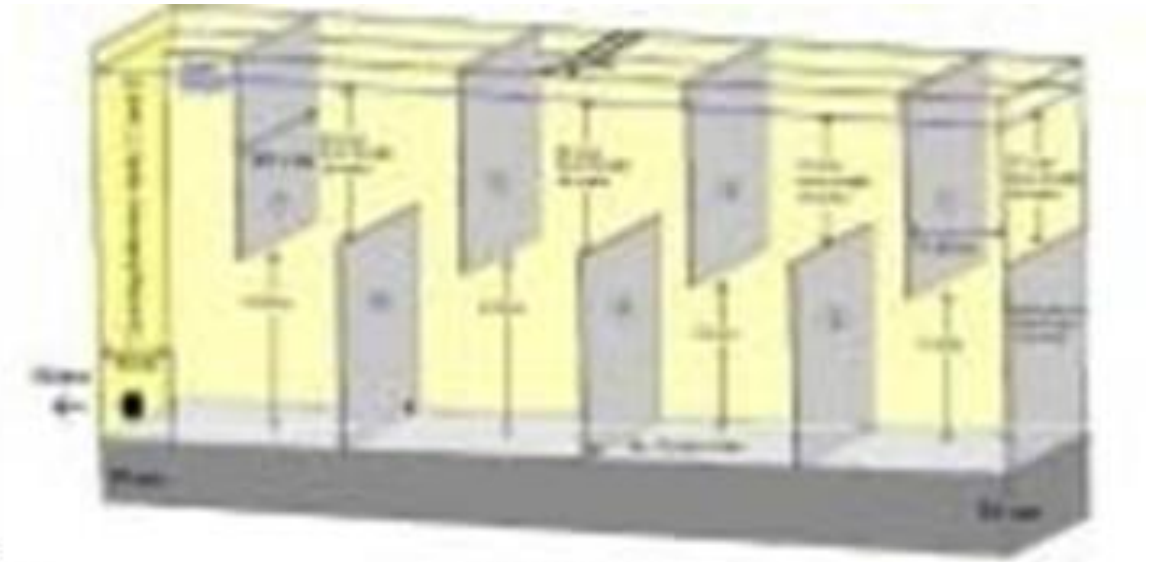
Baffled Channel Flocculators

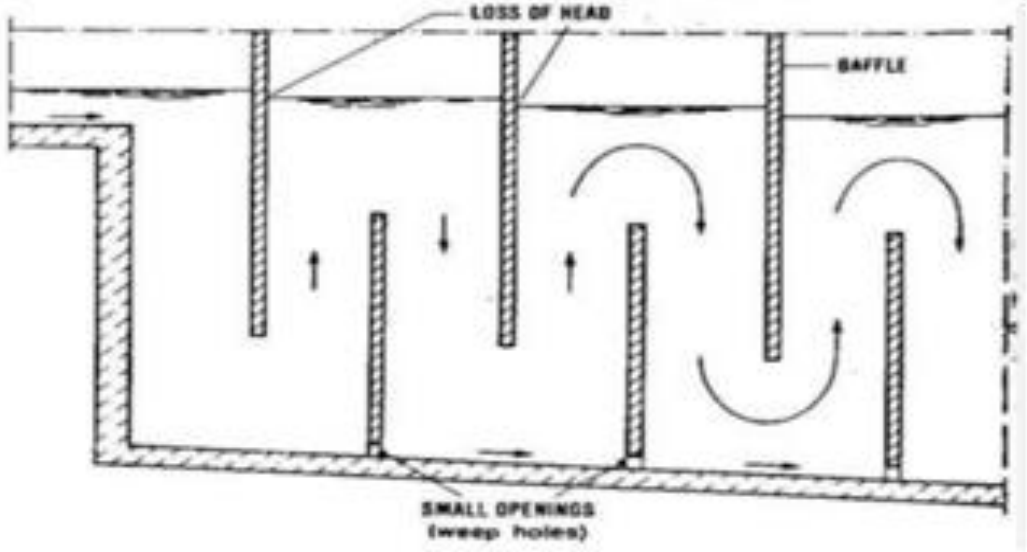
Horizontal Flow (around the end)





Vertical Flow (over and under)



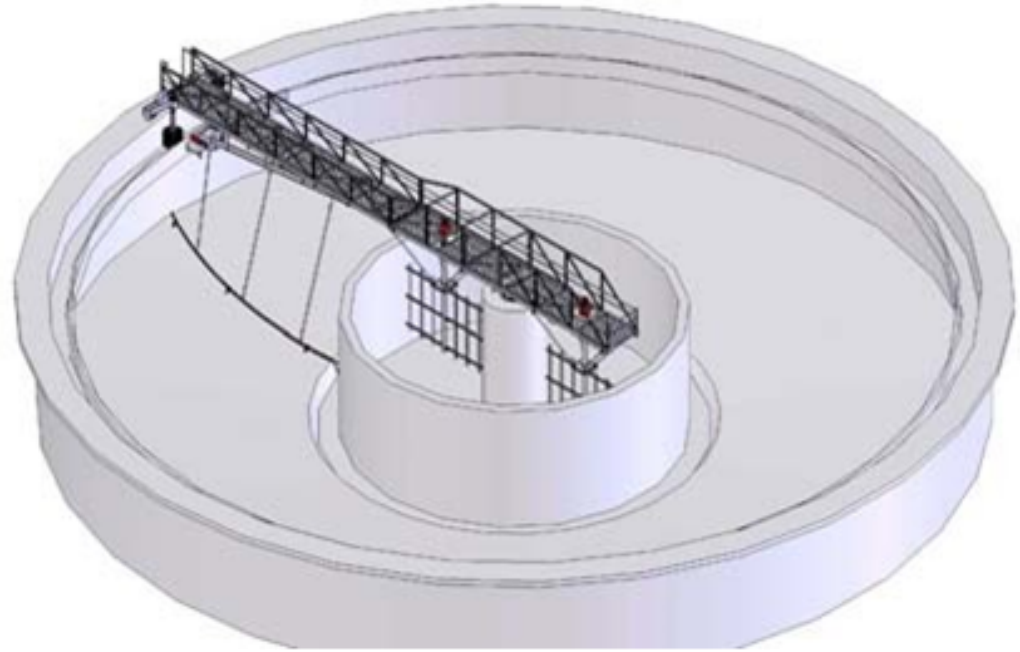


Baffled Channel Flocculators

- Generally used for large treatment plants (> 10 MLD) where the flowrates can
  maintain head-losses in the channels for slow mixing without spacing baffles too close (which may result in operation and maintenance difficulties).
- Baffled flocculators always operate under plug flow conditions and thus are free from short circuiting problems.
- Tapered energy flocculation in baffled channels generally is achieved by varying the spacing of the baffles.
  - Close spacing of baffles High velocity gradients
  - Wider spacing of baffles Lower velocity gradients

Solids Contact Flocculator Clarifier (Clariflocculator)

- Clariflocculator is a combination of flocculation and clarification in a single tank, where inner tank serves as a flocculation basin and the outer tank serves as a clarifier.
- In the Clariflocculator, the water enters the flocculator, where the flocculating paddles enhance flocculation of the feed solids. As heavy particles settle to the bottom, the liquid flows radially upward in the clarifier zone.
- The clarified functions as a circular settling basin, and water is discharged over a peripheral weir into the peripheral launder.



- The deposited sludge is raked to the bottom near the central weir from where it is discharged.
- As the settling zone receives flocculated particles, itfollows Type II (flocculant type) settling.

Zones in a Clariflocculator



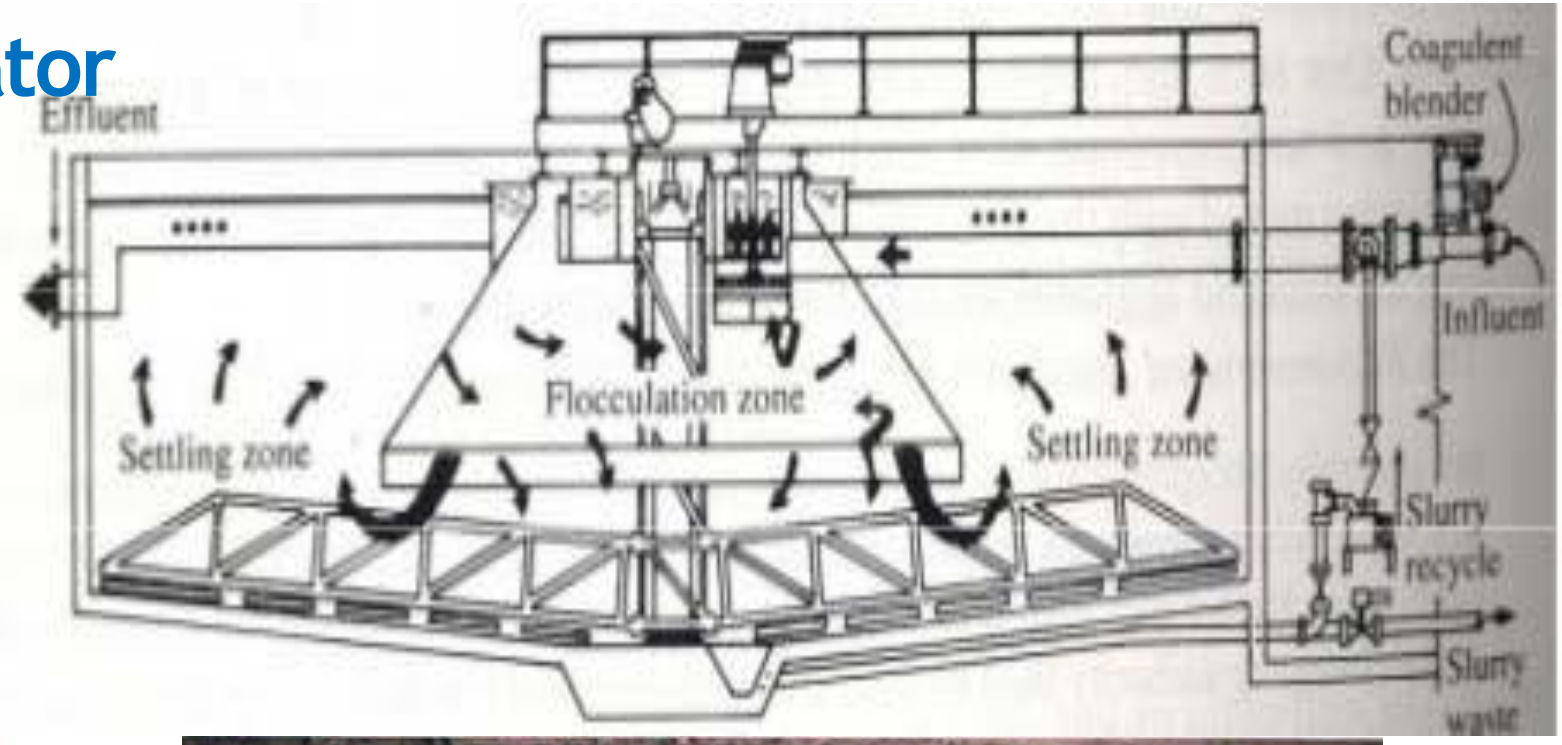
Water enters from here

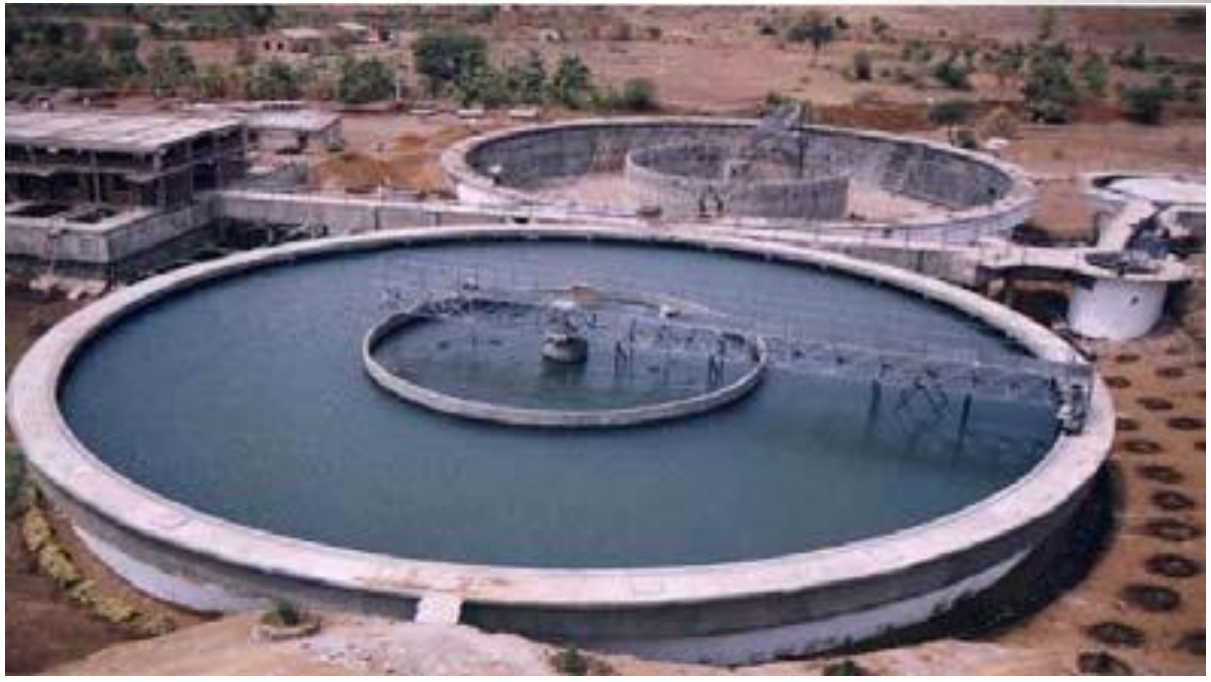
Water is gently mixed to make flocs

Flocs are given time to settle down

Settled flocs are gathered here and sent for sludge treatment

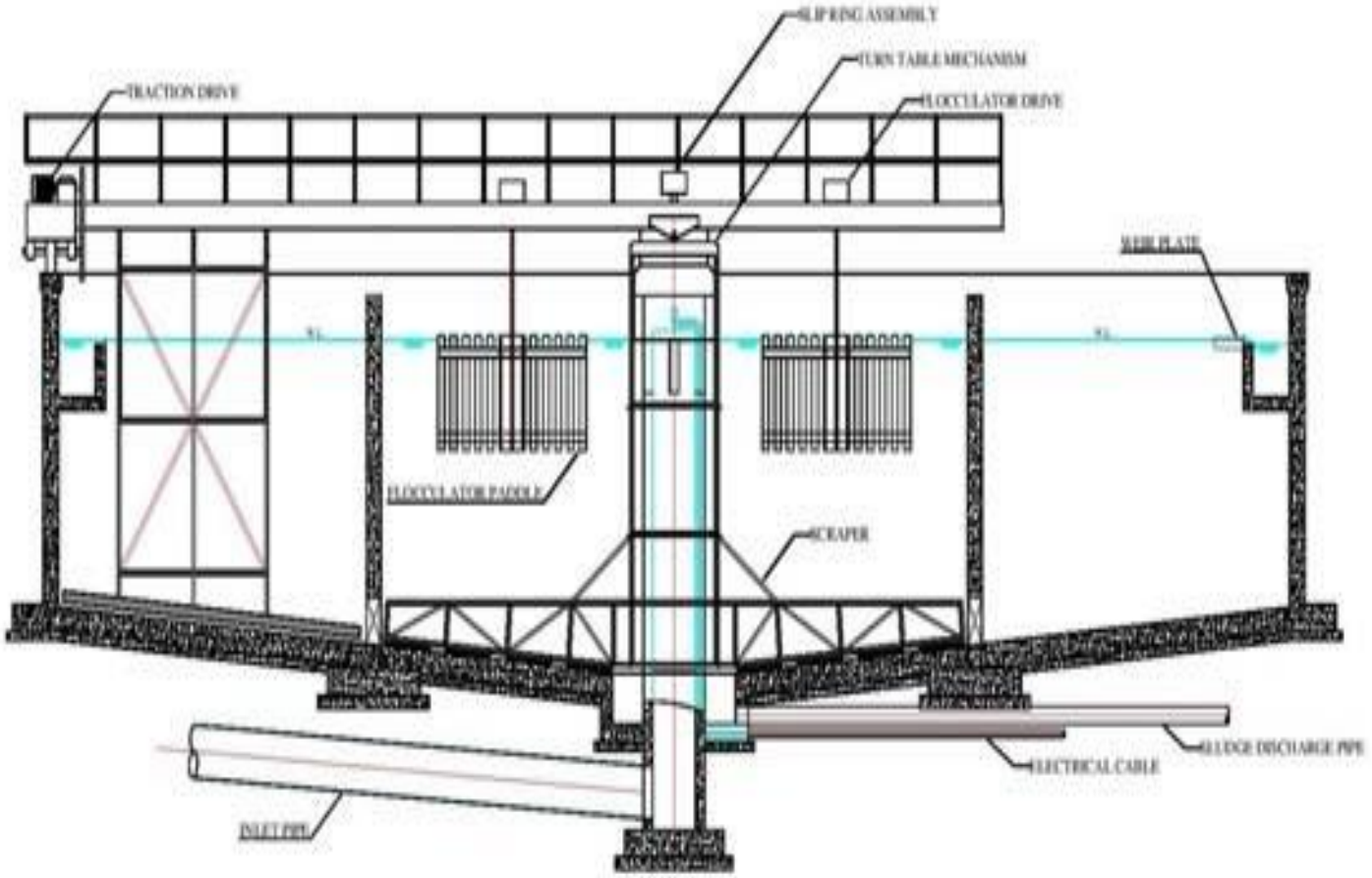
Water is cleanest at the top and is taken out.



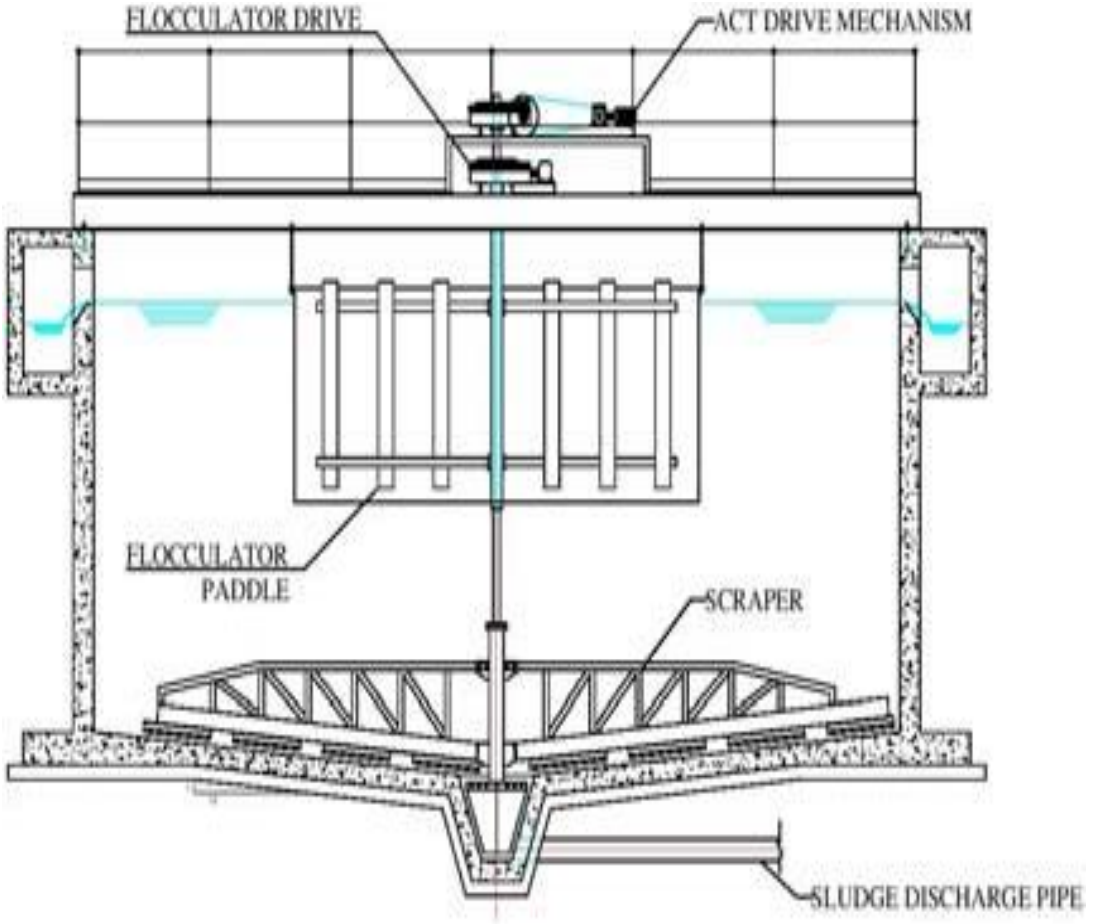


Clariflocculator Types: Depending on driving mechanism Clariflocculators could be *"central driven"* or *"end (or peripheral) driven"* and depending bridge length, it could be *"half bridge"* or *"full bridge"* Clariflocculator:

Peripheral Driven Clariflocculator



Centrally Driven Clariflocculator



1. Aluminum Sulphate (Alum):

"Universal Coagulent"

It is very cheap (economic) easily to handle & store and produce effective floc. It is improve taste & water. remove odour & colour.

Alum + Alkalinity → NCH + floc + CO_2 gas + Water

$$Al_2(SO_4)_3 \cdot 18H_2O + 3Ca(HCO_3)_2 \rightarrow 3CaSO_4 + 2Al(OH)_3 \downarrow + 6CO_2 + 18H_2O$$

— pH = 6.5 to 8.5

— Mol. wt. of Alum

$= 2(Al) + 3[S + 4(O)] + 18[2(H) + O]$

$= 2 \times 27 + 3[32 + 4 \times 16] + 18[2 \times 1 + 16]$

$= 666$

— $3Ca(HCO_3)_2 \rightarrow 3CaCO_3 + 3CO_2 + 3H_2O$

6-11-14

Mol. wt. of Alkalinity = $3CaCO_3 = 3 \times 100$

Alkalinity req in water for Alum dose of x mg/l $= \frac{3 \times 100}{666} \times x$ (mg/l) as $CaCO_3$

Alkalinity deficiency in water (y mg/l) = (Alkalinity)$_{req}$ − (alkalinity)$_{pr}$

To increase alkalinity in water Lime as CaO added to water.

$$CaCO_3 \longrightarrow CaO + CO_2$$

Lime as CaO req. To raise alkality in water $= \frac{56}{100} * y$ (mg/l)

b) Alkalinity deficiency = $(A)_{req} - (A)_{present}$

$= 9 - 6$

$4\,(mg/l) = 3\,mg/l$

∴ dose of Lime req to meet Alk. deficiency $= \frac{56}{100} \times 3$ as CaO

$= 1.68\,mg/l$ as CaO

∴ Total Lime req = Q × dose of Lime

$= 10 \times 1.68$

$= 16.8\,kg/day$

$* 10^6$

∴ Annual Lime req $= 16.8 \times 365$

$= 6132\,kg/yr$

$= 6132\,(*10^6\,mg/yr)$

A water treatment plant treating 10 ~~m~~ MLD of water requires 20 mg/l of filter Alum. The water has 6mg/l of alkalinity as $CaCO_3$. (Al = 26.97, S = 32, O = 16, H = 1, Ca = 40, and C= 12)

(GATE-03)

02. Total alkalinity requirement (10^6 mg per day as $CaCO_3$) matching filter Alum, shall be

(a) 180 (b) 120 (c) 90 (d) 60

03. Quantity of Quick Lime required (10^6 mg per ~~year as~~ CaO) shall be

(a) 2132 (b) 3000 (c) 4132 (d) 6132

02

Alum = 10 mg/l

water = 10 MLD

alkalinity req. $= \frac{18 * 18}{666} \times 6$

Alkalinity req $= \frac{3 \times 100}{666} \times 20 = 9.0\,mg/l$ as $CaCO_3$

in water for alum

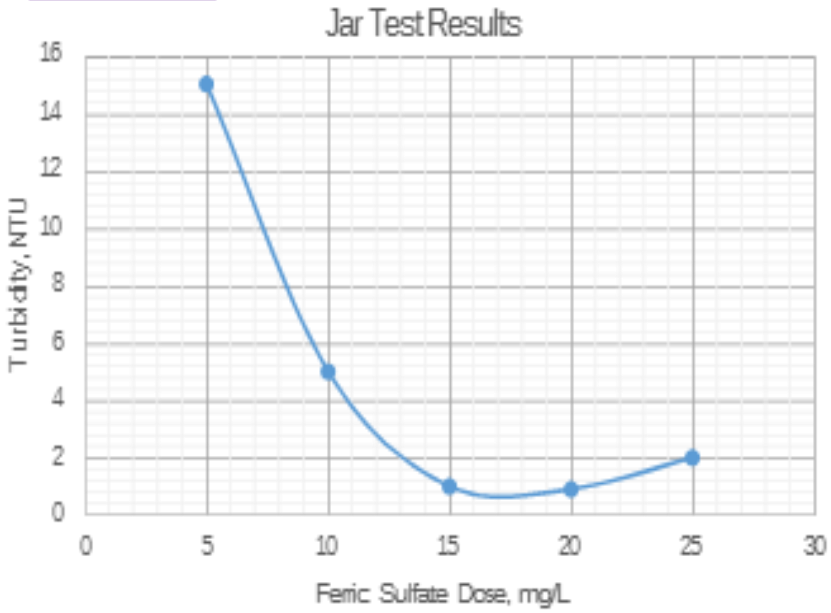
Practice Problem 1: Coagulant Dose

Results of a jar test to determine the optimal Ferric sulfate dose as coagulant are provided below. The initial water sample has a pH = 6.5, turbidity of 30 NTU, and alkalinity of 50 mg/L as $CaCO_3$. Determine the daily (optimal) amount of ferric sulphate required to treat 10 MLD water to a turbidity below 1 NTU. Also, suggest if any alkalinity addition in the system is required. The governing chemical reaction is as under:

$Fe_2(SO_4)_3 + 3Ca(HCO_3)_2 \rightarrow 2Fe(OH)_3\,(s) + 3CaSO_4 + 6CO_2$

Ferric sulfate dose, mg/L	5	10	15	20	25
Turbidity, NTU	15	5	1	0.9	2

Solution:



From Jar Test Results, optimum dose may be considered as 17 mg/L

Coagulant required for 10 MLD flow = 10×10^6 L/d * 17 mg/L = 170 Kg/d

From reaction, we can see that 1 mole of $Fe_2(SO_4)_3$ requires 3 moles of $Ca(HCO_3)_2$.

$[(56*2)+\{32+(4*16)\}*3]$ g $Fe_2(SO_4)_3$ would requires $3*[40+\{1+12+(16*3)\}*2]$ g $Ca(HCO_3)_2$

= 400 g = 486 g

So, 17 mg/L $Fe_2(SO_4)_3$ would requires 486*17/400 mg/L $Ca(HCO_3)_2$

= 20.655 mg/L $Ca(HCO_3)_2$

20.655*M.W. of $CaCO_3$/M.W. of $Ca(HCO_3)_2$ = 12.75 mg/L as $CaCO_3$

As system is already having alkalinity higher than stoichiometric requirement, no further alkalinity addition is required.

Practice Problem 2: Coagulant Dose Calculations

Liquid alum of specific gravity 1.33 is being used as a coagulant. One 25 L container of alum weighs 25 Kg and contains 47.619% of dry alum (w/w). Determine: (a) the alum concentration, (b) mL of liquid alum required to prepare a 100 mL solution of 20,000 mg/L alum concentration, (c) the dosage concentration of 1 mL of stock solution in a 2 L jar sample.

Solution

(a) Alum concentration = 47.619% (w/w) of 25 Kg in 25 L= 0.47619*25/25 = 0.47619 Kg/L = ***476.19 g/L***

(b) Alum concentration required = 20,000 mg/L = 20 g/L; Volume of Stock Sol. Required = 100 mL

So, Total alum needed = 20 g/L * 0.1 L = 2 g

1 L (1000 mL) of liquid alum has 476.19 g alum

So, amount required for 2 g alum = (1000/476.19)*2 = ***4.2 mL***

Alternatively; $C_1V_1=C_2V_2$

$476.19*V_1=20*100$

$V_1= 2000/476.19 = 4.2$ mL

(c) The alum concentration in stock solution = 20,000 mg/L = 20 mg/mL

This means 1 mL of stock solution is equivalent to 20 mg alum

If 1 mL stock solution is added in 2L jar, the concentration would be = 20mg/2 L = ***10 mg/L***

Practice Problem 3: Flash Mix Sizing and Power Requirement

A rapid mixer is to be used for coagulation of surface water with high turbidity. If the flow is 720 m^3/h, find the volume and dimensions of the tank and the power requirements. Assume that the detention time is 20 seconds and G=1000 s^{-1}, and μ = 0.00115 N s/m^2 at T = 15^0C

Solution:

Q = 720 m^3/h = 0.20 m^3/s
Detention time = 20 s
G=1000 s^{-1}

Tank Volume, V = Flow x Detention time = 0.2*20 = 4 m^3

Assume the tank cross section is square, and depth of tank as 1.5 time the width:
$V = W*W*1.5W = 1.5W^3$
$W = (V/1.5)^{1/3} = (4/1.5)^{1/3} = 1.387\ m \approx 1.4\ m$
d = 1.5*1.4 = 2.1 m
Tank Dimensions (Square cross section) = ***1.4m x 1.4m x 2.1m***

Power Requirement $P = \mu V G^2 = 0.00113*4*1000^2$ = 4600 Watt = ***4.6 kW***

Practice Problem 4: Baffled Floculation

A baffled flocculation basin of size 5x4x60 m^3 is divided into 16 channels by 15 around-the-end baffles. The velocities at the channels and at the slots are 0.18 and 0.6 m/s, respectively. The flow rate is 0.34 m^3/s. Find (a) the total head loss neglecting channel friction; (b) the power dissipated; (c) the mean velocity gradient (at 10^0C; μ = 0.00113 $N.s/m^2$); (d) the *Gt* value, if the detention time is 30 min, and (e) the loading rate.

Solution

(a) Head loss in each channel = $v^2/2g = 0.18^2/(2*9.81) = 0.001651$ m

Head loss in each slot = $v^2/2g = 0.6^2/(2*9.81) = 0.018349$ m

Total head loss (neglect friction loss), H = 16*0.001651 + 15*0.018349 ***= 0.30 m***

(b) Power Required P = γQH = 9807 N/m^3 x 0.34 m^3/s x 0.3 m ***= 1005.82 Watt***

(c) Mean velocity gradient $G = \sqrt{(P/\mu.V)} = \sqrt{[1005.82/(0.00113*5*4*60)]}$ ***= 27.24 s^{-1}***

(d) Gt = 27.24 * 30*60 ***= 4.6 x 10^4***

(e) Loading rate = Q/V = (0.34 m^3/s* 3600*24) / (5*4*60 m^3) ***= 24.48 $m^3/d/m^3$***

Practice Problem 5: Clariflocculator Sizing

A clariflocculator is to be designed for a flow of 50 MLD. Assuming, inlet velocity 1.2 m/s, and detention time in flocculator and clarifier as 30 and 120 mins, respectively, surface overflow rate is clarifier as 40 $m^3/m^2/d$, G in flocculator 40 s^{-1}, and depth of flocullator basin as 2.5 m. Determine diameter of inlet pipe, flocculator and clarifier.

Solution:

Q = 50 MLD = 50000 m^3/d = 2083.33 m^3/h = 0.5784 m^3/s

Inlet Pipe:

Area of the pipe required = Q / velocity in pipe = 0.5787 / 1.2 = 0.482 m^2

Dia of pipe required = $\sqrt{(4*0.482/3.14)}$ = 0.782 m ***≈ 800 mm dia pipe (say d_i)***

Flocculator:

Volume Required V_f = Flow x Detention time = 0.5784*30*60 = 1041.667 m^3 Effective area required = Volume / depth = 1041.667/2.5 = 416.667 m^2

= Total flocculator area – inlet pipe area = $\pi d_f^2/4 - \pi d_i^2/4$ Dia of flocculator = $\sqrt{[(4*416.667/3.14) + 0.8^2]}$ = 23.05 m ***≈ 23 m dia (say d_f)***

Clarifier:

Effective Clarifier Area Required = Flow / overflow rate = 50000/40 = 1250 m^2

= Total clariflocculator area – flocculator area = $\pi d^2/4 - \pi d^2/4$

Dia of Clariflocculator = $\sqrt{[(4*1250/3.14) + 23^2]}$ = 46.06 m ***≈ 46 m dia (say d_c)***

P-32
5

Q = 12 MLD

dose of ferrous sulphate = 10 mg/L

∴ Total qty of F.S. = 10 × 12

= 120 kg/day

Mol. wt. of F.S. = Fe + S + 4×(O) + 7[2×(H) + O]

= 278

$Ca(OH)_2 \Longrightarrow CaO + H_2O$

F.S	CaO
278	→ 56
1	→ $\frac{56}{278} \times 1$

Mol. wt. of CaO = 56

dose of Lime req. daily = $\frac{56}{278} \times 10$ mg/l of Lime

= 2.01 mg/l

Total Lime as CaO req. = 12 × 2.01 = MLD × mg/l

= 24.17 kg/day

At a water treatment plant 12MLD is treated daily using ferrous sulphate and lime. If the dosage of ferrous sulphate is 10mg/L, determine the total quantity of ferrous sulphate and lime required daily. Use the following chemical equation

$FeSO_4.7H_2O + Ca(OH)_2 = Fe(OH)_2 + CaSO_4 + 7H_2O$

2. A surface water treatment plant operates round the clock with a flow rate of 3.5 m3/min. The water temperature is 15°C and jar testing indicated an alum dosage of 25 mg/l, Gt = 4×104 for producing optimum result. The alum request for 30 days (in kg) is____ **(GATE-14)**

$Q = 3.5\ m^3/min = \frac{3.5}{60} = 0.058\ m^3/s$

$T = 15°C$

x = alum dose = 25 mg/l

$Gt = 4 \times 10^4$

∴ alum required for 1 day $= 0.058 \frac{m^3}{s} \times 25\ mg/L \times 24 \times$

$= \frac{3.5 \times 24 \times 60 \times 1000}{10^6} \times 25$

MLD mg/L

$= 126\ kg/day$

∴ alum req. for 30 day $= 126 \times 30$

$= 3780\ kg$

3. A flash mixer of 2 m^3 with a velocity gradient of mixing mechanism equal to 600 S^{-1} and fluid absolute viscosity of 1.0×10^{-3} N-sec/m^2 is continuously operated. What is the power input per unit volume? **(IES – 05)**

(a) 360W (b) 720W

(c) 1440W (d) 300W

$V = 2m^3$, $G = 600 s^{-1}$, $\mu = 1 \times 10^{-3}$ Ns/m^2

$$G = \sqrt{\frac{P}{V\mu}}$$

$$600 = \sqrt{\frac{P}{2 \times 1 \times 10^{-3}}}$$

$$\therefore \boxed{P = 720\ W} = \frac{N\text{-}m}{s}$$

$Q = 28800\ m^3/d$

$\rho = 1000\ kg/m^3$ $\quad \nu = \frac{\mu}{\rho} \therefore \mu =$

$\nu = 10^{-6}\ m^2/s$

$G = 900\ s^{-1}$

$x = 35\ mg/l$

$D.T = 2\ minutes$

$\therefore Vol^m = D.T \times Q$

$= \frac{22800}{24 \times 60} \frac{m^3}{min} \times 2\ min = 40m$

$\mu = \nu . \rho$

$= 10^{-6} \times 1000 \quad m^2/s \times \frac{kg}{m^3} = \frac{m^2}{s} \times \frac{N\text{-}s^2}{m^4}$

$\mu = 10^{-3} \frac{N\text{-}s}{m^2} \quad = \frac{N.s}{m^2}$

$\therefore G = \sqrt{\frac{P}{V.\mu}}$

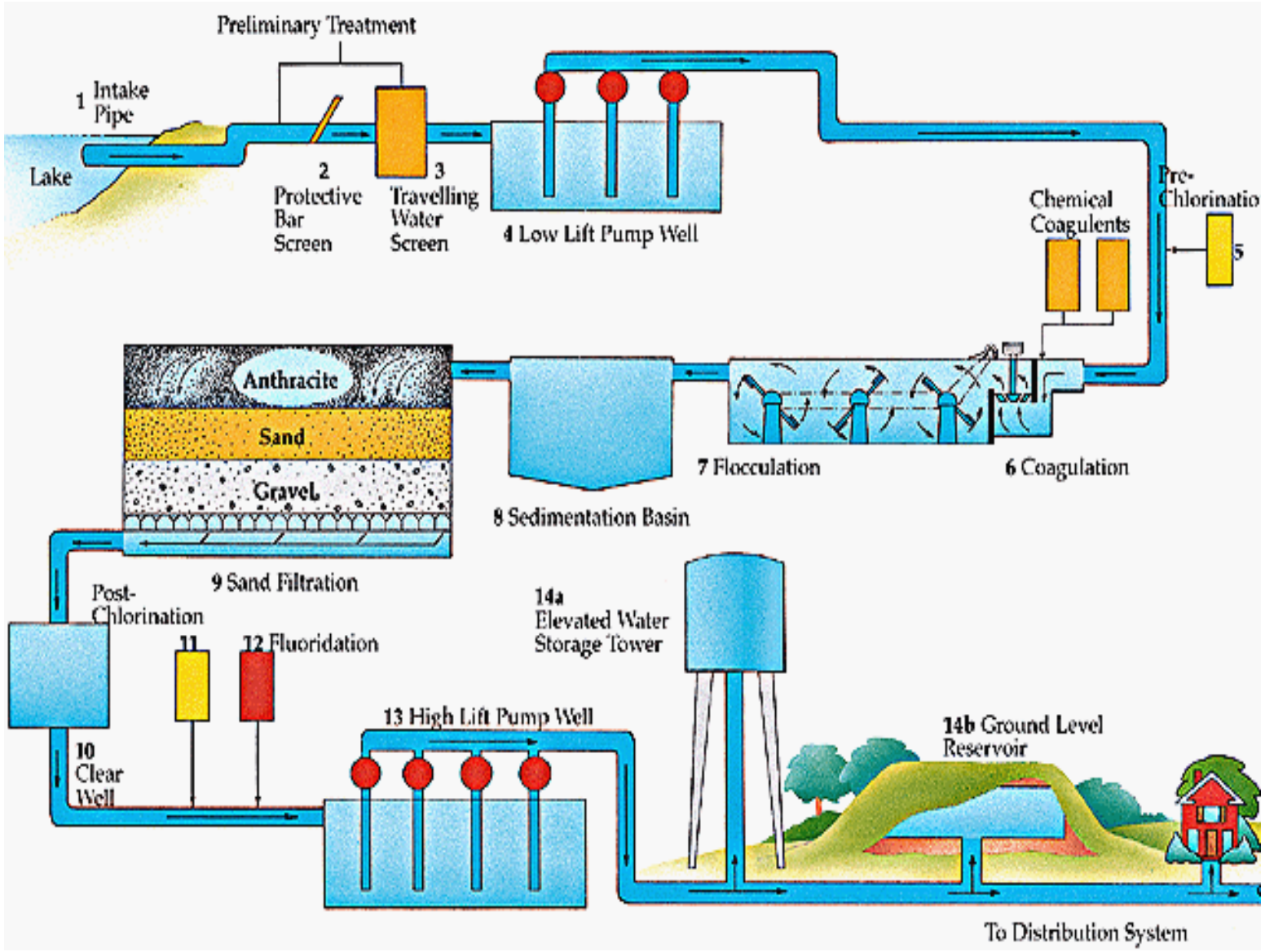
$900 = \sqrt{\frac{P}{40 \times 10^{-3}}} \quad \therefore \boxed{P = 32400\ W}$

28800 m^3/d of raw water (density = 1000 kg/m^3, kinematics viscosity = $10^{-6} m^2/s$). The rapid mixing tank imparts a velocity gradient of $900 s^{-1}$ to blend 35 mg/*l* of alum with the flow for a detention time of 2 minutes. The power input (W) required for rapid mixing is

(a) 32.4 (b) 36 (c) 324 (d) 32400

Filtration in Conventional Drinking Water Treatment Systems

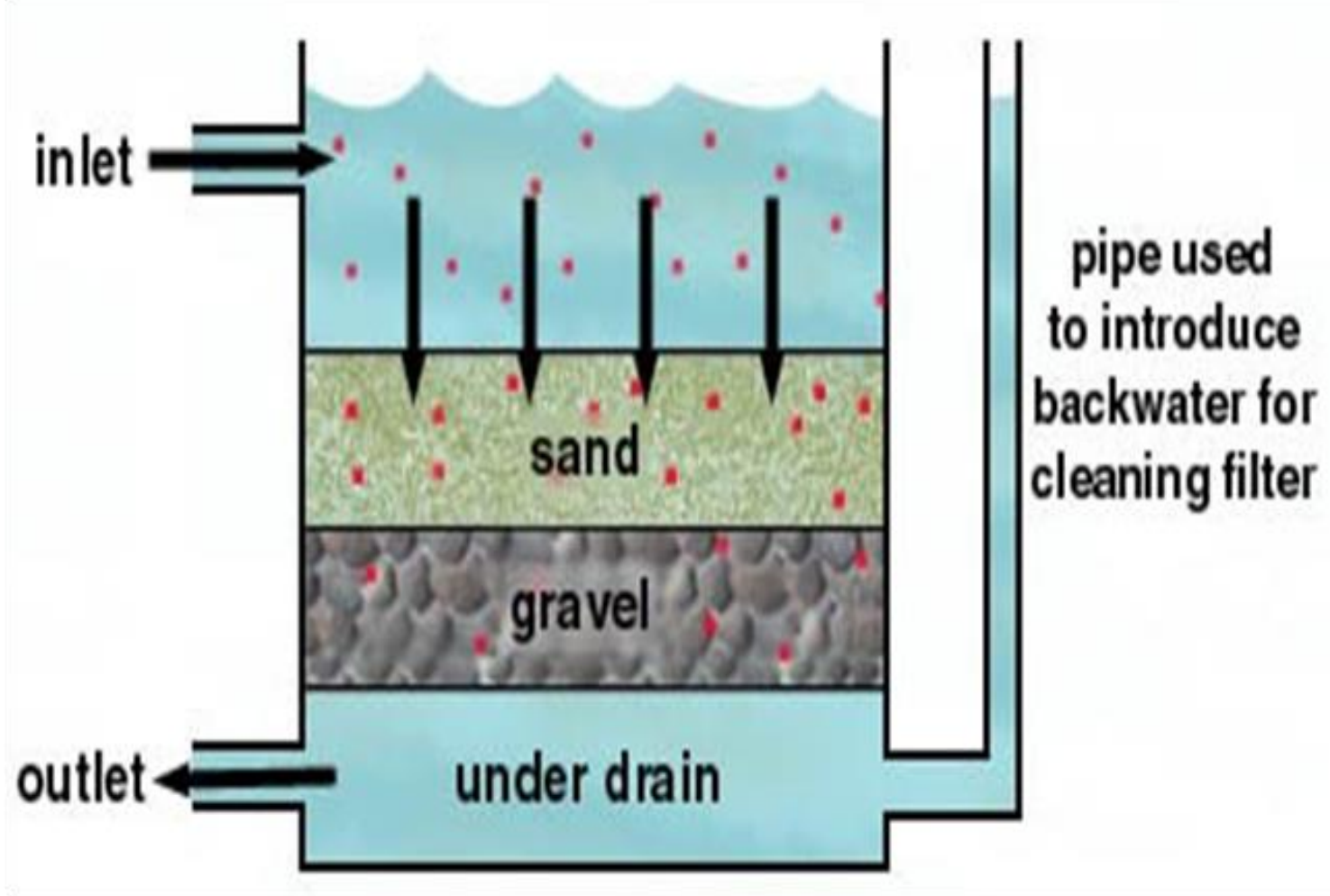
- After water passes through Flash Mixer (coagulant addition) and Flocculation Basin, it is treated in Sedimentation Basin for the removal of flocs. Alternately, after passing through Flash Mixer, it may be sent through Clariflocculator (for flocculation and Clarification).
- After water comes out of clarifier (settling basin), it has lost most of the suspended and colloidal particles, however, usually it still contains very fines sediments and some micro-flocs (as clarifier may not remove all suspended particles and flocs with 100% efficiency).



- **Therefore, water is passed through sand or multi-media filters for the removal of left-over suspended solids and micro-flocs (known as filtration process).**

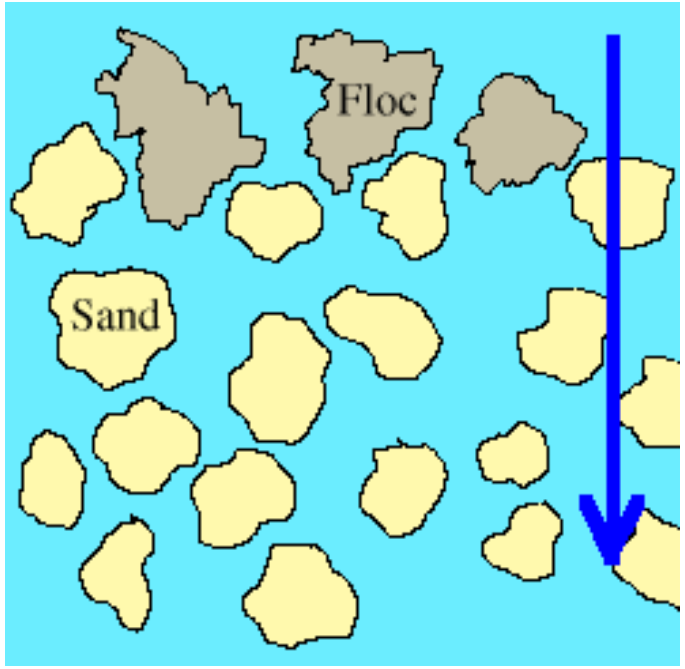
Filtration Objectives and Process

- The water passes through filters, generally made of layers of sand and gravel (in case of sand filtration), and additional layer of anthracite / charcoal that help in removal of even smaller particles (in Dual or Multimedia filters).
- The primary objective of Sand Filtration is to remove the left-over suspended solids and micro-flocs in water. Filtration also removes pathogenic contaminants to an extent. In addition, some organic compound, organically bound nitrogen and phosphate, and undissolved metals are also removed in Multimedia Filters.

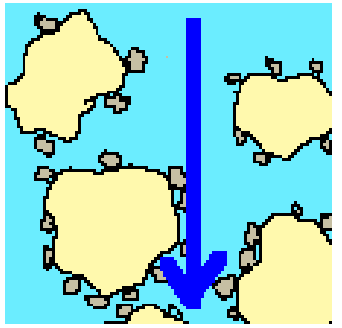


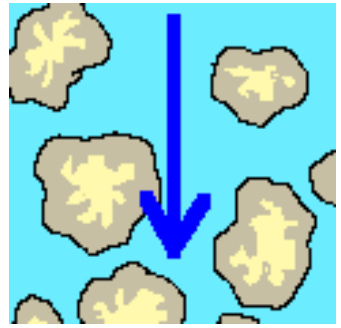
Mechanism of Removal in Filtration

- *Physical Straining:* Capture of particles larger than the pore sizes (Major Mechanism)
- *Sedimentation:* Gravity settling of particle on the surface of filter media.
- *Adsorption:* Particles becoming attracted to and "sticking" to the surface of filter media particles. Adsorption can remove very small particles, even dissolved pollutants from water (Major Mechanism, especially in Multimedia Filters).



- *Biological Actions:* Bacteria grown or caught in the voids decompose organic impurities in water through biochemical reactions. These then gets deposited on the surface of sands forming jelly like layer which again absorbs and strain out impurities (Important for slow sand filters, while, in most other filters the water passes through the filter too quickly for much biological action to occur).





- *Absorption:* Some liquids gets soaked up into the sand grains.
- *Electrolytic Action:* Some charged and polar sand particles may neutralize oppositely charged suspended or dissolved ions in water.

Mechanism of Removal in Filtration

Filtration involves transport of particles through:

- Gravitational settling
- Diffusion
- Interception
- Hydrodynamics

Filtration involves attachment of particles through:

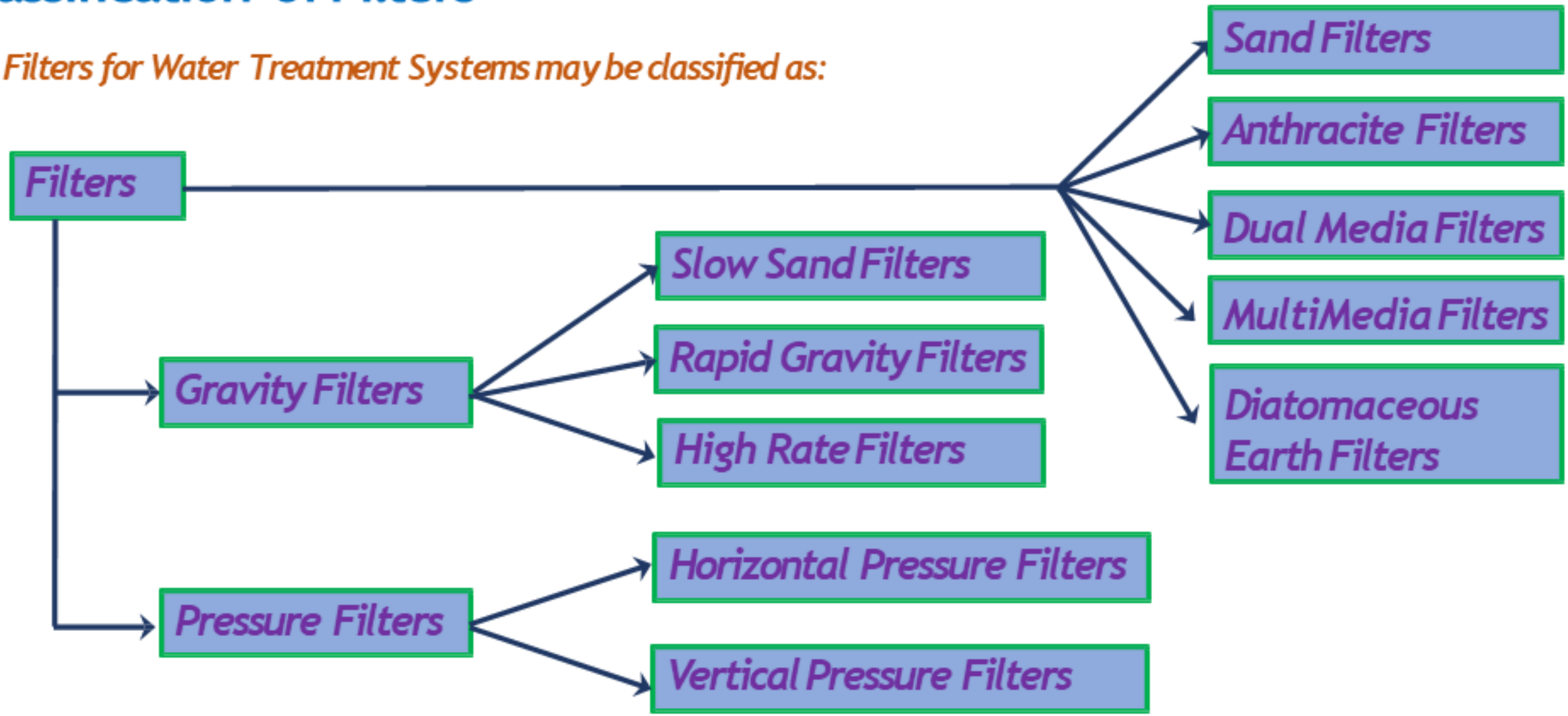
- Electrostatic interactions
- Chemical bridging
- Specific adsorption

Important forces for Filtration:

- Inertia
- van der Waals
- Gravitational
- Electrostatic
- Viscos
- Thermal

Classification of Filters

- *Filters for Water Treatment Systems may be classified as:*



Rate of Filtration

- Rate of Filtration (Filter Loading Rate) is the flow rate of water applied per unit surface area of the filter. It is the same value as flow velocity of the water approaching the surface of the filter. Rate of filtration (v) may be given as:

$$v = Q/A$$

where, v = face velocity, m/d = loading rate, $m^3/d.m^2$

Q = flow rate onto filter surface, m^3/d

A = surface are of filter, m^2

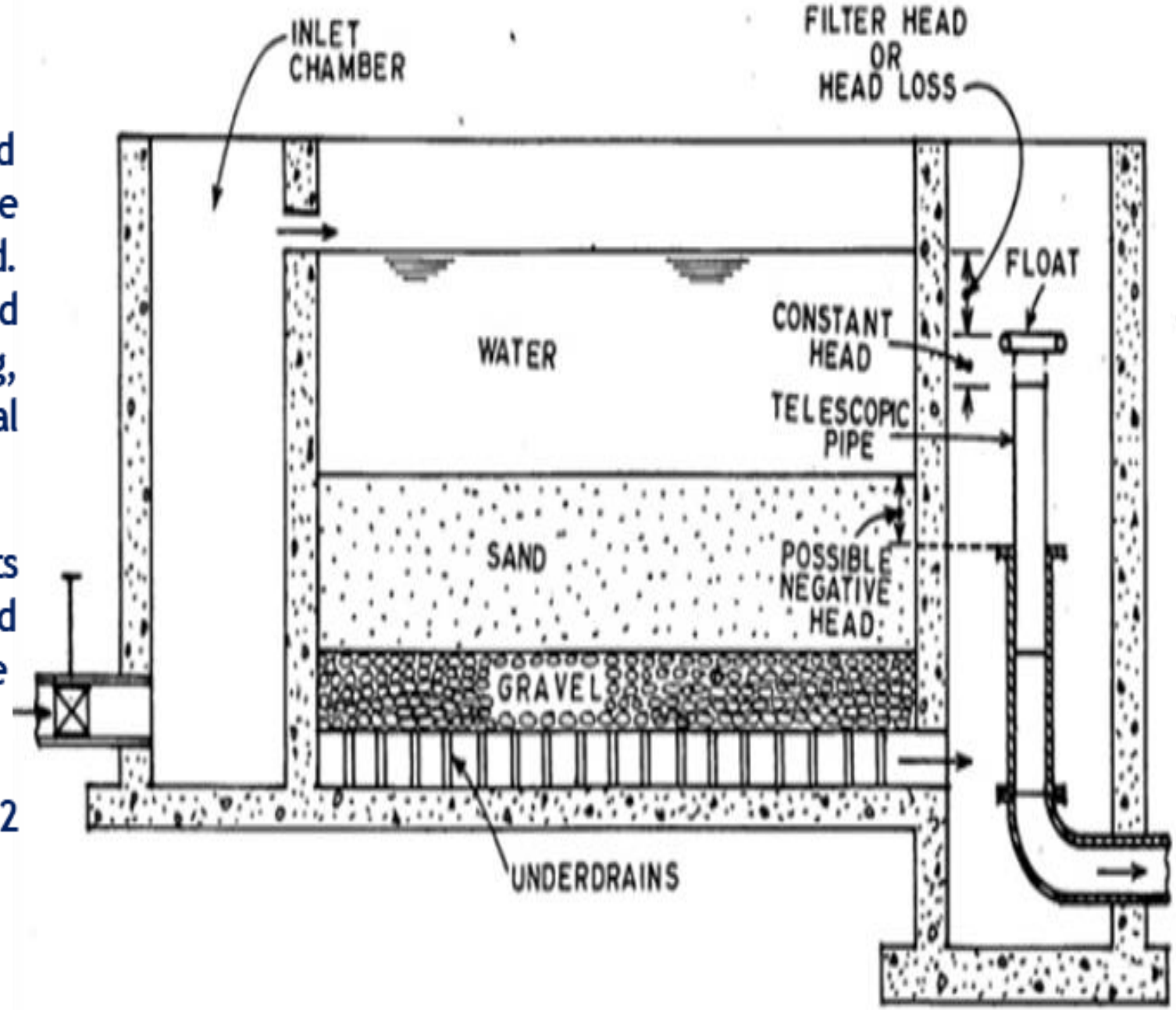
- Typical rate of filtration for:
  - Slow Sand Filters → 0.1 - 0.2 m/h
  - Rapid Sand Filters → 3 - 6 m/h
  - High Rate Sand Filters → 8 - 12 m/h (upto 14 m/h)

Slow Sand Filters

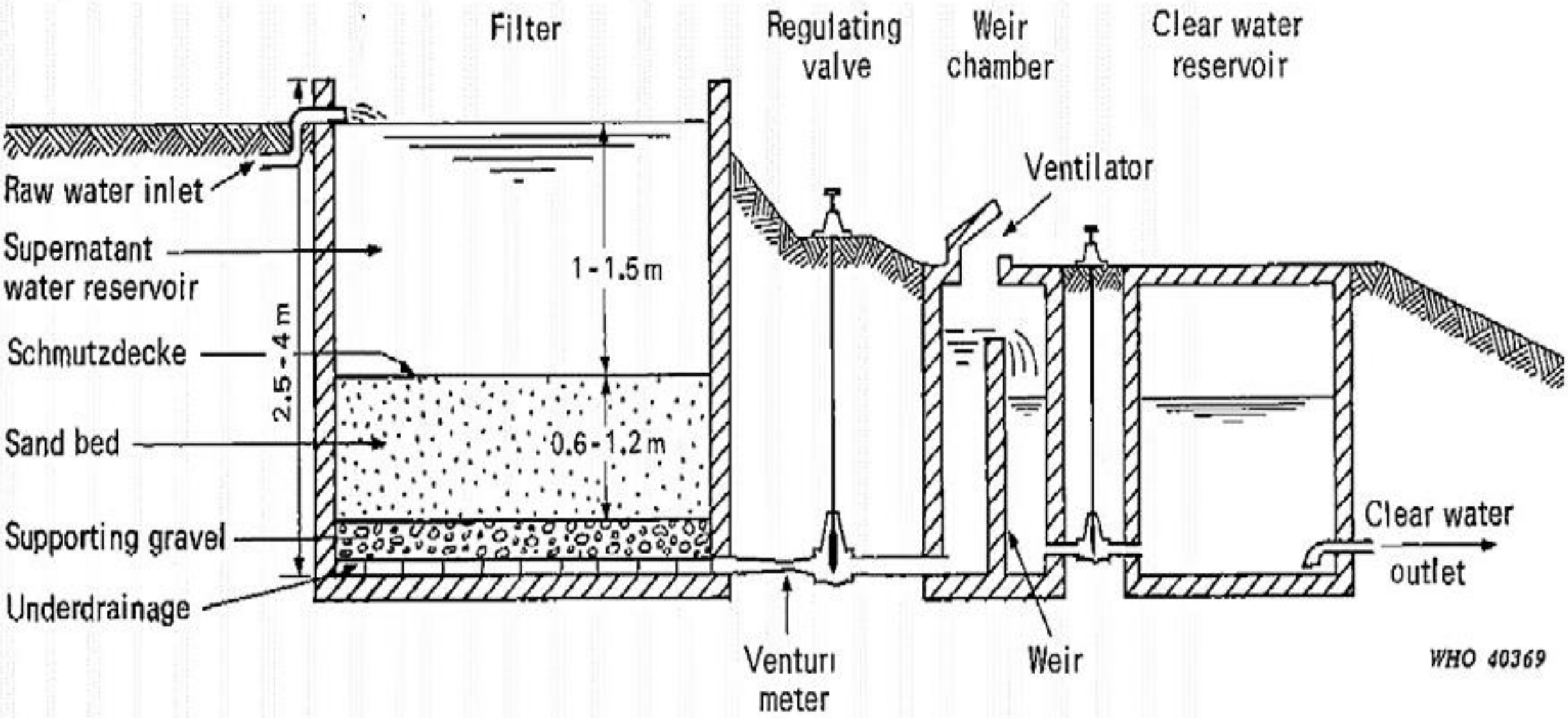
- Water is allowed at a slow rate through a bed of sand, so that coarse suspended solids are retained on or near the surface of the bed. Further, impurities in the water are removed by a combination sedimentation, straining, adsorption, and chemical and bacteriological action.

- Typically used for smaller flow requirements (rural water supply schemes) and lesser turbid raw waters. The raw water turbidity should be < 50 NTU.

- Usually, loading rate is kept at 0.1 - 0.2 $m^3/h.m^2$, however loading rates up to 4 $m^3/h.m^2$ is reported.



A Typical Slow Sand Filter



Functioning of Slow Sand Filter

- During the first few days, water is purified mainly by mechanical and physicochemical processes.
- The resulting accumulation of sediment and organic matter forms a thin layer on the sand surface, which remains permeable and retains particles even smaller than the spaces between the sand grains.
- As this layer (referred to as "*Schmutzdecke*") develops, it becomes living quarters of vast numbers of micro-organisms which break down organic material retained from the water.
- Most impurities, including bacteria and viruses, are removed from the raw water as it passes through the filter skin and the layer of filter bed sand just below.
- The purification mechanisms extend from the filter skin to approx. 0.3-0.4 m below the surface of the filter bed, gradually decreasing in activity at lower levels as the water becomes purified and contains less organic material.
- When the micro-organisms become well established, the filter will work efficiently and produce high quality effluent which is virtually free of disease carrying organisms and biodegradable organic matter.

General Design Criteria for Slow Sand Filter

IS 11401 (Part 2) : 1990

Design Criteria	Recommended Value
Design period	15 to 30 years
Period of operation	24 hours per day
Water demand	40 litre per capita per day for rural water supply
Filtration rate	0·1 — 0·2 m/h
Filter bed area	200 m^2 per filter, minimum of 2 units
Depth of filter sand:	
a) Initial	0·8 — 0·9 m
b) Minimum	0·5 —0·6 m
Specification of sand:	
a) Effective size	0·15 — 0·30 mm
b) Uniformity coefficient	*Max* 5, preferably below 3
Height of underdrain including gravel layer	0·3 — 0·5 m
Height of supernatant water	1 m

General Design Criteria for Slow Sand Filter: MoUD Guidelines

Description	Recommended Design value
Design period	10 years
Filteration rate	
Normal Operation	0.1m/hr
Max. overload rate	0.2m/hr
Depth of Supernatant water	1.0m
Free Board	0.2m
Depth of filter sand Initial	1.0m
Final (Minimum)	0.4m

Description	Recommended Design value
Number of filter beds	2
Minimum	
Areas upto 20 m^2	2
Areas upto 20-249 m^2	3
Areas upto 250-649 m^2	4
Areas upto 650-1200 m^2	5
Areas upto 1201-2000 m^2	6

Description	Recommended Design value
Size of sand	0.2-0.3mm
Effective size	
Uniformity coefficient (U.C)	5
Gravel (3-4 layers) depth	0.3m
Underdrains (made of bricks or perforated pipes)	0.2m
Depth of filter box	2.7m
Effluent weir level above sand bed	20-30mm

Slow Sand Filter Shape and Dimensions

- When filter beds are of an area less than about 125 m^2 and they are to be constructed fully or almost fully below ground, circular shape may be more economical than rectangular shape.
- For larger installations or when small filters are fully or partly above ground, rectangular beds in a battery may be more economical.
- The economical length (L) and width (B) of rectangular filters in a battery are given by:

$$L = \sqrt{2A/n + 1} \text{ and } B = \frac{(n+1) \times L}{2n}$$

Where, A = total area of filter beds, and n= no. of beds

A minimum of two working filter units must be provided. No need to provide any standby unit.

Slow Sand Filter Maintenance

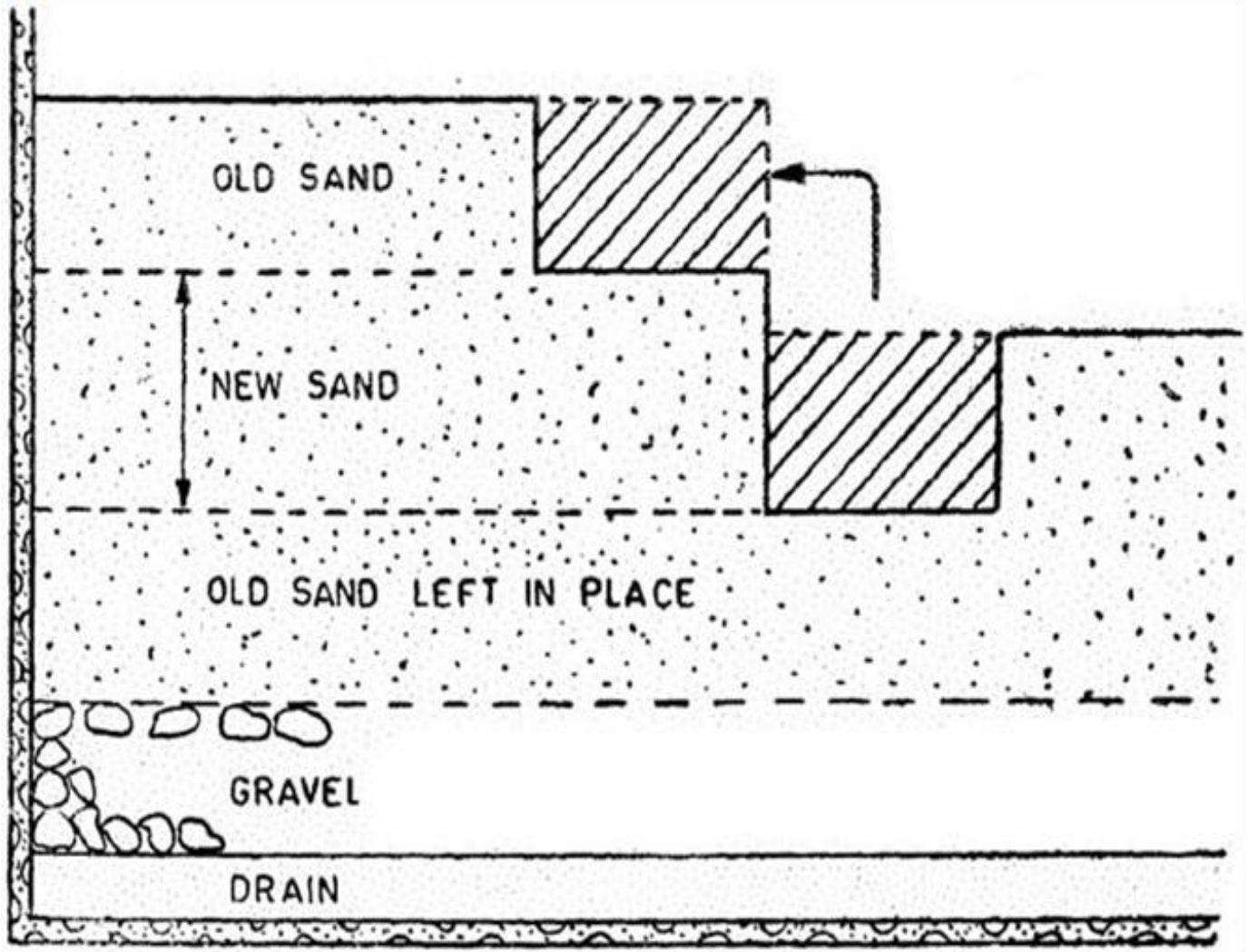
- Slow sand filters slowly lose their performance as the *Schmutzdecke* grows and thereby reduces the rate of flow through the filter requires refurbishing.
- The common methods used for refurbishing Slow Sand Filters including Scrapping and Wet Harrowing.

  - *Scrapping:* The top few mm of sand is carefully scraped off using mechanical plant and this exposes a new layer of clean sand. Water is then decanted back into the filter and re-circulated for a few hours to allowa new *Schmutzedecke* to develop. The filter is then filled to full depth and brought back into service.

  - *Wet Harrowing:* Lower the water level to just above the *Schmutzdecke*, stirring the sand and thereby suspending any solids held in that layer and then running the water to waste. The filter is then filled to full depth and brought back into service.

Slow Sand Filter Maintenance: Resanding

- When the sand level in a filter has been reduced to the minimum permissible level of 0·5 to 0·6 m, the filter bed should be resanded.
- Throwing over method:
  - In this process, the bottom layer of sand is moved to one side, new Sand is placed to make up the level. The old sand is then put back on top of the new sand.
  - Throwing over ensures that the bottom sand layer is also subjected to scraping in due course. It also ensures the presence of biologically active sand at the top after resanding, which will help in quick re-ripening of the filter.



Design of Slow sand filter:

Filters are based on ROF.
designed

- Total Surface area of slow sand filter req. = $\frac{Q}{ROF}$

Fix Size of each filter	Fix no. of filter 'n'
L & B	find size of each filter
Find no. of filters = ?	L = ?
	B = ?

Note: No. of filters in operation = n
No of filters provided = n+1 (or) n+2
+1 — act as Stand by filter.

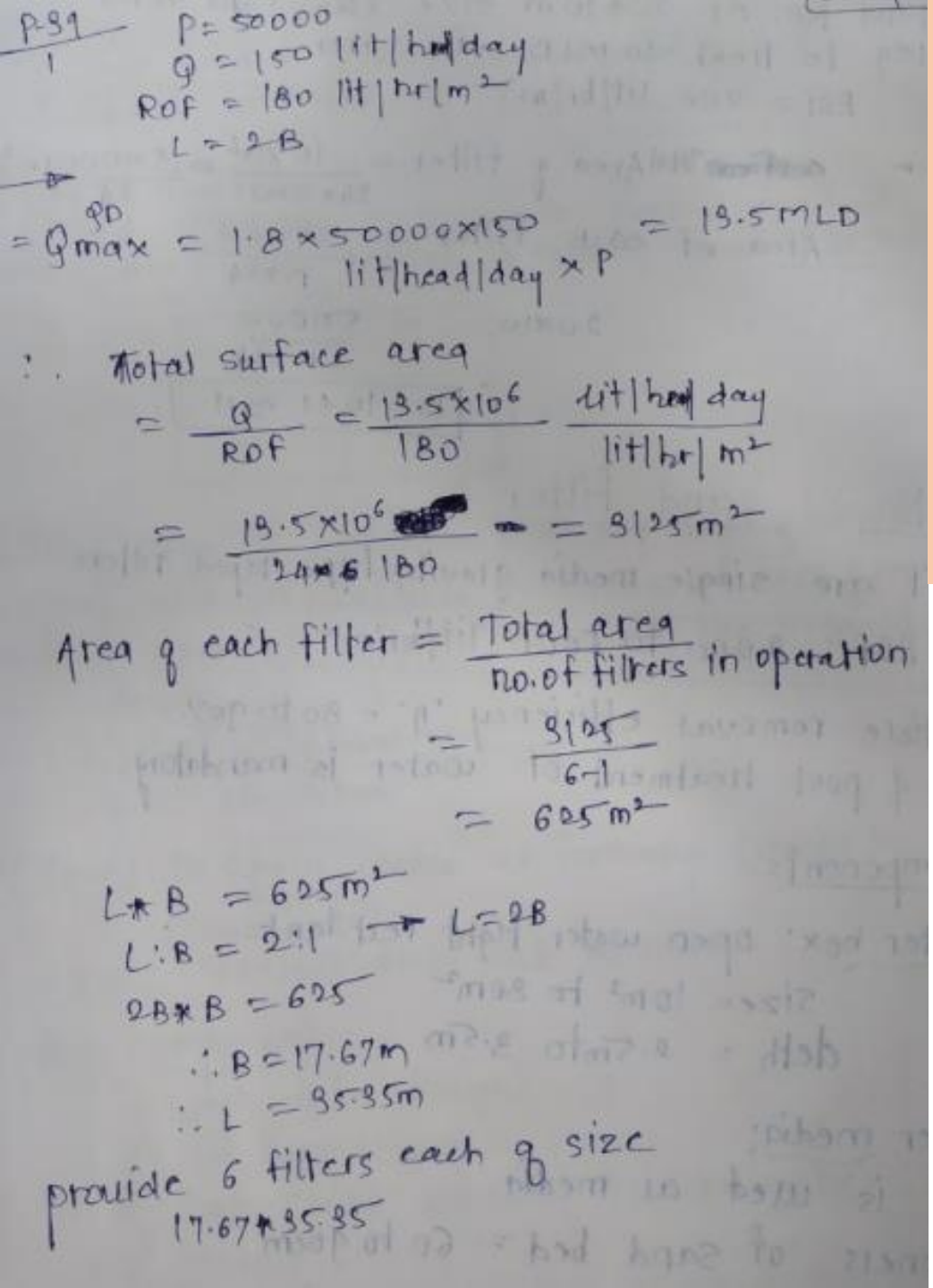
Advantages and Disadvantages of Slow Sand Filter

Advantages

- Simple to construct and supervise. Local materials can be used for construction
- Suitable where land is readily available
- Effective in bacterial removal
- Preferable for uniform quality of treated water
- Long lifespan (estimated >10 years)

Disadvantages

- Large area is required
- Unsuitable for treating highly turbid waters
- Less flexibility in operation due to seasonal variations in raw water quality
- Lower efficiency in colder temperatures, due to a decrease in biological activity



Design six slow sand filter beds from following data :

Population = 50000, Precipatation Demand = 150 lit/head/day, Rate of filteration = 180 lit/hr/m2 , L=2B . Assume max demand as 1.8 times average daily demand . Also assume that one unit out of six will be kept as standby

Slow Sand Filter

Design a slow sand filter to treat a flow of 900 m^3/d, assuming a filtration rate of 0.15 m/h.

Solution Given: Q = 900 m^3/d = 37.5 m^3/h

Filtrate rate v = 0.15 m/h

Required tank area = Q/v = 37.5/0.15 = 250 m^2

Consider 4 beds, each of 62.5 m^2 area [MoUD & CPHEEO guidelines recommend 4 beds for area between 250 to 649 m^2]

The Economic length and width of the filter could be determined as:

L= $\sqrt{2A/(n+1)}$ = $\sqrt{2*250/(4+1)}$ = 10 m; and W = (n+1)L/2n = (4+1)*10/(2*10) = 6.25 m

Assume depth of filter sand = 1 m Assume gravel depth = 0.3 m

Assume system underdrain depth = 0.2 m Assume free board = 0.2 m

Assume depth of supernatant water = 1 m

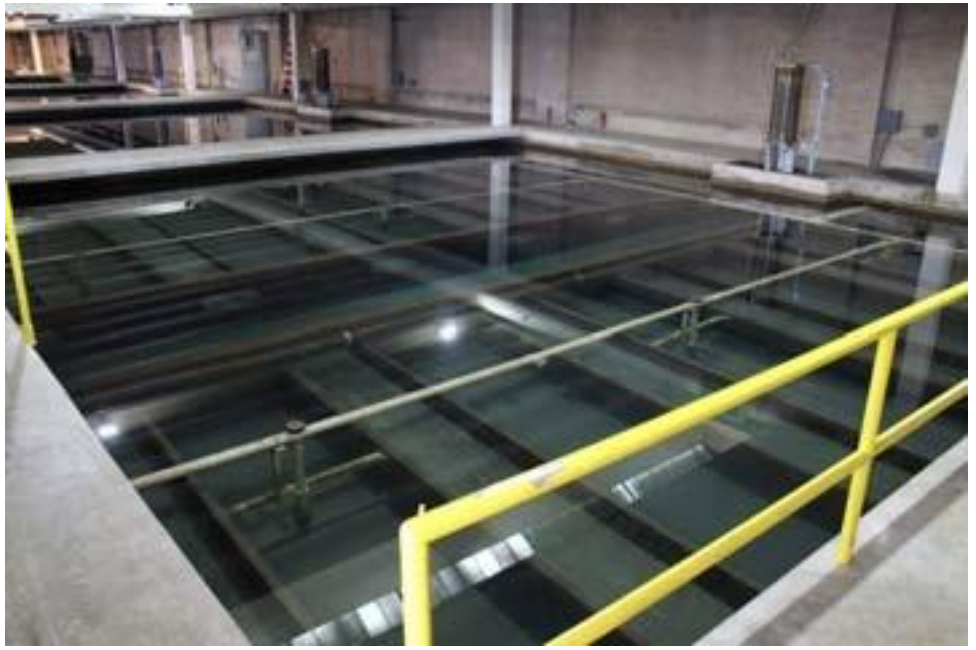
Therefore, total tank height = 1 + 0.3 + 0.2 + 1 + 0.2 = 2.7 m

So, final tank dimensions = ***4 Nos tanks @ 10 m long x 6.25 m wide x 2.7 m deep***

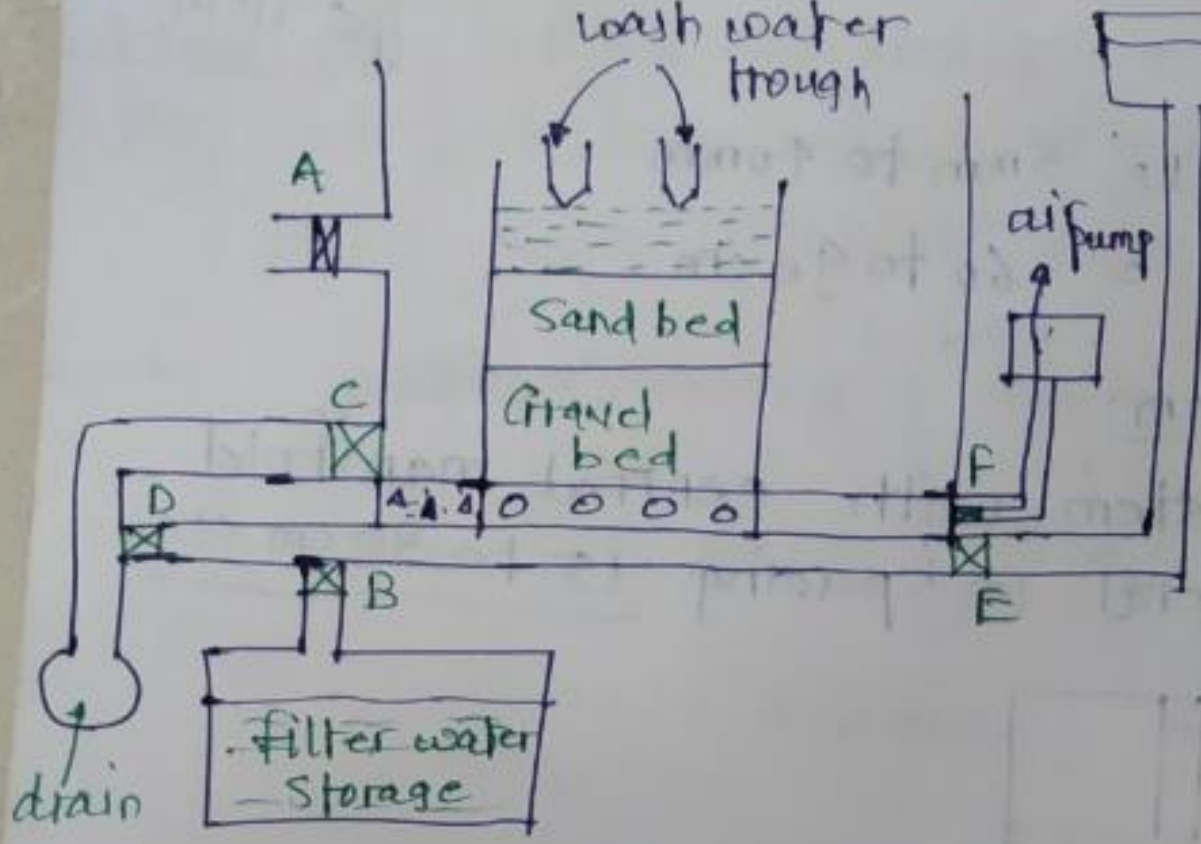
Rapid Gravity Filter



- The most common type of filter for treating municipal water supplies, especially for large flow requirements.
- Its based on purely physical or physico-chemical processes, mostly straining and adsorption. During filtration, water flows downward through the bed under gravity.



- Based on filter material, three types:
  - *Single-media filters:* Only one type of media, usually sand, sometimes crushed anthracite coal.
  - *Dual-media filters:* Two layers of two different types of media, usually crushed anthracite coal and sand.
  - *Multi-media filters:* Three or more layers of different types of media, usually anthracite, sand,



A: Influent Valve
B: Filtered Water Valve
C: Wash water valve
D: filtered wash water valve
E: Back wash water valve
F: Air valve

Initial head loss = 10 to 15cm
Terminal head loss = 3m
Turbidity brake through = 2.5 NTU

If any of these condi
occur, indicated filters
exhausted no more
working with rated
capacity badly need
"cleaning & washing".

Rate of Filtration in Rapid Gravity Filters

- The rate of filtration in sand filters depends on the effective sand sizes. Coarser sand particles results in wider pore spaces thus higher rate of filtration, while finer sand particles results in smaller pore opening and lesser rate of filtration. The standard rate of filtration through rapid sand filters is usually
  4.8 to 6 m/h (as per CPHEEO Manual).
- However, various other literature sources claims different ranges for the rate of filtration for Rapid Sand Filters, and these filters can be designed and operated for the rates ranging from 2.4 to 12 m/h.
- The lower rates usually provide better quality effluent (more time spend in passing through filter media) thus higher factor of safety, but naturally lower rates result in the need for larger filter area.
- As getting larger land in cities becoming increasingly difficult, filtration practice trend is towards higher rates of filtration up to 10 m/h with the use of coarser sand (effective size 1 mm) and adequate pre-treatment of water.
- Choice of the design rate of filtration requires the use of judgment considering various factors affecting filter design, construction and operation.

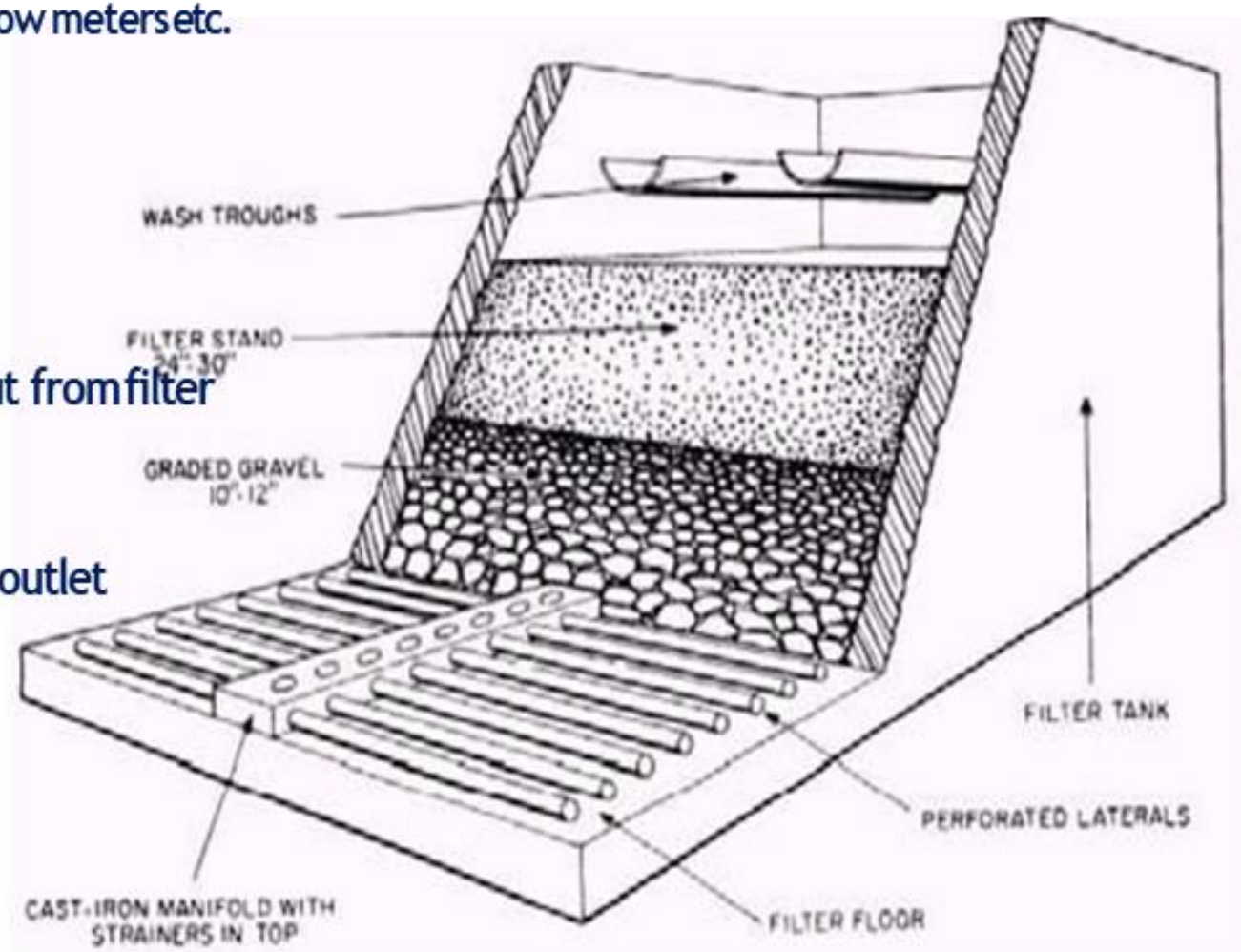
Filter Capacity and Sizing

- Selecting an appropriate rate of filtration, the size (are of the filter) could be determined.
- The capacity of the rapid sand filters should be such that the number of units can take care of the total quantity of water to be filtered, and is optimum to keep the filters working without undue overloading at any time.
- The smaller the number of units, the fewer the appurtenances but the larger the wash-water equipment that will be required. Thus while designing large size filters one must consider the rate at which wash-water must be supplied and the hydraulic problems for securing uniform distribution of wash-water due to the large area.
- A maximum area of 100 m^2 for a single unit is recommended for plants of greater than 100 MLD consisting of two halves each of 50 m^2 area.
- Also for flexibility of operation a minimum of four units should be provided which could be reduced to two for smaller plants.

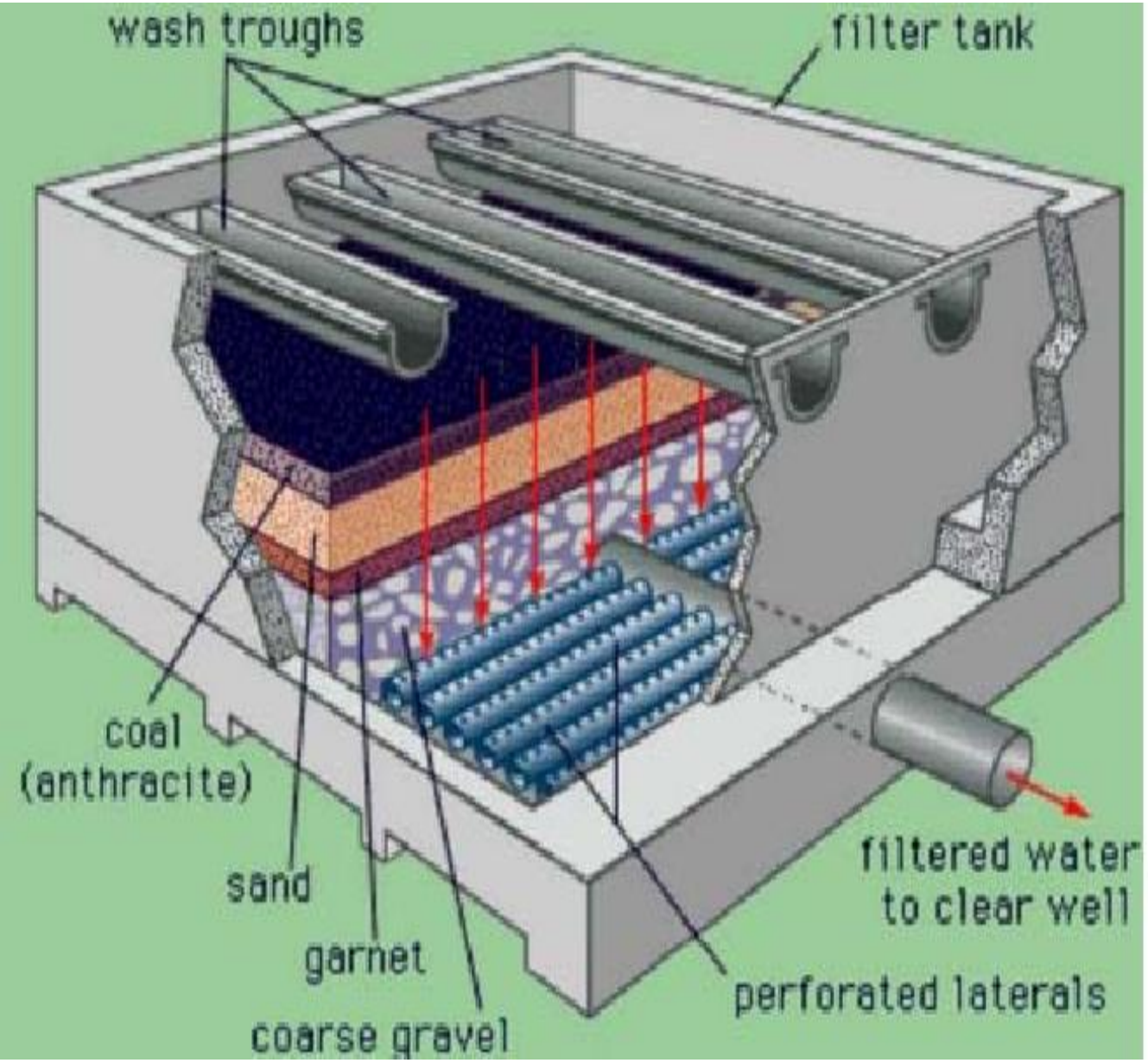
Components in a Rapid Sand Filter (Rapid Gravity Filter)

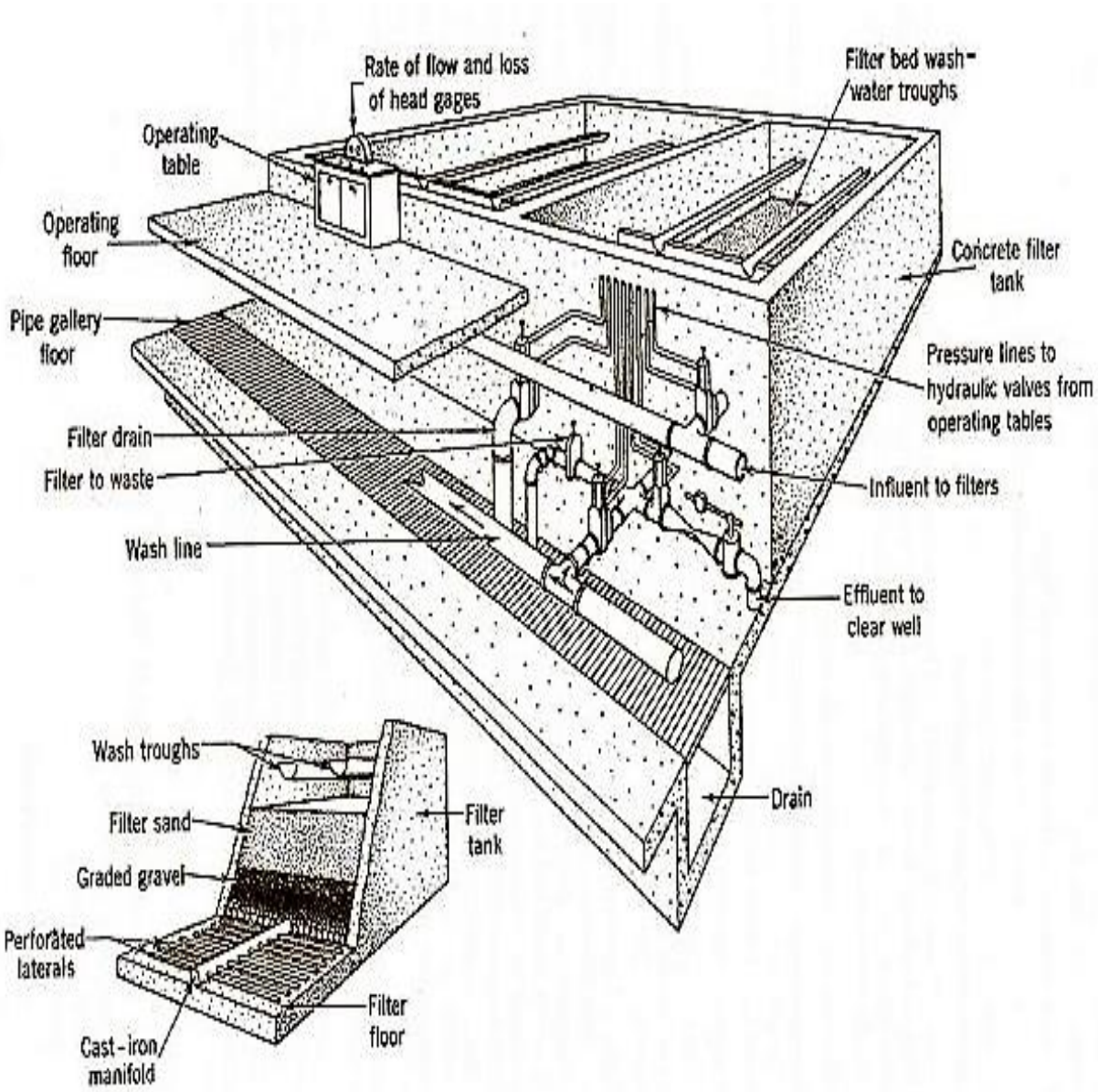
- *Chamber: Filter Tank or Filter Box*
  - To house the other components of the filter
- *Filter media (sand)*
  - To remove the target impurities from water
- *Gravel support*
  - To support the filter media and preventing its washout from filter
- *Under drainage system*
  - To collect the filtered water and channelize it to filter outlet
- *Wash water troughs*
  - To collect the wash water after filter backwashing

Appurtenances:
Air compressors, Rate control devices, Head loss indicators, Flow meters etc.



Components in a Rapid Sand Filter (Rapid Gravity Filter)





Major Components in Rapid Sand Filters

- *Chamber (Filter Tank or Filter Box):* Open water tight tank, usually rectangular or square in shape, constructed of reinforced concrete. The filter box must be strong enough to support weight of under drain system, filter medium and water column.

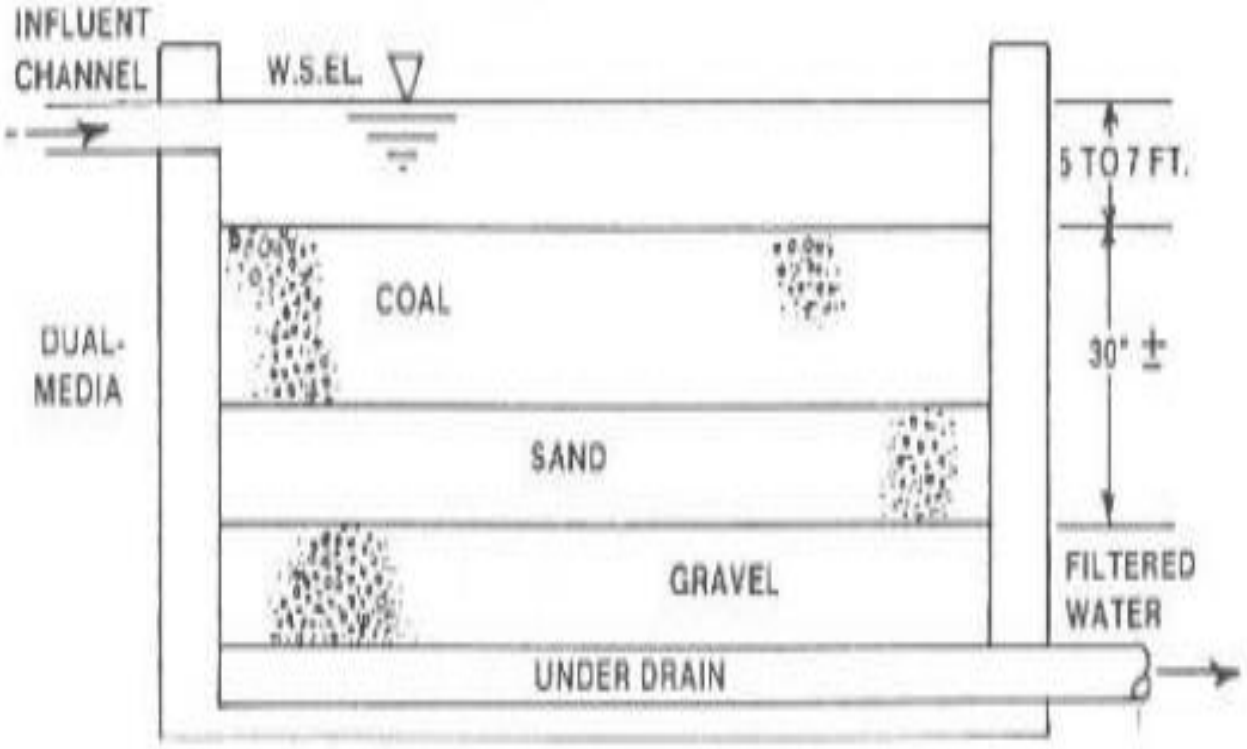
- *Filter Media (sand):* Most commonly silica sand is used. Modern filter applications make use of anthracite coal, garnet sand, in place of or in combination with silica sand. Smaller the media size, smaller filter pore openings, hence better filtration efficiency. But smaller pore size will result in increased headloss through the medium resulting in dimnished flow rate. Larger media increase pore size, reduce head loss and increase flow rate but filtration efficiency will be affected.

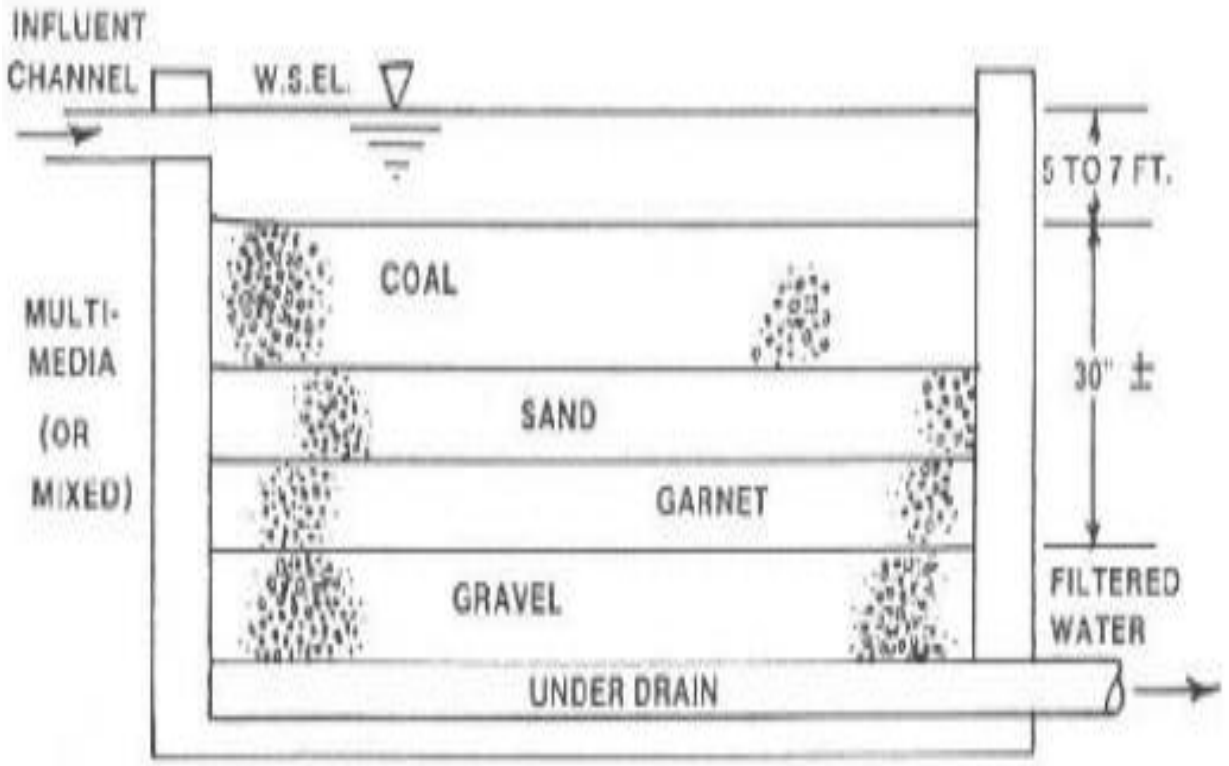
- *Gravel Support:* It serves as a base material and is placed between sand and under drainage system to prevent sand washout. It also helps in even distribution of wash water during backwashing. Graded arrangement of gravel with smaller size gravel at the top, and bigger gravels at the bottom is common.

- *Under drainage System:* Consisting of main and lateral pipe arrangements, it is system for collecting and removing filtered water, and to disperse backwash water. Hydraulically, it must be designed to handle backwash flow rates.

High Rate Filters

- High rate filters, which operate at three-to-four times the rate of rapid sand filters, use a combination of filter media, not just sand. Media combinations are generally sand and anthracite coal. Multi-media or mixed-media filters generally use sand, anthracite coal, and garnet.
- Filter media used in a high rate filter depends on the raw-water quality, raw-water variations, and chemical treatment used. Pilot studies help the operator evaluate which material, or combination of materials. will give the best result.





Common Filter Media and Base Materials

Removal object: Turbidity, SS, Fe

Filter Sand
(for rapid and slow filtration)

To use Filter Sand is a basic filtration method of water purification. Importance of Filter Sand, to work for physical treatment (catching the turbidity component) in rapid filtration and for biological treatment in slow filtration, is widely well-known. For the purpose of making the filtration rapider and water collection ratio more in recent years, the filter layer component and flow direction have been improved, the improved result and further improvement in quality are desired. We, as a manufacturer of Filter Sand, have investigated the quality control and manufacturing technology thoroughly since 1965.



Removal object: Turbidity, SS, Fe

Anthracite Filter Media

Anthracite Filter Media are filtration materials for turbidity and SS removal. Since the specific gravity is lighter than that of Filter Sand, it is much used in combination with Filter Sand in dual-media filtration, to consist reverse grain layers after backwashing. In addition, because of high content carbon of this product, it has high resistance to various chemicals and high physical strength so that it is widely used for city and industrial water purification.



Removal object: Turbidity, SS, Fe

Garnet

As filtration technology has made progress, the method has been improved from "Mono-medium filtration" by use of Filter Sand to "Dual-media filtration" by use of Anthracite Filter Media together. In addition, there is "Multi-media filtration" by use of Garnet together, as an advanced filtration method for taking measures against Cryptosporidium, for pretreatment of pure water making, etc. Garnet is a reddish brown mineral having approx. 4g/cm3 of greater specific gravity and exercises it's performance by using smaller size compared with Filter Sand.



Removal object: support

Gravel

Gravel is used for supporting granular filter media such as Filter Sand and Anthracite Filter Media that filter water. Gravel fulfills an important role not only to prevent filter media from flowing out of collecting equipment of filter in filtration process but also to distribute washing water evenly flowing into the filter in backwashing process.
Make the layer with 3 to 6 kinds of size in order from the big particles of Gravel, to prevent one-sided flow of filtration and unevenness of layer surface.



Filter Media Materials

Filter Media are materials that allow the passage of raw water but retains the particulate matters. Choice of material for filter media depends on their specific gravity, particle sizes, and contaminant retention abilities.

- *Sand* is one of the oldest and most commonly used filter media owing to its low cost, easy availability, and flexibility in operation. Sand filters can effectively remove fine particles and undissolved impurities.

- *Anthracite* is commonly used filter media in drinking or industrial water treatment, mostly used in combination with sands. The special shape of Anthracite grains allows a higher flow, less pressure drop and a better and faster backwash than plain sand filters.

- *Garnet* is a chemical and abrasive resistant granular filter media having high hardness and high density. It is considered vey effective as the lower strata in dual/multi media filter beds.

- *Activated Carbon/Charcoal* is increasingly getting popular due to its potential of effectively removing many dissolved organics and toxins from the water.

Filter Media Characteristics

The effective size (ES) of filter media is considered as d_{10}, defined as the sieve size opening where 90 % of particles, by weight, are retained and only 10 % passes (only 10 % particles are of smaller dia than d_{10}).

The larger the grain size, the faster the water moves through the sand (higher rate of filtration). However, if the grain size is too large, treatment efficiency will be reduced.

The uniformity coefficient (U_c) is a numeric estimate of how sand is graded, and is a dimensionless number. Sand with all the particles in similar size ranges would be defined as narrowly graded sand and would have a low U_c. Sand with near equal proportions in all the fractions would be defined as widely graded sand and would have a high U_c value.

The U_c is calculated as d_{60}/d_{10}

(d_{60} sieve size that passes 60% by weight, and d_{10} is sieve size that passes 10% by weight)

The larger the U_c the less uniform the sand. It is important that the sand grains all be about the same size; i.e. relatively uniform. A uniformity coefficient of 4 or less is recommended for all filter media.

Typical Filter Media Characteristics

Single-Media Filter Characteristics

CHARACTERISTIC	VALUE Range	Typical
Sand medium:		
Depth		
in.	24–30	27
(mm)	(610–760)	(685)
Effective size, mm	0.35–0.70	0.60
Uniformity coefficient	<1.7	<1.7
Anthracite medium:		
Depth		
in.	24–30	27
(mm)	(610–760)	(685)
Effective size, mm	0.70–0.75	0.75
Uniformity coefficient	<1.75	<1.75
Filtration rate:		
gpm/ft^2	2–5	4
(ℓ/s-m^2)	(1.36–3.40)	(2.72)

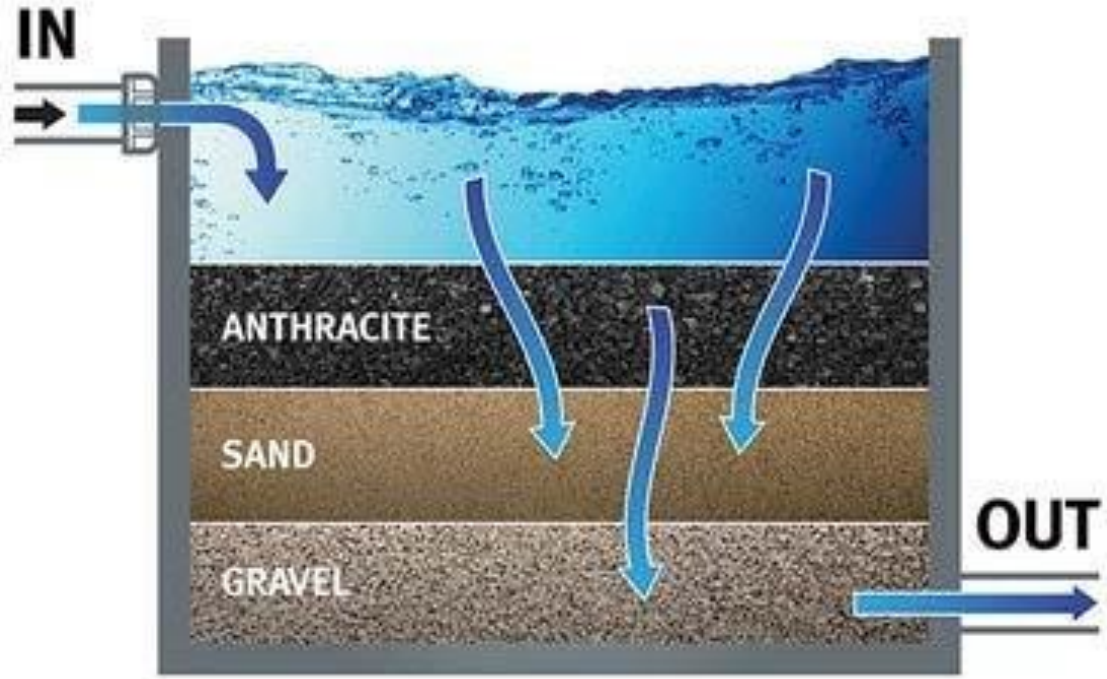
Dual-Media Filter Characteristics

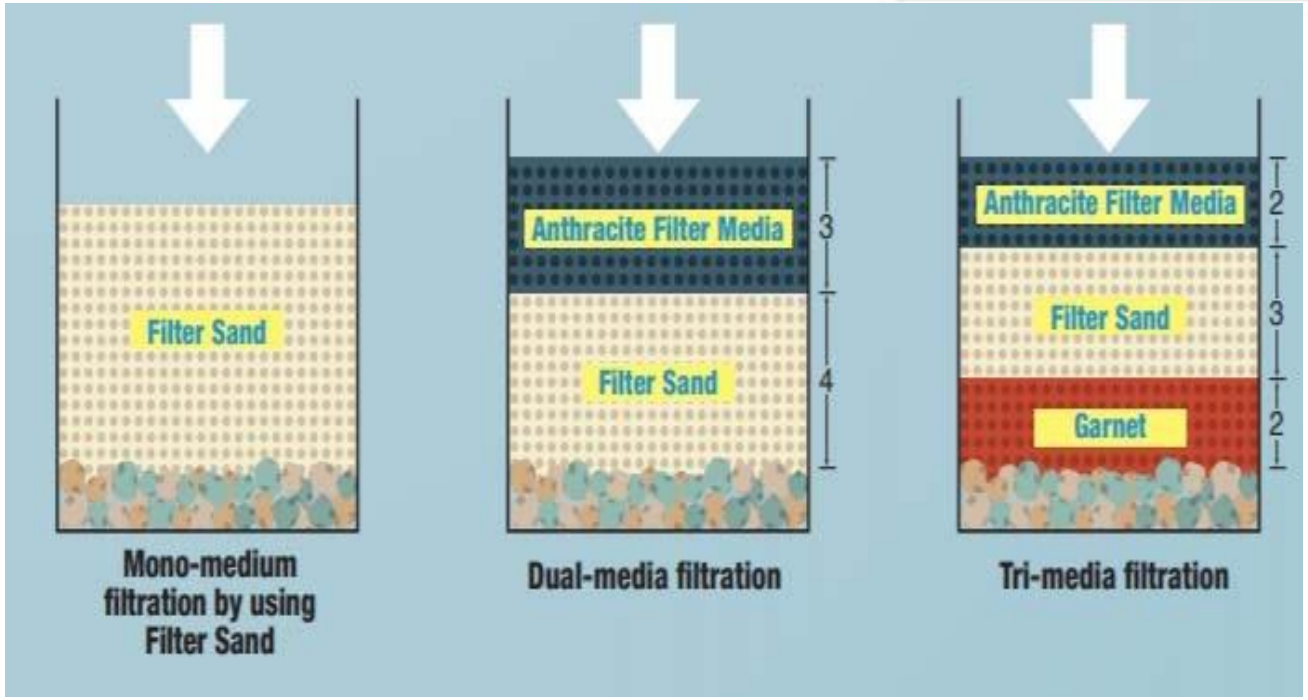
CHARACTERISTIC	VALUE Range	Typical
Anthracite:		
Depth		
in.	18–24	24
(mm)	(460–610)	(610)
Effective size, mm	0.9–1.1	1.0
Uniformity coefficient	1.6–1.8	1.7
Sand:		
Depth		
in.	6–8	6
(mm)	(150–205)	(150)
Effective size, mm	0.45–0.55	0.5
Uniformity coefficient	1.5–1.7	1.6
Filtration rate:		
gpm/ft^2	3–8	5
(ℓ/s-m^2)	(2.04–5.44)	(3.40)

Multi-Media Filter Characteristics

CHARACTERISTIC	VALUE Range	Typical
Anthracite:		
Depth		
in.	16.5–21	18
(mm)	(420–530)	(460)
Effective size, mm	0.95–1.0	1.0
Uniformity coefficient	1.55–1.75	<1.75
Sand:		
Depth		
in.	6–9	9
(mm)	(150–230)	(230)
Effective size, mm	0.45–0.55	0.50
Uniformity coefficient	1.5–1.65	1.60
Garnet:		
Depth		
in.	3–4.5	3
(mm)	(75–115)	(75)
Effective size, mm	0.20–0.35	0.20
Uniformity coefficient	1.6–2.0	<1.6
Filtration rate:		
gpm/ft^2	4–10	6
(ℓ/s-m^2)	(2.72–6.80)	(4.08)




IN
ANTHRACITE
SAND
GRAVEL
OUT


Filter Sand
Mono-medium filtration by using Filter Sand
Anthracite Filter Media
3
Filter Sand
4
Dual-media filtration
Anthracite Filter Media
2
Filter Sand
3
Garnet
2
Tri-media filtration

Filter Sand Specifications in Rapid Sand Filters (CPHEEO Manual)

- Usually the sand layer has a depth of 0.60 to 0.75 m, but for higher rate filtration when the coarse medium is used, deeper sand beds are suggested.
- The standing depth of water over filter varies between 1 and 2m.
- The free board above the water level should be at least
  - 0.5 m, so that when air binding problems are encountered, it will facilitate the additional levels of
  - 0.15 to 0.30 m of water being provided to overcome the trouble.

Shape, size and quality of filter sand shall satisfy the following norms:

(a) Sand shall be of hard and resistant quartz or quartzite and free of clay, fine particles, soft grains and dirt of every description.

(b) Effective size shall be 0.45 to 0.70 mm.

(c) Uniformity coefficient shall not be more than 1.7 nor less than 1.3.

(d) Ignition loss should not exceed 0.7 per cent by weight.

(e) Soluble fraction in hydrochloric acid shall not exceed 5.0% by weight.

(f) Silica content should be not less than 90%.

(g) Specific gravity shall be in the range between 2.55 to 2.65.

(h) Wearing loss shall not exceed 3%.

IS:8419 (Part I) 1977 entitled Filtration Media sand and Gravel may be referred to for details.

Gravel Specifications in Rapid Sand Filters

- Usually, the conventional rapid and filters consists of 45 to 60 cm thick gravel bed, graded and laid in layers, having smaller size gravels at the top most layer, and bigger size gravels at the bottom most layer.

	Thickness / depth	*Size*
Top layer	15 cm	2 to 6mm
Intermediate layers	15 cm	6 to 12 mm
	15 cm	12 to 20 mm
Bottom layer	15 cm	20 to 50 mm
Total	60 cm	

- The depth of gravel layer varies according to the type of the filter bottom and under-drainage system used. CPHEEO recommends the size of the gravel and depth of gravel layer as:
  - *For strainer or wheeler type underdrain system:*
    Gravel shall be of 2 mm minimum size, 50 mm maximum size and 0.30 to 0.50 m deep
  - *For perforated pipe underdrain system:*
    Gravel shall be 2 mm minimum size, 25 mm maximum size and 0.5 m, in depth.

Types of Under-Drainage System

Manifold with perforated lateral pipes (oldest type)

- Perforated pipe laterals are located at frequent intervals along manifold
- Perforation in laterals (6-13 mm) located at 8-30 cm spacing (Ref: AWWA, 1990)
- Openings of underdrain system is larger than the filter medium to be supported
- To prevent the medium from leaking downward into the underdrain system, several layers of graded gravel between the underdrain openings and filter medium is necessary

Fabricated self-supporting underdrain systems [Grouted to the filter floor with top openings of about 6 mm)

False-floor underdrain with nozzles

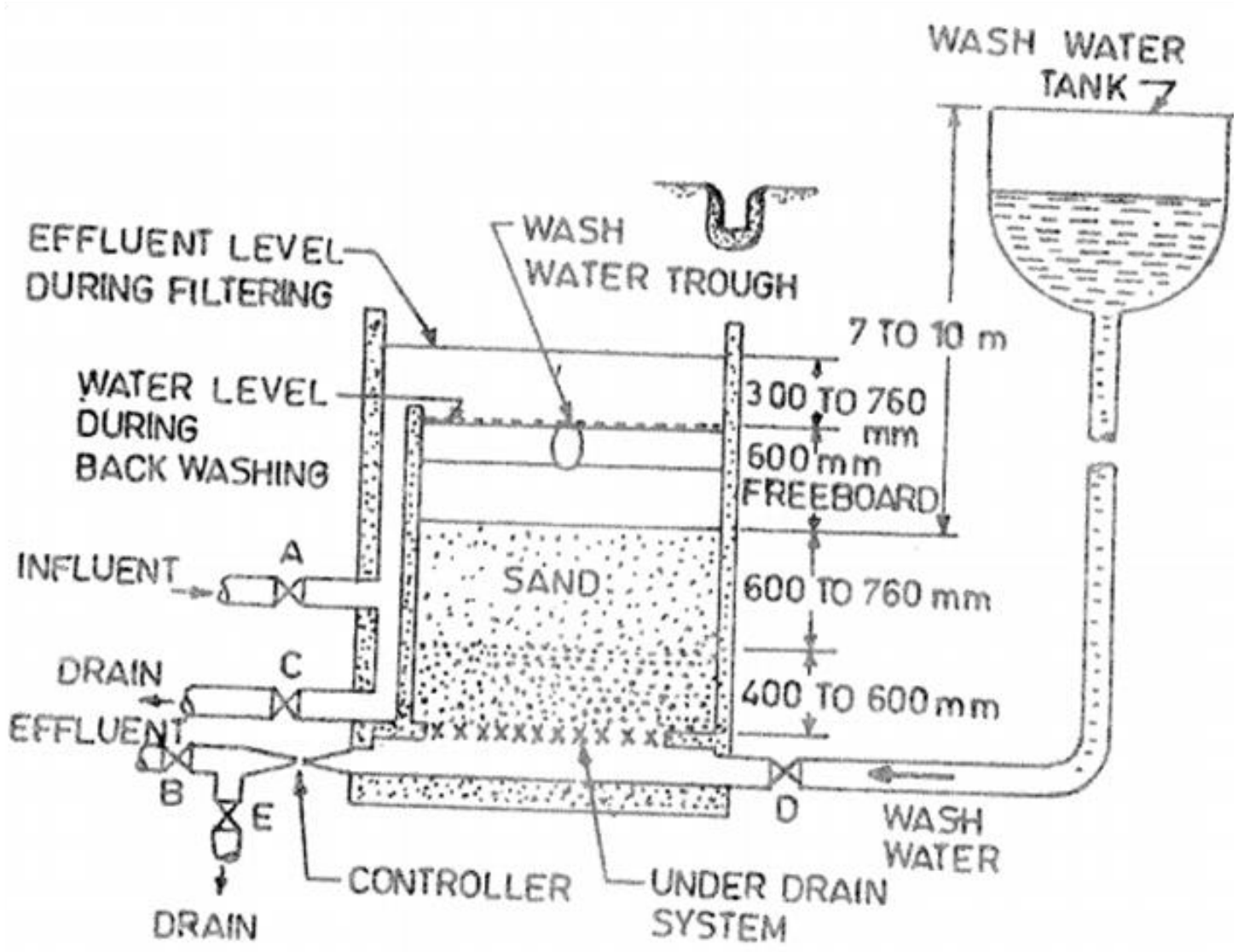
- A false-floor slab is located 0.3–0.6 m above the bottom of filter, providing an underdrain plenum below the false floor
- Nozzles to collect the filtrate and distribute the backwash water are located at 3-20 cm centers
- Openings of nozzles may be coarse (about 6 mm), or may be very fine (to retain the filter medium)
- Usually gravel layer is not required for such underdrain system

Rapid Sand Filter Operation

Usually operated under *Constant Head Variable Flow* or *Constant Flow Variable Head*, or combination of the two.

- *Constant Head Variable Flow:* Water level above the filter is kept at a preselected level. Since there is limited head loss using clean filters, there will be large flow rate initially. As the filters get clogged, head loss increases and filtration rate decreases. When the flow reaches design minimum, filters are back-washed.

- *Constant Flow Variable Head:* Water is introduced into filter at a constant rate, and an equilibrium is established between height of water column and the application rate. In the beginning, head is kept low (less head loss), however as filters get clogged and head loss increases the head level must increase to provide the necessary driving force. When water level reaches predetermined level, filter is backwashed and cycle repeated.

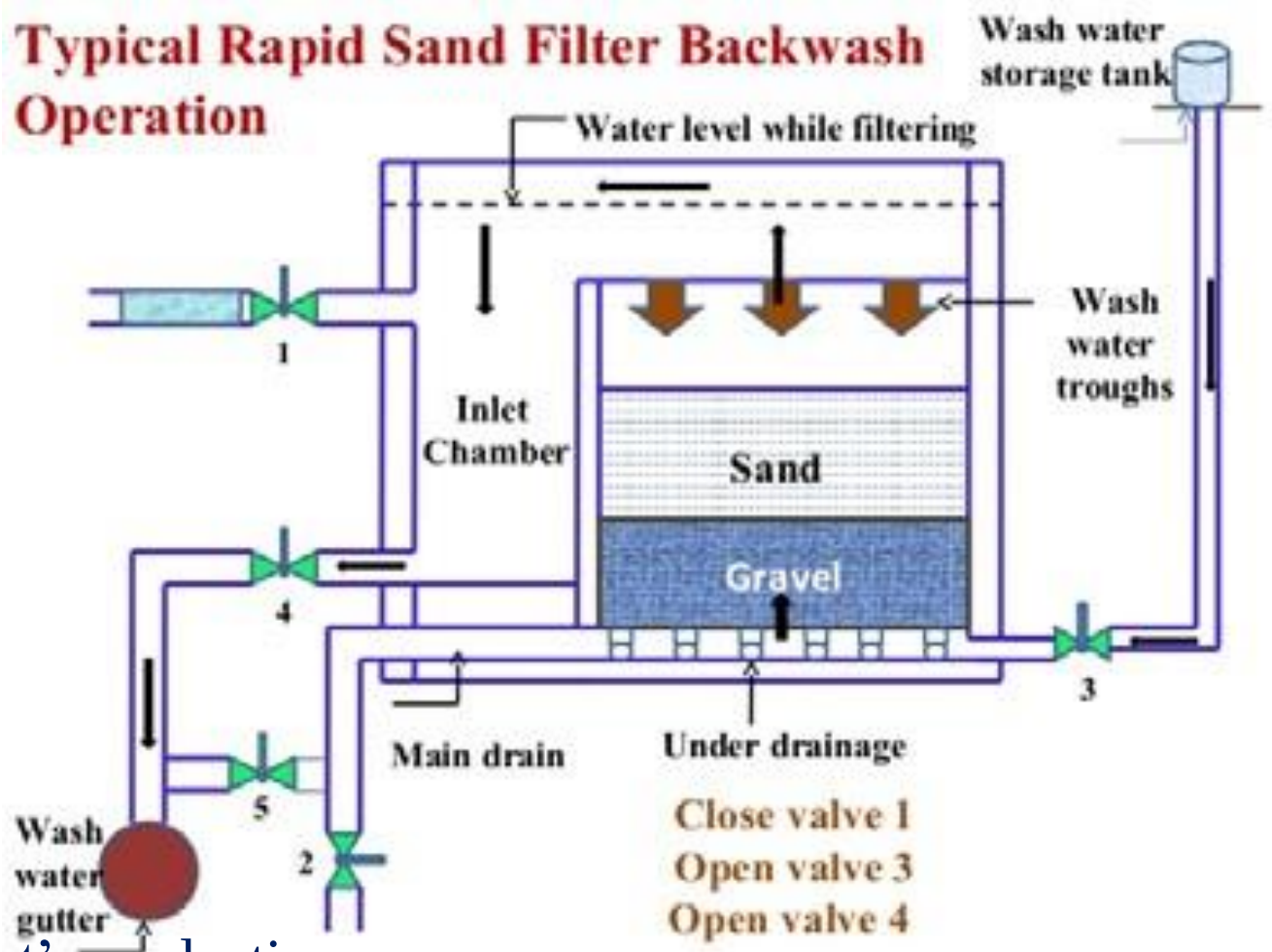


Rapid Sand Filter Operation

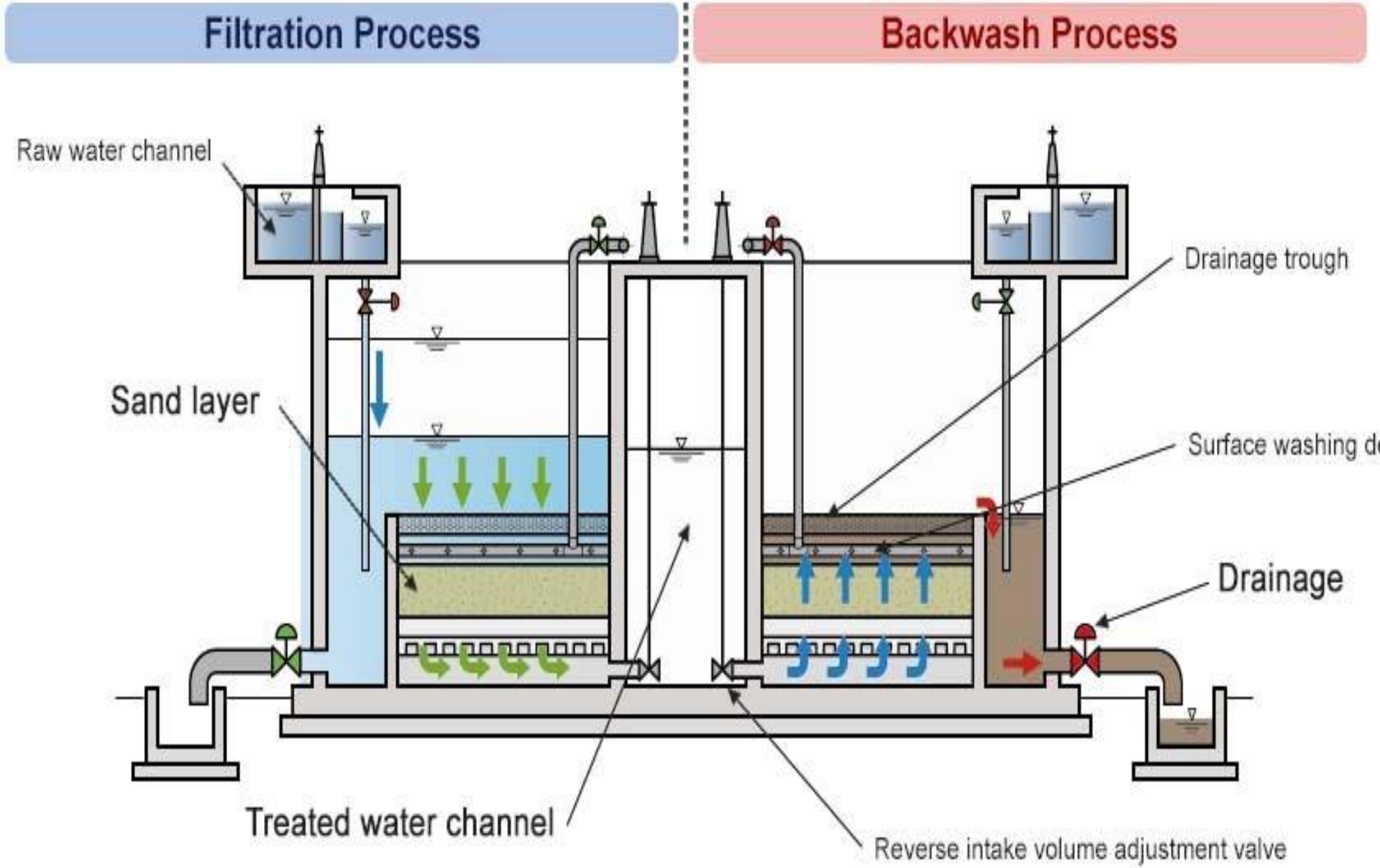
- More recent filters use a combination of the two operation methods.
  - A constant flow is delivered to several filters through common header and allowed to distribute itself according to operating rate of each filter.
  - Height of water column is same above all filters with cleanest filter accepting greatest flow.
  - When flow rate through 1 unit decreases to a predetermined level, it is taken off line and back washed. This causes an increase in head and flow rate through each filter.
  - Once the backwash is complete, the newly cleaned filter is returned to service which will accommodate a larger flow rate resulting in drop of water level and flow rate in other filters.
  - Regardless of operating mode a uniform flow is essential for best working of a granular medium filter.
  - Any variation in flow rate must occur gradually. Else the quality of effluent will deteriorate.

Rapid Sand Filter Backwashing

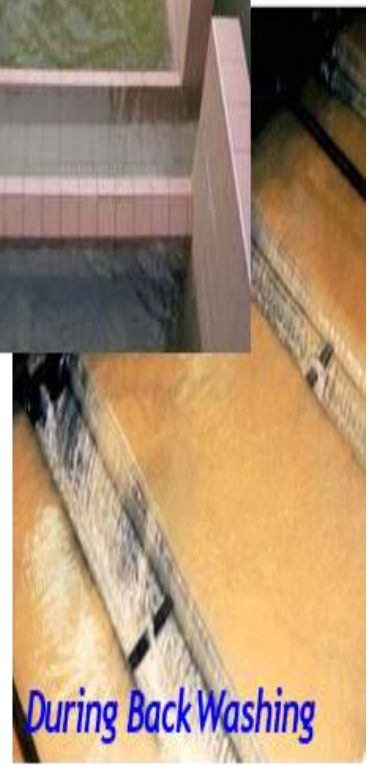
- After prolonged operation, when filter gets chocked or flow rate is substantially reduced (high head-loss, or beyond cut-off level) or treatment efficiency decreases (high turbidity in the effluent), the flow of water through the filter is reversed at higher flow-rates for cleaning out trapped particles. This process is called backwashing.
- When the filter is washed, clean water (usually filter effluent) is pumped upward through underdrainage system, slightly expanding the filter bed to "fluidized" state, and carrying away the accumulated impurities along with wash water through wash water troughs.
- Dirty backwash water is pumped into a settling pond, and usually recycled back into plant.
- Backwashing can consume 1% to 5% of a plant's production.



Rapid Sand Filter Backwashing







Operational Troubles in Rapid Gravity Filters: *Air Binding*

- When the filter is newly commissioned, the loss of head of water percolating through the filter is generally very small. However, the loss of head goes on increasing as more and more impurities get trapped into it.

- A stage is finally reached when the frictional resistance offered by the filter media exceeds the static head of water above the and bed.

- Most of this resistance is offered by the top 10 to 15 cm sand layer. The bottom sand acts like a vacuum, and water is sucked through the filter media rather than getting filtered through it. The negative pressure so developed, tends to release the dissolved air and other gases present in water. The formation of bubbles takes place which stick to the sand grains.

- This phenomenon is known as *Air Binding* as the air binds the filter and stops its functioning.

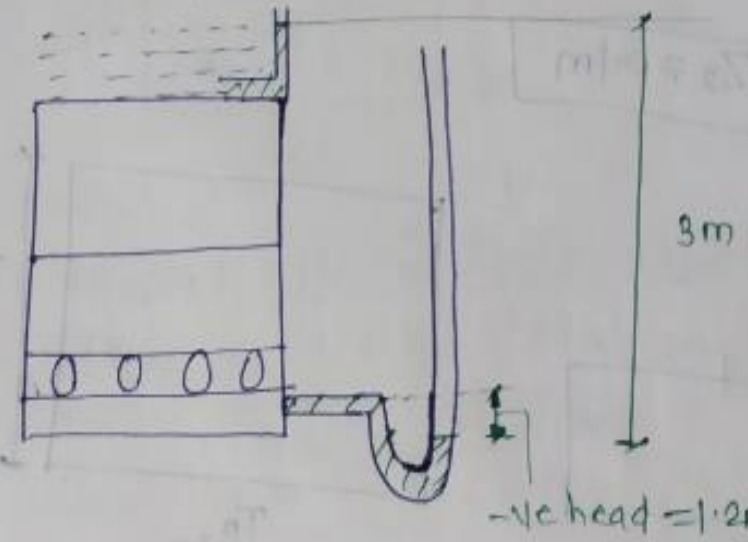
Remedial Measure:

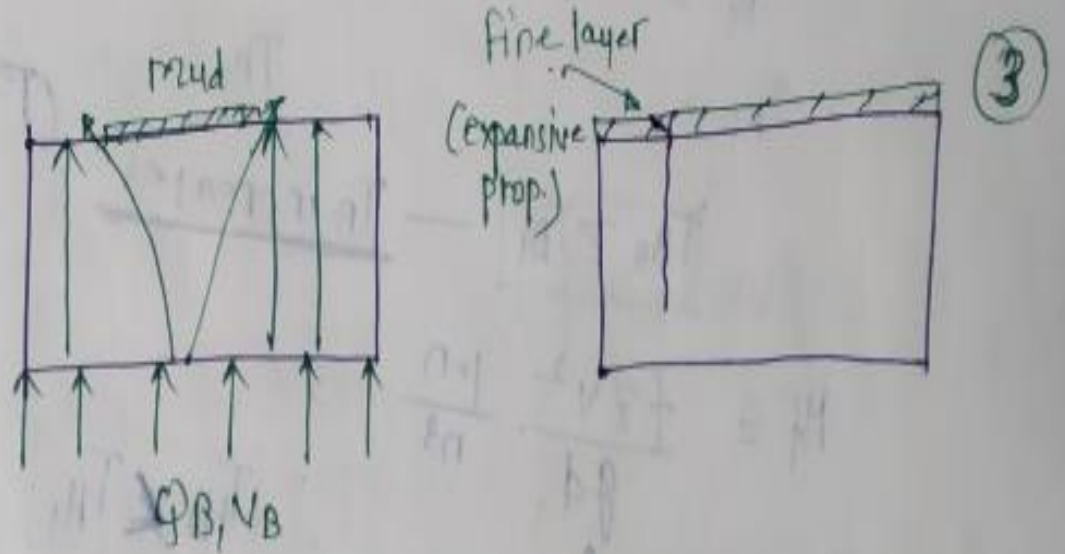
o The filters are cleaned as soon as the head loss exceeds the optimum allowable value.

Operational Troubles:

- *Formation of Mud Balls:* The mud from the atmosphere usually accumulates on the sand surface to form a dense mat. During inadequate washing this mud may sink down into the sand bed and stick to the sand grains and other arrested impurities, thereby forming mud balls.
- *Cracking of Filters:* The fine sand contained in the top layers of the filter bed shrinks and causes the development of shrinkage cracks in the sand bed. With the use of filter, the loss of head and, therefore, pressure on the sand bed goes on increasing, which further goes on widening these cracks.

1. Air Binding (when filter operated above -ve head of 1·2m)
2. Mud Ball formation (inadequate backwash of filter)
3. Cracking of filters





Remedial Measures:

- Breaking the top fine mud layer with rakes and washing off the particles.
- Washing the filter with a solution of caustic soda.
- Removing, cleaning and replacing the damaged filter sand.

Filter Hydraulics

- Water flow through filter are subjected to head losses, both during normal service run, as well as during backwashing.
- For laminar type of flow *Carman-Kozeny equation* or *Rose equation* is used, while *Ergun equation* or *Rose equation* may be used for transitional or turbulent flow regime.
- The laminar or transient flow condition may be established based on the Reynolds Number

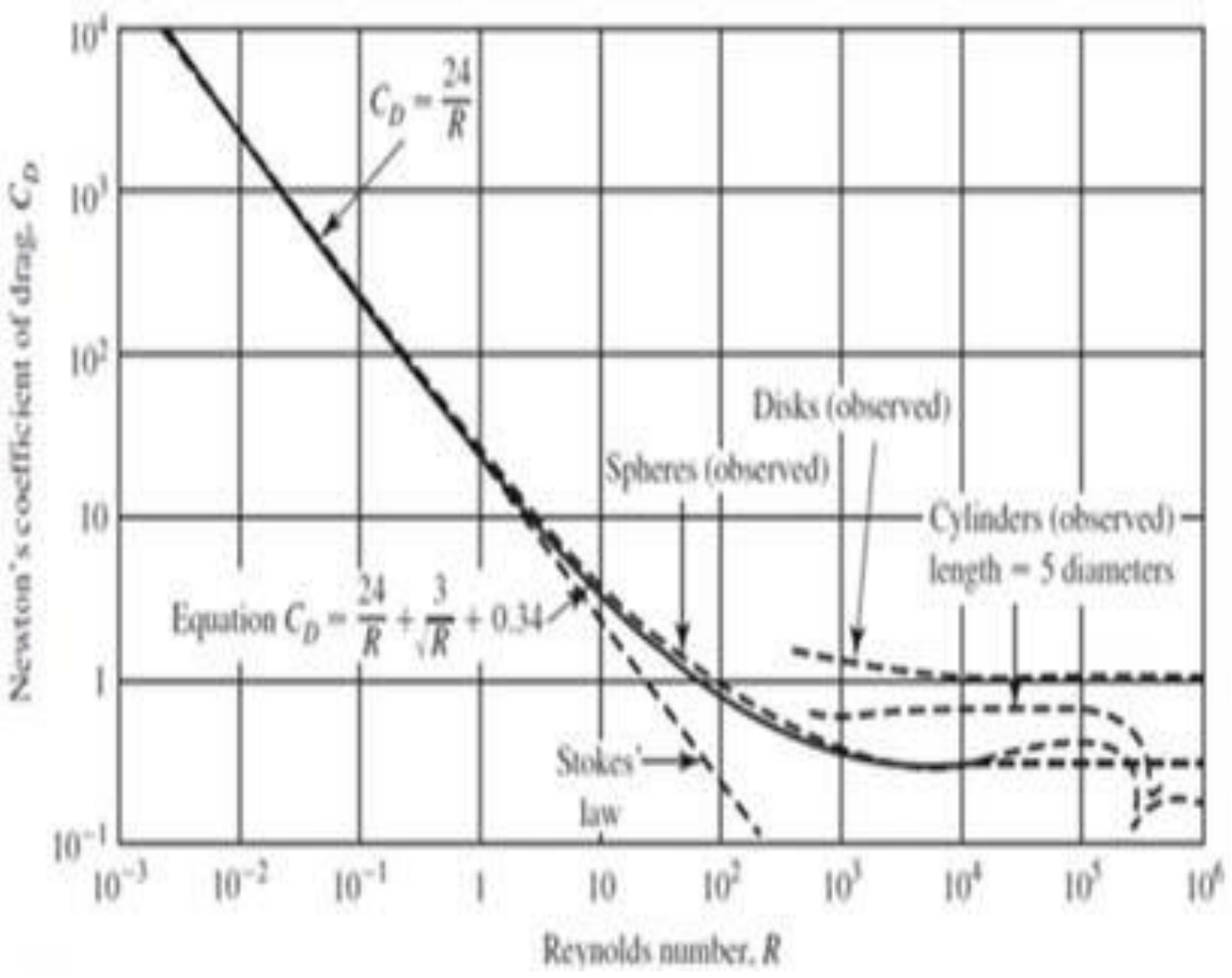
$$Re = (\psi d)\rho v/\mu$$

where,

ψ=Shape factor (to account for non-sphericity of the particles, [ψ= 1 for spherical grains, 0.98 for rounded grains, 0.94 for worn grains, 0.81 for sharp grains, 0.78 for angular grains, and 0.70 for crushed grains for sand medium.]

d= grain diameter, v = filtration velocity,

ρ = density, and μ = dynamic viscosity



Backwash Hydraulics

- During backwash, when water flows through the bed from the bottom towards top, filter grains are lifted and expansion of sand bed takes place.

- To hydraulically expand the bed, head-loss must at least equal to the buoyant weight of the particles in the fluid.

$$h_{fb} = L(1-e)\frac{\rho_m - \rho_w}{\rho_w}$$

Where, h_{fb} - head loss required to initiate expansion (m) L- depth of bed (m)
e - Porosity of fixed bed
ρ_w- density of water (kg/m^3)
ρ_m- density of medium (kg/m^3)

- The head-loss through an expanded bed is essentially unchanged because the buoyant weight of the bed is constant

Backwash Hydraulics

Weight of packed bed = Weight of fluidized bed

$$h = L(1-e)\frac{\rho_m - \rho_w}{\rho_w} = L_{fb}(1-e_{fb})\frac{\rho_m - \rho_w}{\rho_w}$$

Where, L_{fb}- depth of fluidized bed(m)=$L(1-e)/(1-e_{fb})$
e_{fb}- porosity of fluidized bed

$e_{fb}[= (v_B/v_t)^{0.22}]$ is a function of terminal settling velocity of particles and backwash velocity.
Where, v_t- terminal settling velocity, and v_B- backwash velocity = $Q_{backwash}/A$

For Stratified bed of non-uniform particles: Total expansion= ∑ expansion of individual layers

For uniform media $L_{fb} = L(1-e) / [1-(v_B/v_t)^{0.22}]$

For non-uniform media $L_{fb} = L(1-e) \sum [x_{ij} / \{1-(v_B/v_{t,ij})^{0.22}\}]$

Design of Rapid Sand filter:

Total Surface Area of RSF $= \frac{Q}{ROF}$

Fix size of each filter L * B no. of filter = n = ?	Fix no. of filter 'n' Find size of each filter L & B = ?

** Percentage of filter used in back washing
$= 100 * \frac{\text{Vol}^m \text{ of filtered water used in backwash}}{\text{Vol}^m \text{ of water filtered betn backwashing.}}$

Vol^m of filtered water betn backwashing
= ROF * Duration of filtration * Area of each filter

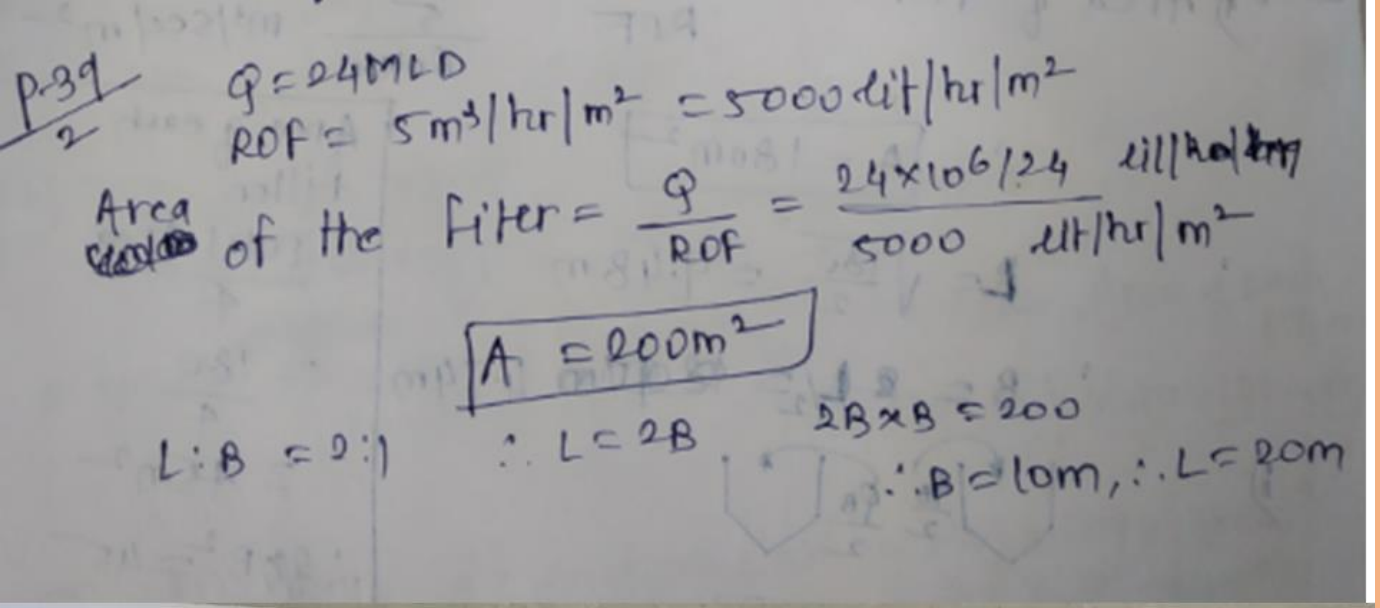
Vol^m of filtered water used in backwashing
= ROB * DOB * Area of each filter.

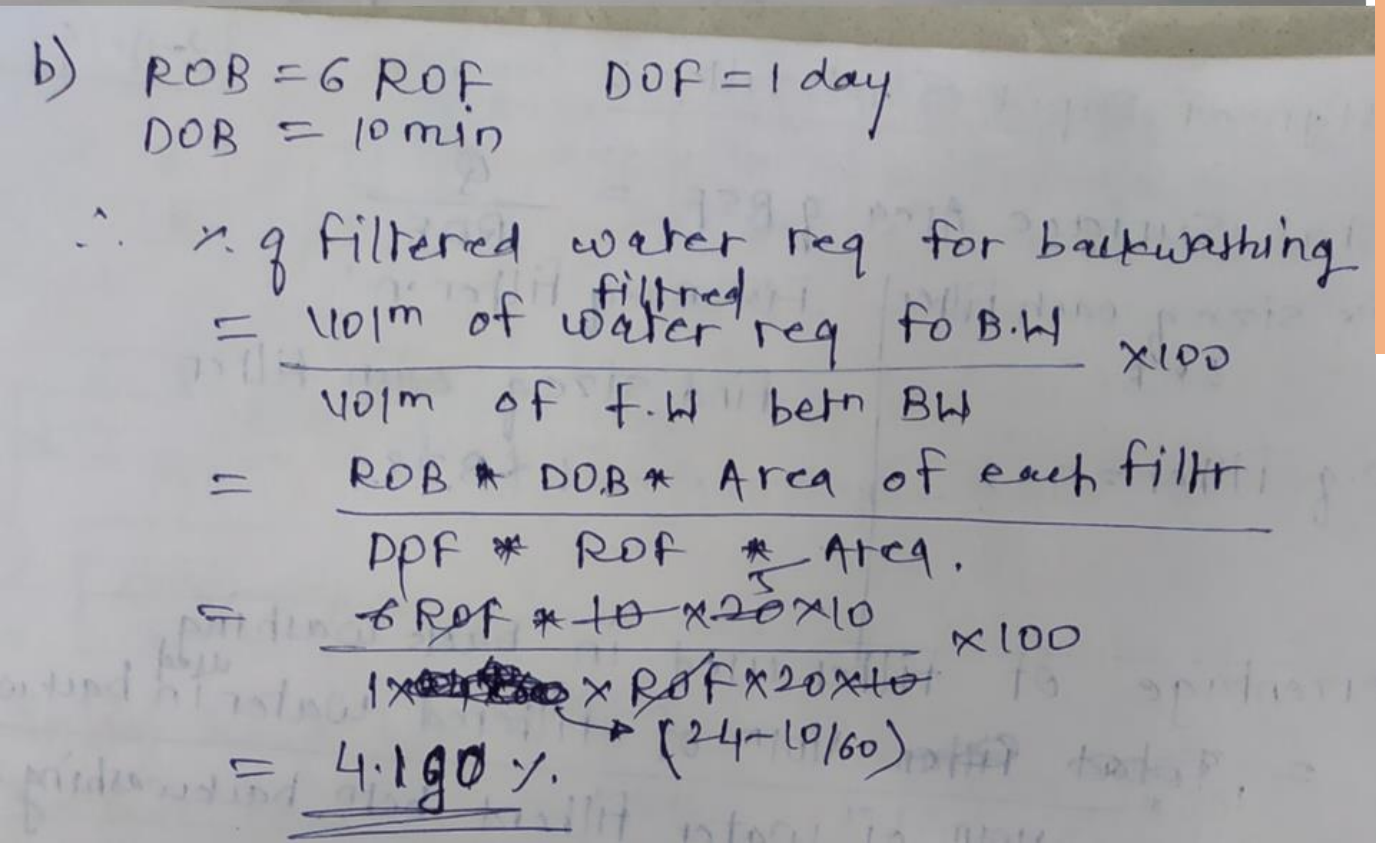
ROB = rate of backwash

A city has to treat 24 MLD of turbid water using rapid sand filter with a filtration rate of 5m3/hr/m2.

a) Determine the size of filter bed if L:B::2:1, only one unit of filer is provided.

b) Determine the percentage of filtered water required for backwashing for filter , if the rate of back wash is 6 times the rate of filtration and duration of back wash is 10 min. Back washing is done once in a day.





P-39

(9)

$Q = 0.25\ m^3/sec$

No. of filter units = 4

ROF = $5\ m^3/m^2/hr = 5000\ l/hr/m^2$

ROB = $10\ l/m^2/sec = 10 \times 24$

L:B = 2:1 , 2 trough / filter

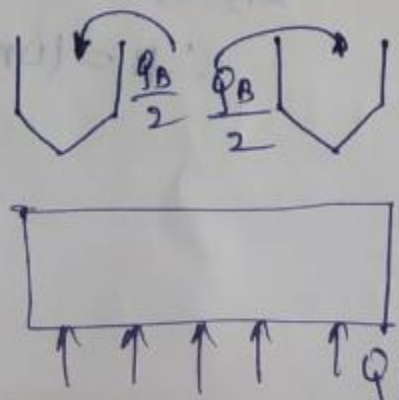
a) Total Area of Filter $= \frac{Q}{ROF} = \frac{0.25\ m^3/sec}{\frac{5}{60\times60}\ m^3/sec/m^2}$

$\therefore A = 180\ m^2$

$\therefore B = \sqrt{\frac{180}{2}} = 9.48\ m$

$\therefore B = L/2 = 4.74\ m$

b)



Area of each filter = $\frac{\text{Total Area}}{4} = \frac{180}{4} = 45\ m^2$

$\therefore 2 \times B^2 = 45$

$B = 4.74\ m$

$L = 9.48\ m$

A water treatment plant is designed to treat 0.25 m3/sec of turbid water. Number of filter units is equal to 4 , rate of filtration is 5m3/m2/hr, Back wash rate = 10 l/m2/sec . L:B::2:1 .

Determine

a)length and breadth of filter unit

b) Flow rate in wash water trough if two trough are provided .

ROB = $V_B = 10\ l/m^2/sec$

$= 10 \times 10^{-3}\ m^3/m^2/sec$

$\therefore V_B = 10 \times 10^{-3}\ m/sec$

Q_B = ROB * Area of each filter

$= 10 \times 10^{-3} \times 45$

$= 0.45\ m^3/sec$

Flow rate in each wash water trough = $\frac{Q_B}{2}$

$= \frac{0.45}{2} = 0.225\ m^3/sec.$

P-40
q

Surface Area req. for filter = $\frac{Q}{ROF}$

$= \frac{0.5 \text{ m}^3/\text{s}}{200 \text{ m}^3/\text{m}^2/\text{day}}$

$= \frac{0.5 \times 24 \times 3600 \text{ m}^3/\text{s}}{200 \quad \text{m}^3/\text{s}/\text{m}^2}$

$A = 216 \text{ m}^2$

(10) no. of filters = $\frac{216}{50} = 4.32 \approx 5.00 \approx 6$ (Assume higher level digit) +1

P-40
15

$Q = 1 \text{ m}^3/\text{s}$
$n = 14$
$A = 50 \text{ m}^2$

Loading rate of back washing with 2 filter
Out of service = 14 − 2 = 12 used.
ROF = ?

Total area of filter = $12 \times 50 = 600 \text{ m}^2$

$ROF = \frac{Q}{A} = \frac{1}{600} \text{ m}^3/\text{m}^2/\text{s} = \frac{1 \times 24 \times 3600}{600} \text{ m}^3$

A city is going to install the rapid sand filter after the sedimentation tanks.
Use the following data.
Design loading rate of the filter = 200m3/m2/d
Design flow rate = 0.5 m3/s
Surface area per filter box= 50 m2

1) Surface are required for rapid sand filter will be
2) The number of filters required shall be

A water treatment plant is designed to treat 1 m3/s of raw water .It has 14 sand filters. Surface area of each filter is 50 m2. What is loading rate in m3/day.m2 with two filters out of service for routine backwashing?

15

$Q = 1 m^3/s$

$n = 14$

$A = 50 m^2$

Loading rate of backwashing with 2 filter out of service $= 14 - 2 = 12$ used.

ROF = ?

Total area of filter $= 12 * 50 = 600 m^2$

$$ROF = \frac{Q}{A} = \frac{1}{600} m^3/m^2/s = \frac{1 \times 24 \times 3600}{600}$$

$$= 144 \frac{m^3}{m^2 day}$$

Practice Problem 7: Sizing Rapid Sand Filter

A city is to install rapid sand filters for a design loading rate of 160 $m^3/(m^2.d)$ for a water works flow of 0.4 m^3/s. Determine the number and size of filters considering maximum surface per filter is limited to 50 m^2. What would be the normal filtration rate?.

Solution Given: Q = 0.4 m^3/s = 34560 m^3/d

Loading Rate v = 160 $m^3/m^2/d$

Required tank area = Q/v = 34560/160 = 216 m^2

Nos of beds required (each bed of 50 m^2) = 216/50 = 4.32 [Min 5 beds will be provided]

Provide 5 beds, each of area = 216/5 = 43.2 m^2 ***[8 m x 5.4 m]*** *Assuming L:W = 1.5*

Or, Provide 6 beds, each of area = 216/6 = 36 m^2 ***[7.5 m x 5 m]***

Normal Filtration Rates = Q/A

If 5 beds of size is 8x5.4 m^2 are used: Q/A = (34560/5)/(8x5.4) = ***160 $m^3/m^2/d$***

If 6 beds of size is 7.5x5 m^2 are used: Q/A = (34560/6)/(7.5x5) = ***153.6 $m^3/m^2/d$***

Rapid Sand Filter Backwash

A filter unit has surface area of 5x10 m^2. After filtering 12,000 m^3 for 50 h, the filter is backwashed for 15 min at a rate of 0.7 m/min. Find: (a) the average filtration rate, (b) the quantity of washwater, (c) percent of washwater to treated water, and (d) the flow rate to each of the four troughs.

Solution

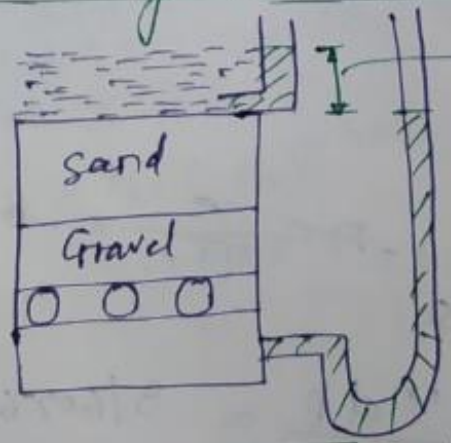
(a)Average filtration rate = 12000 m^3 per 50 h per 50 m^2 area = 12000/(50*50) ***= 4.8 $m^3/h/m^2$ = 4.8 m/h***

(b)Backwash water quantity = 0.7 m/min x 15 min x 50 m2 area ***= 525 m^3***

(c)Percentage of washwater to treated water = 525*100/12000 ***= 4.375 %***

(d)Total washwater flow to each trough = (525 m^3 / number of troughs)/15 min = 525/(4*15) ***= 8.75 $m^3/min \approx 0.146 m^3/s$***

Head loss during filtration :



Head loss (Hf)

Head loss during filtration can also be worked out to be by Carmen & Kozney eqn:

where

$$\therefore \quad H_f = \frac{f z v^2}{g \cdot d} \cdot \frac{(1-n)}{n^3}$$

where,

$$f = \frac{150(1-n)}{Re} + 1.75$$ = friction factor

d = grain size

$$Re = \frac{\rho v \cdot d}{\mu} \cdot \phi = \frac{v \cdot d \phi}{\nu}$$

ϕ = shape factor

n = porosity of filter media

v = ROF (Loading rate)

z = thickness of filter bed

Head loss during backwash :

$$H_b = z(1-n)(s-1)$$

s = sp. gravity of sand

$$ROB = V_B = V_s \cdot (n_e)^{4.5}$$

V_s = settling velo. of sand = $\frac{g}{18} \frac{(s-1) \cdot d^2}{\nu}$

n_e = porosity of expanded bed

$$\frac{z_e}{z} = \frac{(1-n)}{(1-n_e)}$$

Head Loss Through Filter: Carman-Kozeny Equation

➢Carman-Kozeny Equation (for uniform media):

$$h = \frac{k\mu}{\rho g}\frac{(1-e)^2}{e^3}\left(\frac{a}{\forall}\right)^2 VL\,(k \approx 5)$$

Specific surface area $= \frac{6}{d}$ (for spherical)

$= \frac{6}{\psi d}$ (for irregular)

Where,

L =Depth of filter bed, m d =Grain diameter, m

e =porosity ψ =Shape factor V=Filtration velocity, m/sec

k = residual dimensionless constant (normally ≈ 5)

➢For non-uniform media, each sieve fraction is considered as a distinct layer (with assume uniform porosity). Carman-Kozeny Equation (for non-uniform media):

$$h = \frac{k\mu}{\rho g}\frac{(1-e)^2}{e^3} v\left[\frac{6}{\psi}\right]^2 \sum \frac{\chi_{ij}}{(d_{ij})^2} L$$

Where,

χ_{ij} = *% of particles (or fraction) remaining within adjacent sieves*

d_{ij} = *average particle size* $[(d_1 + d_2)/2$ *or* $\sqrt{d_1.d_2}]$

Head Loss Through Filter: Ergun Equation

- For uniform media:

$$\frac{h}{L} = \underbrace{\frac{4.17\mu}{\rho g}\frac{(1-e)^2}{e^3}\left(\frac{a}{v}\right)^2 v}_{\text{For laminar flow}} + \underbrace{k_2\frac{(1-e)}{e^3}\left(\frac{a}{v}\right)\frac{v^2}{g}}_{\text{For transitional or turbulent flow}} \qquad \frac{a}{v} = \frac{6}{\psi d}$$

$k_2 = 0.29$

later reported as

$k_2 = 0.48$ for crushed porous solids

- For non-uniform media, same equation is valid with a/V to be replaced as:

$$\frac{a}{\forall} = \Sigma \chi_i \left(\frac{a}{\forall}\right)_i = \Sigma \chi_i \frac{6}{\psi . d_i}$$

Practice Problem 9: Head Loss in Rapid Sand Filter

A dual medium filter is composed of 0.3 m anthracite (mean size of 1.5 mm) that is placed over a 0.6 m layer of sand (mean size 0.8 mm) with a filtration rate of 10 m/h. Assume the grain sphericity as 0.75 with porosities of 0.4 and 0.55, respectively. Estimate the head loss of the filter at 15^{0}C [Kinematic viscosity = 1.131x10^{-6} m^2/s at 15^{0}C].

Solution Head loss in each layer may be computed using

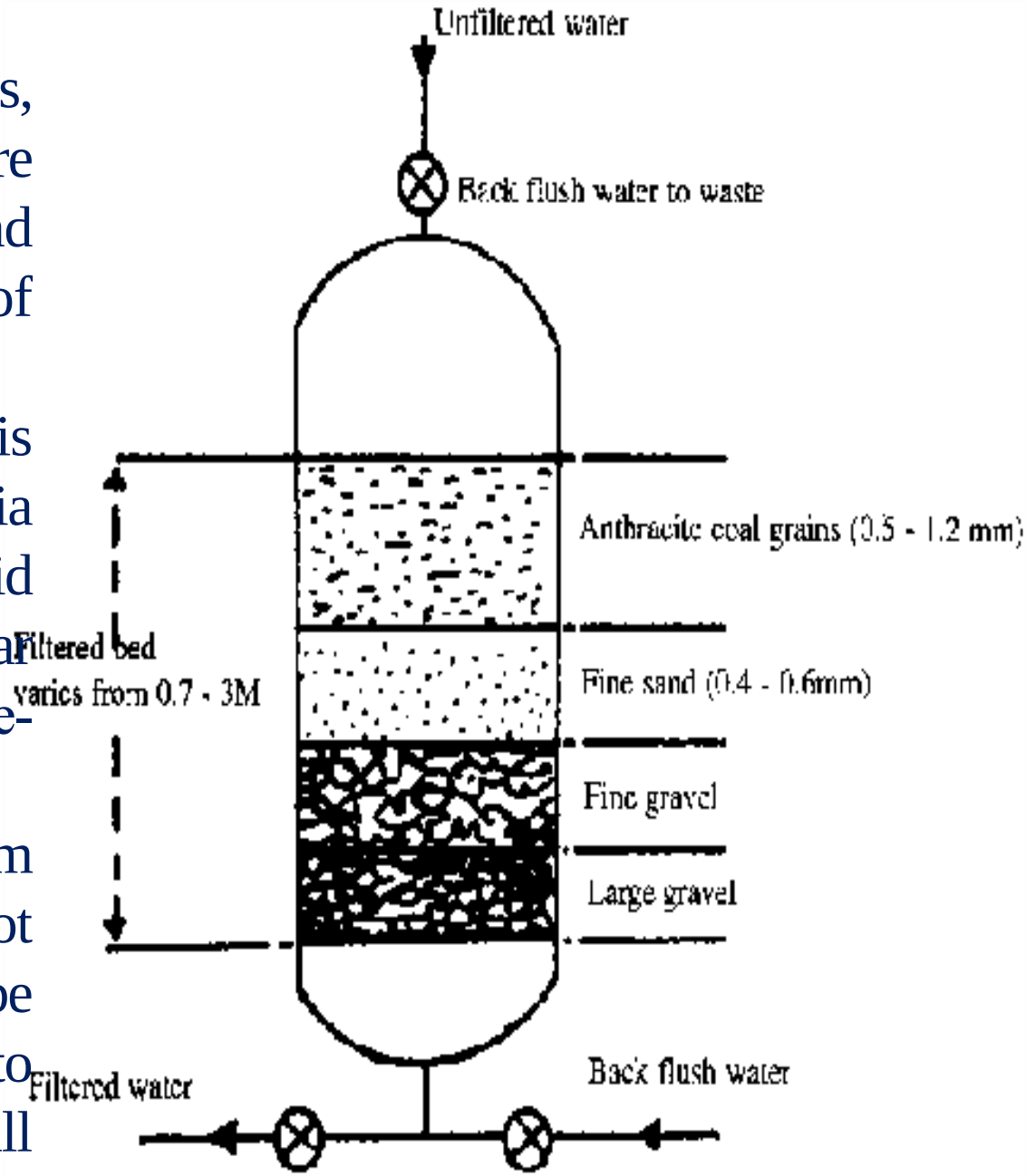
$$h = \frac{k\mu}{\rho g}\frac{(1-e)^2}{e^3}\left(\frac{a}{\forall}\right)^2 VL\,(k \approx 5)$$

	Anthracite Layer	***Sand Layer***
$a/V = 6/(\psi d)$	$= 6/(0.75*1.5*10^{-3}) = 5333$ m	$= 6/(0.75*0.8*10^{-3}) = 10000$ m
V = 10 m/h	= 0.002778 m/s	= 0.002778 m/s
L	= 0.3 m	= 0.6 m
$\nu = \mu/\rho$	$= 1.131x10^{-6}$ m^2/s	$= 1.131x10^{-6}$ m^2/s
e	= 0.4	= 0.55
h (calculated)	= 0.077 m	= 0.117 m

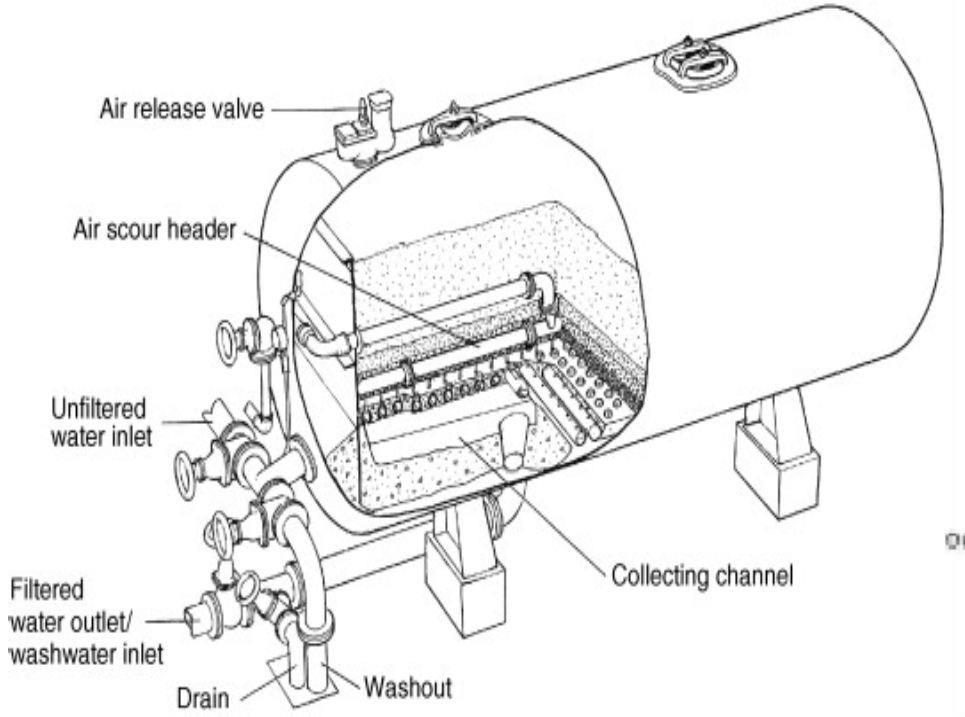
Total Head Loss = 0.077+0.117 = ***0.194 m***

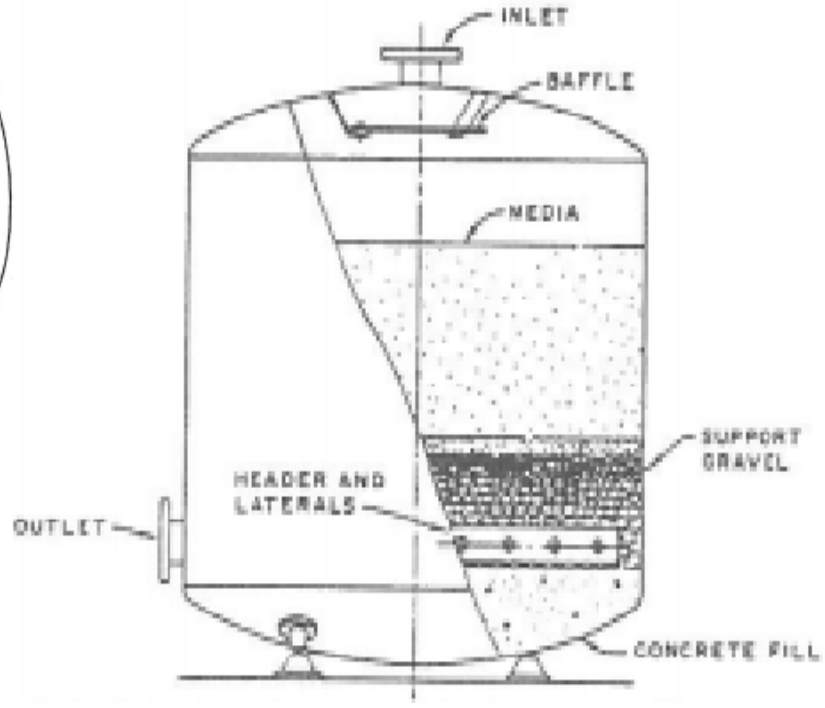
Pressure Filters

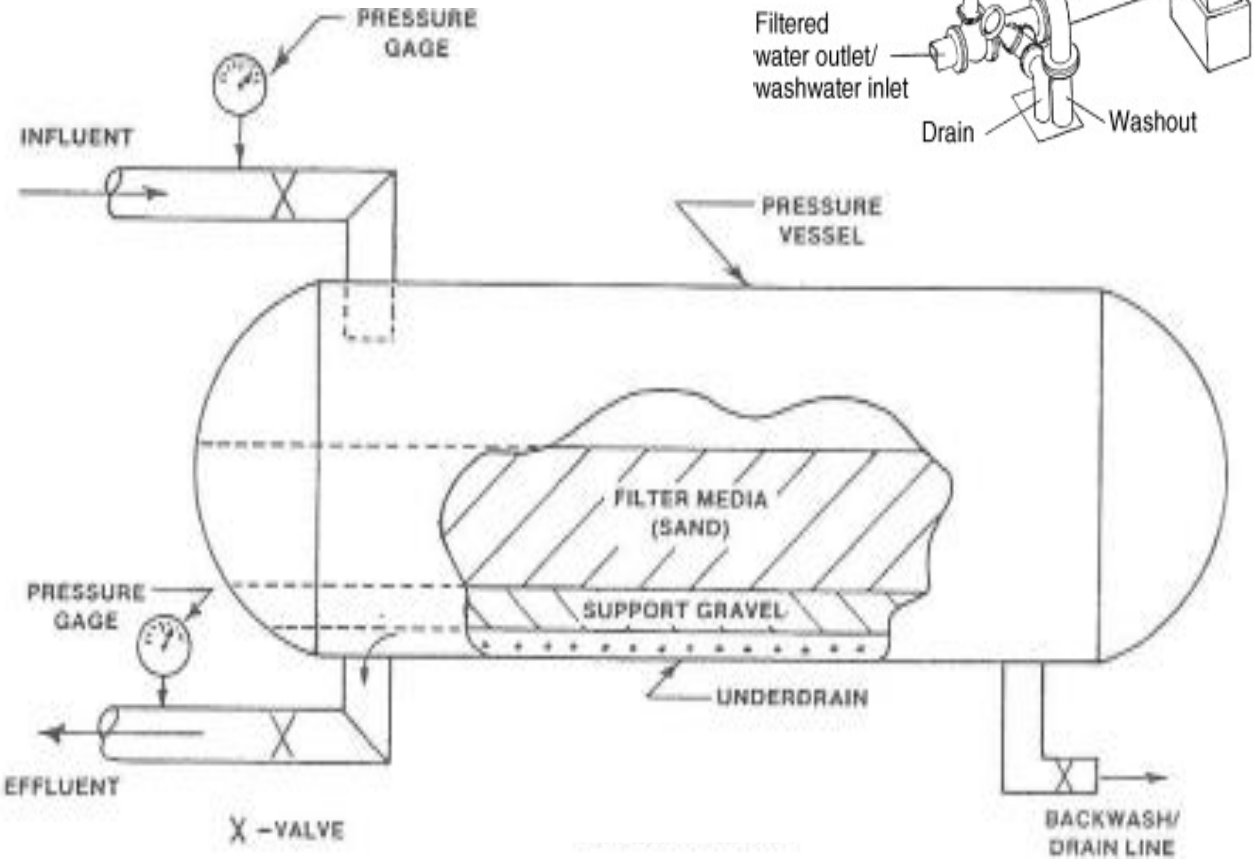
- Pressure filters are similar to conventional rapid filters, however the media, gravel and underdrains are contained in a closed vessel (steel or cast iron tank) and the filter is operated under pressure head of 30-70 m of water.
- Similar to gravity filters, the media in pressure filter is usually sand or a combination of media (in multimedia filters). Filtration rates (6-15 m/h) are similar to rapid gravity filters, and these also require regular backwashing. These are compact, and can be pre-fabricated and moved to site.
- Typically, pressure filters are used for the filtration from groundwater or high quality surface water that does not require clarification. In groundwater systems they can be used for the removal of Fe and Mn in addition to turbidity. Pressure filters are used primarily in small plants, industries, swimming pools etc.

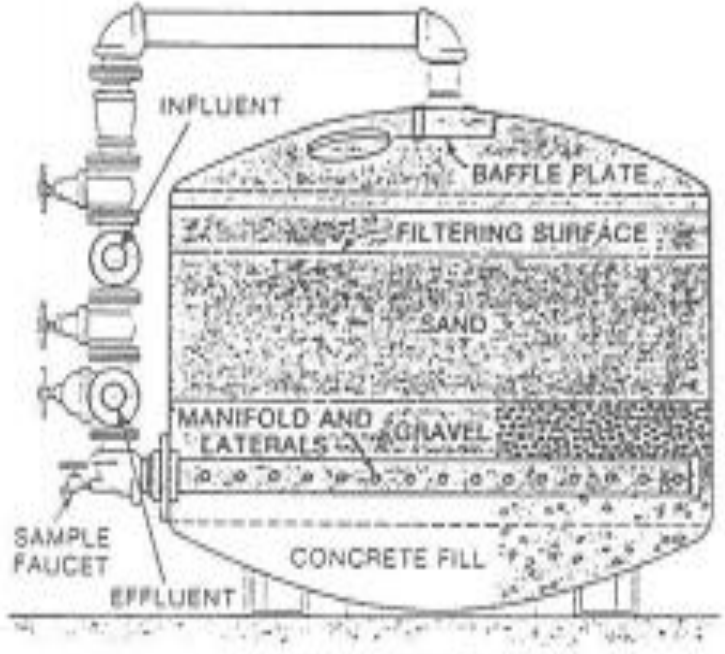


Pressure Filters Arrangements









Vertical Pressure Sand Filter

Pressure Filters Disadvantages

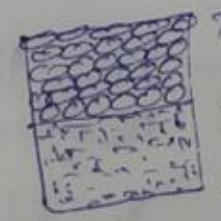
The use of pressure filters for public water supplies is unusual because of cost, operation problems, and difficulty in maintenance. As per CPHEEO Manual, Pressure filters suffer from the following disadvantages:

- Treatment of water under pressure seriously complicates effective feeding, mixing and flocculation of water to be filtered
- In case of direct supply from pressure filters, it is not possible to provide adequate contact time for chlorine.
- Water under filtration and the sand bed are out of sight and it is not possible to observe the effectiveness of the backwash or the degree of agitation during washing process;
- It is difficult to inspect, clean and replace the sand, gravel and under-drains of pressure filters
- Because the water is under pressure at the delivery end, on occasions when the pressure on the discharge main is released suddenly the entire sand bed might be disturbed violently with disastrous results to the filter effluent.

* Dual media filter (Double media filter):

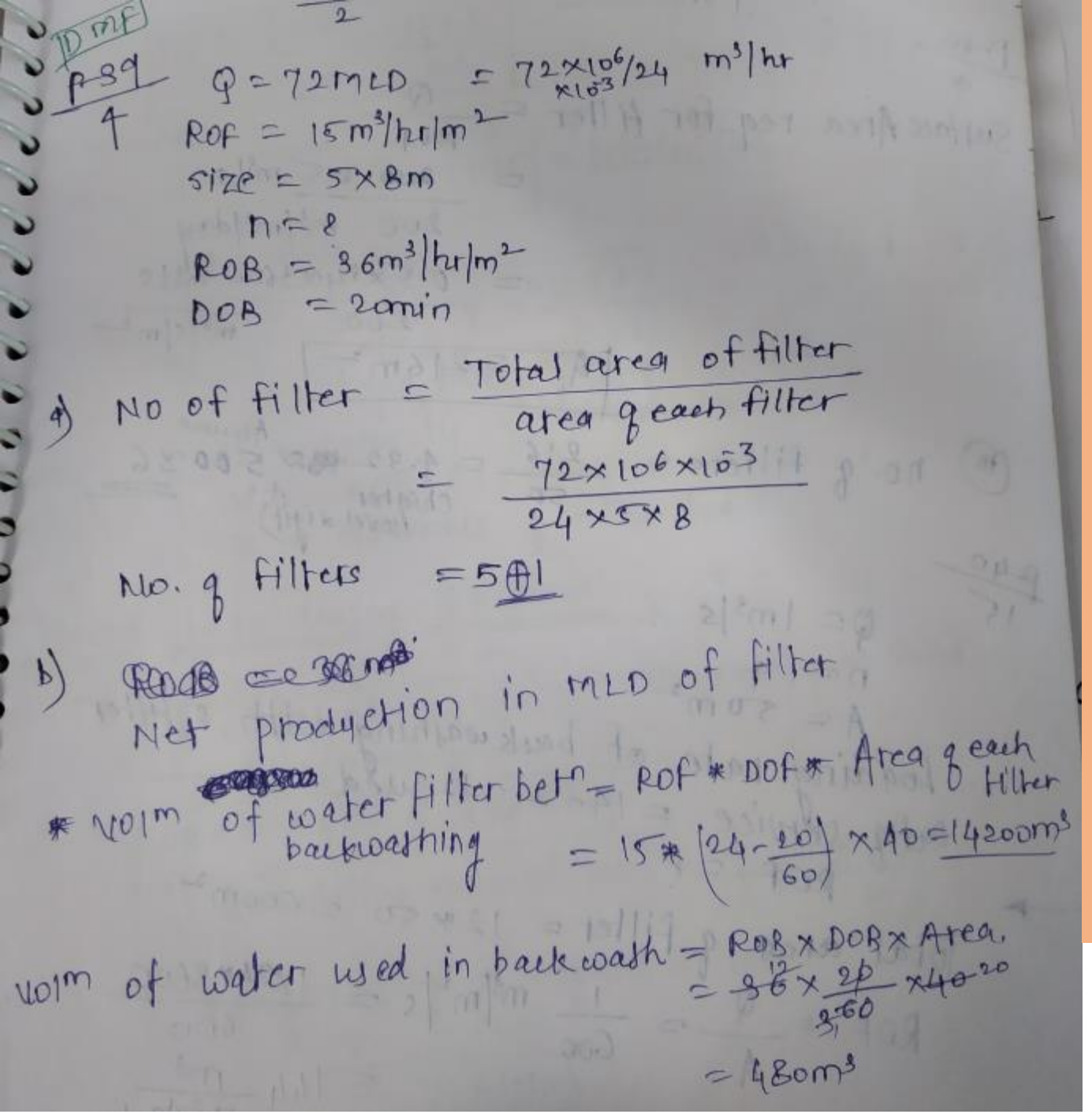
— Dual media filter is a RSF where two filter media packed in single filter.

1. Anthracite Coal ---- $S = 1.4$ to 1.5 --- $D_{10} = 0.9$-$1mm$
2. Silica Sand ---- $S = 2.65$ --- $D_{50} = 0.5mm$



— 1) DMF improve RoF
2) Increase filter run time interval betn backwash to backwash increases)

{ RoF = 10000 to 15000 $lit/hr/m^2$
$\eta = 90\%$
Filtration is from Coarse to fine



DMF

P-89

Q = 72 MLD = $72 \times 10^6 / 24 \times 10^3$ m³/hr

ROF = 15 m³/hr/m²

size = 5 × 8 m

n = 8

ROB = 36 m³/hr/m²

DOB = 20 min

a) No of filter = Total area of filter / area of each filter

= $\frac{72 \times 10^6 \times 10^3}{24 \times 5 \times 8}$

No. of filters = 5 + 1

b) Net production in MLD of filter

Volm of water filter betn backwashing = ROF * DOF * Area of each filter

= $15 * \left(24 - \frac{20}{60}\right) \times 40 = 14200 m^3$

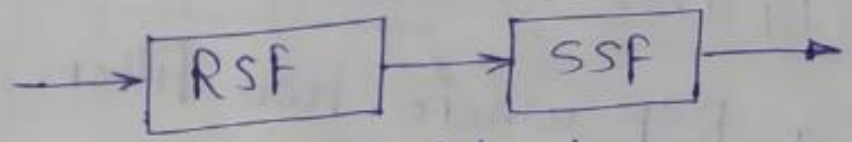
Volm of water used in backwash = ROB × DOB × Area.

= $\frac{36}{1} \times \frac{20}{60} \times 40$

= 480 m³

A Dual media rapid sand filter plant is to be constructed for treatment of 72 MLD of water . A pilot plant study indicated that filtration rate of 15m3/hr/m2 would be acceptable . Allowing one unit out of service for back washing , how many 5 x 8 filter units will be required ? Determine the net production in MLD of each filter unit if back washing is done at 36m3/hr/m2 for 20 min and the water is wasted for the first 10 min of each filter run.

Roughening Filters :-



(Roughning filter)
pre filter

— prefilter which reduces load on SSF.

multimedia filter : (mixed media filter)

It is RSF provided with 3 filter media

1) Anthracite — $s = 1.4$ to 1.5 — $D_{10} = 9$ to 1 mm
2) Silica Sand — $s = 2.65$ — $D_{10} = 0.5$ mm
3) Garnet — $s = 4.2$ — $D_{10} = 0.1$ mm
(green Sand)

$R_{of} = 15000$ to 20000 Lit/hr/m^2

$\eta = 90\%$

Comparison: Slow Sand Filter *vs* Rapid Sand Filter

Item	Slow Sand Filter	Rapid Sand Filter
Pre treatment	Not required except plain sedimentation	Coagulation, Flocculation and Sedimentation
Base materials	Gravel base of 30 to 75 cm depth with 3 to 65mm size graded gravel.	Gravel base of 45 to 50 cm depth with gravel size varies from 3 to 50 mm in 4 or 5 layers
Filter sand ▪ Effective size ▪ Uniformity coefficient ▪ Thickness of sand bed	 ▪ 0.25 to 0.35 mm ▪ 3 to 5.0 ▪ 80 to 100 cm	 ▪ 0.45 to 0.70 mm ▪ 1.2 to 1.7 ▪ 60 to 75 cm
Under drainage system	Open jointed pipes or drains covered with perforated blocks	Perforated pipe laterals discharging into main header
Size of each unit	50 to 200 sq.m	10 to 100 sq.m
Rate of filtration	100 to 200 Lph/sq.m	4800 to 7200 Lph/sq.m
Cost ▪ Installation ▪ O&M	 ▪ High ▪ Low	 ▪ Low ▪ High

Item	Slow Sand Filter	Rapid Sand Filter
Efficiency ▪ Turbidity of feed water ▪ Removal of bacteria	 Low; < 30 NTU 98 to 99%	 Any level of turbidity of feed water; (with pre-treatment) 80 to 90%
Suitability	For water supply to rural areas and small town	For public water supply to towns and cities
Post treatment	Slight disinfection	Complete disinfection is a must
Ease of construction	Simple	Complicated;
Skilled supervision	Not essential	Essential
Loss of head ▪ Initial ▪ Final	 ▪ 10c m ▪ 80 to 120 cm	 ▪ 30 cm ▪ 250 to 350 cm
Method of cleaning	▪ Scrapping and removing *Schmutzedecke* and 1.5 to 3 cm thick sand layer ▪ Laborious	▪ Back washing with or without compressed air agitation ▪ Simple and easy
Quantity of wash water required	0.2 to 0.5% of total water filtered	1 to 5% of the total water filtered
Cleaning Interval	Three to four months	One to two days

Comparison: Slow Sand Filter *vs* Rapid Sand Filter

	Item	*Slow Sand Filter*	*Rapid Gravity Filter*
1.	Area	Requires very large area	Requires small area
2.	Quantity of sand	Requires Considerable quantity of sand	Requires less quantity of sand
3.	Quality of sand	Finer filter media of 0.2 to 0.4 effective size and 2 to 4 uniformity coefficient	Slightly coarser filter media of 0.36 to 0.6 effective size and 1.2 to 1.8 uniformity coefficient
4.	Quality of raw water	It may not be treated with chemicals, but should not have turbidity more than 50 p.p.m.	Treatment with chemicals is essential.
5.	Flexibility in operation	Not possible	Possible
6.	Rate of filtration	100-180 *litres/m²/hour*	4,000-5,000 *litres m²/hour*
7.	Size of one unit	30 *m* × 60 *m*	6 *m* × 8 to 8 *m* × 10 *m*
8.	Distribution	Uniform	Smaller at top and coarser in bottom
9.	Underdrainage system	Open jointed pipes or drains covered with blocks	Manifold and pipe laterals. Vitrified tile Blocks ; the Wheeler Filter Bottom ; the Porous Plate Bottom Concrete Ridge and Valley Bottom etc.
10.	Period of Cleaning	1 to 3 months	24 to 48 hours
11.	Method of cleaning	Scrapping 2-3 *cm* sand from the surface and replacing it with new sand.	By back washing with water under pressure, with or with-out compressed air agitation before washing with water
12.	Skilled Supervision	Not required	Most essential
13.	Loss of head	15 *cm* to 75 *cm*	2 *m* to 4 *m*
14.	Penetration of suspended impurities	Very small, only a dirty layer is formed at the surface	2 *m* to 4 *m* Very deep
15.	Amount of water required for washing	0.2 to 0.6% of filtered water	2 to 4% of filtered water
16.	Overall cost of unit	More, because large land and much quantity of materials are required	Cheap and economical
17.	Cost of Maintenance	Small	More
18.	Efficiency	Efficient in removal of bacteria and suspended matter	Cannot remove all bacteria, disinfection necessary. Removes colour, odour and taste.

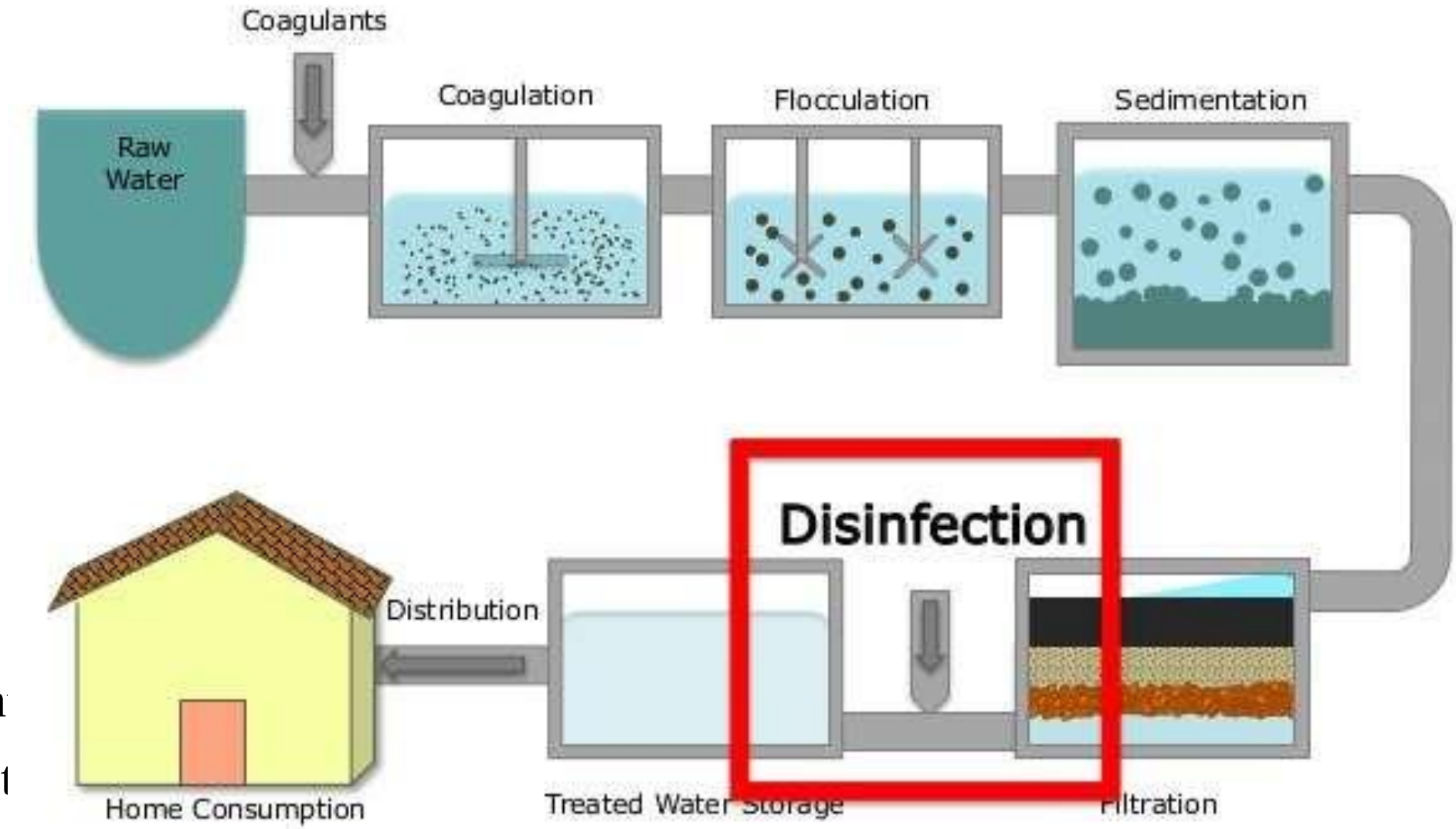
Comparison of Filters for Water Treatment

Characteristic	Traditional Slow Sand Filter (TSSF)	BioSand Water Filter (BSF)	Rapid Sand Filter (RSF)	Pressure Sand Filter (PSF)	Low Op. Head Polishing Sand Filter (LHPF)
Effectiveness in removing:					
Pathogens					
Parasites	Very effective	Very effective	Possible	Possible	Very effective
Bacteria	Very effective	Very effective	Not effective	Not effective	Very effective
Viruses	Very effective	Very effective	Not effective	Not effective	Very effective
Particulates					
Silt Clay Organic	Very effective and practical at low turbidity.	Very effective and practical at all turbidities. Pre-treatment may be useful.	Effective as part of conventional treatment systems. (These include use of coagulants and clarification prior to filtration.)	Effective as part of conventional treatment systems. (These include use of coagulants and clarification prior to filtration.)	Very effective and practical at all turbidities. Pre-treatment may be useful.
Oxidized					
Iron Manganese	Effective but not usually practical.	Very effective and practical.	Not sufficiently effective or normally used.	Not sufficiently effective or normally used.	Very effective and practical.
Arsenic	Not used because pre-treatment impractical	Very effective and practical with required pre-treatment.	Not sufficiently effective or normally used	Not sufficiently effective or normally used	Very effective and practical with required pre-treatment
Fluoride	Not used because pre-treatment impractical	Very effective and practical with required pre-treatment	Not sufficiently effective or normally used	Not sufficiently effective or normally used	Very effective and practical with required pre treatment
Dissolved organics	Not used because pre-treatment impractical	Very effective and practical with required pre-treatment	Very effective and practical with required pre-treatment	Very effective and practical with required pre-treatment	Very effective and practical with required pre-treatment
Opportunity for Breakthrough	Not possible.	Not possible.	Normal. Used to indicate need to clean.	Normal. Used to indicate need to clean.	Not possible.
Structural Issues					
Relative surface area	Very large.	Large.	Small.	Very Small.	Large.
Relative height	Deep.	Shallow	Very deep.	Shallow.	Shallow.
Piping requirements	Minimal.	Minimal.	Extensive.	Extensive.	Minimal.

Typical Water Treatment Processes

Major/Minor Steps:

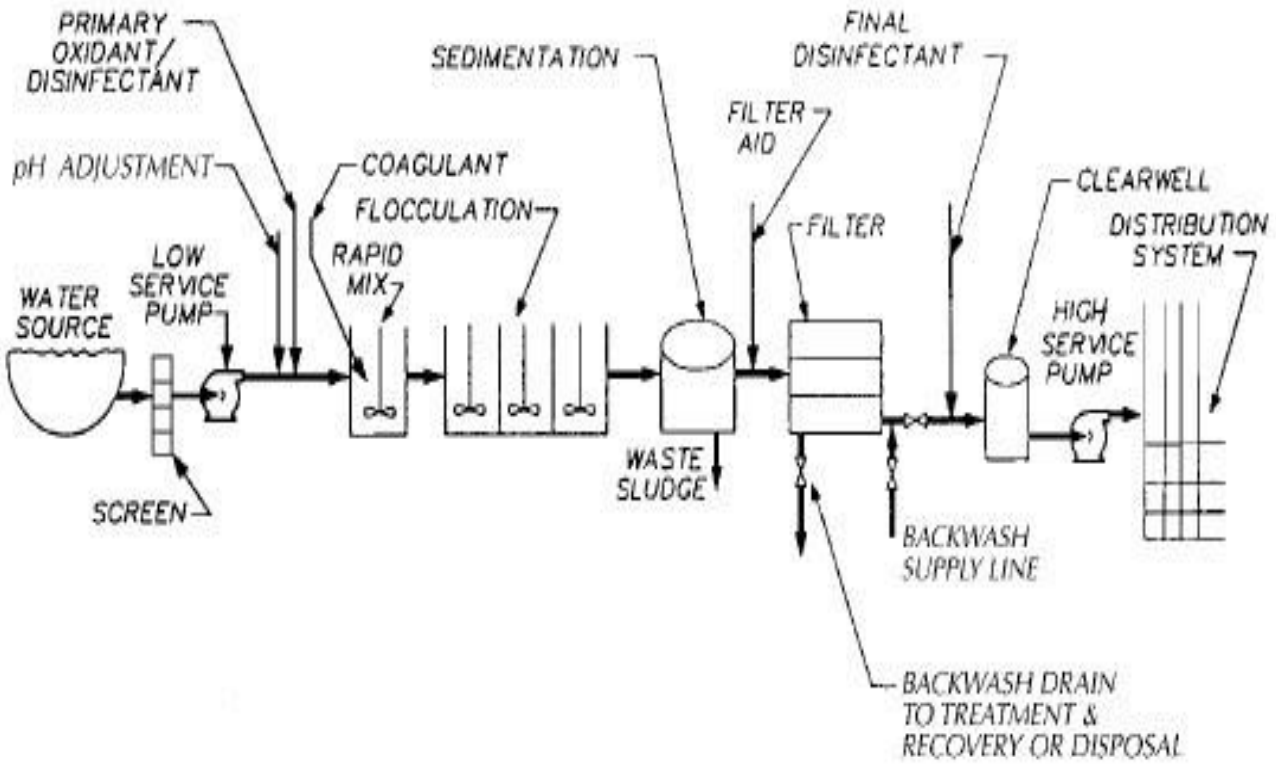
- Screening
- Aeration
- Coagulation and Flocculation
- Sedimentation
- Filtration
- *Disinfection*
- Advanced Treatmen
- Sludge Managemen



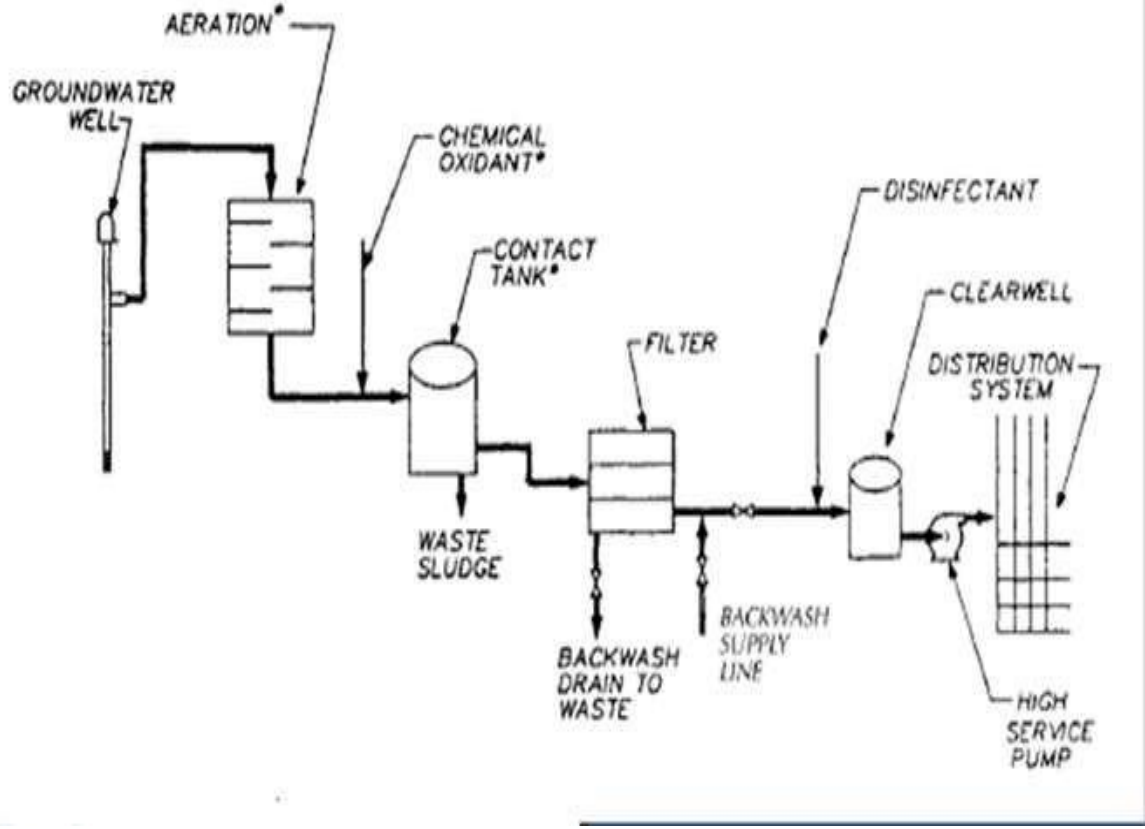
Disinfection in Surface Water and Groundwater Treatment Systems

One of the prime objectives of water treatment is the removal or inactivation of pathogenic micro- organisms to preven
the spread of waterborne disease. Disinfection is the process to remove, deactivate or kill pathogenic microorganisms
so that they are not infectious to humans and animals.

It is the "most important" treatment steps for both, surface waters as well as groundwater treatment.



Typical Surface Waters Treatment Scheme



Typical Groundwater Treatment S

Water Related Diseases

- *Water-borne diseases:* infections spread through contaminated drinking water
- *Water-washed diseases:* diseases due to the lack of proper sanitation and hygiene
- *Water-based diseases:* infections transmitted through an aquatic invertebrate organism
- *Water-related vector-borne diseases:* transmitted by insects that depend on water for their propagation

Disease group	*Disease*	*Estimated infection rate (1'000/year)*	*Estimated morbidity (1'000/year)*	*Estimated mortality (1'000/year)*
WATER-BORNE DISEASES	Diarrhoeal Diseases	not available	1'000'000[1]	5'000[1]
	Typhoid Fever	1'000	500	25
WATER-WASHED DISEASES	Ascariasis (=roundworm infection)	800'000-1000'000	1'000	20
	Ancylostomiasis (=hookworm infection)	700'000-900'000	1'500	50-60
WATER-BASED DISEASES	Schistosomiasis (Bilharzia)	200'000	?	500-1'000
WATER-RELATED VECTOR-BORNE DISEASES	Malaria	240'000	100'000	not available
	Lymphatic filariasis	90'200	2'000-3'000	low
	Onchocerciasis	17'800	340	20-50
	Japanese encephalitis	not available	20-40	case fatality ratio between 10-30%

Need for Disinfection of Water

Pathogen	Disease Caused
Bacteria:	
Anthrax	anthrax
Escherichia coli	*E. coli* infection
Myobacterium tuberculosis	tuberculosis
Salmonella	salmonellosis, paratyphoid cholera
Vibrio cholerae	
Viruses:	
Hepatitis Virus	Hepatitis A
Polio Virus	polio
Parasites:	
Cryptosporidium Giardia lamblia	cryptosporidiosis giardiasis

- Microorganisms in water are most often associated with several waterborne disease like typhoid, cholera, paratyphoid, bacillary dysentery, poliomyelitis etc.
- Conventional filtration and excess-lime softening (if used) may provide removal of microorganism only to a certain degree, but cannot not ensure the complete removal of disease- causing microbes.
- Therefore, additional treatment process, Disinfection, is administered after filtration for the complete removal or inactivation of harmful micro-organisms in water.
- The treated water must meet the microbial contamination standards set by the regulatory authorities.

Disinfection *verses* Sterilization

- ***Disinfection is different from sterilization***, which is the complete destruction of all microorganisms found in water.

- **Sterilization is usually much more expensive and unnecessary for water treatment which simply aims to just ensure safety from the risk of waterborne disease.**

- **Disinfection (adequate cleanliness) does not ensure the same security level as that of sterilization (extreme cleanliness), and does not necessarily inactivate all forms of microorganisms, for instance, bacterial spores.**

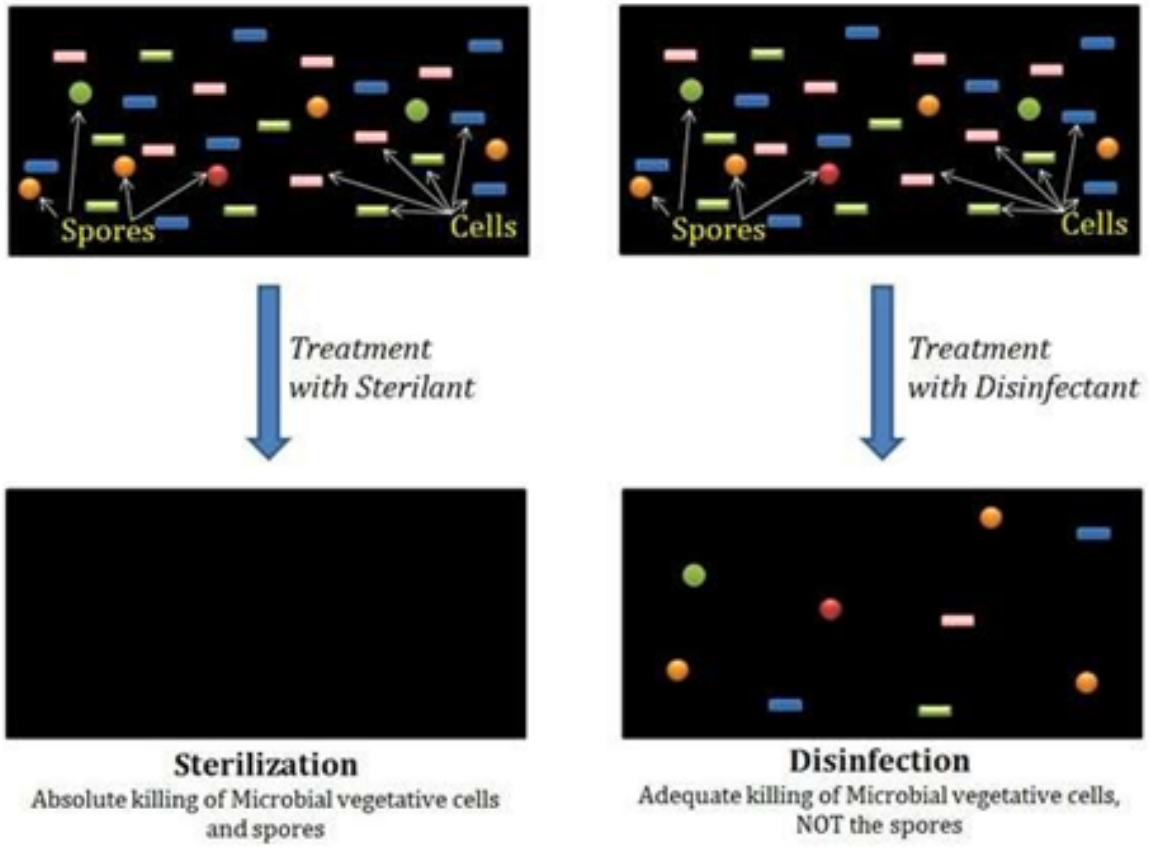


Image Source: https://www.easybiologyclass.com/difference-between-sterilization-and- disinfection-comparison-table/

Disinfection Objectives and Approaches

- *Objective of disinfection is to produce microbially safe water* by removing, inactivating or killing pathogens. It primarily aims for safety from the risk of waterborne diseases.

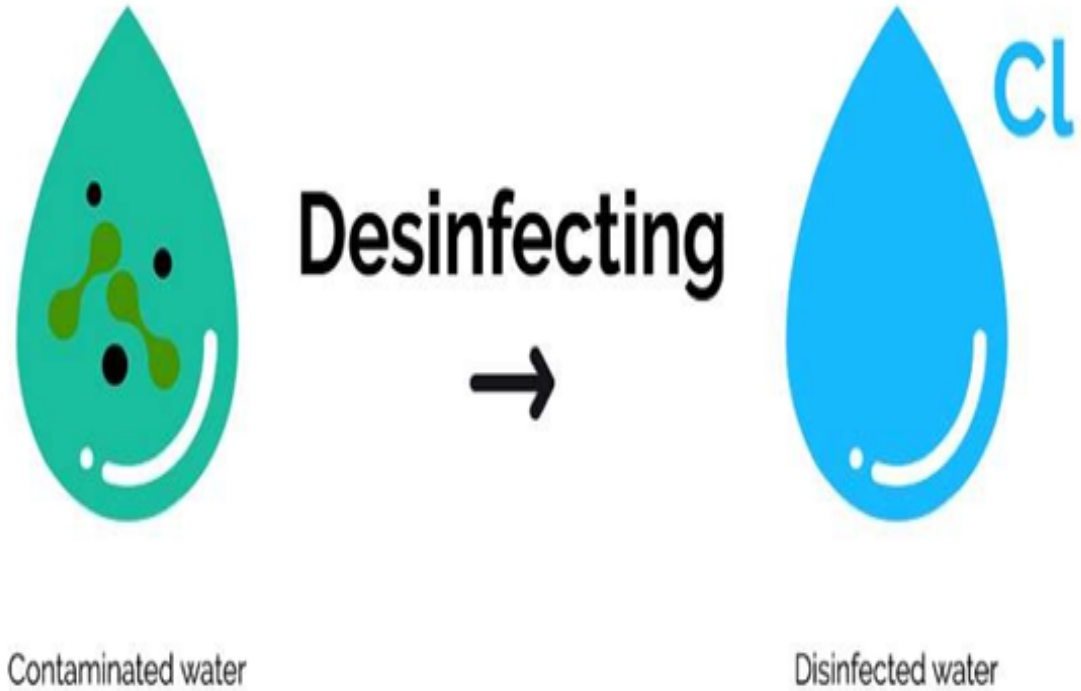
- *Approaches:* Disinfection of water can be achieved by:
  - *Removal of pathogens:* Filtering out all the pathogenic microorganisms
  - *Inactivation or killing:* Destroying the cellular structure of the micro-organisms or disrupting its metabolism, biosynthesis or ability to grow/reproduce.
    - *In the case of bacteria*, inactivation describes the subsequent inability of the microorganism to divide and form colonies.
    - *For viruses*, inactivation measures the inability of the microorganism to form plaques in host ce
    - *For protozoan Cryptosporidium oocysts*, it measures the inability of the microorganism to multiply, thereby preventing consequent infection of a host by *Cryptosporidium*.

Disinfection Mechanisms

- *Straining or physical removal* by nano or finer-grade filtration
- *Altering / destroying structures or functions of essential components within the pathogens* including *Proteins (structural proteins, enzymes, transport proteins, etc), Nucleic acids (genomic DNA or RNA, mRNA, tRNA, etc), and Lipids (lipid bi-layer membranes, other lipids)*
  - *Alteration of cell permeability:* Agents such as phenolic compounds and detergents destroys the selective permeability of the membrane.
  - *Damage to cell wall:* Radiation, heat and highly acidic and alkaline agents alter the protoplasm.
  - *Inhibition of enzyme activity:* Heat coagulates the cell protein, acids or bases denatures proteins resulting in a lethal effect.
  - *Alteration of colloidal natute of protoplasm:* Oxidizing agents alter the chemical arrangement of the enzymes and deactivating them.

Typical Methods for Disinfection

- *Physical methods:* Filtration or Boiling
- *Radiation methods:* Ultrasonic, Electronic, or UV irradiation
- *Chemical methods:* Chlorination, Ozonation, or Other chemical agents
  - The most common disinfectant used is chlorine, which is used in different forms including Chlorine gas, Hypochlorite, Chloramines and Chlorine dioxide.
  - Other chemical agents which have been used for disinfection include iodine, bromine, hydrogen peroxide, ozone, copper/silver ions.



Properties of an Ideal Disinfectant

- *Broad spectrum: active against all microbes*
- *Fast kinetics: produces rapid inactivation*
- *Non-toxic; Non-flammable; Non-explosive*
- *Effective in the presence of interfering constituents*
- *Compatible with various materials/surfaces*
- *Stable for the intended exposure period*
- *Provides a residual, if desirable*
- *Easy to generate and apply*
- *Economical*

Properties of Common Disinfectants

- *Free chlorine:* Produces strong oxidant HOCl (hypochlorous acid) and OCl^- (hypochlorite ion) in water, which are stable and may provide a residual ensuring future protection as well. HOCl, which is the more effective disinfectant of the two, dominates at low pH, while OCl^- at higher pH.
- *Chloramines:* The disinfectants (mostly NH_2Cl and some $NHCl_2$) are weak oxidant, however provides a stable residual.
- *Chlorine dioxide, ClO_2:* Very strong oxidant but unstable (dissolved gas).
- *Ozone, O_3:* Very strong oxidant, but provides no residual (too volatile, reactive)
- *UV radiation:* Very effective disinfectant that primarily damages nucleic acids. Works at low intensity monochromatic radiation at 254 nm usually produced by low pressure mercury lamp, or higher intensity polychromatic radiations produced by medium pressure mercury lamp.
- *Boiling:* Effective in killing microorganisms, but associated energy and environmental cost is high. Also boiling does not leaves any residual protection.

Factors Affecting Disinfection

- *Microbe Types and their resistance to disinfectants:*
  - Vegetative bacteria like Salmonella, coliforms, etc. have low resistance
  - Enteric viruses like coliphages, HAV, Noroviruses etc. have moderate resistance
  - Bacterial and Fungal Spores, Protozoan, cysts, helminth ova, etc. [*e.g. Cryptosporidium, Giardia lamblia cysts, Ascaris lumbricoides ova*], and Acid-fast bacteria like *Mycobacterium sp.* have high resistance
- *Type of disinfectant and their mode of action:*
  - Strong oxidants like HOCl, OCl^-, ClO_2, O^3 etc. oxidizes various protein sulfhydryl groups, alters membrane permeability, and oxidize/denature nucleic acid components, etc.
- Ultraviolet radiation primarily causes nucleic acid damage by thymidine dimer formation, strand breaks, etc.
- *Water Quality:* Suspended and dissolved matters, pH, ORP, Turbidity, Temperature, Hardness, Salts and Ions etc.
- *Reactor Design, Mixing & Hydraulic Conditions:* Plug flow (better kinetics) *or* Mixed reactor systems.

Disinfection Kinetics

- Death of microorganisms is usually considered as the first order with respect to time
- Thus, the remaining number of viable microorganisms, N, decreases with time, t, according to:

  $dN/dt = -kN$

  where k is an empirical constant descriptive of the microorganism, pH and disinfectant used.
- Integrating with respect to time, and replacing limits $(N = N_0 \text{ at } t = 0)$ yields:

  $ln(N/N_0) = -kt$ or $N = N_0 e^{-kt}$
- The is known as Chick's Law, which assumes:
  - All organisms are identical
  - Death (inactivation) results from a first-order or "single-hit" reaction.
- Other models used to study disinfection kinetics (when not first-order):
  - Multi-hit-hit or concave up: initial slow rate, increases over time
  - Concave down or retardant kinetics: initial fast rate, decreases over time

The *Ct* Concept

- The disinfection is usually not instantaneous, and inactivation is a function of time, concentration of disinfectant (especially for chemical disinfectants), pH and Temp.
- If pH and temperature variations are less, inactivation typically increases with increased exposure (or contact time), as well as increased level (concentration) of disinfectant.
- Therefore, disinfection can be expressed at the product of disinfectant concentration (*C*) and contact time (*t*).
- This applies best when disinfection kinetics is first order.

Inactivation of *Cryptosporidium Oocysts* in Water by Chemical Disinfectants

Disinfectant	CT_{99} (mg-min/L)	Reference
Free Chlorine	7,200+	Korich et al., 1990
Monochloramine	7,200+	Korich et al., 1990
Chlorine Dioxide	>78	Korich et al., 1990
Mixed oxidants	<120	Venczel et al., 1997
Ozone	~3-18	Finch et al., 1994 Korich et al., 1990 Owens et al., 1994

C. parvum oocysts inactivated by low doses of UV radiation: <10 mJoules/cm^2

- **Chlorination for Disinfection**
- **Chlorine Reactions in Water**
- **Chlorine Demand**
- **Free Chlorine Residual and Combined Chlorine Residual**
- **Breakpoint Chlorination**
- **Types of Chlorination**
- **Issues with Chlorination: DBPs and Odour**

(8) Halogen group of Chemicals:

Iodine
Bromine } → very costly (non availability) available in the form of pallets or tablets. (Army uses only)
Chlorine

Chlorine:

Disinfection of water chlorine (or) its compds is known as chlorination.

— when chlorine is added to water, chlorine hydrolises then ionizes.

$Cl_2 + H_2O \longrightarrow HOCl + HCl$ → Hydrolization

$HOCl \overset{PH}{\rightleftharpoons} H^+ + OCl^-$ → Ionization

HOCl & OCl^- related in water, together it is known as "free Chlorine" (or) freely Available Chlorine.

But HOCl is more powerful than OCl^-.
It is 80 time more destructive than OCl^-.

$HOCl \rightleftharpoons H^+ + OCl^-$

$$\frac{[H^+][OCl^-]}{[HOCl]} = K$$

$$\therefore \boxed{\frac{K}{H^+} = \frac{OCl^-}{HOCl}}$$

$\downarrow OCl^- \propto \frac{1}{H^+}$ ↑

$\uparrow HOCl \propto H^+ \uparrow$

$\uparrow H^+ \propto \frac{1}{pH}$ ↓

pH = 0 to 5.5 → only HOCl form in water

(but chlorine remains as elemental Cl_2 only)

pH = 5.5 to 7.5 → Both OCl^- & HOCl form but $HOCl >> OCl^-$

pH = 7.5 to 9.5 → Both HOCl & OCl^- form but $OCl^- >> HOCl$

pH = 9.5 to 14 → only OCl^- form in water.

% HOCl in water $= \frac{HOCl}{HOCl + OCl^-} * 100$

$$\% HOCl = \frac{1}{1 + \frac{OCl^-}{HOCl}} * 100$$

$$* \boxed{\% HOCl = \frac{1}{1 + \frac{K}{H^+}} * 100}$$

Concn of HOCl = dose of Cl_2 * % of HOCl

Chlorine Reactions in Water: Free Chlorine Residual

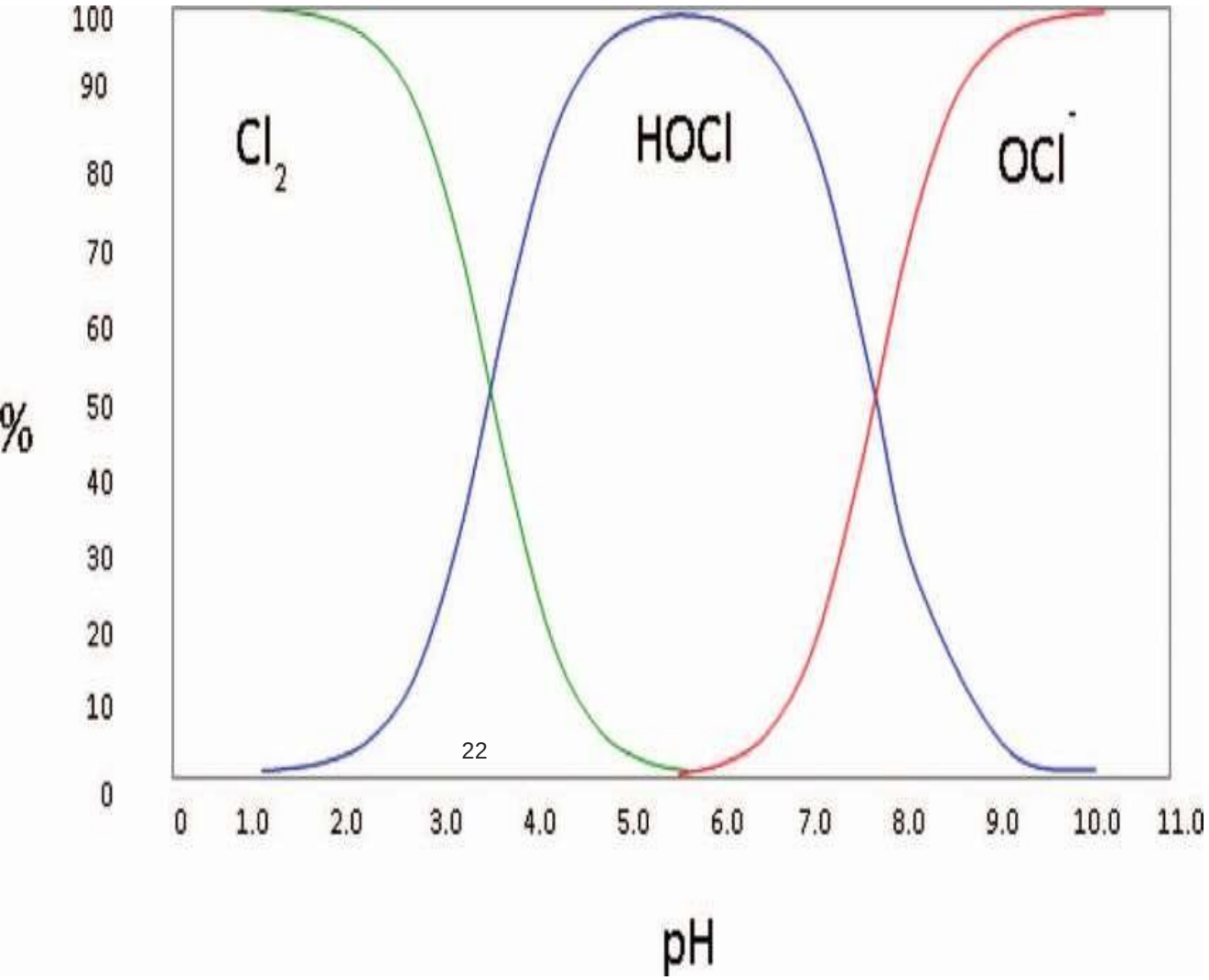
- Usually, after satisfying the "Chlorine Demand" of water, chlorine reacts with water to form *HOCl (hypochlorous acid)* and *OCl^- (hypochlorite ions)* in water. These are known as *"Free Chlorine Residuals"* and have good disinfection abilities due to their strong oxidizing potential.
- Chlorine may be added in the form of chlorine gas, hypochlorite, or chlorine dioxide.
- Chlorine gas is the most pure form of chlorine, but is difficult to handle in gaseous form as it is toxic, heavy, corrosive, and an irritant. Usually, it is compressed into a liquid and stored in metal cylinders.

$$Cl_2 + H_2O \rightarrow HOCl + HCl \qquad HOCl \leftrightarrow H^+ + OCl^- \text{ (pH dependent)}$$

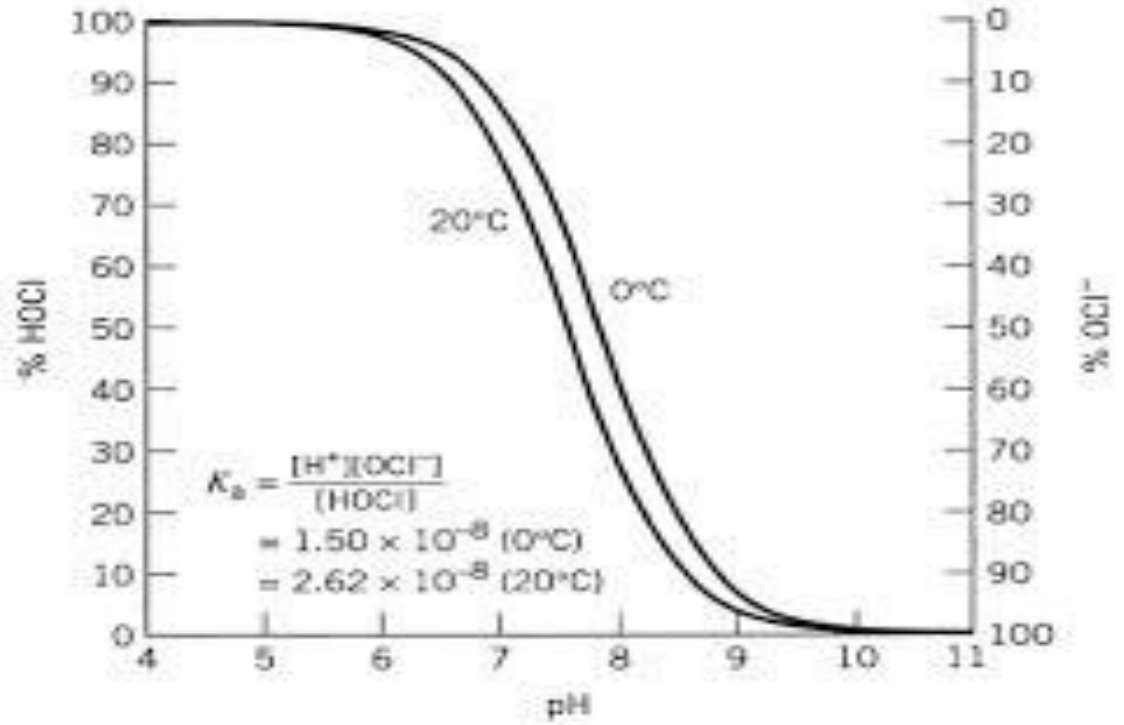
- The interconversion of hypochlorous acid and hypochlorite ions are pH dependent, as HOCl dominates at lower pH (<7.5), while OCl^- dominates at higher pH (>7.4).
- Between the two, hypochlorous acid has much higher disinfection abilities than hypochlorite ions. Therefore, disinfection with chlorine is more efficient at a low pH than at a high pH.

Chlorine Reactions in Water: Free Chlorine Residual

Effect of pH and Temperature on Cl dissociation in water



pH	% HOCl 0°C	% HOCl 20°C	pH	% HOCl 0°C	% HOCl 20°C
4	100	100	8	40.1	27.6
5	99.9	99.7	8.5	17.4	10.8
6	98.5	97.4	9	6.3	3.7
7	86.9	79.2	10	0.7	0.4
7.5	67.9	54.7	11	0.07	0.04



P.46

(4) $k = 2.7 \times 10^{-8}$

$$\frac{k}{H^+} = \frac{OCl^-}{HOCl}$$

$$\% HOCl = \frac{1}{1 + \frac{k}{H^+}} \times 100$$

$$\frac{90}{100} = \frac{1}{1 + \frac{2.7 \times 10^{-8}}{H^+}} \times 100$$

$$H^+ = 2.43 \times 10^{-7} \text{ mol/L } H^+$$

$$\therefore pH = \log_{10} \frac{1}{H^+}$$

$$= \log_{10} \frac{1}{2.43 \times 10^{-7}}$$

$$\boxed{pH = 6.6}$$

P.47

14 $pH = 7.0$

$H^+ = 1 \times 10^{-7}$ mol/L H^+

$k = 2.5 \times 10^{-8}$ mol/L

$$\% \text{ of } HOCl = \frac{1}{1 + \frac{k}{H^+}} \times 100$$

$$= \frac{1}{1 + \frac{2.5 \times 10^{-8}}{1 \times 10^{-7}}} \times 100$$

$= 80\%$ i.e. 0.8 fraction HOCl

Chlorine Reactions in Water: Hypochlorites

- *Hypochlorites (also known as bleach)* are less pure than chlorine gas, thus, are also less dangerous. Further, they decompose in strength over time while in storage. Typically, these are of three types - sodium hypochlorite, calcium hypochlorite, and commercial bleach:
  - *Sodium hypochlorite (NaOCl):* Usually comes in solution form, containing up to 12% chlorine.
  - *Calcium hypochlorite ($Ca(OCl)_2$):* Comes in both, liquid and solid (powder/granules) forms, and contains around 65-70 % chlorine. This is also know as *"HTH" (High Test Hypochlorite)*.
  - *Commercial Bleach:* Commonly available in grocery stores in both liquid and powder forms, with chlorine concentration ranging from under 10 % to over 40 %, depending on the brand.
- Disinfection through hypochlorites occurs similar to that with chlorine. However, the differences lie in quantities to be used (due to different chlorine concentrations), and feeding, storage and handling procedures.

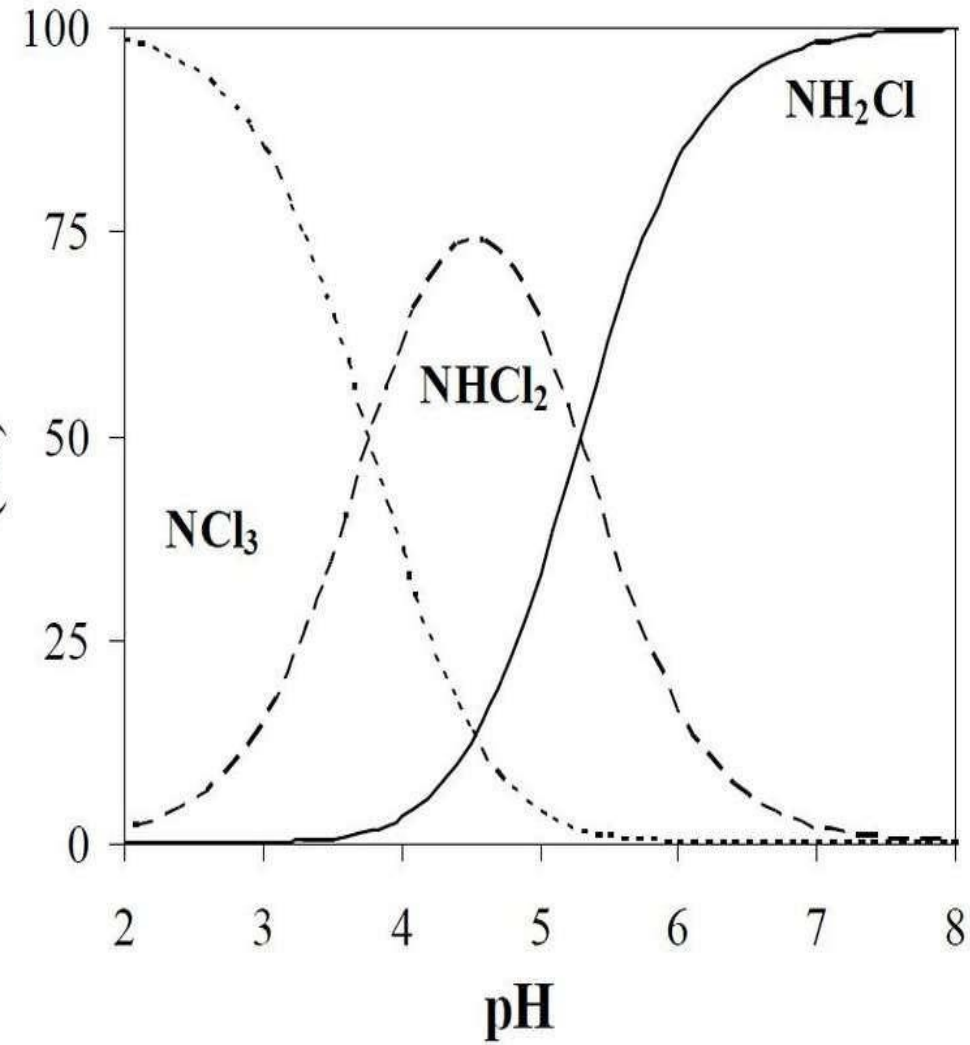
$Ca(OCl)_2 + 2\,H_2O \rightarrow HOCl + Ca(OH)_2$ $NaOCl + H_2O \rightarrow HOCl + NaOH$ $HOCl \leftrightarrow H^+ + OCl^-$

Chlorine Reactions in Water: Chloramines [Chloramination]

- Some plants use chloramines for disinfection of water. Chloramines are chlorine and nitrogen based chemical compounds *formed during a reaction between chlorine (Cl_2) and ammonia (NH_3).* These are produced when ammonia (either already present or added) reacts with free chlorine (HOCl or OCl^-, depending on the pH, ideal pH is 8.4) to produce *Monochloramine (NH_2Cl)*, which may then react with more hypochlorous acid to form a *Dichloramine ($NHCl_2$)*, and subsequently *Trichloramine (NCl_3)*.
- The extent of these reactions depends on the pH of the medium. Organic chloramines can also be formed during these reactions. Monochloramines and dichloramines can both be used as a disinfecting agent (called *"Combined Chlorine Residual"*).
- Chloramines are weaker oxidant than free chlorine residuals, so have lesser disinfection potential. Among chloramines, monochloramine has the highest disinfection abilities. These are effective at killing majority of pathogenic bacteria and some protozoans, but are not very effective at killing viruses.
- However, chloramines are relatively more stable, so many times preferred as the disinfectant in the distribution lines of water treatment systems.

Chlorine Reactions in Water: Chloramines

Formula	Name	Molecular Weight	Preferred pH Biocidating Value	Effect
NH_2Cl	Monochloramine	52		
$NHCl_2$	Dichloramine	85	> 7	good
NCl_3	Trichloramine	119	4 - 7	tolerable
			1 - 3	average
R-NHCl	Organic chloramines	varies		
			unknown	bad



- When chloramines are used as a disinfectant, ammonia is added to chlorine treated water.
- Ammonia is added after chlorine, because this causes *Ct* values to be lower than when ammonia is added primarily.

Chlorine Reactions in Water: Chlorine dioxide

- High solubility in water – 5 times of free chlorine
- High oxidative potentials – 2.63 times of free chlorine (only 20% at neutral pH)
- Effective at a higher pH than other forms of chlorination.

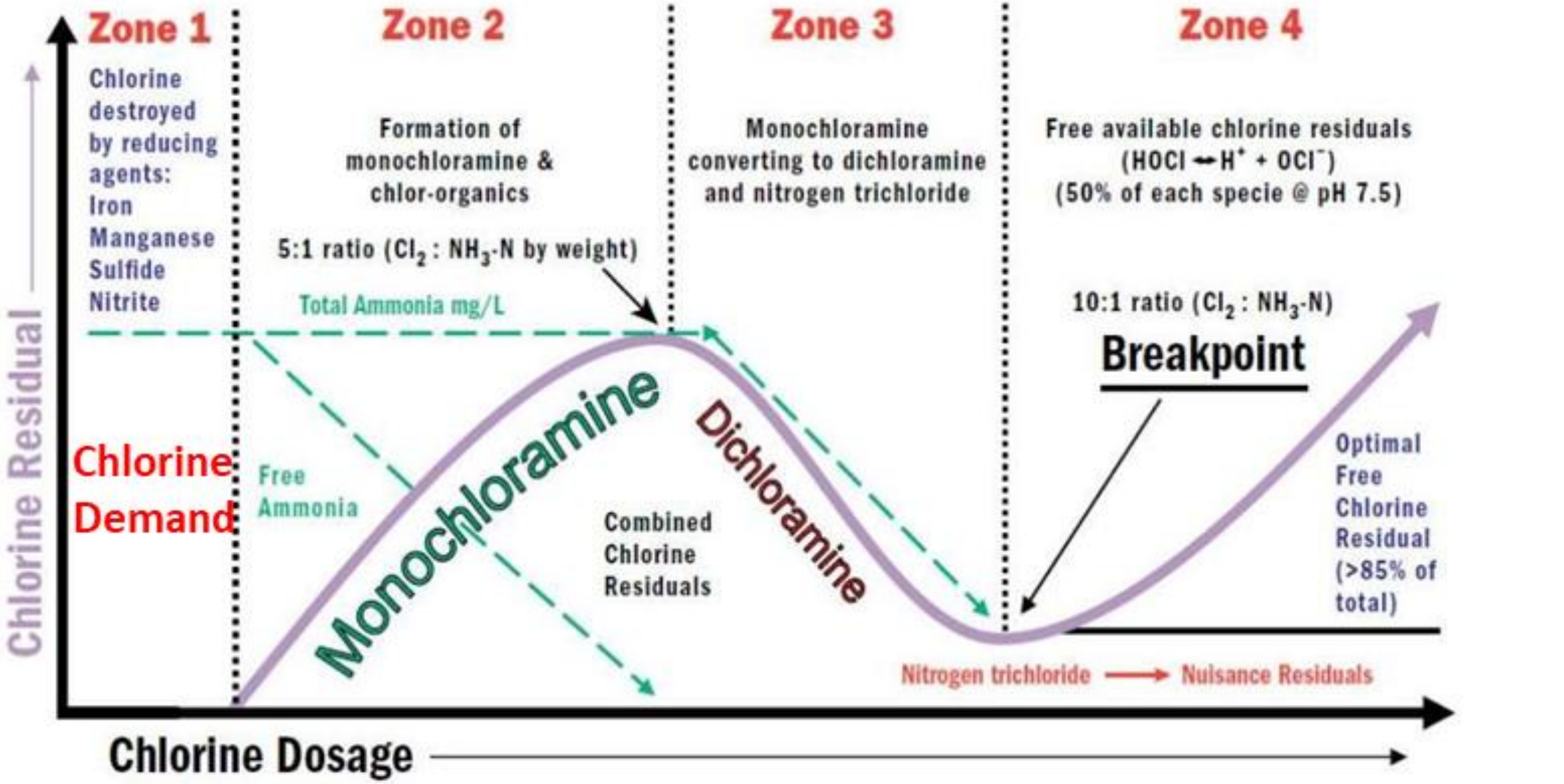
Chlorine dioxide is the most effective form of chlorination since it will kill protozoans, *Cryptosporidium*, *Giardia*, and viruses that other systems may not kill. In addition, chlorine dioxide oxidizes all metals and organic matter, converting the organic matter to carbon dioxide and water. Chlorine dioxide can be used to remove sulfide compounds and phenolic tastes and odors. Contrary to chlorine, chlorine dioxide does not react with ammonia nitrogen (NH_3).

So why isn't chlorine dioxide used in all systems?

- Chlorine dioxide must be generated on site, which is a very costly process requiring a great deal of technical expertise.
- Unlike chlorine gas, chlorine dioxide is highly combustible and care must be taken when handling the chlorine dioxide.

Break Point Chlorination

Chlorine Dose = Chlorine Demand + Chlorine Residual



Residual Chlorine

- Although, the evidences of microbs regrowth in treated water is inconclusive, residual chlorine in water aims at protection against any regrowth or recontamination, by preventing microbs growth in the treated water, specifically when the water is biologically unstable.
  - Water temperatures and nutrient concentrations are not generally elevated enough within the distribution system to support the growth of *E. coli (or enteric pathogenic bacteria)* in biofilms. Thus, the presence of E. coli should be considered as evidence of recent faecal contamination.
  - Viruses and the resting stages of parasites (cysts, oocysts, ova) are also unable to multiply in treated water.
  - However, biofilms and coatings of organic and inorganic materials in pipes that can harbour, and allow the proliferation of few bacterial pathogens, including *Legionella* and *Mycobacterium avium complex* (MAC).
- Residual chlorine in water can also control non-pathogenic slime-forming organisms

Effect of Chlorination: Bacteria

Bacteria	Cl_2 Concentration (mg/l)	Time (min)	Ct Factor (mg-min/l)	Reduction(%)	Reference
Campylobacter jejuni	0.1	5	**0.5**	**99.99**	Blaser et al, 1986
Escherichia coli	0.2	3	**5**	**99.99**	Ram and Malley, 1984
Legionella pneumophila	0.25	60-90	**18.75**	**99**	Kuchta et al, 1985
Mycobacterium chelonei	0.7	60	**42**	**99.95**	Carson et al, 1978
Mycobacterium fortuitum	1.0	30	**30**	**99.4**	Pelletier and DuMoulin, 1987
Mycobacterium intracellulare	0.15	60	**9**	**70**	Pelletier and DuMoulin, 1987
Pasteurella tularensis	0.5-1.0	5	**3.75**	**99.6-100**	Baumann and Ludwig, 1962
Salmonella typhi	0.5	6	**3**	**99**	Korol et al, 1995
Shigella dysenteriae	0.05	10	**0.5**	**99.6-100**	Baumann and Ludwig, 1962
Staphylococcus aureus	0.8	0.5	**0.4**	**100**	Bolton et al, 1988
Vibrio cholerae **(smooth strain)**	1.0	< 1	**< 1**	**100**	Rice et al, 1993
Vibrio cholerae **(rugose strain)**	2.0	30	**60**	**99.999**	Rice et al, 1993
Yersinia enterocolitica	1.0	30	**30**	**92**	Paz et al, 1993

Chlorination Risks: Taste Issues

- Chlorine compounds present in water may induce some specific taste in the water.

Compounds	Order of the taste threshold (mg/L)
HOCl and OCl^-	~0.5
NH_2Cl	~0.8
$NHCl_2$ and Chloroform	~0.1
Carbon tetrachloride	~0.3
Dichlorophenol and Dichlorobenzene	~ 0.0005

- Residual chlorine species add to some taste in water, which may prevent some consumers from using treated water. Specially, introduction of chlorine into a community that has always drunk water without the addition of chemicals can be difficult.

- Need to convince consumers that the chemical taste is healthy. With less chlorine, there is less breakthrough of chlorinated compounds into the finished water, and therefore less of a chemical taste. Reducing the amount of chlorine added might increase the social acceptance of chlorination.

Types of Chlorination

Plain chlorination

- Only chlorination for raw water contains turbidity <10 NTU
- The dosage of chlorine for plain chlorination is about 0.5 mg/L.

Pre-chlorination

- May be applied to the water before coagulation for heavily contaminated waters.
- The chlorine dose for pre-chlorination should be 0.1 to 0.5 mg/L.
- Always followed by post chlorination to ensure final safety of water.
- It has become much less common in the western countries due to risks of DBPs.

Post chlorination

- Normal process of applying chlorine in the end.
- Intended to leave a residual-chlorine of about 0.1-0.2 mg/L after 20-30 min contact.

Types of Chlorination

Break point chlorination

- To get an idea of the amount of chlorine to be added to water for desired residual chlorine, especially when intention is to leave free chlorine residual in treated water.

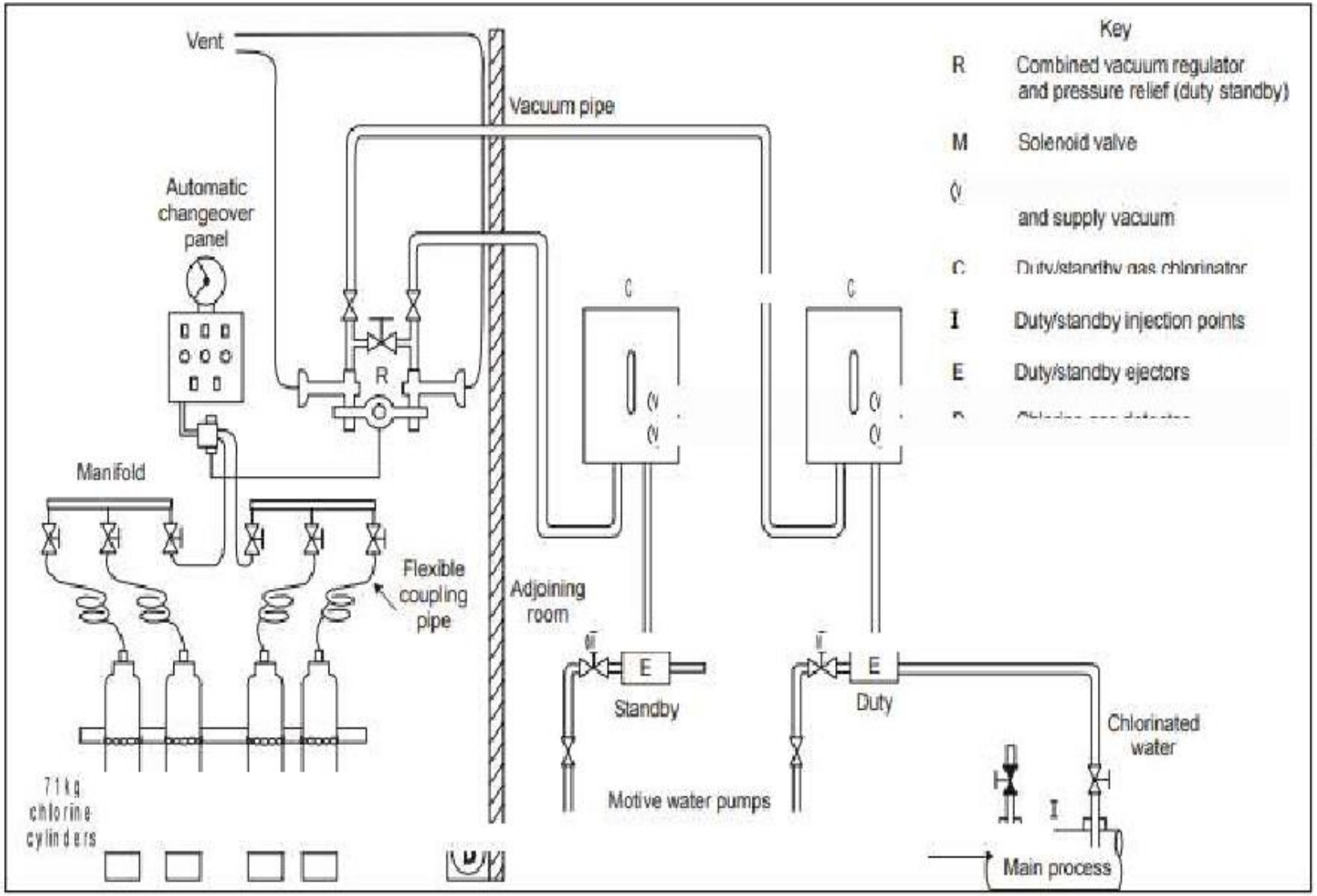
Super chlorination

- Addition of excessive chlorine (i.e., 5 to 15 mg/L) to give about 1 to 2 mg/L
- May be required in some special cases when the water is highly polluted, or during epidemics of water borne diseases.

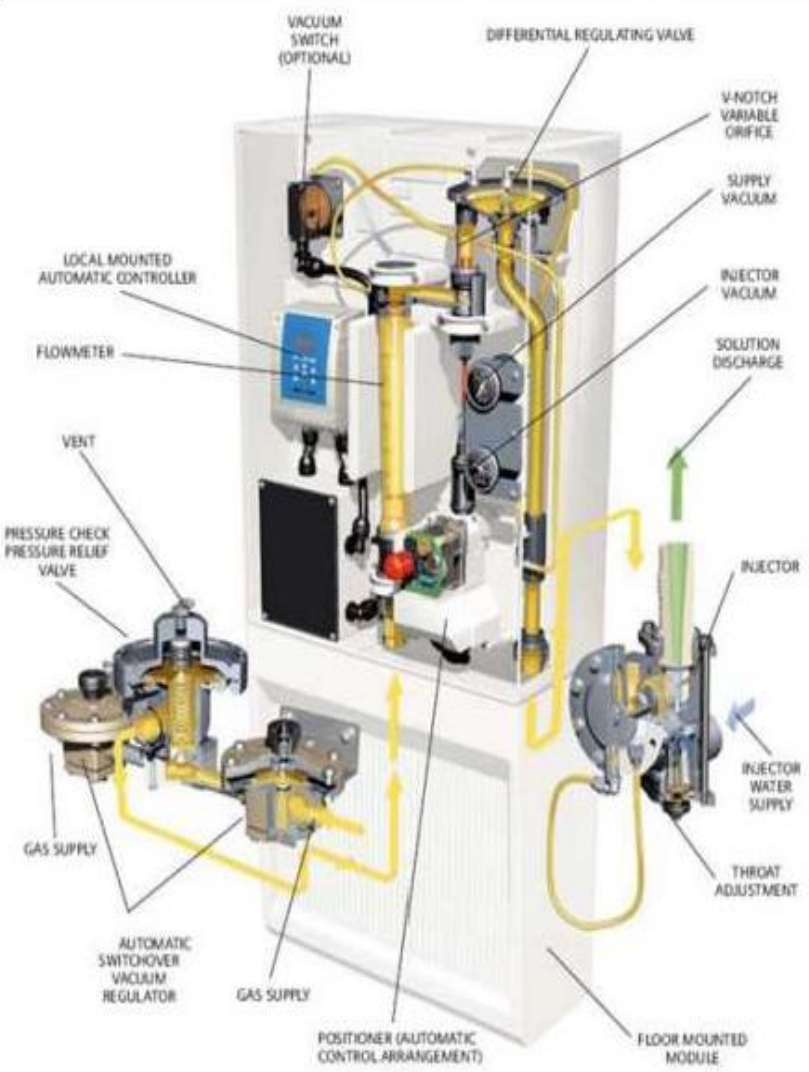
De-chlorination

- Removing chlorine using de-chlorinating agents such as sulphur dioxide gas, activated carbon, sodium thiosulphiate, sodium metasulphate, and ammonia.
- Generally required post super-chlorination, to reduce residual chlorine to ~ 0.2 mg/L

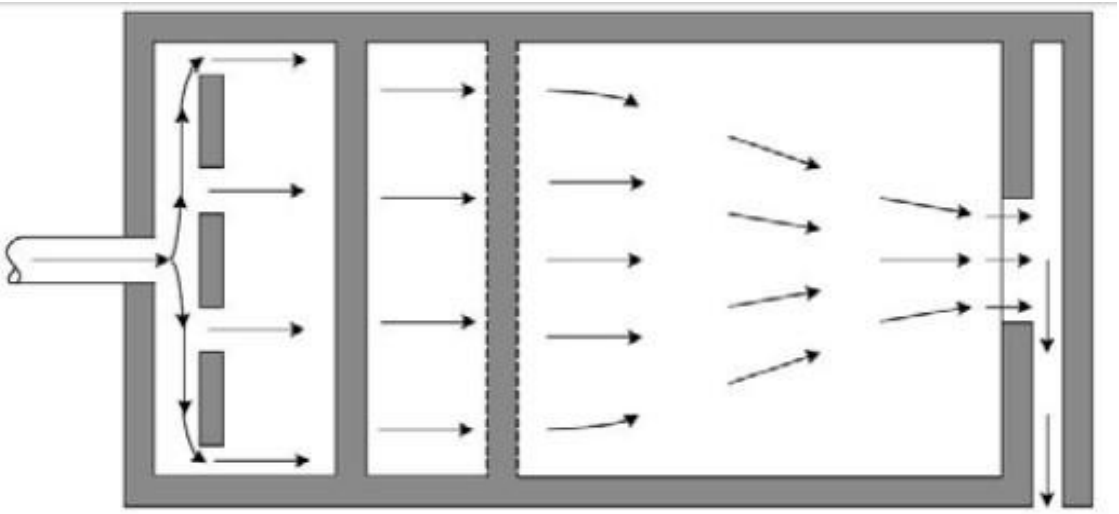
Chlorination Systems: Chlorine Gas



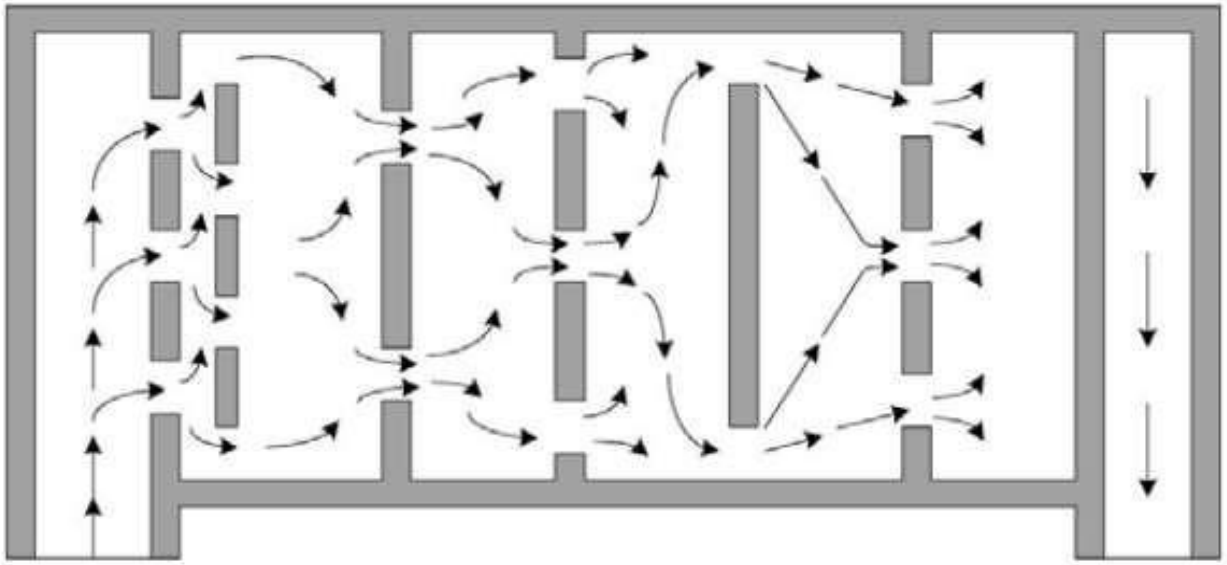
Typical Gas Chlorinator Equipment



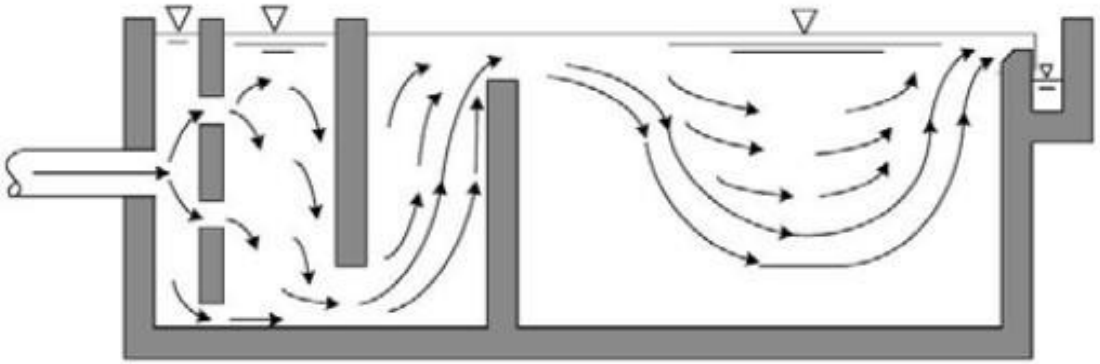
Chlorination Systems: Chlorine Contact Tanks



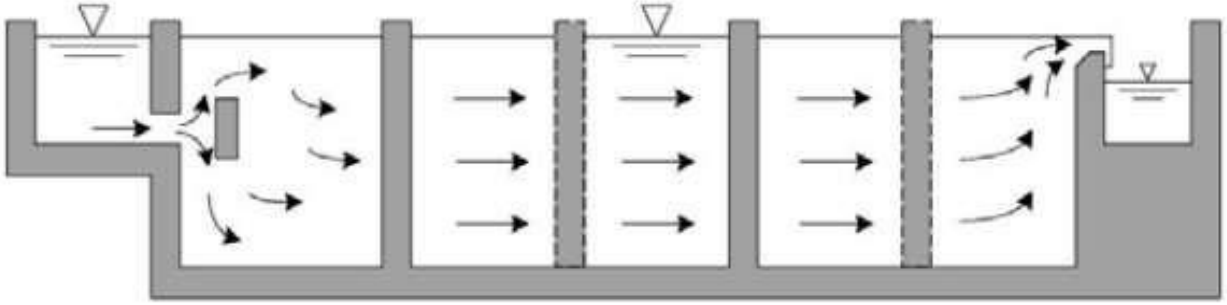
Plan View



Plan View



Section View



Section View

Chlorine demand of water :-

Chlorine react first with iron & magnese & then react with ammonia present in water. Then Cl_2 distroys pathogens present ——.
In doing above 3 jobs some Cl_2 consumes & if any excess present it appears at residual Cl_2.

Cl_2 demand of water = Cl_2 dose added to water − Residual Cl_2 to water

→ **Cl_2 demand = Cl_2 dose − Residual Cl_2**

p.46
2

Usage
Cl_2 ~~demand~~ = 8 kg/day

$Q = 20000\ m^3/day = \frac{20000 \times 1000}{10^6} = 20 mLD$

Residual aftr 10min = 0.15 mg/l

Total Cl_2 consume/day = Q * dose of Cl_2

8 = 20 * dose of Cl_2

(kg/day) (mLD) (mg/l)

$\therefore$ dose of Cl_2 = 0.4 mg/l

$\therefore$ Cl_2 demand = 0.4 - 0.15 = 0.25 mg/l

Practice Problem 1: Chlorine Dose and Demand

A water treatment plant uses 50.0 kg/d of chlorine to treat 40 MLD of water. The residual chlorine after 30 minutes contact time is 0.15 mg/L. Determine the chlorine dosage and chlorine demand of the water

Solution:

Chlorine Dose = 50 kg in 40 ML water

$= 50*10^6$ mg / $40*10^6$ L

= 1.25 mg/L

Chlorine Residual = Chlorine Dose – Chlorine Demand Or, Chlorine Demand = Chlorine Dose – Chlorine Residual

= 1.25 mg/L – 0.15 mg/L

= 1.1 mg/L

Practice Problem 2: Chlorine Dose Calculations

A water sampled from the filter outlet has resulted in the following data for Chlorine dose-residual curve after 20 mins contact time. What are the predominant forms of chlorine constituting residual chlorine at chlorine dose of 1 mg/L and 2.5 mg/L. Further, estimate the followings:

(a)chlorine demand (in mg/L) of water,

(b)chlorine dose (in mg/L) for achieving breakpoint chlorination,

(c)chlorine dose (in mg/L) for ensuring 0.2 mg/L free residual chlorine in treated water,

(d)daily requirement of chlorine for treating a flow of 10 MLD, ensuring 0.2 mg/L free residual chlorine in treated water.

Solution:

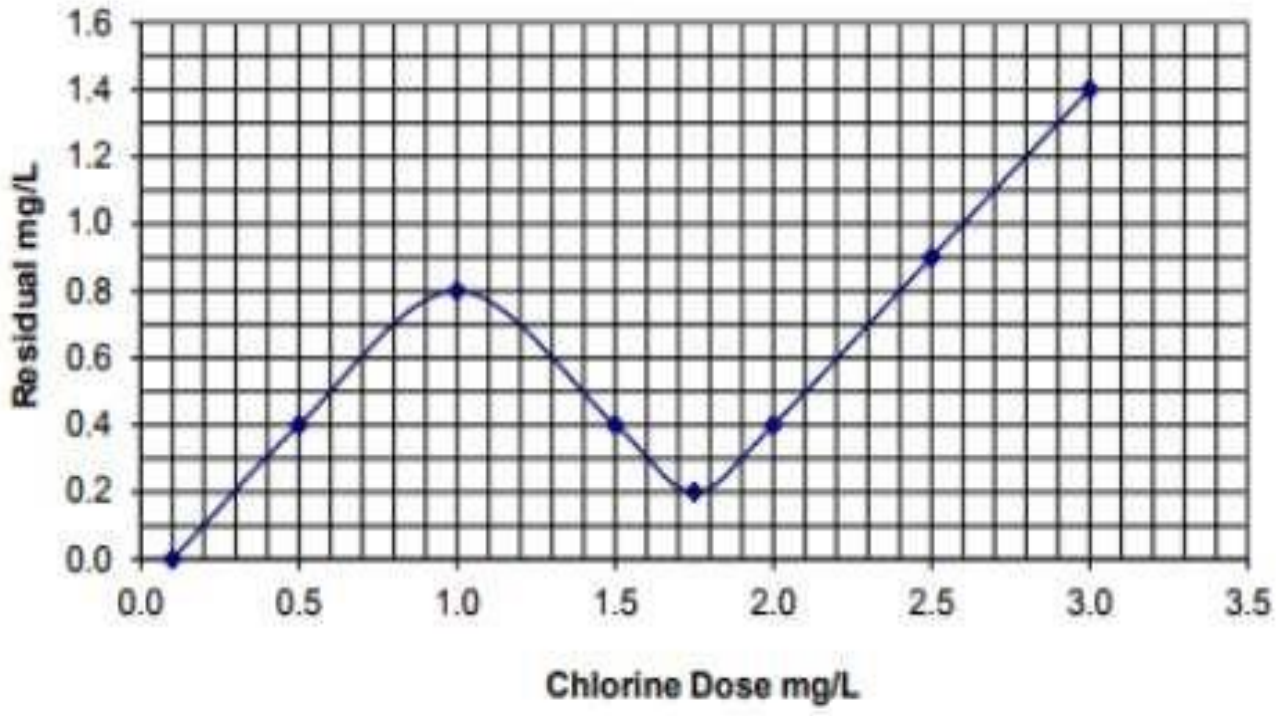
(a)Chlorine Demand ***= 0.1 mg/L***

(b)Chlorine dose for breakpoint chlorination ***= 1.75 mg/L***

(c)Chlorine dose for 0.2 mg/L free residual chlorine ***= 2 mg/L***

(d)Daily requirement of Chlorine for treating 10 MLD

= 2 mg/L*10*10^6 L/d = 20*10^6 mg/d ***= 20 kg/d***

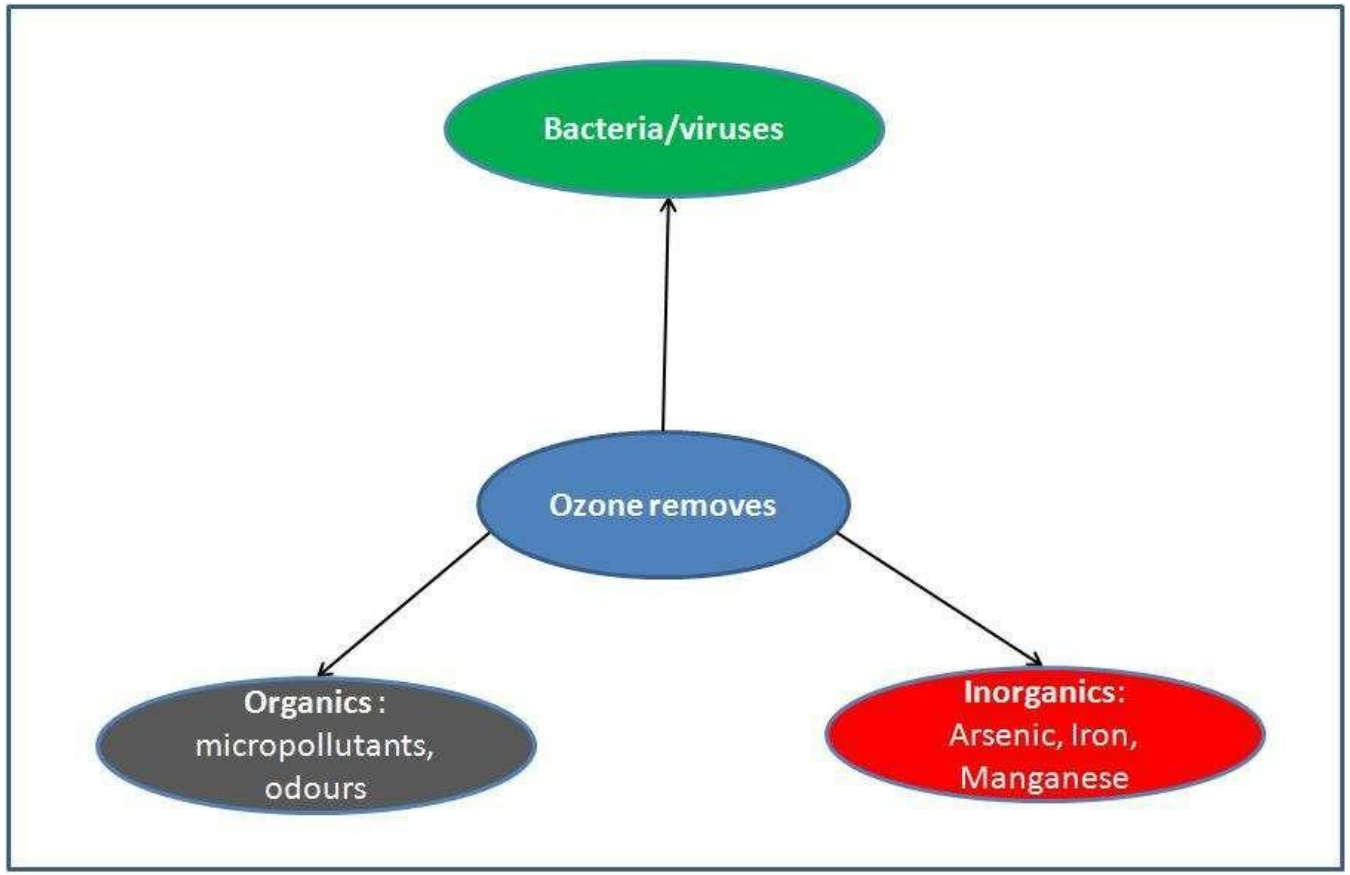


Disinfection with Ozone

- Ozone (O_3) is one of the strongest disinfectants and oxidants available in drinking water treatment. It's a colorless gas, relatively unstable, especially at higher pH, as it reacts with itself and with OH^- in water.
- Ozone leads to enhanced disinfection with reduced and lesser toxic DBPs (mainly include aldehydes, ketones and carboxyl acids).
- Ozone is also helpful in removal of taste and odour, organic and inorganic matter, micro-pollutants such as pesticides and pharmaceuticals etc. It also helps in enhancing the flocculation/coagulation-decantation process.
- Ozone must be generated onsite and used immediately, as it has very short half-life (typically < 30 min). Thus, it can only be used as a primary disinfectant, and a residual can not be maintained in downstream processes. A secondary disinfectant

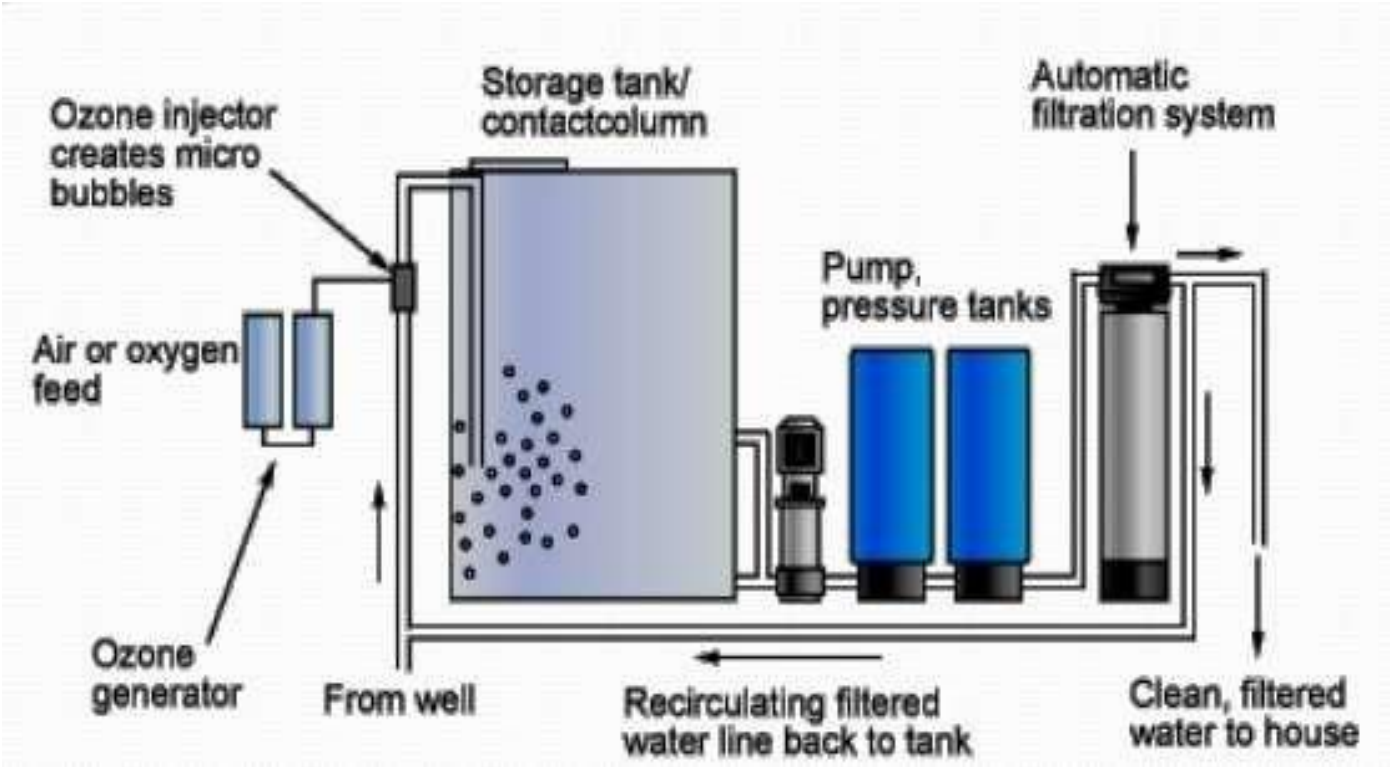
such as chlorine or chloramine must be added to maintain

disinfectant residual within the distribution system, if needed.



Ozone Treatment Process

- Ozone is generated onsite by an ozone generator that uses either dried air (requiring air dryers and compressors) or liquid oxygen (preferred, as it produces higher a percent weight concentration of ozone.
- The solubility of ozone in water depends on temperature and its concentration in the feed gas. Ozone contactors (diffused bubble or in-line injection systems) are used to dissolve ozone in water.
- Diffused bubble systems with countercurrent flow are more common in drinking water treatment, as they can be better controlled to ensure desired contact time. If needed, more (than one) chambers are provided to ensure sufficient contact time between ozone and water.
- The exhaust gas from ozone contactor must be recycled or destroyed to minimize exposure risks (as all the applied ozone may not get transferred into water). Ozone destructors usually use heat or a combination of heat and a catalyst to remove ozone from the air.



Ozone Treatment: Applications, Advantages and Disadvantages

- Ozone application for drinking water treatment is more popular in European countries with over 1500 installations, while it has not had similar acceptance elsewhere including US due to its higher installation and operational cost, and the fact it does not remain present for long in water.

Advantages

- Rapidly reacts with bacteria, viruses and protozoa over a wide pH range
- Stronger germicidal properties than chlorination
- No chemicals are added to water
- Also efficient for organics degradation and inorganics removal
- Removes colour, taste and odour

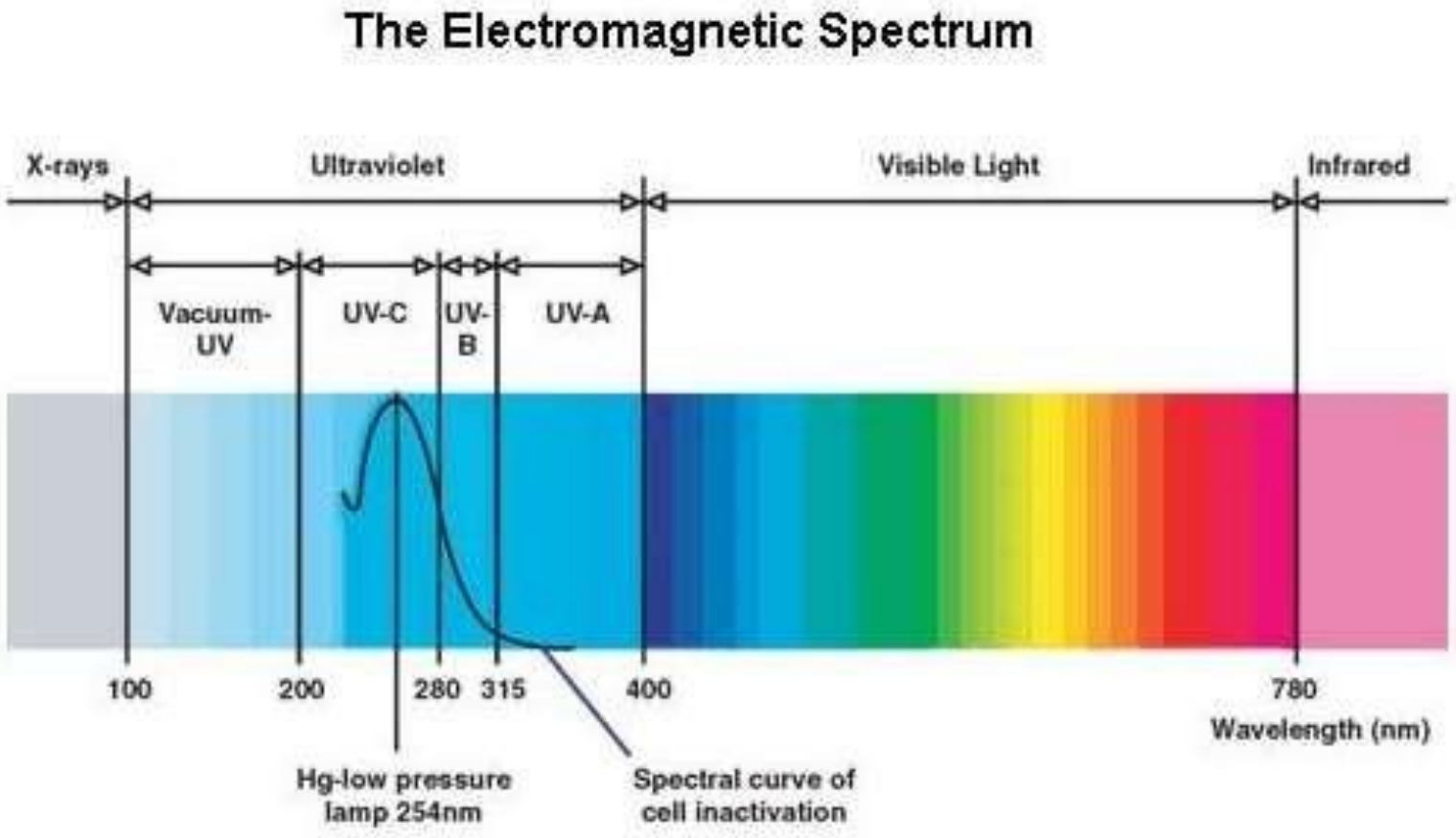
Disadvantages

- Relatively high equipment costs
- Requires large amounts of energy
- Qualified professionals required for design and system maintenance
- Formation of potentially harmful disinfection by-products (DBPs) in the case of brome existence in water
- No residual effect is present in the distribution system
- Potential fire hazard and toxicity associated with ozone generation

Disinfection with Ultraviolet Radiation (UV)

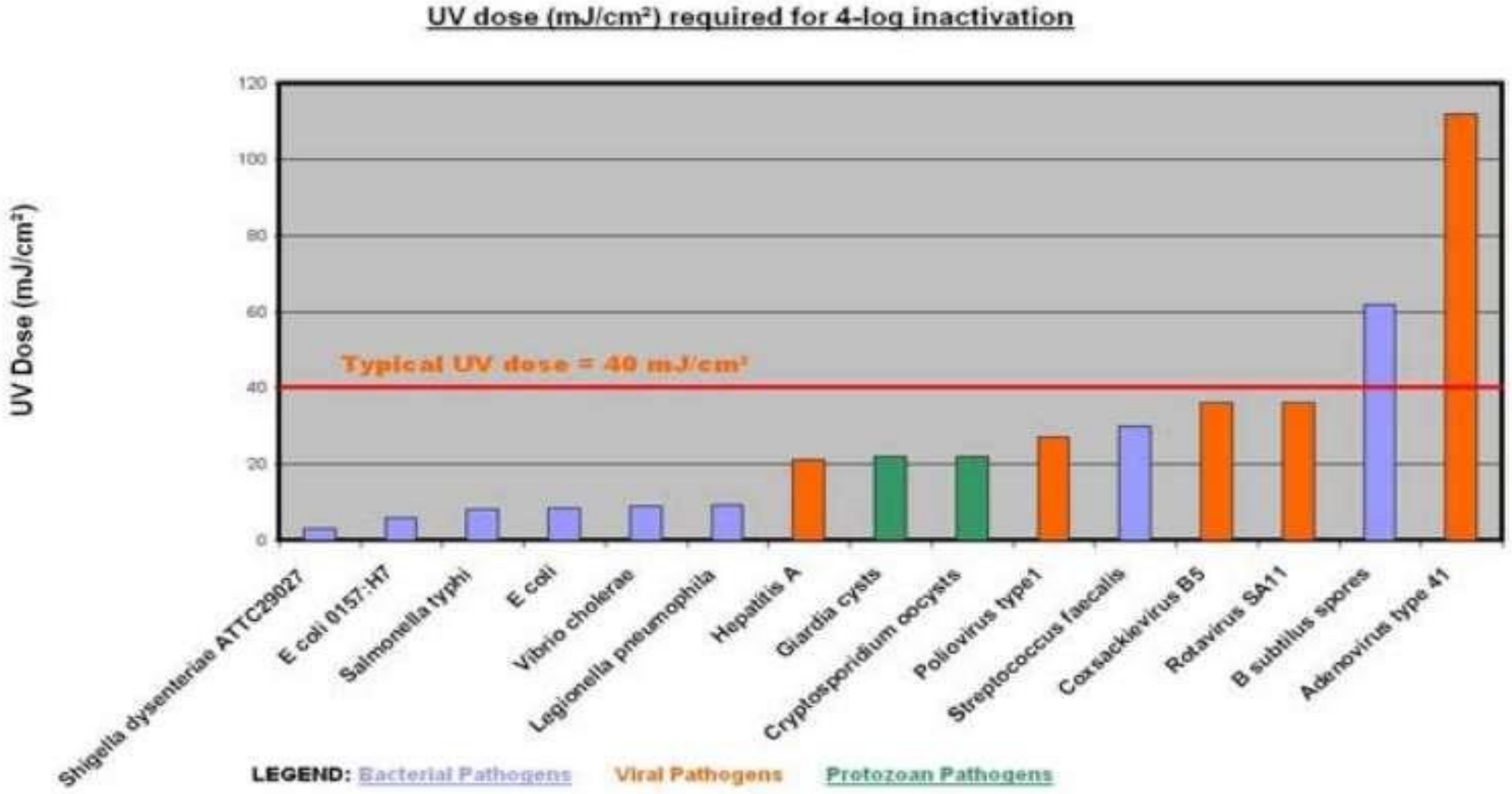
- The UV spectrum is higher in energy (higher frequency and lower wavelength) than visible light. It is known to be an effective disinfectant due to its strong germicidal (inactivating) ability. UV disinfects water containing bacteria and viruses and can be effective against protozoans like, Giardia lamblia cysts or Cryptosporidium oocysts.

- When UV light enters a microorganism, its energy leads to damage the microorganism's cellular function by physical damages including strand breaks. The energy is absorbed by DNA, and inhibits cell replication.



UV Disinfection Abilities

Required UV dose for 4-log inactivation of common waterborne pathogens



UV Disinfection Systems

Skokie Water Treatment Facility, Chicago



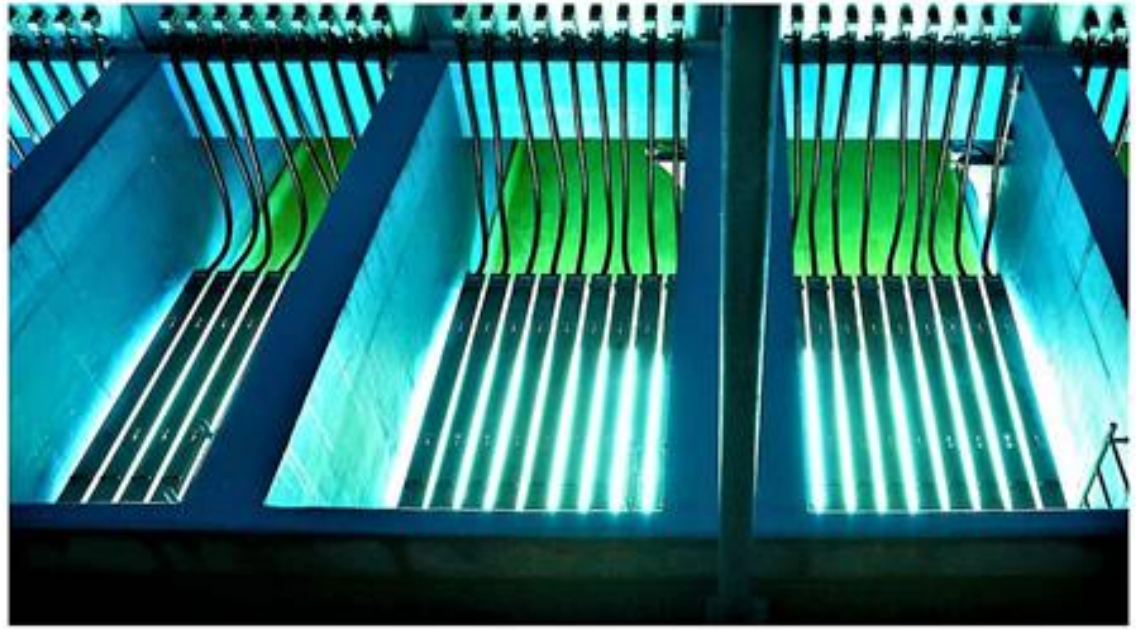


Image Source: https://news.wttw.com/2016/03/23/worlds-largest-ultraviolet-disinfection-facility-tackles-chicago-river; http://archive.sswm.info/print/8007?tid=3147

UV Disinfection: Advantages and Disadvantages

Advantages:

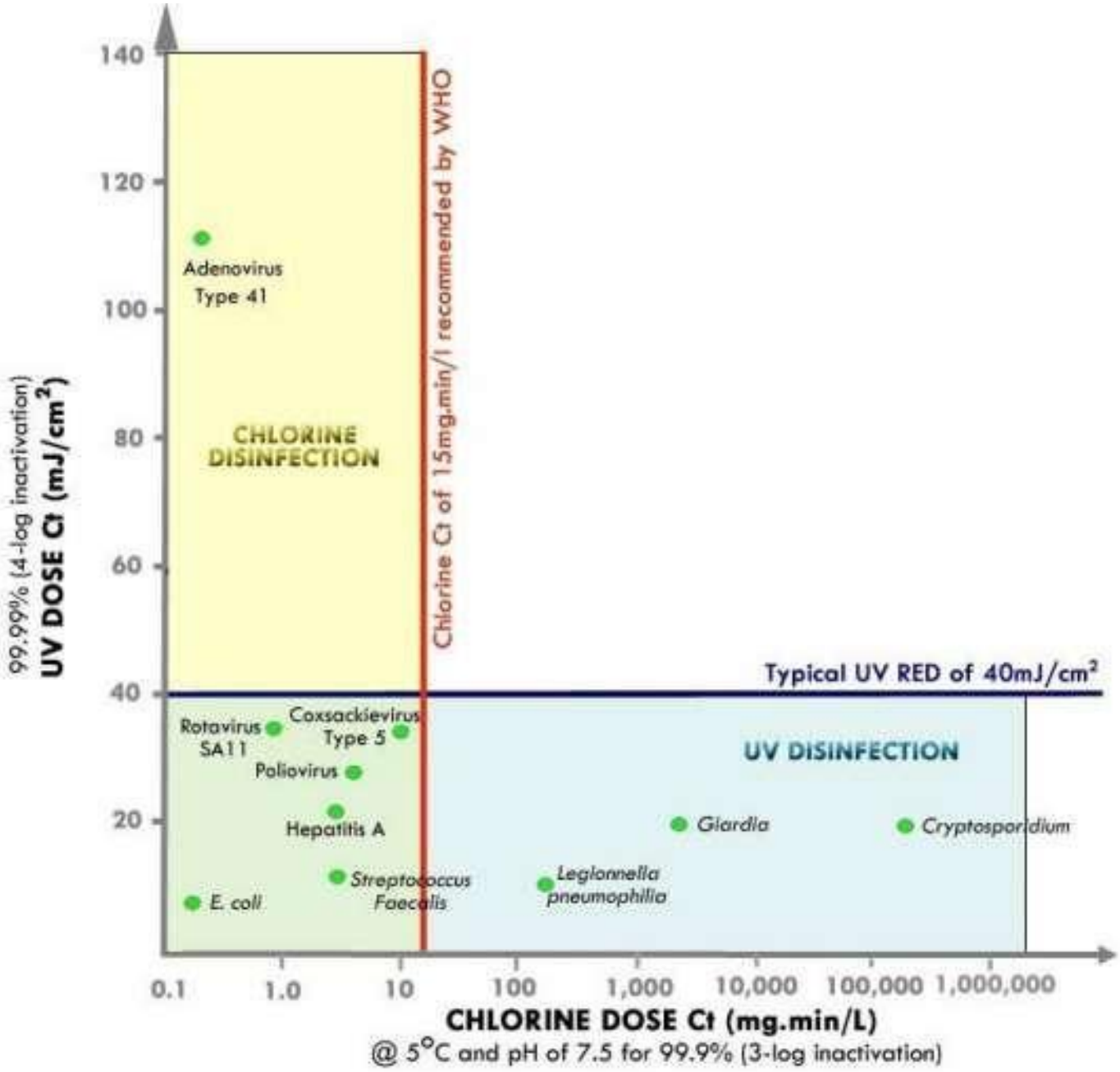
- UV disinfection is effective at inactivating most viruses, spores, and cysts.
- Does not alter taste, odour, colour or pH of the water.
- Does not require the addition of chemicals.
- Does not impart toxic by-products into the water
- Systems are compact and easy to install

Disadvantage:

- Low dosages may not effectively inactivate some viruses, spores, and cysts.
- Organisms can sometimes repair and reverse the destructive effects of UV through a "repair mechanism," known as photoreactivation, or in the absence of light known as *"dark repair."*
- Turbidity and total suspended solids (TSS) in the wastewater can render UV disinfection ineffective.
- UV disinfection is not as cost-effective as chlorination, but costs are competitive when chlorination- dechlorination is used and fire codes are met.

Combinations of Disinfectants

- There are published reports from laboratory tests of synergistic benefits from using two or more disinfectants,
  i.e. the overall inactivation is greater than the sum of the inactivation achieved for each disinfectant individually.
- For example, one benefit from ozonation before UV treatment is that ozone can degrade natural organics which cause UV absorption thereby allowing the UV dose to be a more effective disinfectant and more energy efficient.
- Chlorine dioxide also shows a synergistic effect when combined with other disinfectants such as ozone, chlorine, and chloramines. Combination of disinfectants is known to lead to greater inactivation when the disinfectants are added in series rather than individually.)



By-products Implication of Different Disinfectants

Process	By-product issues
Chlorination	Trihalomethanes, trihaloacetic acids are formed by reaction with natural organic matter in water. Where chlorine is obtained from hypochlorite, chlorate and bromate formation can be an issue depending on bromide content of salt used in manufacture and subsequent conditions of storage of hypochlorite. Can be controlled by appropriate product specification and management of storage.
Chloramination	No significant by-product issues. Nitrite formation in distribution has been an indirect issue.
Ozone	Bromate formation in waters with high concentration of bromide.
Chlorine dioxide	Dosage rates in the future are likely to be limited by consideration of inorganic by products (chlorate and chlorite) in accordance with current international practice.
UV	No significant by-product issues.

Advantages and Limitations of Common Disinfectants

Process	Advantages	Limitations
Chlorination	Well understood disinfectant capability. Established dosing technology.	Chlorination by-products and taste and odour issues can affect acceptability. Ineffective against *Cryptosporidium*.
Chloramination	No significant by-product issues. Generally less taste and odour issues than chlorine.	Considerably less effective than compared with chlorine. Not usually practical as a primary disinfectant.
Ozone	Strong oxidant and highly effective disinfectant compared with chlorine. Benefits of destruction of organic micropollutants (pesticides, taste and odour compounds).	Bromate by-product and increased assimiable organic carbon (AOC) can impact on re-growth in distribution. Complex, energy intensive and expensive equipment compared with chlorination. Residual insufficiently long lasting for distribution.
Chlorine dioxide	Can be more effective than chlorine at higher pH, and less taste and odour and by-product issues.	Weaker oxidant than ozone or chlorine. Dose limited by consideration of inorganic by products (chlorate and chlorite).
UV	Generally highly effective for protozoa, bacteria and most viruses and particularly for *Cryptosporidium*. No significant by-product implications.	Less effective for viruses than chlorine. No residual for distribution.

Image Source: Water Treatment Manual: Disinfection (2011). The Environmental Protection Agency (EPA)

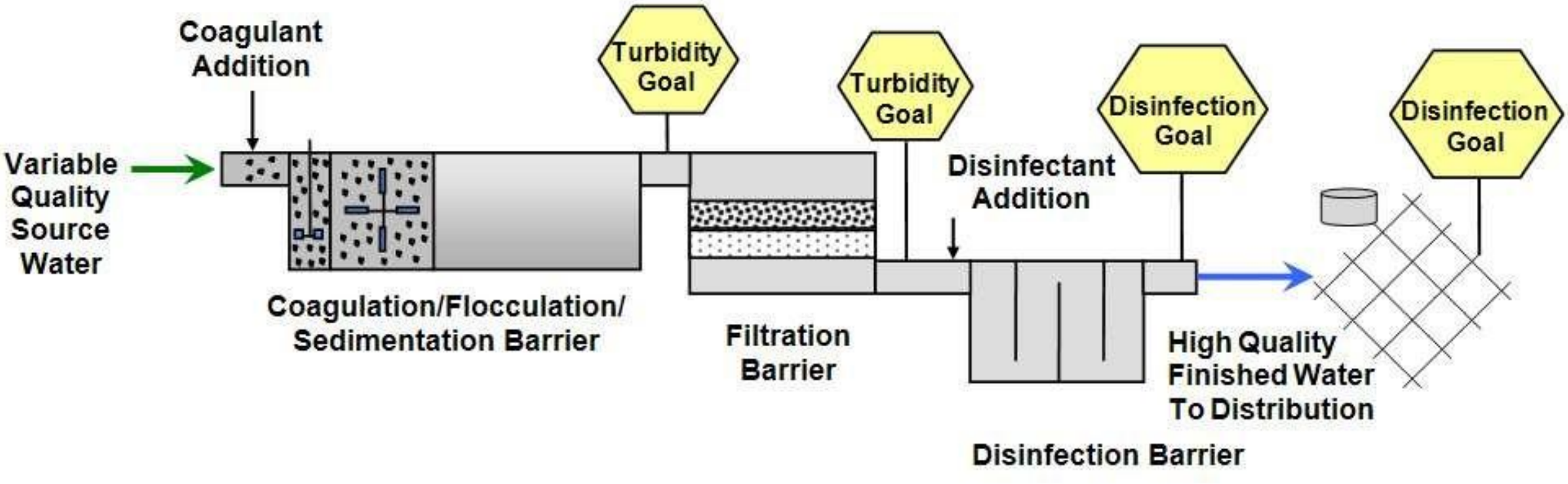
Choice for Disinfectants

	Chlorine (Gas or Hypochlorite)	Chlorine Dioxide	Chloramine	Ozone	Ultraviolet
Produces trihalomethanes?	yes	no	yes	sometimes	no
Produces other troublesome byproducts?	yes	yes	yes	yes	sometimes
Impacted by lime softening?	yes	no	yes	no	yes
Impacted by turbidity?	somewhat	somewhat	somewhat	somewhat	yes
Meets Giardia removal standards?	no	yes	no	yes	no
Meets Cryptosporidium removal standards?	no	no	no	yes	no
Meets virus removal standards?	yes	yes	no	yes	yes
Operator skill level	low	high	low/medium	high	medium
Applicable to large utilities?	yes	yes	yes	yes	no
Applicable to small utilities?	yes	yes	yes	yes	yes

Risks with Conventional Water Treatment

- The conventional water treatment processes are not able to address adequately the removal of a wide spectrum of toxic chemicals, nutrients, salts and pathogenic microorganisms such as Cryptosporidium in raw water.

- The new/emerging micro contaminants (e.g., endocrine disrupting compounds (EDCs), pharmaceuticals, personal care products etc) are usually not removed through conventional processes.



Advanced/Alternate Water Treatment Processes

Processes

Non-Membrane Systems

- Lime Soda Softening or Precipitation
- Alum or Iron compound co-precipitation
- Dissolved Air Flotation
- Ion Exchange
- Granular Activated Carbon
- Advanced oxidation processes
- UV / Ozone disinfection

Membrane Processes

- Microfiltration
- Ultrafiltration
- Nanofiltration
- Reverse Osmosis
- Electrodialysis

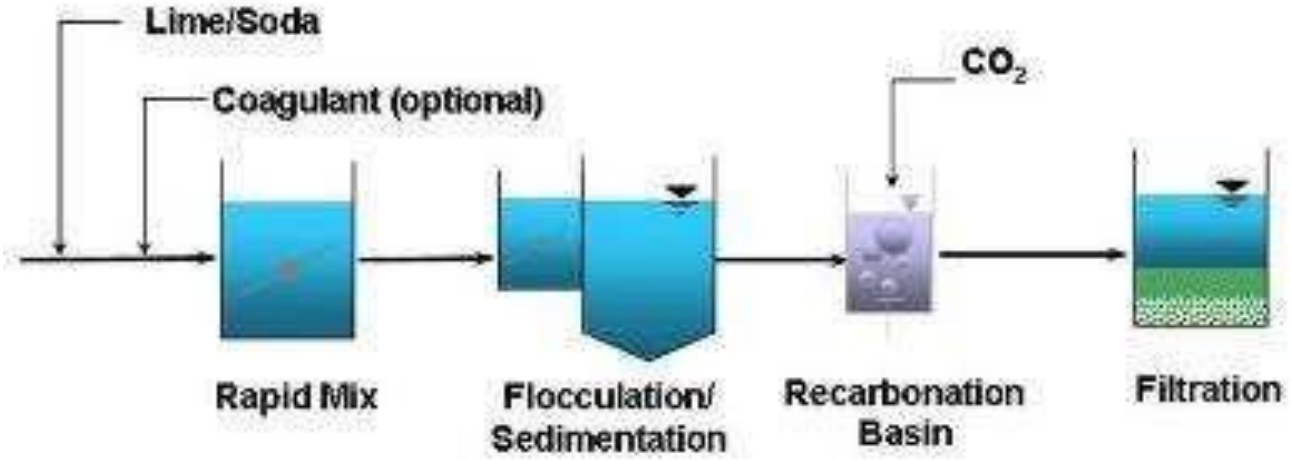
Objectives

- Water softening
- Heavy metals removal
- Nutrients removal
- Emerging contaminants removal
- Advanced microbiol controls

Lime/Soda Softening: Hardness Removal

Cold Lime Softening: When hydrated lime, Ca(OH)2, is added to the water being treated at ambient temperatures, the following reactions occurs:

Hardness		Lime		Precipitate
CO_2.	+	$Ca(OH)_2$	->	$CaCO_3 + H_2O$
$Ca(HCO_3)_2$	+	$Ca(OH)_2$	->	$2CaCO_3 + 2H_20$
$Mg(HCO_3)_2$	+	$Ca(OH)_2$	->	$CaCO_3 + MgCO_3 + 2H_20$
$MgCO_3$	+	$Ca(OH)_2$	->	$CaCO_3 + Mg(OH)_2$

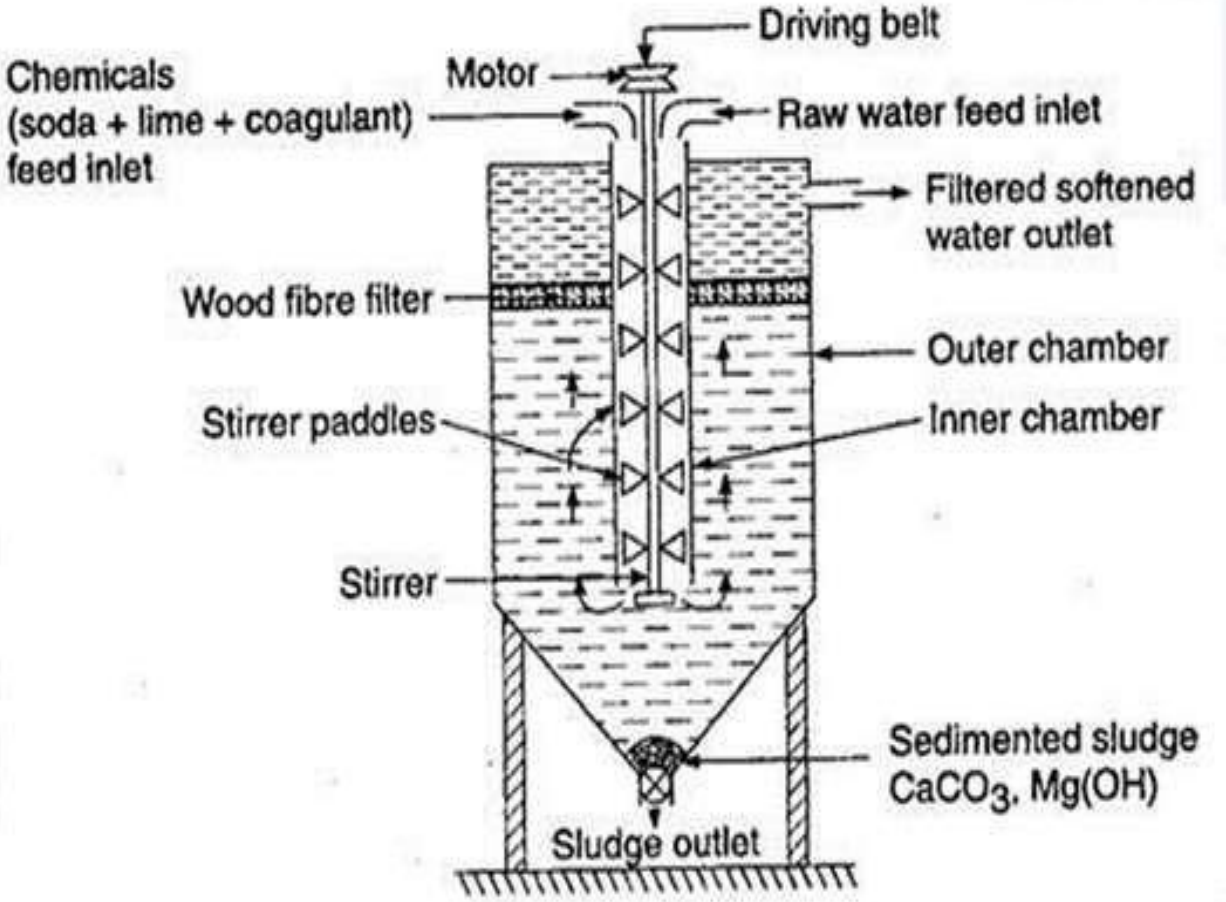


Hot Lime Softening: Hot process softening is usually carried out under pressure, and at elevated temp. of 227-240°F (108-116°C), leading to softening reactions go essentially to completion. It involves same reactions, except that raw water CO_2 is vented and does not react. The use of lime and soda ash permits hardness reduction down to about 8 ppm as $CaCO_3$. Magnesium is reduced to 2-5 ppm because of the lower solubility of magnesium hydroxide at the elevated
temperatures

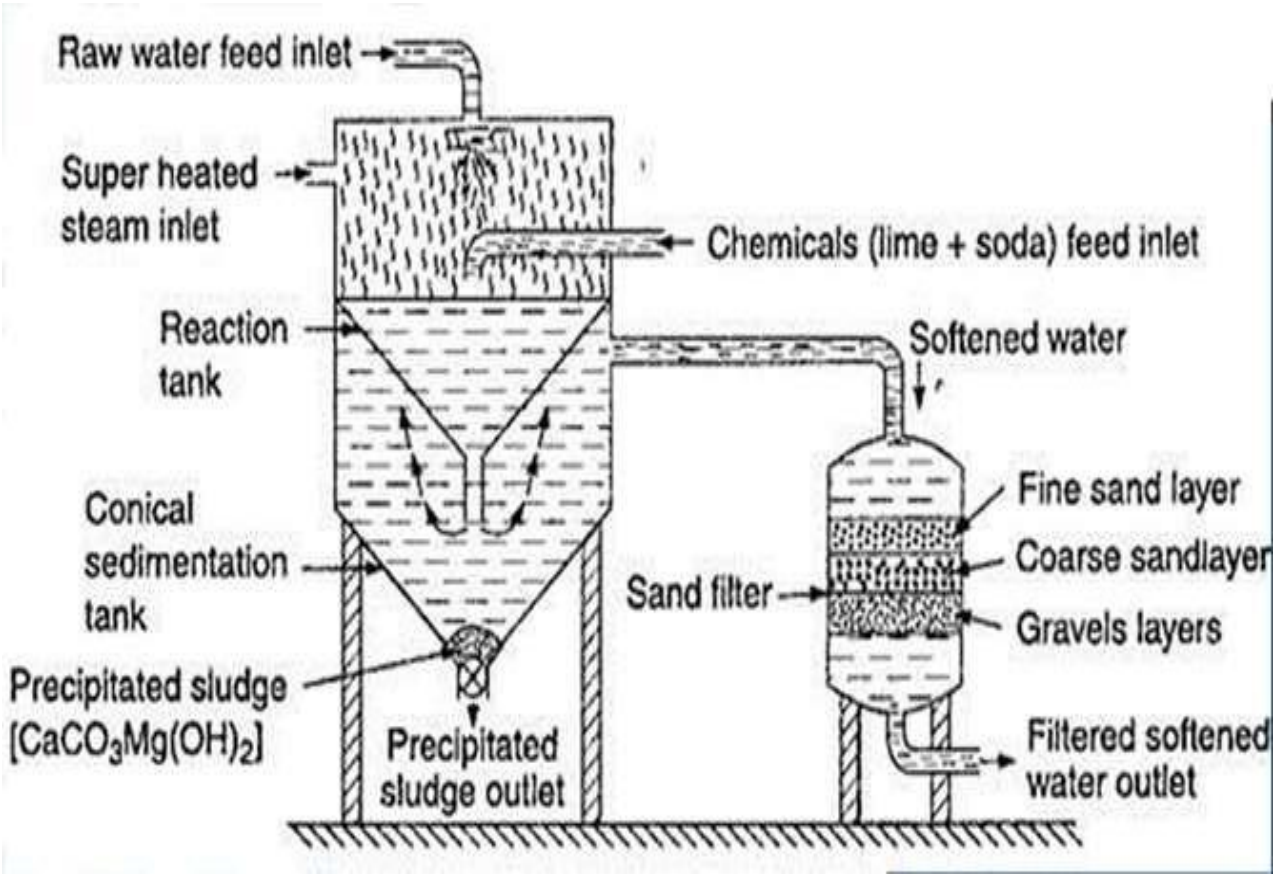
Lime/Soda Softening: Design Consideration

- In lime soda-ash softening plants, the softening process may be carried out by a sequence of rapid mix, flocculation, and sedimentation or in a solids contactor.
- In the solids contactor the rapid mix, flocculation, and sedimentation occur in a single unit.
- Detention time in the flocculator is important to allow particles to come in contact with each other. The minimum time recommended is 30 minutes for water softening.

Hot Lime Softening



Solid Contactor



Lime/Soda Softening: Performance

Typical softener effluent analyses

	Raw Water	Lime-soda Softening (Cold)	Lime-soda Softening (Hot)
Total Hardness (as $CaCO_3$), ppm	250	81	20
Calcium Hardness (as $CaCO_3$), ppm	150	35	15
Magnesium Hardness (as $CaCO_3$), ppm	100	46	5
"P" Alkalinity (as $CaCO_3$), ppm	0	37	23
"M" Alkalinity (as $CaCO_3$), ppm	150	55	40
Silica (as SiO_2), ppm	20	18	1-2
pH	7.5	10.6	10.5

Adsorption

- Adsorption is a mass transfer process which involves the accumulation of pollutant at the solid surfaces thereby leaving the water phase, and is typically used for removing a wide range of compounds including various micropollutants, organic compounds, and a few metals from the source waters.

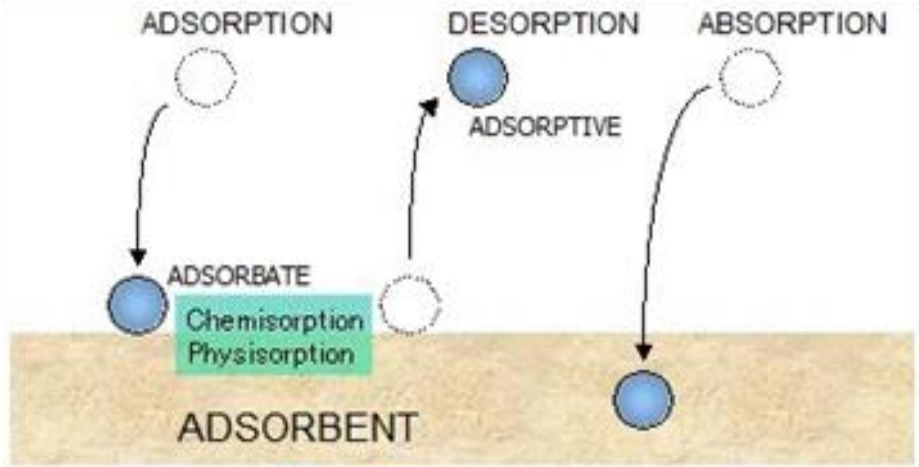


Image Source : https://www.microtrac-bel.com/en/tech/bel/entry40.html

Physical / Physisorption (*van der Waals adsorption*):

- Weak bonding
- Exothermic (of the order of 0.1 Kcal/mole)
- Reversible

Chemical / Chemisorption:

- Chemical bonding by reaction
- Exothermic (of the order of 10 Kcal/mole)
- May be irreversible

More popular for wastewater treatment

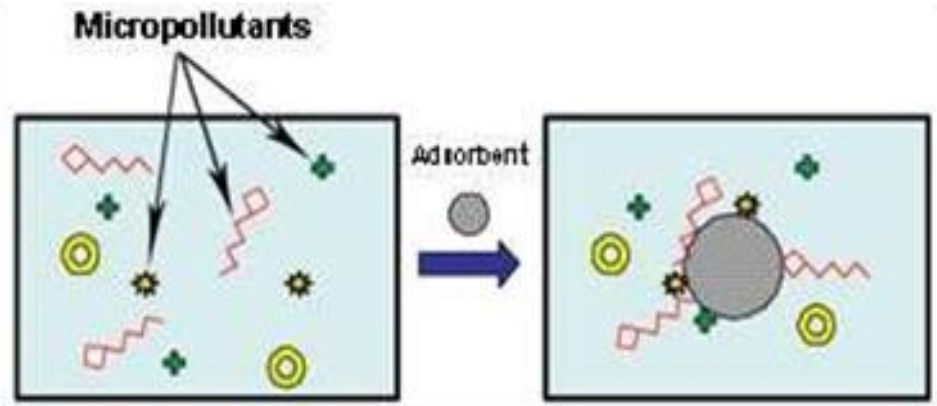
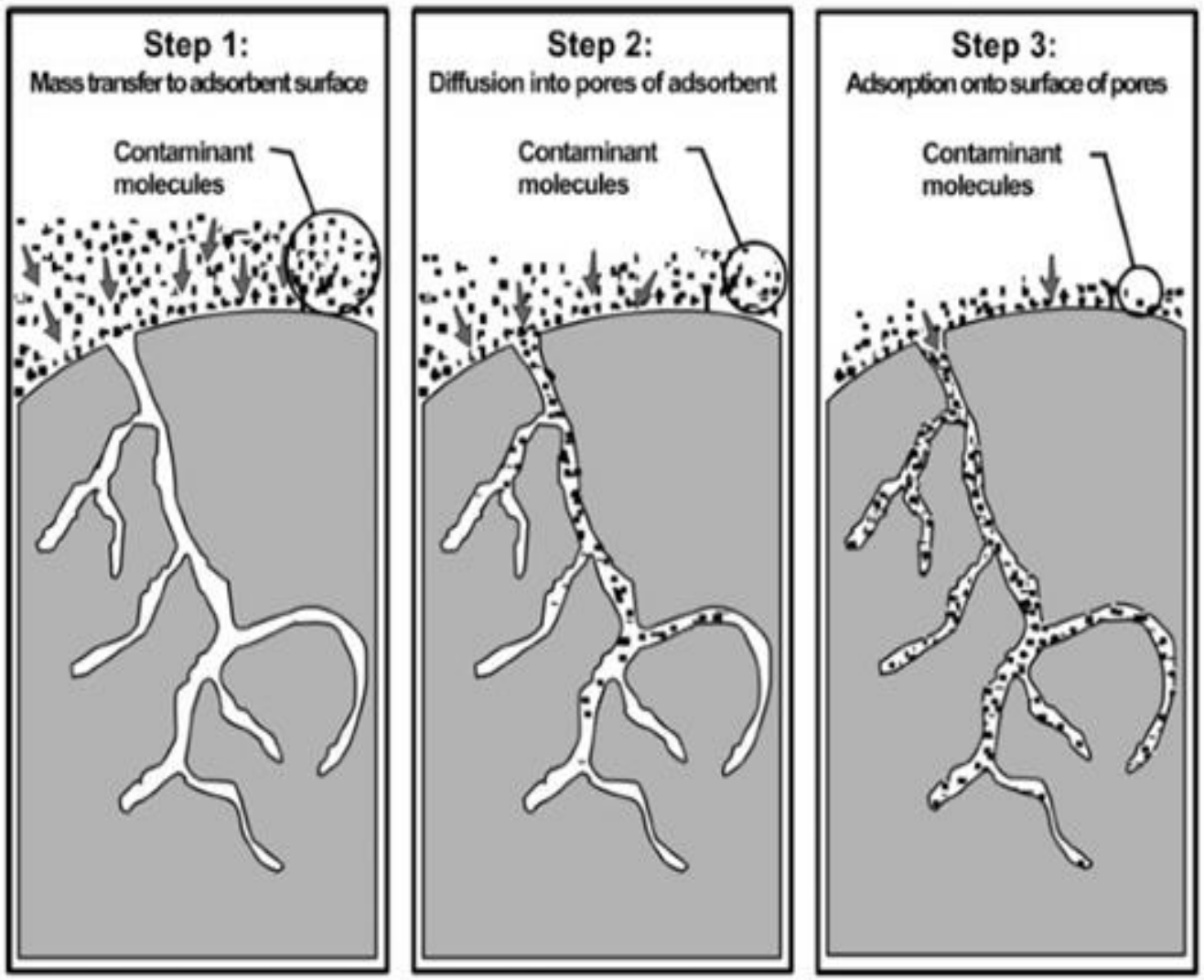
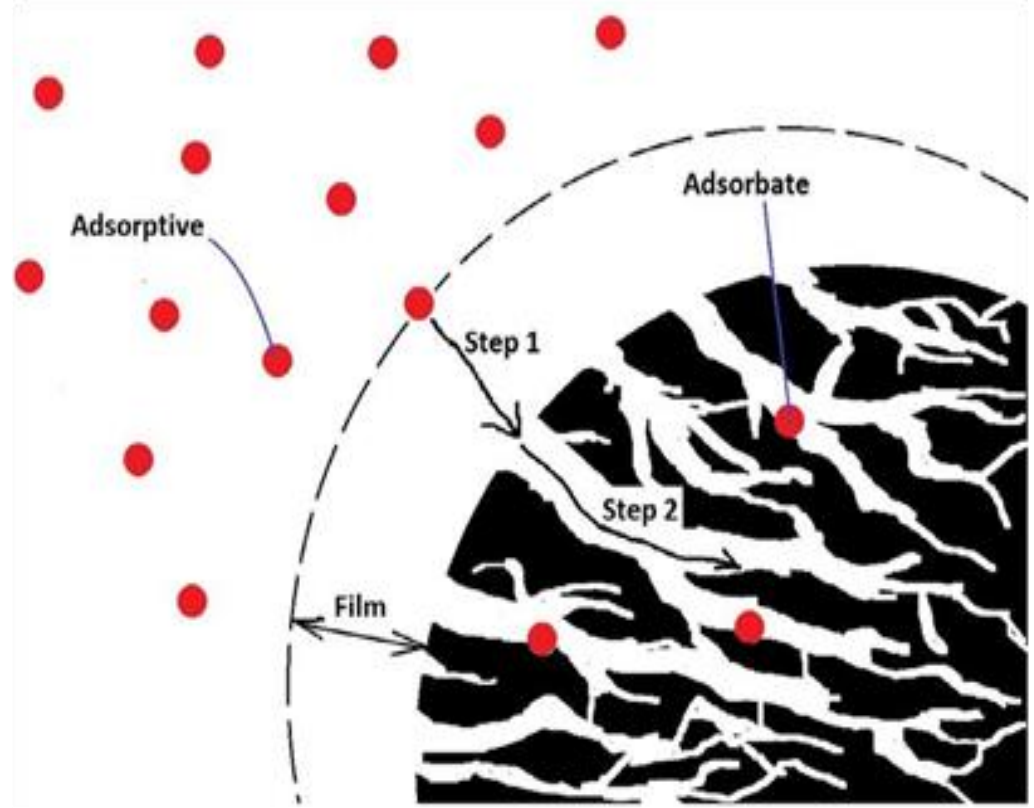


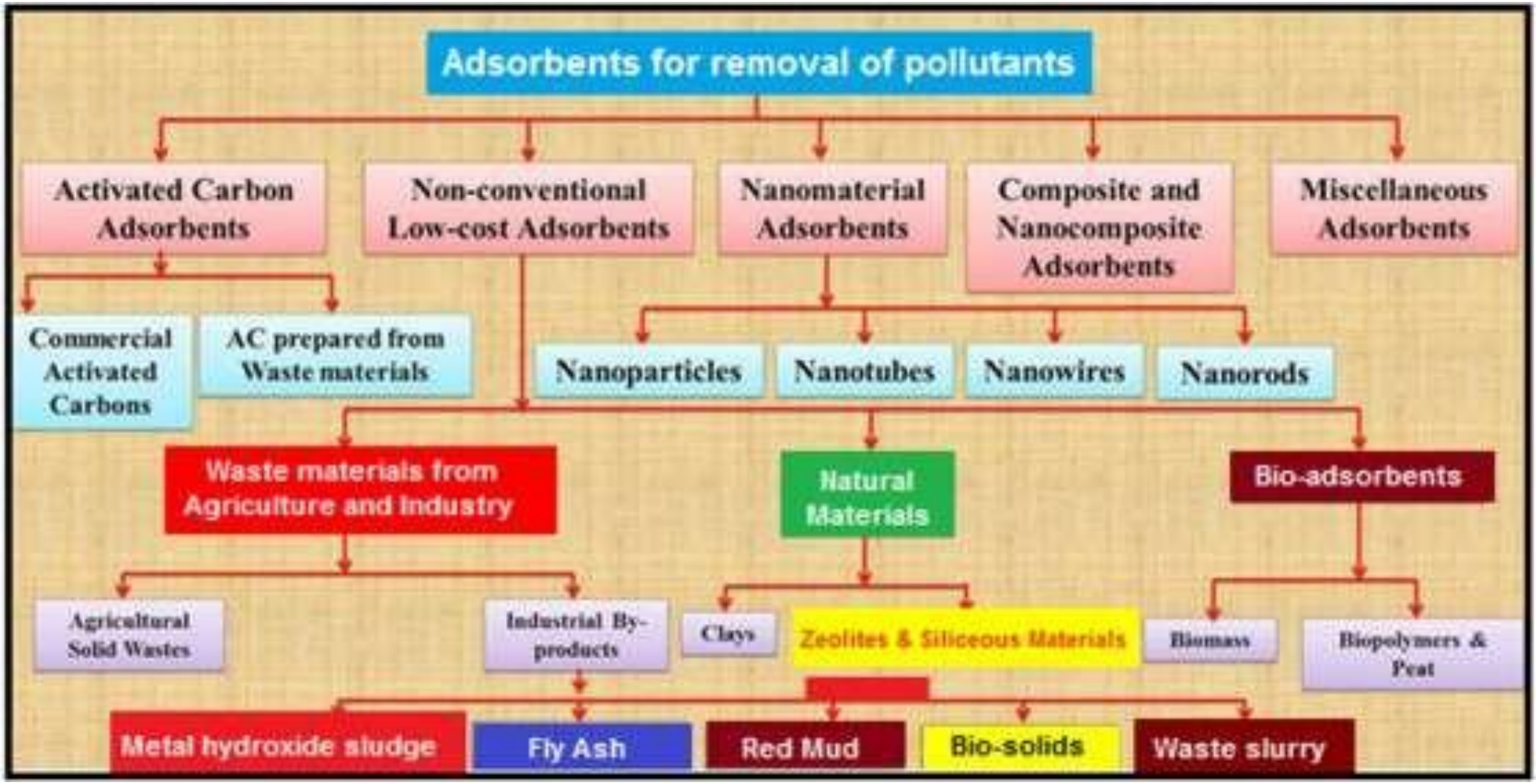
Image Source : /

Adsorption Mechanism



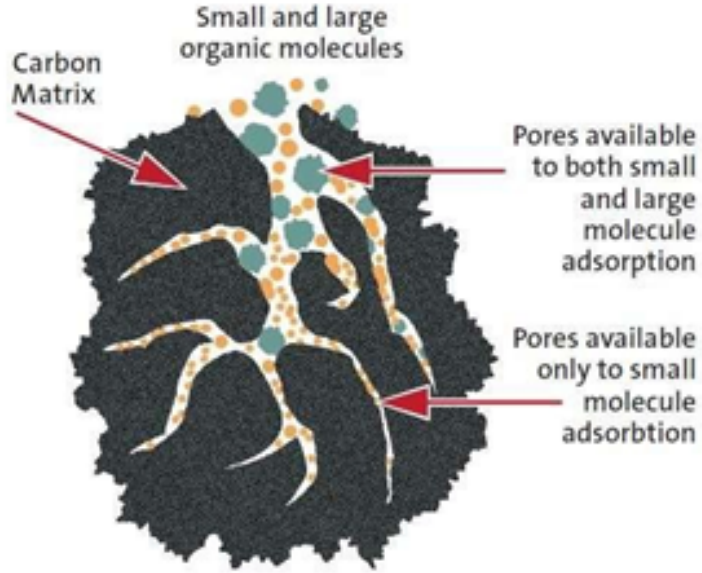


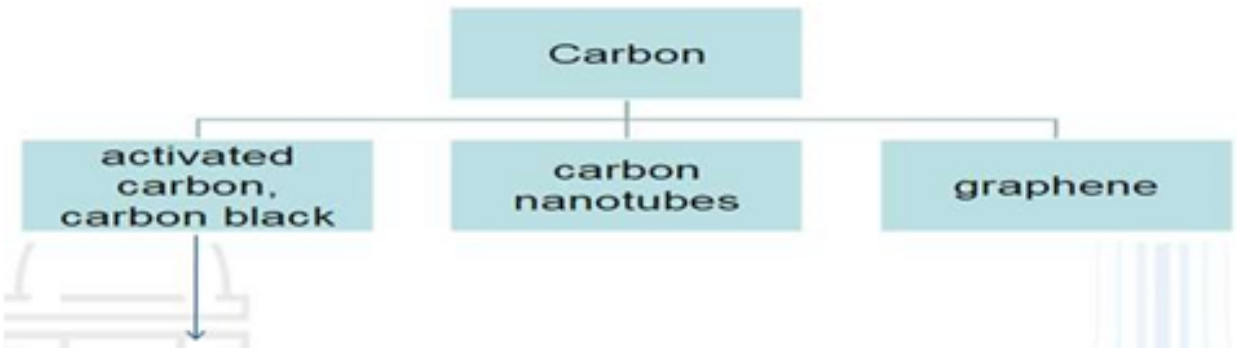
Common Adsorbents in Water Treatment

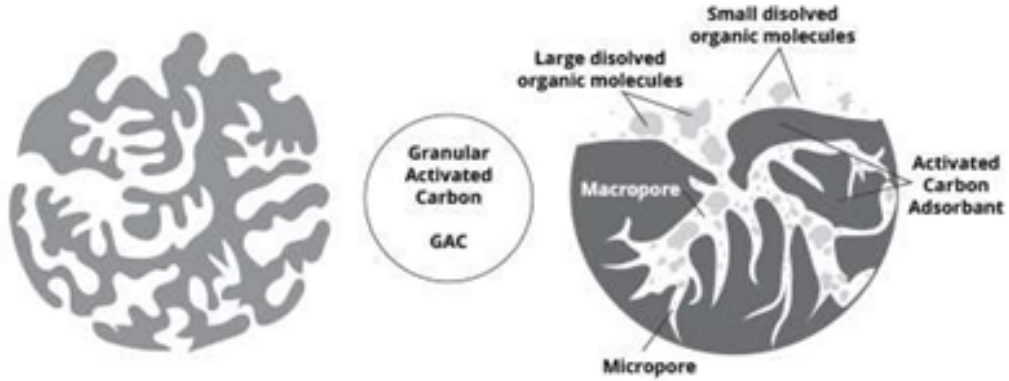


Carbon Based Adsorbents

- A typical carbon particle has numerous pores that provide a large surface area for water treatment. Activated carbon is prepared to exhibits a high degree of porosity and an extended surface area.
- Contaminants adhere to the surface of these carbon granules or become trapped in the small pores of the activated carbon. These are efficient to remove various organics, metals, chlorine, fluorine or radon and other micro pollutants from water (not very effective for microbial contaminants, some metals, nitrates and other inorganic contaminants.







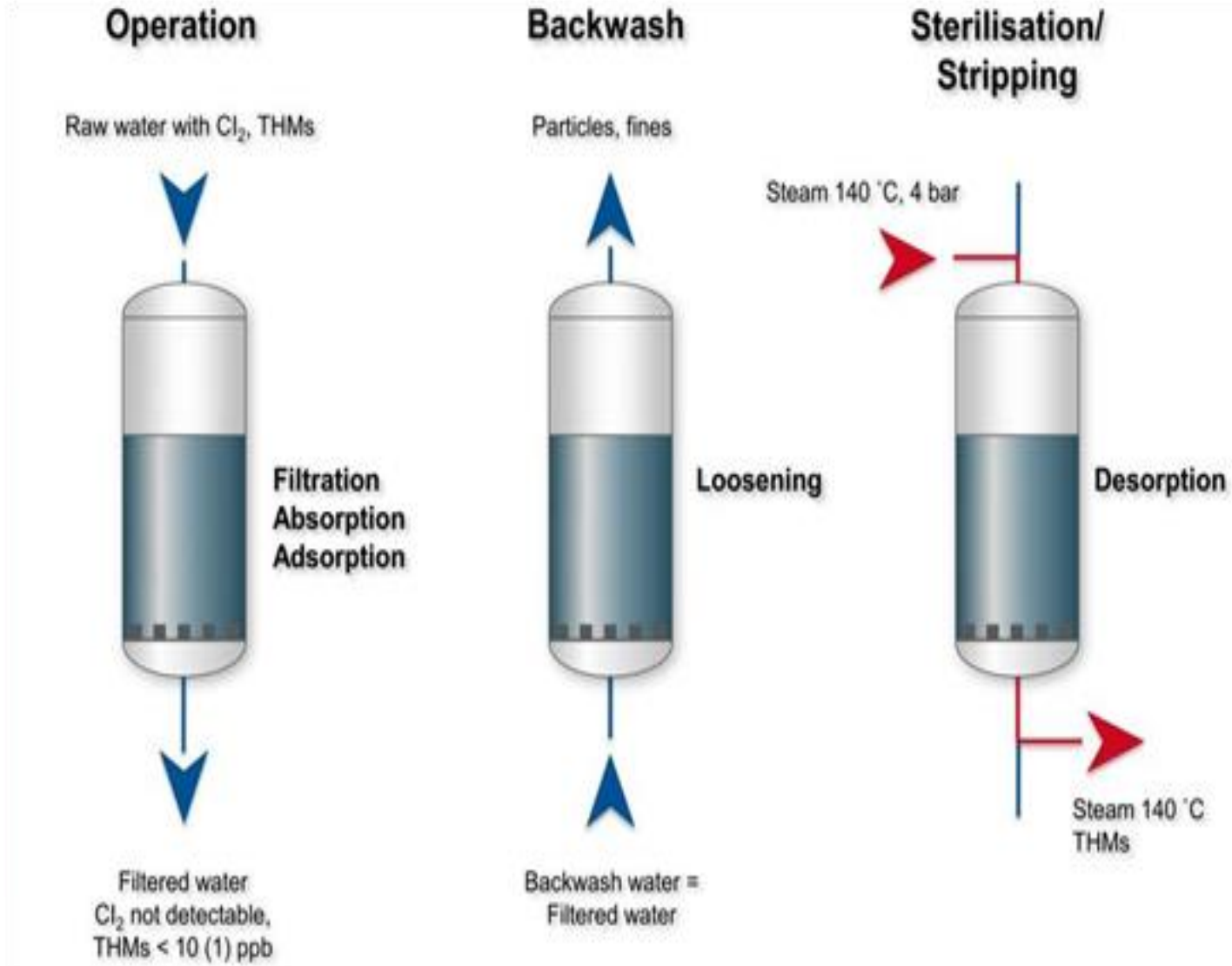
PAC - Powdered activated carbon; particle size 1-150 µm
GAC - Granular activated carbon, particle size 0.5-4 mm EAC - Extruded activated carbon, particle size 0.8-4 mm

Image Source : http://aqua-cache.com/components/wpu, https://www.elgalabwater.com/technologies/activated-carbon

Activated Carbon Filters



1. Carbon
2. Gravels & Pables
3. Strainer Plate
4. Strainers
5. Hand hole
6. Service Inlet
7. Service Outlet
8. Air vent
9. Davit Arm
10. Backwash Inlet
11. Air Scoring



Activated Carbon Filter Performance

Active carbon adsorption shows very good removal efficiencies for organic contaminants and moderate to good efficiency with select metals and ions. Reported removal efficiency for various organics are as under:

- *BTEX (80-99,9 %);*
- *PAH (98-99,8 %);*
- *AOX (>90 %);*
- *COD;*
- *Colorants;*
- *Alcohols, xylenes;*
- *phenols (60 – 90 %);*
- *NH_4-N (flows with maximum 40 mg/l,*
- *Zeolite adsorption;*

Source : https://emis.vito.be/en/techniekfiche/adsorption-techniques

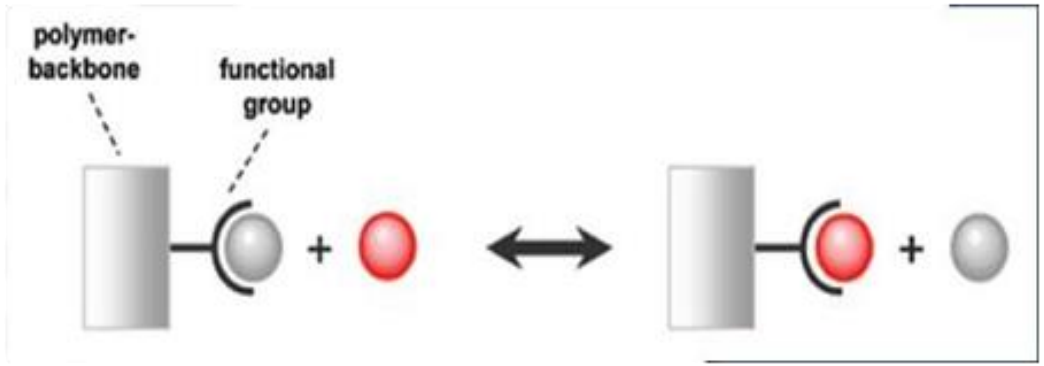
five commercially available types of activated carbon *Source:* Pavonia et al 2006

Removal efficiency (%) of chlorinated compounds from wastewater by

Substance	Precenage of adsorption efficiency (%)
Dichloromethane	98.3
Trichloromethane	98.8
1,1,1- Trichloromethane	99.0
Carbon tetrachloride	99.0
1,2-Dichloroethane	82.8
Trichloroethylene	94.7
1,1,2- Trichloroethane	86.3
Tetrachloroethylene	91.6
1,1,1,2- Tetrachloroethane	87.3
Trans 1,4-dichloro-2-butene	94.2
1,2,4-Trichlorobenzene	99.2
1,2,3-Trichlorobenzene	90.5
Hexachloro-1,3-butadiene	99.4
Hexachlorobenzene	95.1

Ion-Exchange

- Ion Exchange (IX) process is typically used for dissolved ionic impurities and hardness. Contaminant ions in water is removed by exchange with another non-objectionable ionic substance. Ion exchangers hold the retained ion temporarily, and then release it to a regenerant solution.
- It is a physicochemical process in which ions are swapped between a solution phase and solid resin phase.
- The solid resin is typically an elastic three- dimensional hydrocarbon network containing a large number of ionizable groups electrostatically bound to the resin. A synthetic resin of relatively well defined ion exchange capacity exchanges ions held electrostatically on the surface with ions of similar charge dissolved in water.



- The target ion removal is accomplished by continuously passing water under pressure through one or more columns packed with exchange resin.

Depends on Selectivity of Ions

$Cu^{++} > Hg^{++} > Pb^{++} > Ni^{++} > Zn^{++} > Cd^{++} > Co^{++} > Fe^{++} > Mn^{++}$
$>> Ca^{++} > Mg^{++} > Sr^{++} > Ba^{++} >>$ Alkalis

Adsorption and Ion-Exchange Limitations

- Exhaustion of capacity of adsorbent or Ion-Exchange resin
- Differential pressure development in the unit
- Temperature affects the efficiency of adsorption and Ion-Exchange: Less effective at higher temperature
- Competitive solutes or ions present in the water reduces the efficiency of the systems
- Requires good process control
- Moderate cost (cheaper than membrane processes, but costlier that conventional systems)